

Awesome Video

A curated list of awesome video tools, frameworks, libraries, and learning resources

1814 resources — exported 7/19/2026

Protocols & Transport

100ms: RTMP vs WebRTC vs HLS - Live Video Streaming Protocols Compared

<https://www.100ms.live/blog/rtmp-vs-webrtc-vs-hls>

A detailed comparison of RTMP, WebRTC, and HLS protocols, highlighting their differences in latency, scalability, quality, and reach. This resource assists developers in choosing the appropriate protocol based on their specific live video streaming requirements.

Tags: RTMP, WebRTC, HLS

507_hls_authoring_for_airplay_2.

https://devstreaming-cdn.apple.com/videos/wwdc/2019/507fk9wyls0np6piwk/507/507_hls_authoring_for_airplay_2_video.pdf

Apple WWDC presentation on HLS authoring best practices specifically for AirPlay 2 compatibility and long-form video delivery.

510_validating_http_live_streams.

https://devstreaming-cdn.apple.com/videos/wwdc/2016/510ndmh9wkcqvneegv2/510/510_validating_http_live_streams.pdf

Apple WWDC presentation covering validation techniques and tools for HTTP Live Stream (HLS) compliance and quality assurance.

515_hls_authoring_update

https://devstreaming-cdn.apple.com/videos/wwdc/2017/515vy4sl7iu70/515/515_hls_authoring_update.pdf

WWDC 2017 presentation covering updates and best practices for authoring HLS streams for Apple devices.

AWS Stale Playlist Detector

<https://github.com/awslabs/aws-stale-playlist-detector>

The Stale Playlist Detector (SPD) is a tool to monitor live HLS origin endpoints for changing playlists. The Stale Playlist Detector (SPD) will use data in the top-level playlist, the child playlis...

Tags: HLS, monitoring, playlist, streaming, origin, endpoint

Ad Creative Signaling in DASH and HLS

<https://www.svta.org/project/ad-creative-signaling-in-dash-and-hls/>

This project by the Streaming Video Technology Alliance defines a data scheme for signaling advertisement creative assets within DASH manifests and HLS playlists, facilitating better ad integration in streaming workflows.

Tags: advertising, DASH, HLS, streaming, metadata

Adaptive Streaming

<https://github.com/moorthy-g/adaptive-streaming>

A comprehensive overview and experimental project focusing on adaptive streaming technologies, specifically HLS and DASH. It covers content generation, playback across various browsers, and includes a Node.js module for creating multi-bitrate DASH and HLS content.

Tags: HLS, DASH, adaptive streaming, Node.js, video

Adaptive Streaming with MPEG-DASH and HLS using AWS

<https://gist.github.com/PaulRosset/7bf10bf53895b1a53630bc947c8b1834>

A comprehensive guide detailing the process of setting up adaptive streaming using MPEG-DASH and HLS with AWS services. It covers S3 bucket configuration, Elastic Transcoder usage, and integration with Video.js for playback.

Tags: adaptive streaming, MPEG-DASH, HLS, AWS, Video.js

Advances in HTTP Live Streaming - WWDC 2017 - Videos - Apple Developer

<https://developer.apple.com/videos/play/wwdc2017/504/>

HTTP Live Streaming allows you to stream live and on-demand content to global audiences. Learn about great new features and enhancements...

Apple HLS - IETF rfc8216bis (HTTP Live Streaming 2nd Edition draft)

<https://datatracker.ietf.org/doc/html/draft-pantos-hls-rfc8216bis>

Defines both the metadata structure and some client/server requirements

Apple HLS Tools

<https://developer.apple.com/documentation/http-live-streaming/using-apple-s-http-live-streaming-hls-tools>

Apple's HLS Tools assist developers in creating and validating HTTP Live Streaming (HLS) content for Apple devices.

Apple Reference Streams

<https://developer.apple.com/streaming/examples/>

Apple provides example streams demonstrating the features of HTTP Live Streaming (HLS) for testing and development purposes.

Apple Technical Note TN2224 - HLS Best Practices

https://developer.apple.com/library/archive/technotes/tn2224/_index.html

An Apple Technical Note detailing best practices for HTTP Live Streaming. It covers how to encode media, segment files, set up playlists, and deliver HLS streams efficiently.

Tags: hls, streaming, apple, technical-note, best-practices, encoding, streaming-protocols

Auto Dash HLS

<https://github.com/majamee/auto-dash-hls>

Automated toolchain for generating DASH/ HLS compatible files from all videos inside a root folder

Awesome DASH

<https://github.com/yuan-jinghao/awesome-dash>

A curated list of resources, tools, and information related to MPEG-DASH. This repository serves as a comprehensive guide for developers interested in DASH technology, offering insights into its implementation, best practices, and related research.

Tags: DASH, video streaming, resources, adaptive streaming, MPEG-DASH

BBC R&D Adaptive Bitrate Test Media

<https://rdmedia.bbc.co.uk/>

The BBC Research & Development department offers MPEG-DASH test media for developers and researchers.

BOLA: Near-Optimal Bitrate Adaptation for Online Videos

<https://arxiv.org/pdf/1601.06748.pdf>

Academic paper presenting the BOLA algorithm for adaptive bitrate selection in video streaming players.

Bitmovin Player

<https://github.com/bitmovin/bitmovin-player-web-samples>

A comprehensive adaptive streaming player supporting HLS and MPEG-DASH, offering features like low latency, DRM integration, and customizable UI. It is designed for high-quality video playback across various devices and platforms.

Tags: adaptive streaming, HLS, MPEG-DASH, DRM, customizable UI

BlendVision: Top 6 Live Streaming Protocols Comparison

<https://web.archive.org/web/20251116092923/https://blendvision.com/en/blog/live-streaming-protocols-comparison>

A comprehensive comparison of six common live streaming protocols: RTP, SRT, RTMP, RIST, Zixi, and WebRTC. The article explores their technical characteristics, application scenarios, and differences, aiding developers in selecting the most suitable protocol for their live streaming services.

Tags: RTP, SRT, RTMP, RIST, WebRTC

CMAF for Low Latency Streaming – Bitmovin

<https://bitmovin.com/cmaf-low-latency.html>

A detailed blog explaining how CMAF enables low-latency streaming. Bitmovin covers the technical aspects of chunked CMAF, the changes required in players and CDNs, and results from tests using their encoder and player in low-latency mode.

CTA WAVE Low Latency Service Guidelines

<https://cta.tech/Research-Standards/Standards/CTA-WAVE>

Guidelines from the CTA WAVE (Web Application Video Ecosystem) project focusing on low-latency streaming service implementation, including Low-Latency DASH and HLS considerations.

Combine Mpd

<https://github.com/nickdesaulniers/combine-mpd>

Combine MPEG DASH MPD manifest files.

Comparison of Streaming Media Software

https://en.wikipedia.org/wiki/Comparison_of_streaming_media_software

This resource provides a comprehensive comparison of various streaming media software, detailing the protocols they support, including HTTP, MPEG DASH, WebRTC, RTSP, and more. It's a valuable reference for developers seeking to understand the capabilities and protocol support of different streaming solutions.

Tags: streaming media, protocol comparison, software, video streaming, protocols

Comparison of Streaming Protocols

<https://gist.github.com/travelhawk/89c17064afcedc9b919d985da37f72fd>

A comprehensive GitHub Gist that provides a detailed comparison of various streaming protocols, including RTMP, WebRTC, SRT, and others. It outlines the supported codecs, latency, pros, and cons of each protocol, serving as a valuable resource for developers seeking to understand the differences and applications of these protocols.

Tags: streaming protocols, RTMP, WebRTC, SRT, video streaming

Content Protection for HLS with AES-128 Encryption

<https://www.theoplayer.com/blog/content-protection-for-hls-with-aes-128-encryption>

We will outline the most popular method for content protection with the HTTP Live Streaming (HLS) protocol: AES-128 content encryption.

Create DASH HLS

<https://github.com/matmoi/create-DASH-HLS/>

A tutorial to generate fMp4 files compatible with dash and HLS

Creating a Master Playlist | Apple Developer Documentation

https://developer.apple.com/documentation/http_live_streaming/example_playlists_for_http_live_streaming/creating_a_master_playlist#overview

Apple's developer documentation explaining how to construct HLS master playlists that reference multiple variant streams for adaptive bitrate video delivery.

DASH IF Conformance

<https://github.com/Dash-Industry-Forum/DASH-IF-Conformance>

This repository provides the source code for MPEG-DASH/DASH-IF Conformance Software/Validator. It has been extended according to further standards, such as CMAF, DVB-DASH, HbbTV, and CTA WAVE. - Da...

DASH-IF Conformance Tool

<https://github.com/Dash-Industry-Forum/Conformance-Software>

An official tool by the DASH Industry Forum to validate MPEG-DASH MPD files and segment formats. It checks for spec compliance and common issues in DASH content, helpful for anyone generating DASH streams.

Tags: validation, conformance, MPEG-DASH, specification, tool

DASH-IF IOPs

<https://dashif.org/guidelines/>

DASH Industry Forum guidelines providing interoperability points and implementation recommendations for DASH streaming systems.

DVBlast

<https://www.videolan.org/projects/dvblast.html>

A simple and powerful MPEG-2/TS demux and streaming application. DVBlast can stream digital TV (DVB) channels over a network, often used for IPTV or SAT-IP solutions.

Dash

<https://github.com/djvergad/dash>

An MPEG/DASH client-server module for simulating rate adaptation mechanisms over HTTP/TCP.

Dash Adaptation Simulator

https://github.com/pokey909/dash_adaptation_simulator

Simulate bitrate switching algorithms based on real data traces

Dash DRM

<https://github.com/shengbinmeng/dash-drm>

Example implementations demonstrating MPEG-DASH playback with DRM protection schemes.

Dash IF Test Assets Database

<https://testassets.dashif.org/#testvector/list>

A database of test assets provided by the DASH Industry Forum for validating DASH implementations.

Dash JS

<https://github.com/Dash-Industry-Forum/dash.js>

A reference client implementation for the playback of MPEG DASH via Javascript and compliant browsers.

Dash JavaScript Player

<https://reference.dashif.org/dash.js/latest/samples/dash-if-reference-player/index.html>

Reference implementation player for MPEG-DASH using dash.js library, useful for testing and compliance verification.

Dash Live Source Simulator

<https://github.com/Dash-Industry-Forum/dash-live-source-simulator>

DASH live source simulator providing reference live content.

Dash Proxy

<https://github.com/Viblast/dash-proxy>

Easy downloading and mirroring of MPEG-DASH streams

Dash Proxy (tchakabam)

<https://github.com/tchakabam/dash-proxy>

Experimental MPEG-DASH media gateway - proxy on-the-fly modified MP4 segment metadata

Dash Sequences GPAC Wiki

<https://github.com/gpac/gpac/wiki/dash-sequences>

The GPAC Wiki provides DASH sequences for conformance testing and validation.

Dash Server

<https://github.com/theolampert/dash-server>

Small, command-line HTTP/2 file server for serving MPEG-DASH content.

Dash Validator JS

<https://github.com/Eyevinn/dash-validator-js>

MPEG DASH validator JS library.

Dashencrypt

<https://github.com/castlabs/dashencrypt>

DASH fragmenter/segmenter and encrypter. Contribute to castlabs/dashencrypt development by creating an account on GitHub.

Dashif Conformance

<https://conformance.dashif.org/>

The DASH Industry Forum provides a conformance tool to validate MPEG-DASH content against the DASH standard.

DI

<https://github.com/schedules/dl>

Node.js DASH and HLS downloader.

Tags: downloader, DASH, HLS, Node.js, streaming

Docker 2dash

<https://github.com/Eyevinn/docker-2dash>

A Docker container to pre-package MPEG DASH on demand content

Docker Dash Packager

<https://github.com/Eyevinn/docker-dash-packager>

Open source MPEG DASH packager for live and VOD.

Eyevinn SRT to RTMP

<https://github.com/Eyevinn/srt-rtmp>

This project by Eyevinn Technology provides a tool to ingest MPEG-TS streams over SRT and output them as RTMP streams. It's particularly useful for streaming SRT content to platforms like Twitch or YouTube Live, facilitating integration between different streaming protocols.

Tags: SRT, RTMP, video streaming, protocol conversion, live streaming

Fmp4 JS

<https://github.com/Eyevinn/fmp4-js>

A Javascript library to parse ISO Base Media File Format (MPEG-4 Part 12)

Generate MPEG DASH content encrypted with MPEG CENC ClearKey ·

Dash-Industry-Forum/dash.js Wiki

<https://github.com/Dash-Industry-Forum/dash.js/wiki/Generate-MPEG-DASH-content-encrypted-with-MPEG-CENC-ClearKey>

A reference client implementation for the playback of MPEG DASH via JavaScript and compliant browsers.

Go Dash

<https://github.com/zencoder/go-dash>

A Go library for generating MPEG-DASH manifests.

Go Ts Segmenter

<https://github.com/jordicenzano/go-ts-segmenter>

Live TS segmenter and HLS manifest creation in Go.

GoSRT

<https://github.com/eyevinn/gosrt>

GoSRT is a pure Go implementation of the SRT protocol with minimal dependencies. Designed with live video/audio streaming in mind, it supports features like handshake, message mode, timestamp-based packet delivery, and encryption, making it suitable for developers working with Go.

Tags: SRT, Go, video streaming, protocol implementation, live streaming

Guide to Mobile Video Streaming with HLS

<https://mux.com/blog/mobile-hls-guide/>

HTTP Live Streaming, also known as HLS, is the most common format used today for streaming video. If you're building a video streaming application today, you should probably use HLS. Apple created the HLS standard in 2009, and it is the required streaming format for iOS devices. Since then, Android

HEVC in HLS: 10 Key Questions for Streaming Video Developers

<https://www.streamingmedia.com/Articles/ReadArticle.aspx?ArticleID=122637&PageNum=2>

Many who heard that Apple is adding support for HEVC playback in HTTP Live Streaming were left with more questions than answers. Here's what developers need to know.

HLS (HTTP Live Streaming)

<https://developer.apple.com/streaming/>

HLS is a streaming protocol developed by Apple that delivers video content over HTTP. It is widely supported across various devices and platforms, making it a popular choice for delivering live and on-demand video content. HLS supports adaptive bitrate streaming, allowing for dynamic adjustment of video quality based on network conditions.

Tags: HLS, HTTP Live Streaming, video streaming, adaptive bitrate, Apple

HLS Analyzer

<https://github.com/epiclabs-io/hls-analyzer>

Analyzer for HTTP Live Streams (HLS) content.

Tags: hls, analyzer, streaming, protocol, stream-analysis

HLS Authoring Specification for Apple Devices | Apple Developer Documentation

https://developer.apple.com/documentation/http_live_streaming/hls_authoring_specification_for_apple_devices

Apple's official specification for creating HLS streams compatible with Apple devices and players.

HLS Cutsegment

<https://github.com/Eyevinn/hls-cutsegment>

A web app that lets you insert a cut into a segment, which is then cut into two new segments.

Tags: hls, segmentation, web-app, video-editing, streaming-tools

HLS Download

<https://github.com/Eyevinn/hls-download>

Download HLS and convert to MP4.

HLS Downloader

<https://github.com/denex/hls-downloader>

Download all video files from HLS (HTTP Live Streaming) VoD (Video on Demand) m3u8 playlist for local playback

Tags: downloader, m3u8, video-streaming, hls, vod

HLS Downloader Chrome Extension

<https://github.com/puemos/hls-downloader-chrome-extension>

Google Chrome Extension for sniffing and downloading HTTP Live streams (HLS)

Tags: chrome-extension, streaming, downloader, hls, video-tools

HLS Fetch

<https://github.com/oskili/hls-fetch>

Download and decrypt videos served by the HTTP Live Streaming (HLS) protocol.

HLS Get

<https://github.com/SoulMelody/hls-get>

An asynchronous terminal-based hls video stream (m3u8) downloader & combiner, with AES-128 decryption support.

Tags: downloader, m3u8, streaming, decryption, async, terminal

HLS JS

<https://github.com/video-dev/hls.js>

JavaScript HLS client using Media Source Extension

HLS Origin Scripts

<https://github.com/Eyevinn/hls-origin-scripts>

Scripts to manipulate HLS manifests at origin or edge server

HLS Playlist Parser

<https://github.com/Eyevinn/hls-playlist-parser>

A Javascript library to parse Hls playlists.

HLS Relay

<https://github.com/Eyevinn/hls-relay>

Script to pull HLS stream from one origin and push to another origin

HLS Server

<https://github.com/t-mullen/hls-server>

Middleware for serving HTTP Live Streaming (HLS) compatible media streams.

HLS Streaming

<https://github.com/tozastation/HLS-Streaming>

HLS streaming implementation or example project by tozastation demonstrating HTTP Live Streaming delivery.

HLS To MP4

<https://github.com/egg-bread/hls-to-mp4>

Download HLS videos as MP4 (WebVTT for caption download optional)

HLS Ts Analyzer

<https://github.com/Eyevinn/hls-ts-analyzer>

Example implementation of hls-ts.js library.

HLS Ts JS

<https://github.com/Eyevinn/hls-ts-js>

HLS MPEG-TS parser library in Javascript.

HLS VOD

<https://github.com/mifi/hls-vod>

HTTP Live Streaming with on-the-fly encoding of any video file for Web/Apple TV/iPhone/iPad/iPod

Tags: streaming, encoding, apple-devices, web-streaming, mobile-streaming

HLS and Fragmented MP4

<https://hlsbook.net/hls-fragmented-mp4/>

At WWDC 2016, Apple announced support for fragmented MP4 (fMP4) as an alternative to MPEG-TS, which prior to their announcement was the only supported format. So why use fragmented MP4 files? Well,...

HLS | Bento4

<https://www.bento4.com/developers/hls/>

Bento4 developer documentation for HLS-specific tools, including utilities for creating, analyzing, and manipulating HTTP Live Streaming playlists and segments.

HLSCachingReverseProxyServer

<https://github.com/StyleShare/HLSCachingReverseProxyServer>

A simple local reverse proxy server for HLS segment cache

HLSCore

<https://github.com/fcanas/HLSCore>

A collection of Swift packages for working with HLS

HLSDownloader (qi-shun-wang)

<https://github.com/qi-shun-wang/HLSDownloader>

Download Crypted HLS with server key and play video as local playing on iOS device.

HLSion

<https://github.com/r-plus/HLSion>

HTTP Live Streaming (HLS) download manager to offline playback.

HTTP Streaming

<https://github.com/videojs/http-streaming>

HLS, DASH, and future HTTP streaming protocols library for video.js

Tags: HLS, DASH, streaming, video.js, adaptive bitrate, HTTP streaming

HTTPLiveStreamingTools

<https://github.com/creeveliu/HTTPLiveStreamingTools>

Latest Apple HLS tools copy from Apple Developer Center

Tags: apple, streaming, tools, http-live-streaming, development

Haivision SRT Gateway

<https://www.haivision.com/products/srt-secure-reliable-transport/>

A cloud-based service by Haivision that acts as a video router, protocol converter, and stream duplicator. The SRT Gateway facilitates secure and reliable video transport across different networks and protocols, enhancing the flexibility of video streaming workflows.

Tags: SRT Gateway, video transport, protocol conversion, Haivision, cloud service

Haivision SRT Protocol Technical Overview

<https://www.haivision.com/resources/white-paper/srt-protocol-technical-overview/>

This comprehensive technical overview provides in-depth information about the SRT protocol, including its architecture, features, and implementation details. It serves as a valuable resource for developers looking to understand and implement SRT in their video streaming applications.

Tags: SRT, technical overview, video streaming, protocol, Haivision

Haivision SRT Solutions

<https://www.haivision.com/products/srt-solutions/>

A suite of products by Haivision leveraging the SRT protocol, including encoders, decoders, gateways, and mobile transmitters. These solutions provide secure, high-quality, and low-latency video streaming over IP networks, suitable for broadcasters, enterprises, and government agencies.

Tags: SRT, video streaming, Haivision, encoders, decoders

Hlsclient

<https://github.com/globocom/hlsclient>

Python-based HLS client library from Globo for downloading and processing adaptive streaming video content.

Tags: python, streaming, client, protocol

Hlsdl

<https://github.com/selsta/hlsdl>

C program to download VoD HLS (.m3u8) files.

Hlsdownloader

<https://github.com/nmrony/hlsdownloader>

Downloads HLS Playlist file and TS chunks.

Hlsdownloader CLI

<https://github.com/nmrony/hlsdownloader-cli>

Downloads HLS Playlist file and TS chunks using Terminal

Tags: cli, download, streaming, hls, playlist, terminal

Hlsq

<https://github.com/soldiermoth/hlsq>

A CLI for adding some color to your HLS manifests along with some basic filtering.

Hltools

<https://github.com/muxinc/hltools>

Tools for working with HLS streams, including parsing, inspection, and manipulation of HTTP Live Streaming manifest and segment files.

Tags: hls, streaming, tools, adaptive-streaming

How to generate a fmp4 hls live stream with FFMPEG

<https://nomadyun.wordpress.com/2018/04/12/how-to-generate-a-fmp4-hls-live-stream-with-ffmpeg/>

ffmpeg -re -stream_loop -1 -i voweib.mp4 -hls_fmp4_init_filename init.mp4 -vf

"setb=AVTB,setpts='trunc(PTS/1K)*1K+st(1,trunc(RTCTIME/1K))-1K*trunc(ld(1)/1K)',\ drawtext=fontfile=...

ISO/IEC 23009-1:2019

<https://www.iso.org/standard/75485.html>

ISO/IEC 23009-1:2019 (MPEG-DASH Part 1) — the standard defining the media presentation description (MPD) and segment formats for dynamic adaptive streaming over HTTP.

ISO/IEC 23009-1:2019 (MPEG-DASH)

<https://www.iso.org/standard/75440.html>

The ISO/IEC standard for Dynamic Adaptive Streaming over HTTP (DASH). It defines the MPD manifest format and segment formats for adaptive bitrate streaming.

Tags: adaptive streaming, ISO standard, HTTP streaming, manifest, bitrate adaptation

ISOSegmentValidator

<https://github.com/Dash-Industry-Forum/ISOSegmentValidator>

Contribute to Dash-Industry-Forum/ISOSegmentValidator development by creating an account on GitHub.

Implementation and Analysis of Real-Time Streaming Protocols

<https://pmc.ncbi.nlm.nih.gov/articles/PMC5424723/>

A study analyzing the most used video streaming protocols, focusing on RTSP and WebRTC, with developed platforms to compare their performance in Android and web real-time video streaming applications.

Tags: RTSP, WebRTC, real-time streaming, protocol analysis, Android

Ingest

<https://github.com/Dash-Industry-Forum/Ingest>

Dash-Industry-Forum/Ingest - A GitHub repository for video/multimedia development.

Introducing Low-Latency HLS - WWDC 2019 - Videos - Apple Developer

<https://developer.apple.com/videos/play/wwdc2019/502>

Since its introduction in 2009, HTTP Live Streaming (HLS) has enabled the delivery of audio and video...

Larix Broadcaster

<https://softvelum.com/srt/>

A mobile application by Softvelum that enables live streaming from Android and iOS devices using the SRT protocol. Larix Broadcaster supports various streaming modes and provides a user-friendly interface for high-quality, low-latency video broadcasting.

Tags: Larix Broadcaster, SRT, mobile streaming, low latency, Softvelum

Libdash (GitHub)

<https://github.com/bitmovin/libdash>

MPEG-DASH Access Library - Official ISO/IEC MPEG-DASH Reference Implementation

Live555 Streaming Media

<http://www.live555.com/liveMedia/>

Live555 is an open-source set of C++ libraries for multimedia streaming, supporting open standards such as RTP/RTCP, RTSP, and SIP. It is widely used for RTSP server and client implementations and provides a solid foundation for developing streaming applications that require support for these protocols.

Tags: RTSP, RTP, streaming media, C++ library, open source

Low Latency Live Streaming

https://docs.google.com/presentation/d/1ZwqWcweR5SqeMBRmJjSukWaHbpdPy-EPYvJCS23_n3U/edit?usp=sharing

Low Latency Live Streaming Apple LLHLS / CMAF Kevin Staunton-Lambert Solutions Architect R&D (July 2019) @kevleyski
www.switch.tv

M3U8

<https://github.com/globocom/m3u8>

Python library from Globo for parsing and generating M3U8 playlists used in HLS streaming workflows.

Tags: m3u8, playlist, streaming, library, adaptive-streaming

M3U8 (grafov)

<https://github.com/grafov/m3u8>

Parser and generator of M3U8-playlists for Apple HLS. Library for Go language.

Tags: library, go, playlist, streaming, parser, generator

M3U8 DI

<https://github.com/excalibur-kvrv/m3u8-dl>

A CLI m3u8-downloader program to parse.m3u8 playlist file and download mpeg2-ts video files, concat them and convert it to mp4 using FFMPEG

M3U8 Downloader

<https://github.com/lcy0321/m3u8-downloader>

Download the ts files according to the given m3u8 file.

M3U8 Downloader (huzhenjie)

https://github.com/huzhenjie/m3u8_downloader

Tool for downloading video streams from m3u8 playlists. Utility for HLS content retrieval and testing.

M3U8 Parser

<https://github.com/carlanton/m3u8-parser>

HLS compliant m3u8 parser for Java.

M3U8 Parser (videojs)

<https://github.com/videojs/m3u8-parser>

M3U8 playlist parser from the Video.js project. Core utility for handling HLS manifests in JavaScript.

M3U8Parser

<https://github.com/M3U8Kit/M3U8Parser>

A light weight M3U8 parser. Support X-Key & X-Session-Key.

Tags: m3u8, parser, streaming, hls, playlist

MP4 To Mpegdash Py

<https://github.com/stultus/mp4-to-mpegdash-py>

Python Script to convert a MP4 file into onDemand MPEG-DASH

MPEG DASH - ISO/IEC 23009-01

<https://www.iso.org/standard/83314.html>

Defines both the metadata structure and associated media format(s)

MPEG DASH Downloader

<https://github.com/mahbubcseju/MPEG-DASH-Downloader>

Contribute to mahbubcseju/MPEG-DASH-Downloader development by creating an account on GitHub.

MPEG-DASH (Dynamic Adaptive Streaming over HTTP)

<https://www.dashif.org/>

MPEG-DASH is an international standard for streaming multimedia content over the internet. It enables adaptive bitrate streaming, allowing for high-quality streaming experiences across various network conditions and devices. MPEG-DASH is codec-agnostic and supports a wide range of media formats.

Tags: MPEG-DASH, adaptive streaming, video streaming, HTTP, standard

MPEG-DASH MPD XML Schema (DASHSchema)

<https://github.com/MPEGGroup/DASHSchema>

Official XML Schema (XSD) definitions and example MPD files for the MPEG-DASH Media Presentation Description, maintained by MPEG's DASH working group for validating standards-compliant manifests.

Tags: mpd, xml, xsd, schema, validation, manifest, mpeg-dash

MPEGDASH iOS Player

<https://github.com/MPEGDASHPlayer/MPEGDASH-iOS-Player>

The MPEG-DASH Player iOS Application.

Mamba

<https://github.com/Comcast/mamba>

Mamba is a Swift iOS, tvOS and macOS framework to parse, validate and write HTTP Live Streaming (HLS) data.

Manifestparser

<https://github.com/Eyevinn/manifestparser>

JavaScript library from Eyevinn for parsing HLS and DASH manifest files in video applications.

Media Tools (Dash-Industry-Forum)

<https://github.com/Dash-Industry-Forum/media-tools>

A collection of tools for analyzing, handling, and creating media and media containers

MediaFlow Proxy

<https://github.com/mhdzumair/mediaflow-proxy>

MediaFlow Proxy is a high-performance proxy server for streaming media, supporting HTTP(S), HLS, and MPEG-DASH with real-time DRM decryption. It can convert MPEG-DASH DRM-protected streams to decrypted HLS live streams in real-time, making it one of the fastest live decrypter servers available.

Tags: HLS, MPEG-DASH, DRM, proxy, streaming

MediaMTX SRT

https://github.com/dm-dma/mediamtx_srt

MediaMTX (formerly rtsp-simple-server) is a versatile media server and proxy that supports multiple protocols, including SRT, WebRTC, RTSP, RTMP, and LL-HLS. It allows for publishing, reading, proxying, recording, and playback of video and audio streams, making it a comprehensive solution for media routing.

Tags: SRT, media server, streaming, protocol support, media routing

Mediasegmenter

<https://github.com/iliya-gr/mediasegmenter>

Command-line tool for segmenting media files into HLS-compatible chunks for adaptive streaming delivery.

Tags: hls, segmenter, streaming, media, adaptive-streaming

Middleman SCTE35 104 Parser

<https://tools.middleman.tv/scte35-parser>

A web-based tool to parse and decode SCTE-35 markers for ad insertion and broadcast signaling analysis.

Mock HLS Server

<https://github.com/tjenkinson/mock-hls-server>

Fake a live/event HLS stream from a VOD one. Useful for testing.

Morsel

<https://github.com/krad/morsel>

Ø=ÜÇ Swift library for creating HLS playlists and fragmented mp4 files. Works on Linux

Mpd Parser

<https://github.com/videojs/mpd-parser>

Contribute to videojs/mpd-parser development by creating an account on GitHub.

Mpd Tools

<https://github.com/carlanton/mpd-tools>

Java utilities for parsing and manipulating DASH Media Presentation Description files.

Mux HLS Streams

<https://test-streams.mux.dev/>

Mux offers a set of test HLS streams, primarily used by hls.js and hosted by Mux, for testing and development.

NGINX Audio Track For HLS Module

<https://github.com/flavioribeiro/nginx-audio-track-for-hls-module>

Nginx module that generates audio track for HTTP Live Streaming (HLS) streams on the fly.

Tags: nginx, audio, streaming, module, hls

NGINX HLS

<https://github.com/zhaiweiwei/nginx-hls>

Contribute to zhaiweiwei/nginx-hls development by creating an account on GitHub.

NGINX HLS Analyzer

<https://github.com/openHPI/nginx-hls-analyzer>

Fork of fmsloganalyzer to adapt it for HLS streaming analyzes with nginx

NGINX HLS Module

<https://github.com/rounce/nginx-hls-module>

Smooth Streaming Module fork.

NGINX VOD Module Fmp4 HLS

<https://github.com/flavioribeiro/nginx-vod-module-fmp4-hls>

Play fragmented mp4's on HLS using nginx-vod-module

Tags: nginx, hls, fragmented-mp4, streaming, video-delivery

NGINX-RTMP Module

<https://github.com/arut/nginx-rtmp-module>

An Nginx module that adds RTMP streaming server capabilities to Nginx. It supports streaming live content via RTMP and can also be used to create HLS/DASH streams from RTMP input.

Node Video Lib

<https://github.com/gkozlenko/node-video-lib>

Node.js Video Library / MP4 & FLV parser / MP4 builder / HLS muxer

OBS Studio - Streaming With SRT or RIST Protocols

<https://obsproject.com/wiki/Streaming-With-SRT-Or-RIST-Protocols>

OBS Studio provides comprehensive support for both SRT and RIST protocols, enabling users to stream with enhanced resilience and low latency. This resource offers detailed guidance on configuring OBS Studio for SRT and RIST streaming, making it invaluable for developers and streamers seeking robust streaming solutions.

Tags: OBS Studio, SRT, RIST, video streaming, configuration guide

OBS Studio - Streaming with SRT and RIST

<https://github.com/obsproject/obs-studio/wiki/Streaming-With-SRT-or-RIST-Protocols>

OBS Studio, a popular open-source software for video recording and live streaming, offers support for both SRT and RIST protocols. This integration allows users to leverage these protocols for enhanced streaming performance, including improved resilience to network issues and lower latency. The project's wiki provides detailed instructions on setting up and using SRT and RIST within OBS Studio.

Tags: OBS Studio, SRT, RIST, live streaming, open source

OBS Studio SRT Support

<https://srtlab.github.io/srt-cookbook/apps/obs-studio.html>

OBS Studio, a popular open-source software for video recording and live streaming, introduced support for SRT output starting from version 25.0. This integration allows users to leverage the benefits of SRT, such as low latency and secure streaming, directly within OBS Studio. The SRT CookBook provides detailed guidance on configuring and using SRT with OBS Studio.

Tags: SRT, OBS Studio, live streaming, video recording, open source

Open M3U8

<https://github.com/iheartradio/open-m3u8>

Open-source parser for m3u8 playlist files by iHeartRadio. Essential for HLS streaming implementations.

OpenVidu

<https://openvidu.io/>

An open source platform for building video conferencing or streaming solutions. OpenVidu provides easy-to-use APIs for WebRTC, enabling multi-party video calls and streaming with recording and broadcasting features.

Tags: WebRTC, video conferencing, real-time streaming, multi-party communication

Opus Dash

<https://github.com/webmproject/opus-dash>

Specification for Encapsulating Opus Audio in ISO-BMFF Container

OvenPlayer

<https://github.com/airensoft/ovenplayer>

OvenPlayer is a JavaScript-based player that supports Low-Latency HLS (LLHLS) and WebRTC playback. Designed for integration with OvenMediaEngine, it offers sub-second latency streaming, making it ideal for applications requiring real-time video delivery.

Tags: JavaScript, LLHLS, WebRTC, low-latency, streaming

PeerJS

<https://peerjs.com/>

A simple peer-to-peer JavaScript library that abstracts WebRTC for direct video, audio, and data connectivity between browsers. Provides an easy way to build video chat and P2P apps.

Tags: WebRTC, peer-to-peer, real-time communication, browser technology

Pensieve (Adaptive Bitrate AI Algorithm)

<https://github.com/hongzimaopensieve>

A research system that uses reinforcement learning to generate adaptive bitrate (ABR) algorithms. Pensieve learned an ABR algorithm that can outperform traditional heuristics under certain conditions.

Pion WebRTC

<https://github.com/pion/webrtc>

A pure Go implementation of the WebRTC protocol, enabling developers to build real-time video (and audio) streaming applications without external dependencies.

Protocol Extension for Low-Latency HLS (Preliminary Specification) | Apple Developer Documentation

https://developer.apple.com/documentation/http_live_streaming/protocol_extension_for_low-latency_hls_preliminary_specification#3291001

Apple's preliminary specification for protocol extensions that reduce latency in HTTP Live Streaming, enabling lower-delay video delivery.

Puffer (Stanford Live Streaming Research)

<https://puffer.stanford.edu/>

A free live TV streaming research study by Stanford University that uses volunteer viewers to train and evaluate new adaptive streaming algorithms in real-world conditions.

Tags: research, live streaming, algorithm, stanford, adaptive bitrate

Python Mpegdash

<https://github.com/caststack/python-mpegdash>

MPEG-DASH MPD(Media Presentation Description) Parser

QUIC (Quick UDP Internet Connections)

<https://www.chromium.org/quic/>

QUIC is a transport layer network protocol designed by Google to improve the performance of connection-oriented web applications. It reduces latency compared to TCP and is particularly beneficial for video streaming applications.

Tags: QUIC, transport protocol, low latency, video streaming, Google

RFC 8216: HTTP Live Streaming (HLS)

<https://datatracker.ietf.org/doc/html/rfc8216>

The IETF RFC that specifies HTTP Live Streaming (HLS), defining the playlist format (M3U8) and media segment structure for Apple's streaming protocol.

Tags: streaming, protocol, specification, apple, rfc, m3u8

RTMP (Real-Time Messaging Protocol)

<https://www.adobe.com/devnet/rtmp.html>

RTMP is a protocol developed by Adobe Systems for streaming audio, video, and data over the Internet. It is widely used for live streaming applications and is supported by many media servers and streaming platforms. RTMP provides low-latency streaming and is commonly used for ingesting live streams into media servers, which then distribute the content to viewers using other protocols like HLS or DASH.

Tags: RTMP, video streaming, live streaming, protocol, Adobe

RTMP vs. RTSP: Which Protocol Should You Choose?

<https://www.wowza.com/blog/rtmp-vs-rtsp-which-protocol-should-you-choose>

This article provides an in-depth comparison between RTMP and RTSP protocols, discussing their functionalities, pros and cons, and suitable use cases. It's a useful resource for developers deciding between these protocols for their video streaming applications.

Tags: RTMP, RTSP, protocol comparison, video streaming, transport protocols

RTMP vs. WebRTC vs. HLS - A Comparison of Streaming Protocols

<https://web.archive.org/web/20250904075527/https://dyte.io/blog/rtmp-webrtc-hls/>

An in-depth blog post by Dyte that compares RTMP, WebRTC, and HLS streaming protocols. It discusses their concepts, supported browsers and devices, use cases, pros and cons, and provides a comparison chart. This resource is beneficial for developers evaluating which streaming protocol best fits their project requirements.

Tags: RTMP, WebRTC, HLS, streaming protocols, video streaming

RTMPDump

<https://rtmpdump.mplayerhq.hu/>

A command-line utility to download RTMP streams. RTMPDump can save streaming media content transmitted via Real-Time Messaging Protocol, often used to retrieve videos from sites using RTMP.

Tags: rtmp, streaming, download, protocol, media-retrieval

RTP (Real-time Transport Protocol)

<https://tools.ietf.org/html/rfc3550>

RTP is a network protocol for delivering audio and video over IP networks. It is widely used in communication and entertainment systems that involve streaming media, such as telephony, video teleconference applications, and web-based push-to-talk features.

Tags: RTP, transport protocol, video streaming, real-time, IETF

RTP Benchmarks

<https://github.com/ultravideo/rtp-benchmarks>

A benchmarking suite designed to compare the video streaming performance of various RTP libraries, including uvgRTP, Live555, and FFmpeg. It provides tools for measuring goodput, latency, and CPU usage, aiding developers in evaluating and optimizing RTP-based video streaming solutions.

Tags: RTP, benchmarking, video streaming, performance analysis, protocol comparison

RTSP (Real-Time Streaming Protocol)

<https://tools.ietf.org/html/rfc2326>

RTSP is a network control protocol designed for use in entertainment and communications systems to control streaming media servers. It is used for establishing and controlling media sessions between endpoints. RTSP is commonly used in surveillance systems and supports both TCP and UDP for transport, offering flexibility in streaming applications.

Tags: streaming protocol, media control, TCP, UDP, surveillance

Real-Time Streaming Protocol (RTSP)

<https://www.ietf.org/rfc/rfc2326.txt>

RTSP is a network control protocol designed for use in entertainment and communications systems to control streaming media servers. It is commonly used for establishing and controlling media sessions between endpoints. RTSP is suitable for applications that require real-time streaming capabilities.

Tags: RTSP, streaming protocol, media control, real-time streaming, video streaming

RootEncoder

<https://github.com/pedroSG94/RootEncoder>

RootEncoder is an Android library that enables streaming of video and audio to media servers using protocols such as RTMP, RTSP, SRT, and UDP. Written in Java/Kotlin, it provides a versatile solution for mobile live streaming applications.

Tags: SRT, Android, streaming, library, Java

SRT (Secure Reliable Transport) Protocol

https://en.wikipedia.org/wiki/Secure_Reliable_Transport

SRT is an open-source video transport protocol that utilizes the UDP transport protocol. It provides connection and control, reliable transmission similar to TCP, while maintaining low latency. SRT supports packet recovery and encryption using AES, making it suitable for secure and reliable video streaming applications.

Tags: SRT, video streaming, transport protocol, UDP, low latency

SRT Live Server

<https://github.com/Edward-Wu/srt-live-server>

SRT Live Server (SLS) is a live streaming server that supports the SRT protocol, enabling low-latency and secure video streaming. It allows for efficient transmission of live video content over the internet, even in challenging network conditions.

Tags: SRT, live streaming, server, open-source, low-latency

SRT Protocol Documentation

<https://doc.haivision.com/SRT/1.5.4/Haivision/>

Haivision provides comprehensive documentation for the SRT protocol, offering guidance on setting up and deploying SRT technology. This resource is invaluable for developers looking to implement SRT in their video streaming workflows.

Tags: SRT, documentation, video streaming, protocol, Haivision

SRT Support in VLC and GStreamer

<https://www.srtalliance.org/srt-with-vlc-gstreamer-and-wireshark-support/>

The SRT protocol has been widely adopted by the open-source community, with support integrated into leading technologies like VLC media player and GStreamer multimedia framework. This adoption enables developers to utilize SRT's capabilities for secure and reliable video streaming within these platforms, enhancing their applications' performance over unpredictable networks.

Tags: SRT, VLC, GStreamer, video streaming, open source

Samples players for dash.js

<https://reference.dashif.org/dash.js/latest/samples/index.html>

Collection of sample players demonstrating dash.js features and implementation patterns for MPEG-DASH streaming.

Scte35 JS

<https://github.com/Comcast/scte35-js>

A SCTE 35 Parser for JavaScript.

Secure Reliable Transport (SRT) Protocol Specification

<https://datatracker.ietf.org/doc/html/draft-sharabayko-srt>

An Internet Draft from the IETF detailing the SRT protocol, an open-source video transport protocol that utilizes UDP for low-latency and reliable video streaming. This document is essential for developers seeking an in-depth understanding of SRT's technical specifications.

Tags: SRT, UDP, video transport, low-latency, protocol specification

StreamOptix: A Cross-layer Adaptive Video Delivery Scheme

<https://arxiv.org/abs/2406.04632>

A research project that presents a cross-layer video delivery scheme, StreamOptix, proposing a joint optimization algorithm for video delivery that leverages the characteristics of the physical, medium access control, and application layers. It aims to enhance video streaming performance by addressing challenges in cross-layer design.

Tags: adaptive streaming, cross-layer optimization, video delivery, research, StreamOptix

Streaming 101: Understanding the Differences Between Streaming Protocols

<https://www.cardinalpeak.com/blog/understanding-and-choosing-the-right-streaming-protocol>

An article explaining the differences between various streaming protocols, including RTMP, RTSP, HLS, and DASH, and providing guidance on choosing the right protocol for specific applications.

Tags: RTMP, RTSP, HLS, DASH, protocol selection

Streaming Protocol Comparison: RTMP, WebRTC, FTL, SRT

<https://restream.io/blog/streaming-protocols/>

An in-depth guide by Restream that explains and compares popular live streaming protocols, including RTMP, WebRTC, FTL, and SRT. This resource helps developers understand the strengths and weaknesses of each protocol to make informed decisions for their streaming needs.

Tags: RTMP, WebRTC, FTL, SRT, streaming protocols

TCP Video Streaming Tools

<https://github.com/AgustinSRG/tcp-video-streaming>

A collection of tools aimed at implementing video streaming features over the TCP protocol. It supports sending video streams via RTMP and WebSocket, and receiving streams through HLS, facilitating the development of robust video streaming applications over TCP.

Tags: TCP, video streaming, RTMP, WebSocket, HLS

TCP vs. UDP for Video Streaming

<https://www.geeksforgeeks.org/tcp-vs-udp-for-video-streaming/>

This article explores the differences between TCP and UDP protocols in the context of video streaming. It discusses their respective advantages, disadvantages, and scenarios where one may be preferred over the other, providing valuable insights for developers optimizing their streaming solutions.

Tags: TCP, UDP, video streaming, protocol comparison, transport protocols

THEOplayer

<https://www.theoplayer.com/>

A commercial HTML5 video player that supports both HLS and MPEG-DASH streaming protocols. THEOplayer offers a wide range of features including low latency streaming, DRM support, and extensive customization options, making it suitable for professional-grade streaming applications.

Tags: HLS, DASH, video streaming, DRM, low latency

TSDuck

<https://tsduck.io/>

An open source toolkit for MPEG transport streams. TSDuck provides a rich set of command-line utilities and a C++ library for analyzing, editing, generating, and broadcasting MPEG TS streams.

Tags: transport stream, MPEG, streaming, toolkit, command-line, analysis

Teradek Reliable Transport (TRT)

<https://teradek.com/blogs/articles/reliable-transport-protocol-intro>

TRT is a transport protocol developed by Teradek for streaming video over local (LAN) and wide area networks (WAN). It is designed for resilience and ultra-low latency, delivering audio and video over single and bonded networks with minimal round trip time. TRT is suitable for live production environments requiring real-time point-to-point solutions.

Tags: TRT, Teradek, video streaming, low latency, protocol

The structure of an MPEG-DASH MPD

<https://www.brendanlong.com/the-structure-of-an-mpeg-dash-mpd.html>

The MPEG-DASH Media Presentation Description (MPD) is an XML document containing information about media segments, their relationships and information necessary to choose between them, and other metadata that may be needed by clients. In this post, I describe the most important pieces of the MPD, starting from the top level (Periods) and going to the bottom (Segments).

Tide

<https://github.com/yuhuili-lab/Tide>

Simple m3u8 and MPEG-DASH MPD video downloader using libcurl

UDP vs. TCP and Which One to Use for Video Streaming

<https://www.wowza.com/blog/udp-vs-tcp>

An article by Wowza that explores the differences between UDP and TCP protocols, focusing on their applications in video streaming. It provides a detailed comparison to help developers understand which protocol suits their streaming needs better.

Tags: UDP, TCP, video streaming, protocol comparison, network protocols

UMA Technology: Streaming Protocol Comparison - RTMP, WebRTC, FTL, SRT

<https://umatechnology.org/streaming-protocol-comparison-rtmp-webrtc-ftl-srt/>

A comparative analysis of streaming protocols RTMP, WebRTC, FTL, and SRT, focusing on aspects such as latency, reliability, and error management. This resource is beneficial for developers seeking to understand the performance characteristics of these protocols in various streaming scenarios.

Tags: RTMP, WebRTC, FTL, SRT

Unreal Media Server

<https://www.umediaserver.net/umediaserver/>

Unreal Media Server is a streaming server that supports protocols such as RTMP, RTSP, and WebRTC. It is designed for live and on-demand streaming, offering low latency and support for various media formats.

Tags: streaming server, RTMP, RTSP, WebRTC, low latency

VASTreaming MPEG-DASH Server Library

<https://www.vastreaming.net/dash-library.html>

VASTreaming's MPEG-DASH library enables the creation of MPEG-DASH servers to distribute streams to connected clients. It supports ingesting live streams via protocols like RTMP or RTSP and can serve from hundreds to tens of thousands of simultaneous connections, depending on server hardware.

Tags: MPEG-DASH, server library, video streaming, RTMP, RTSP

VOD To Live

<https://github.com/Eyevinn/vod-to-live>

A python library to generate Live HLS from VOD.

Vast Info

<https://github.com/Eyevinn/vast-info>

Parse a VAST or VMAP to show valuable information in a readable format

Video Downloader

<https://github.com/shrimpgo/video-downloader>

Helper to download HLS videos.

Video Streaming Protocols Comparison

<https://www.wowza.com/blog/video-streaming-protocol-comparison>

A comprehensive article by Wowza Media Systems that compares common protocols used for live streaming, such as RTSP, HLS, WebRTC, and more. It provides insights into the pros and cons of each protocol, aiding developers in selecting the appropriate protocol for their applications.

Tags: RTSP, HLS, WebRTC, streaming protocols, comparison

Videojs Contrib Dash

<https://github.com/videojs/videojs-contrib-dash>

Video.js plugin for supporting the MPEG-DASH playback through a video.js player

Videojs Contrib HLS

<https://github.com/videojs/videojs-contrib-hls>

An HLS library for video.js that enables HLS playback in environments where it's not natively supported. It integrates seamlessly with video.js, providing features like adaptive bitrate selection, AES-128 segment encryption, and support for multiple audio tracks.

Tags: HLS, video.js, adaptive-bitrate, streaming

WISI SRT Implementation

<https://wisigroup.com/us/solutions/low-latency-video-streaming-with-srt/>

WISI has integrated SRT support into its Intelligent Headend Product family, including Inca, Tangram, and Chameleon. This integration allows video operators to utilize SRT for low-latency and reliable video streaming, enhancing the quality and security of video delivery over IP networks.

Tags: SRT, video streaming, WISI, headend, low latency

WebTorrent

<https://webtorrent.io/>

A streaming torrent client for the web. WebTorrent uses WebRTC for peer-to-peer transport and allows in-browser video streaming from torrents without plugins.

Tags: WebRTC, peer-to-peer, streaming, torrent, web-transport

Wireshark

<https://www.wireshark.org/>

A network protocol analyzer that can be used to debug streaming video at the packet level. Wireshark supports protocols like RTMP, HLS (HTTP), and MPEG-TS, allowing developers to inspect segment requests, manifest files, and timing to diagnose streaming issues.

Tags: network analysis, protocol debugging, streaming protocols, packet inspection

Wowza: Streaming Protocols Overview

<https://www.wowza.com/blog/streaming-protocols>

An overview of traditional and emerging video streaming protocols, including RTMP, RTSP, HLS, MPEG-DASH, and WebRTC. The resource discusses the advantages and limitations of each protocol, providing insights into their latency, scalability, and compatibility, which is valuable for developers in the video streaming domain.

Tags: RTMP, RTSP, HLS, MPEG-DASH, WebRTC

YouTube Live Streaming Ingestion Protocol Comparison

<https://developers.google.com/youtube/v3/live/guides/ingestion-protocol-comparison/>

This official guide from YouTube provides a detailed comparison of various ingestion protocols supported by YouTube Live Streaming, including RTMP, RTMPS, HLS, and DASH. It outlines the encryption status, supported video codecs, and suitability for different streaming scenarios, offering valuable insights for developers choosing the appropriate protocol for live streaming.

Tags: YouTube, Live Streaming, Protocol Comparison, RTMP, HLS

aiortc (Python WebRTC)

<https://github.com/aiortc/aiortc>

A Python library for Web Real-Time Communication that implements WebRTC and ORTC, enabling real-time audio/video and data streaming in Python applications.

bash scripts to create VOD HLS stream with ffmpeg almighty (tested on Linux and OS X)

<https://gist.github.com/mrbar42/ae111731906f958b396f30906004b3fa>

bash scripts to create VOD HLS stream with ffmpeg almighty (tested on Linux and OS X) - README.md

coturn

<https://github.com/coturn/coturn>

An open source implementation of TURN and STUN servers, used to facilitate NAT traversal for peer-to-peer video (and audio) communication in WebRTC and other protocols.

dash-mpd-cli

<https://github.com/emarsden/dash-mpd-cli>

Commandline application for downloading media content from a DASH MPD manifest.

dash-mpd-rs

<https://github.com/emarsden/dash-mpd-rs>

Rust library for parsing, serializing and downloading media content from a DASH MPD file.

draft-pantos-hls-rfc8216bis-00 - HTTP Live Streaming 2nd Edition

<https://tools.ietf.org/html/draft-pantos-hls-rfc8216bis-00#section-5.1>

Draft specification for the second edition of HTTP Live Streaming (HLS), documenting the protocol standard for adaptive bitrate video delivery over HTTP.

go-astits

<https://github.com/asticode/go-astits>

A Golang library for parsing and demuxing MPEG Transport Stream (.ts) files natively, enabling inspection and processing of TS files.

gortsplib

<https://github.com/aler9/gortsplib>

A pure Go RTSP server and client library that supports handling RTSP streams without external dependencies, ideal for building streaming servers or proxies.

hls-dash-dl

<https://github.com/ssnangua/hls-dash-dl>

hls-dash-dl is a tool for downloading HLS (.m3u8) and DASH (.mpd) video streams. It supports multiple audio and subtitle tracks, and provides options for quality selection and decryption of encrypted streams.

Tags: HLS, DASH, video downloader, streaming, decryption

libdash

<https://bitmovin.com/libdash/>

libdash is the official reference software of the ISO/IEC MPEG-DASH standard, providing an object-oriented interface to the MPEG-DASH standard. Developed by Bitmovin, it serves as an open-source library for developers implementing DASH streaming in their applications, ensuring compliance with the standard and facilitating adaptive streaming solutions.

Tags: MPEG-DASH, adaptive streaming, video streaming, open-source, Bitmovin

libvds

<https://github.com/Parkour-Labs/libvds>

libvds is a command-line tool designed to download video streams served over HTTP from websites, including HLS (.m3u8) and DASH (.mpd) playlists. It supports capturing network requests, collecting media playlists, muxing streams into a single video container using ffmpeg, and decrypting AES-128 and CENC encrypted playlists. This tool is valuable for developers needing to handle various streaming protocols and encryption methods.

Tags: HLS, DASH, video streaming, ffmpeg, encryption

mediamtx-rtsp-simple-server

<https://github.com/xiejiulong/mediamtx-rtsp-simple-server>

mediamtx-rtsp-simple-server is a ready-to-use media server and proxy that supports various protocols, including SRT, WebRTC, RTSP, RTMP, and LL-HLS. It allows for reading, publishing, proxying, recording, and playback of video and audio streams, offering a flexible solution for developers working with multiple streaming protocols.

Tags: SRT, media server, streaming, proxy, open source

mediasoup

<https://mediasoup.org/>

A modern WebRTC SFU (Selective Forwarding Unit) for Node.js that enables building multi-party video conferencing and streaming applications with low latency.

Tags: webrtc, real-time, transport, video-conferencing, node.js

mp4dash | Bento4

<https://www.bento4.com/documentation/mp4dash/>

Command-line tool from Bento4 for creating DASH-compliant MP4 files and streaming packages.

mpquic-video-streaming

<https://github.com/heavycrystal/mpquic-video-streaming>

A client-server video streaming platform utilizing the MPQUIC protocol, developed in Go. It leverages libVLC and FFMPEG for media processing, offering insights into implementing video streaming over the MPQUIC transport protocol.

Tags: MPQUIC, video streaming, Go, libVLC, FFMPEG

multicat

<https://www.videolan.org/projects/multicat.html>

A set of lightweight tools for multicast and transport stream manipulation (from the VLC project) – used for efficient UDP streaming, recording, and conversion of TS streams.

python-ffmpeg-video-streaming

<https://github.com/quasarstream/python-ffmpeg-video-streaming>

This Python package leverages FFmpeg to package media content for online streaming, supporting both DASH and HLS formats. It offers functionalities to implement DRM for HLS packaging and provides options to handle files from and to cloud storage. This tool is valuable for developers looking to integrate FFmpeg-based streaming solutions into their Python applications.

Tags: HLS, DASH, FFmpeg, Python, DRM

ristreceiver and ristsender

<https://code.videolan.org/rist/librist/-/wikis/LibRIST%20Documentation>

Command-line tools provided by the librist project for sending and receiving video streams using the RIST protocol. These tools serve as practical examples of implementing RIST in applications and are useful for testing and development purposes.

Tags: RIST, video streaming, command-line tools, video transmission

srt-live-transmit

<https://github.com/Haivision/srt/blob/master/docs/apps/srt-live-transmit.md>

A command-line application that demonstrates the usage of the SRT library API. It allows for the transmission of live video streams over the SRT protocol, showcasing the protocol's capabilities in real-world scenarios.

Tags: SRT, live streaming, video transmission, command-line tool

uvgRTP

<https://github.com/ultravideo/uvgRTP>

uvgRTP is an open-source Real-Time Transport Protocol (RTP) library written in C++ that focuses on high-efficiency media delivery over the Internet. It supports various video codecs, including VVC, HEVC, and AVC, and offers features like Secure RTP (SRTP) for encrypted media delivery. Its intuitive API and cross-platform compatibility make it valuable for developers working on real-time video streaming applications.

Tags: RTP, SRTP, video streaming, C++, open-source

Community & Events

2025 NAB Show: CineCentral

<https://www.tvtechnology.com/features/2025-nab-show-cinecentral-fuses-tools-and-technique>

The 2025 NAB Show's CineCentral focuses on the interplay between tools and techniques in storytelling across media formats. It offers hands-on education through live demos, expert-led training, and interactive workshops, covering topics like scene blocking, emotional camera work, and technical skills essential for video production.

Tags: video production, storytelling, workshops, training, media formats

ACM Mile-High Video 2025

<https://www.mile-high.video/technical-program>

The ACM Mile-High Video 2025 conference features a comprehensive technical program covering topics such as CDN-based video streaming, content-adaptive encoding, video enhancement, AI-generated content, and film grain technology. It includes presentations, panels, and discussions led by industry experts.

Tags: ACM, Mile-High Video, conference, video streaming, AI

AOSP and AAOS Meetup Archive

<https://aospandaaos.github.io/meetup.html>

A comprehensive archive of meetings focused on the Android Open Source Project (AOSP) and Android Automotive OS (AAOS). The archive includes slides and videos from various talks, providing valuable insights into Android development and related technologies.

Tags: AOSP, AAOS, Android development, technical talks, developer resources

Apple Developer Forums - Media Technologies

<https://developer.apple.com/forums/topics/media-technologies/streaming>

Apple's Developer Forums offer a dedicated section for Media Technologies, including streaming. Developers can engage in discussions about HTTP Live Streaming (HLS), AVFoundation, and other Apple media technologies, making it a crucial resource for those developing on Apple platforms.

Tags: Apple, developer forum, media technologies, HLS, AVFoundation

Atlanta Tech Resources

<https://github.com/citizen00147/atlanta-tech-resources>

A curated collection of groups, events, and opportunities for developers in the Atlanta, GA area. This repository serves as a comprehensive guide to active tech meetups, affordable conventions, and networking opportunities, making it easier for developers to connect and collaborate.

Tags: developer meetups, Atlanta, tech events, networking, community resources

Awesome Tech Communities

<https://github.com/MartinRosenberg/Awesome-Tech-Communities>

A curated list of tech communities across various regions, including those focused on video technology. This resource helps developers find and connect with local and global tech groups, fostering collaboration and knowledge sharing.

Tags: tech communities, developer meetups, networking, collaboration, resources

BOS Video Technology Meetup

<https://www.meetup.com/bos-video-technology/>

The BOS Video Technology meetup in Boston is for individuals interested in the technology behind video playback, storage, and delivery. Topics include HTML5 video, streaming, transcoding, codecs, advertising, open standards, and CDNs. The group focuses on the engineering aspects of video technology.

Tags: video technology, streaming, transcoding, codecs, engineering

Bay Area Tech Meetups

https://github.com/elizabethsiegle/Bay_Area_tech_meetups

A comprehensive list of tech meetups in the San Francisco Bay Area, covering topics like mobile development, iOS, AI, and more. It provides details on meetup topics, locations, and schedules, aiding developers in finding relevant community events.

Tags: Bay Area, tech meetups, developer community, networking, event listings

Bitmovin Community Site

<https://bitmovin.com/blog/video-technology-developers/>

Bitmovin's community site offers a wealth of public projects, open and solved questions, forums, and hundreds of resources for video technology developers. It's an excellent location to exchange information with peers and share opinions about Bitmovin's products and video streaming technologies.

Tags: video technology, developer community, forums, resources, Bitmovin

Brightcove Global Streaming Technology Summit 2023

<https://webinar.brightcovegallery.com/bcovkorea2023-en>

The Brightcove Global Streaming Technology Summit 2023, held on December 14, 2023, focused on how media companies in South Korea can expand their global footprint using the latest streaming technologies. The event featured discussions on the global FAST market, choosing the right streaming solutions, and the importance of streaming technology for global success.

Tags: streaming technology, global expansion, FAST market, media companies, video streaming

CNCF Community Groups

<https://github.com/cncf/communitygroups>

Managed by the Cloud Native Computing Foundation, this repository organizes community groups (formerly meetups) focused on cloud-native technologies. It offers resources for organizing and participating in community events, including training sessions and support for event management.

Tags: CNCF, cloud-native, community groups, meetups, developer resources

Canon Video Forums

<https://community.usa.canon.com/t5/Video/ct-p/videosolutions>

The Canon Video Forums provide a platform for discussions related to Canon's video products, including camcorders, professional video equipment, and video accessories. Users can seek advice, share experiences, and discuss technical aspects, making it a valuable community for developers and professionals working with Canon's video technology.

Tags: video technology, Canon, developer forum, video equipment, video accessories

Codec Developers - Google Groups

<https://groups.google.com/a/webmproject.org/g/codec-devel>

The Codec Developers Google Group is a mailing list for discussions related to codec development, including topics on libvpx and AV1. It's a platform for developers to share information, ask questions, and collaborate on codec-related projects.

Tags: codec development, libvpx, AV1, developer forum, video encoding

ConferenceXP

<https://github.com/conferencexp/conferencexp>

ConferenceXP is an open-source video conferencing and collaboration platform developed by Microsoft Research. It supports low-latency, high-fidelity conferencing applications and is optimized for multicast networks, making it suitable for academic, government, and research networks.

Tags: video conferencing, collaboration, open-source, Microsoft Research, multicast

Creative COW

<https://creativecow.net/>

Creative COW is a website of support communities for digital video, video editing, and media production professionals in broadcasting, motion graphics, visual effects, and film. It provides over 60 online support discussion forums spanning a wide range of professional video tools and software, along with over 1,000 free text and video tutorials, podcasts, and other resources for video professionals.

Tags: video editing, media production, broadcasting, motion graphics, visual effects

DV Info Net Forums

<https://www.dvinfo.net/forum/>

DV Info Net Forums is a community dedicated to digital video production, covering topics such as camera systems, editing software, and production techniques. It provides a platform for professionals and enthusiasts to share knowledge, ask questions, and discuss the latest developments in video technology.

Tags: digital video, production, camera systems, editing software, video technology

DVXuser Forums

<https://www.dvxuser.com/>

DVXuser is a forum community focused on digital video production, covering topics from camera equipment to editing software. It's a valuable resource for developers interested in video streaming and encoding, offering discussions and insights from industry professionals.

Tags: digital video production, camera equipment, editing software, forums, industry professionals

Demuxed 2017

<https://www.youtube.com/watch?v=PSdhW-R9u6s&list=PLkyaYNWEKcOfntbMd6KtHhF7qpL9hj6of>

2017 Demuxed talks & presentations

Tags: conference, video technology, presentations, talks

Demuxed 2018

https://www.youtube.com/watch?v=bfK_f7GBA8s&list=PLkyaYNWEKcOfARqEht42i1P4kBemzEV2V

2018 Demuxed talks & presentations

Tags: conference, video technology, talks, presentations

Demuxed 2019

https://m.youtube.com/playlist?list=PLkyaYNWEKcOf_C_6W45abNvXMb40xUUqh

2019 Demuxed talks & presentations

Tags: conference, video technology, talks, presentations

Demuxed 2021

<https://www.youtube.com/playlist?list=PLkyaYNWEKcOfD1GYFxFbZXDP03XM-cZPg>

2021 Demuxed talks & presentations

Tags: conference, talks, video technology, presentations

Demuxed 2022

<https://www.youtube.com/playlist?list=PLkyaYNWEKcOf98lZxnCcL6y7ZIVU3oSYO>

2022 Demuxed talks & presentations

Tags: conference, video, talks, technology, streaming

Demuxed 2024

<https://2024.demuxed.com/>

Demuxed is a conference for video engineers focusing on video technology, encompassing topics from encoding to delivery and playback. Originating from the SF Video Technology meetup, it aims to create a community for engineers building the

technology powering video. The event features discussions on various aspects of video engineering and is organized by a community-led group with support from Mux.

Tags: video engineering, conference, encoding, delivery, playback

Demuxed | Heavybit

<https://www.heavybit.com/library/podcasts/demuxed>

Demuxed is a podcast made for and by engineers working with video. Brought to you by Heavybit.

Tags: podcast, engineering, video, community, learning

Denver Video Technology Meetup

<https://www.meetup.com/Denver-Digital-Video/>

The Denver Video Technology meetup is for individuals interested in the technology behind video and video publishing, focusing on software development. Topics include video streaming, transcoding, codecs, and open standards, emphasizing the software aspects of video technology.

Tags: video technology, streaming, transcoding, codecs, software development

Digital Video Forum – The Digital FAQ

<https://www.digitalfaq.com/forum/video/>

The Digital Video Forum on The Digital FAQ website offers discussions on DVD and Blu-ray creation, video hardware repair, videography, and blank media. It includes sub-forums for project planning, workflows, capturing, editing, encoding, and authoring, providing a comprehensive resource for developers and professionals in video streaming and encoding.

Tags: video streaming, video encoding, video hardware, developer forum, video technology

Digital Video Information Network (DVinfo.net)

<https://www.dvinfo.net/>

DVinfo.net is a platform offering forums and resources focused on digital video technology, including discussions on video cameras, editing software, and production techniques. It's a valuable resource for developers seeking information on video streaming and encoding technologies.

Tags: digital video, video technology, forums, resources, community

Digital Video Talk Forum

<https://www.dpreview.com/forums/1045>

Hosted by Digital Photography Review, this forum is dedicated to discussions on digital video capture and processing, encompassing topics from digital still cameras to camcorders. It's a valuable resource for sharing knowledge and troubleshooting video-related issues.

Tags: digital video, video processing, camcorders, video capture, community forum

Digital Video Talk Forum on DPReview

<https://www.dpreview.com/forums/1009>

The Digital Video Talk Forum on DPReview is a community where videographers and video editors discuss topics related to videography, video editing, and equipment. It's a valuable platform for sharing knowledge and seeking advice on video production.

Tags: videography, video editing, forums, video production, community

Doom9's Forum

<https://forum.doom9.org/>

Doom9's Forum is a longstanding community dedicated to digital audio and video manipulation. It hosts discussions on various video encoding tools and codecs, including x264, AviSynth, and MeGUI. The forum is frequented by developers and enthusiasts, making it a rich resource for technical insights and support.

Tags: video encoding, x264, AviSynth, MeGUI, developer forum

Doom9's Forum - Video Encoding Section

<https://forum.doom9.org/forumdisplay.php?f=17>

An online forum dedicated to discussions on digital audio and video manipulation, including video encoding. It serves as a hub for major video encoding tools and technologies, offering a wealth of information and community support.

Tags: video encoding, forum, digital video, codecs, streaming

Future of Video: Business of Streaming

<https://www.parksassociates.com/event/future-of-video>

Scheduled for November 18-20, 2025, in Marina del Rey, California, this conference covers key topics in connected entertainment and video services. It includes an interactive research workshop on consumer viewing habits and offers multiple sponsorship opportunities for brand promotion within the streaming video industry.

Tags: streaming video, business, entertainment, conference, networking

GITEX Asia 2025

<https://gitexasia.com/>

GITEX Asia 2025 is a technology conference and exhibition tailored to the Asian market, bringing together businesses, governments, and innovators to explore new technologies influencing the digital world. The event features over 500 speakers addressing topics such as AI, machine learning, 5G, and smart cities.

Tags: technology conference, AI, machine learning, 5G, smart cities

Google Cloud Next 2025

<https://www.itpro.com/cloud/live/google-cloud-next-2025-all-the-news-and-updates-live>

A conference highlighting Google's advancements in cloud computing and AI, with sessions relevant to video technology developers. The 2025 event introduced new AI tools and infrastructure improvements beneficial for video streaming and encoding applications.

Tags: Google Cloud, AI, cloud computing, video technology, developer conference

Hybrid Meetup Platform

<https://github.com/montevideo-tech/hybrid-meetup-platform>

An open-source project designed to host and run hybrid meetups for the video community. Built by video developers for video developers, it leverages WebRTC technology to facilitate seamless virtual and in-person interactions.

Tags: WebRTC, hybrid meetups, video technology, open-source, developer tools

InfoComm 2025

<https://www.infocommshow.org/infocomm-2025>

InfoComm 2025 is a premier event for audiovisual professionals, featuring a wide range of sessions and exhibits on topics such as automated conference and collaboration solutions, integrating pro video and audio management solutions, and optimizing Zoom Room deployments with Logitech hardware. The event provides opportunities for hands-on education, networking, and exploring the latest innovations in the AV industry.

Tags: audiovisual, conference, collaboration, AV industry, technology

Kaltura Community Forums

<https://forum.kaltura.org/c/community-forums/21>

The Kaltura Community Forums provide a platform for developers and users to discuss various aspects of Kaltura's video technology, including APIs, SDKs, and server management. It's a valuable resource for troubleshooting, sharing best practices, and staying updated on Kaltura's offerings.

Tags: Kaltura, video technology, developer support, APIs, SDKs

Level1Techs Forums - Encoding/Compression Video and Audio Mega Thread

<https://forum.level1techs.com/t/encoding-compression-video-and-audio-mega-thread/210653>

This mega thread on the Level1Techs Forums is dedicated to discussions about video and audio encoding/compression. It covers various topics, including tools, techniques, and best practices, serving as a valuable resource for developers interested in video streaming and encoding technologies.

Tags: video encoding, audio encoding, compression, developer resources, best practices

Los Angeles Video Technology

<https://www.meetup.com/los-angeles-video-technology/>

A Los Angeles-based meetup for those interested in the technology behind video and video publishing, covering topics like HTML5 video, streaming, transcoding, codecs, advertising, open standards, virtual reality, 360-degree video, and HDR. The group focuses on the engineering side, offering a platform for developers to exchange ideas and knowledge.

Tags: video technology, streaming, transcoding, codecs, virtual reality

MacRumors Forums - Digital Video Section

<https://forums.macrumors.com/forums/digital-video.80/>

The MacRumors Forums host a dedicated section for digital video discussions, where users share tips, resources, and experiences related to video technology, including forums for learning about video technologies and recommended forums for digital video.

Tags: digital video, forums, video technology, community, Mac

Microsoft Build 2025

<https://build.microsoft.com/>

An annual conference by Microsoft aimed at software engineers and web developers, covering topics including video technology. The 2025 event offers sessions on Windows, Azure, and other Microsoft technologies pertinent to video streaming and encoding.

Tags: Microsoft Build, developer conference, video technology, Azure, software development

MonteVIDEO Tech Meetup

<https://montevideotech.dev/>

A community of engineers, developers, and enthusiasts passionate about video technology. The meetup offers events like 'Mate Talks' and 'Summer Camp,' focusing on collaboration, knowledge sharing, and cultural exchange within the video tech community. Topics include video engineering, product design, and development.

Tags: video engineering, product design, development, collaboration, knowledge sharing

Montevideo Video Tech Summer Camp

<https://www.qualabs.com/community>

An annual event organized by Qualabs, bringing together video technology professionals for discussions on industry trends, technical challenges, and cultural experiences. The event offers opportunities to learn from experts, share ideas, and build a strong community among video developers.

Tags: video technology, summer camp, Qualabs, networking, industry trends

Movie Studio Zen Forum - Technology Forums

<https://www.moviestudiozen.com/forum/technology-forums>

The Technology Forums on Movie Studio Zen provide discussions about computer hardware and electronic equipment used in video production, such as graphics cards, CPUs, cameras, and microphones. It's a community-driven platform for sharing knowledge and troubleshooting video technology-related issues.

Tags: video production, hardware, forums, community, technology

NAB Show 2025

<https://www.tvtechnology.com/features/welcome-to-the-2025-nab-show-where-the-future-is-now>

The NAB Show 2025, themed 'Welcome to the Future,' is a pivotal event in the media and entertainment industry, showcasing the integration of advanced technologies like AI and cloud computing. It features over 500 speakers and exhibits emphasizing trends such as artificial intelligence, cloud virtualization, the creator economy, sports, and streaming. Highlights include a Sports Summit, hands-on activities with tech giants, and insights into emerging streaming monetization models.

Tags: NAB Show, media technology, AI, cloud computing, streaming

OBS Forums

<https://obsproject.com/forum/>

The OBS Forums are dedicated to discussions around Open Broadcaster Software (OBS), a popular open-source software for video recording and live streaming. The forums include topics on video encoding, hardware compatibility, and troubleshooting, providing a collaborative space for developers and users to share knowledge and solutions.

Tags: OBS, video recording, live streaming, video encoding, open-source software

P2P Audiovisual Resource Pages and Links

https://wiki.p2pfoundation.net/P2P_Audiovisual_Resource_Pages_and_Links

A comprehensive collection of software and resources for audio and video players, aggregators, web TV, codecs, editing tools, and 3D applications. This resource is valuable for developers seeking tools and information on video streaming and encoding technologies.

Tags: video technology, resources, P2P, audiovisual, software

RIST Forum Events

<https://www.rist.tv/events>

The RIST Forum hosts various events showcasing and demonstrating RIST interoperability and benefits. These events include webinars, conferences, and exhibitions focused on the Reliable Internet Stream Transport (RIST) protocol, which is essential for secure and reliable transportation of live video over unmanaged networks.

Tags: RIST Forum, video streaming, RIST protocol, live video, networking

Reddit r/videostreaming

<https://www.reddit.com/r/videostreaming/>

A subreddit dedicated to video streaming technology. Community members share knowledge, ask questions, and discuss streaming protocols, encoding settings, and tools.

Tags: community, forum, discussion, streaming, technology

Resource Pages and Links to the P2P Audiovisual Net

https://wiki.p2pfoundation.net/Resource_Pages_and_Links_to_the_P2P_Audiovisual_Net

An extensive list of tools and resources related to online media, remix culture, and technologies like DRM, P2P, and RSS TV. This resource is beneficial for developers interested in the broader context of video streaming and encoding within the P2P framework.

Tags: P2P, audiovisual, resources, online media, technology

SEGMENTS:2023 by Streaming Video Technology Alliance

<https://web.archive.org/web/20230521170721/https://www.businesswire.com/news/home/20230516005279/en/Streaming-Video-Technology-Alliance-Introduces-New-Industry-Conference-SEGMENTS2023>

SEGMENTS:2023, introduced by the Streaming Video Technology Alliance, was a one-day public conference held on May 16, 2023, in New Orleans. The event provided a comprehensive walkthrough of the streaming video workflow, covering topics from content ingestion to analytics, and featured speakers from over 25 companies discussing challenges and best practices in operating streaming services.

Tags: streaming video, conference, video workflow, analytics, best practices

SF Video Technology Meetup

<https://www.meetup.com/sf-video-technology/>

The SF Video Technology meetup in San Francisco caters to those interested in the technology behind video and video publishing. Discussions cover HTML5 video, streaming, transcoding, codecs, advertising, and open standards, with a focus on engineering rather than creative aspects.

Tags: video technology, streaming, transcoding, codecs, engineering

SIGGRAPH 2025

<https://www.siggraph.org/>

SIGGRAPH 2025 is an annual conference on computer graphics and interactive techniques, featuring presentations on the latest research, technologies, and applications in the field. It includes a trade show, educational courses, and panel discussions relevant to video technology professionals.

Tags: computer graphics, interactive techniques, video technology, conference, trade show

SMPTE Events

<https://www.smpte.org/events>

The Society of Motion Picture and Television Engineers (SMPTE) offers a range of events including webcasts, courses, and conferences focused on media and entertainment technology. These events cover topics like high dynamic range, imaging system fundamentals, and the latest industry standards.

Tags: SMPTE, media technology, courses, conferences, standards

SVTA STREAMups

<https://www.svta.org/streaming-video-alliance-streamups-and-meetups/>

Streaming Video Technology Alliance (SVTA) organizes STREAMups, which are industry meetups bringing together professionals to discuss critical technical challenges in video streaming. These events foster collaboration and innovation within the video technology community.

Tags: video streaming, industry meetups, technical challenges, collaboration, innovation

Silicon Valley Video Summit 2025

<https://svgplay.sportsvideo.org/silicon-valley-video-summit-2025>

The Silicon Valley Video Summit 2025 is an annual exchange of enterprise media know-how, focusing on how video is transforming enterprise communication. The event features sessions on topics like live television production, AI in media creation, virtual production, and immersive experiences. It brings together corporate video producers, technologists, and industry leaders to discuss the latest trends and technologies in video production.

Tags: video summit, enterprise media, AI, virtual production, immersive experiences

Streaming Media Conferences

<https://www.streamingmedia.com/conferences/>

Streaming Media organizes multiple conferences and exhibitions annually, serving as the industry's meeting place for streaming media professionals. These events cover a wide range of topics, including audio and video on the Internet, content delivery, and emerging technologies in streaming media.

Tags: Streaming Media, streaming conferences, online video, content delivery, media technology

Streaming Media Connect 2023

<https://www.streamingmediaglobal.com/Articles/ReadArticle.aspx?ArticleID=161593>

Streaming Media Connect 2023 was a virtual conference held from November 13 to 16, 2023, featuring industry experts discussing topics such as cloud workflows, live streaming optimization, and codec developments. The event provided valuable insights into the latest trends and technologies in the streaming media industry.

Tags: streaming media, virtual conference, video technology, live streaming, cloud workflows

Streaming Media Connect 2025

<https://www.streamingmedia.com/Conferences/Connect2025-May/Default.aspx>

Streaming Media Connect is a free virtual event series offering practical advice and insights from industry leaders on live streaming, OTT, content delivery, and monetization. The next installment is scheduled for May 20-22, 2025.

Tags: streaming, OTT, content delivery, monetization, virtual event

Streaming Media Connect February 2025

<https://www.streamingmedia.com/conferences/connect2025/default.aspx>

A free virtual event from February 25-27, 2025, offering practical advice, thought leadership, and insights from industry leaders. Topics include live streaming, OTT, content delivery, and monetization strategies. The event features live Q&A sessions, expert debates, and exclusive insights to help build stronger streaming solutions.

Tags: streaming media, virtual event, live streaming, OTT, content delivery

Streaming Media University Workshops

<https://www.streamingmedia.com/Conferences/StreamingMediaUniversity/Schedule.aspx>

Streaming Media University offers a series of workshops designed to educate professionals on the latest tools and techniques in the streaming industry. These workshops cover a range of topics, including production gear, software solutions, and best practices for delivering high-quality streaming content. They provide hands-on experience and insights from industry experts, making them valuable for developers and technologists in the video streaming field.

Tags: streaming media, workshops, education, video production, best practices

Streaming Video Industry Events Calendar

<https://www.svta.org/streaming-video-industry-events/>

The Streaming Video Technology Alliance provides a comprehensive calendar of streaming video industry events, including conferences, meetups, and more. This resource helps professionals stay informed about upcoming events related to online video, networking opportunities, and industry trends.

Tags: Streaming Video Technology Alliance, streaming events, video industry, networking, online video

Streaming Video Technology Alliance (SVTA) Events

<https://www.svta.org/svta-calendar/>

The SVTA organizes meetings, webinars, and events focused on improving streaming video technologies. Their calendar includes various sessions on topics like Common Media Library Roadmap and dash.js, providing valuable resources for professionals in the streaming industry.

Tags: streaming video, technology, webinars, meetings, SVTA

Streaming Video Technology Alliance (SVTA) Special Events

<https://www.svta.org/svta-special-events/>

SVTA organizes special events such as networking parties and workshops during member meetings, focusing on streaming video technologies. These events provide opportunities for professionals to network, learn about new technologies, and discuss industry challenges.

Tags: SVTA, streaming video, networking, workshops, technology

Streaming Video Technology Alliance Q1 2025 Member Meeting

<https://www.svta.org/2025/03/07/streaming-video-technology-alliance-tucson-first-quarterly-member-meeting-2025/>

The Streaming Video Technology Alliance (SVTA) convened for its Q1 2025 Member Meeting in Tucson, Arizona. The event featured working group sessions, industry presentations, and networking opportunities. Notably, a special project was launched to support NASA's streaming from the moon and beyond, and new members including BBC, EBU, NASA, and Setplex were announced.

Tags: SVTA, streaming video, NASA, BBC, EBU

Streaming+ at NATPE Global 2025

<https://global.natpe.com/2025/streamingplus/>

Streaming+ is a dedicated conference strand at NATPE Global 2025 that delves into pressing issues and challenges surrounding the OTT business, content, and consumer experience. The event features keynotes, panel discussions, and networking opportunities with thought leaders and changemakers transforming the streaming industry.

Tags: Streaming+, NATPE Global, OTT, content, consumer experience

Tech Meetups

<https://github.com/thisdot/tech-meetups>

A curated list of various tech meetups covering a wide range of topics, including video technology. This resource helps developers find and participate in relevant community events to enhance their skills and network.

Tags: tech meetups, developer community, networking, video technology, resources

The Video Consortium Gatherings and Workshops

<https://videoconsortium.org/events>

The Video Consortium hosts industry screenings, happy hours, and professional development workshops for filmmakers. These gatherings are designed to foster community, provide learning opportunities, and support filmmakers in their

professional growth.

Tags: Video Consortium, filmmakers, workshops, networking, professional development

Twitch Developer Forums

<https://discuss.dev.twitch.com/>

The Twitch Developer Forums provide a platform for developers to discuss and collaborate on Twitch's API, Extensions, and other developer products. It includes categories for announcements, API discussions, and game development integrations, serving as a central hub for developers building on the Twitch platform.

Tags: Twitch, API, developer forum, game development, extensions

TwitchCon

<https://www.twitchcon.com/>

TwitchCon is a semi-annual convention organized by Twitch Interactive, focusing on the culture of livestreaming, entertainment, esports, and internet culture. It provides opportunities for industry professionals, streamers, and fans to meet, discover new products, and trial upcoming games.

Tags: TwitchCon, livestreaming, convention, esports, internet culture

TwitchCon 2025

<https://en.wikipedia.org/wiki/TwitchCon>

TwitchCon is a semi-annual convention organized by Twitch Interactive, focusing on livestreaming, entertainment, esports, and internet culture. The 2025 events are scheduled for May 31 - June 1 in Rotterdam and October 17-19 in San Diego, marking the 10th anniversary of TwitchCon. The convention allows community members to meet streamers, discover new products, and trial upcoming games.

Tags: TwitchCon, livestreaming, esports, internet culture, community

VidCon

<https://vidcon.com>

VidCon is an annual convention for influencers, fans, executives, and online brands, primarily featuring prominent video stars from across the internet. Founded by veteran YouTube creators John and Hank Green, VidCon has expanded internationally with events in Singapore, Abu Dhabi, Mexico City, Australia, and São Paulo. The convention includes panels, workshops, and meet-and-greets, focusing on online video content creation and community building.

Tags: online video, content creation, influencers, convention, community

VidCon 2025

<https://en.wikipedia.org/wiki/VidCon>

VidCon is an annual convention for influencers, fans, executives, and online brands, featuring prominent video stars from across the internet. The 2025 event is scheduled for June 19–21 at the Anaheim Convention Center in California, offering a platform for networking and learning about online video trends.

Tags: online video, influencers, networking, conference, entertainment

Video Tech NYC

<https://www.meetup.com/video-tech-nyc/>

A meetup group in New York City for individuals interested in the technology behind video and online video publishing, covering topics like HTML5 video, streaming, transcoding, codecs, advertising, open standards, video testing pipelines, VR/360, and analytics. The focus is on the engineering aspects, though creatives are also welcome.

Tags: video technology, streaming, transcoding, codecs, analytics

VideoHelp Forum

<https://forum.videohelp.com/>

The VideoHelp Forum is a comprehensive platform covering a wide range of topics related to video technology, including video conversion, editing, authoring, and capturing. It serves as a valuable resource for developers and enthusiasts seeking advice, tutorials, and discussions on various video-related tools and techniques.

Tags: video conversion, video editing, authoring, capturing, developer forum

VideoTech 2023 Conference

<https://vtconf.com/en/archive/2023/>

VideoTech 2023 was a conference on video technologies held online from November 16–17 and in Moscow from November 20–21, 2023. The event featured a diverse lineup of speakers from companies like VK, Yandex, and Huawei, covering topics related to video technology advancements and industry trends.

Tags: video technology, conference, industry trends, VK, Yandex

VideoTech 2025

<https://vtconf.com/en/>

A conference for video technology enthusiasts scheduled for September 2025 in Moscow and online. It covers topics from video production to distribution, including encoding, decoding, delivery, and the application of ML and AI in video technologies. The event aims to develop a professional community of engineers in the field.

Tags: video technology, conference, ML, AI, video production

Videography

<https://www.reddit.com/r/Videography/>

r/Videography is a subreddit dedicated to videography discussions, where members share insights, ask questions, and provide feedback on video projects. It's a dynamic community suitable for developers interested in video streaming and encoding tools, offering real-world experiences and advice.

Tags: videography, video production, community, forums, discussion

Videomaker Community

<https://www.videomaker.com//connect/>

Videomaker's community offers forums and resources for videographers of all skill levels. It provides discussions on video production techniques, equipment reviews, and post-production tips, serving as a valuable platform for learning and networking.

Tags: videography, video production, equipment reviews, post-production, networking

Wowza Community

<https://community.wowza.com/>

The Wowza Community is a comprehensive forum for users of Wowza's streaming solutions, including Wowza Streaming Engine and Wowza Video. It offers sections for troubleshooting, developer discussions, and product announcements, making it a valuable resource for developers working with Wowza's tools.

Tags: Wowza, streaming, developer forum, video encoding, troubleshooting

Intro & Learning

3 Cases from a Video Expert: Encoding Basics

<https://medium.com/vicuesoft-techblog/3-cases-from-a-video-expert-encoding-basics-b1ba1c398af8>

Vicuesoft blog article explaining fundamental video encoding concepts through practical case studies.

A Brief History of Video Coding

<https://docslib.org/doc/10606874/a-brief-history-of-video-coding>

Authored by Marco Jacobs and Jonah Probell, this document provides a detailed account of the history of video coding. It discusses early forms of compression, the development of interlacing, and the progression of video coding standards. This resource is useful for those interested in the technical aspects and historical context of video coding.

Tags: video coding, history, technical, compression, standards

A Brief History of Video Compression Standards: 1929 to 2020

<https://www.mo4tech.com/a-brief-history-of-video-compression-standards-1929-to-2020.html>

An article by Moment For Technology that provides a concise history of video compression standards, covering key developments from 1929 to 2020.

Tags: video compression, standards, history, Moment For Technology, timeline

A Comprehensive Guide To Develop A Live Video & On-demand Video Streaming App

<https://raftlabs.medium.com/the-complete-guide-to-build-a-live-streaming-app-learn-the-best-approach-and-tech-stack-263bd67e2915>

An extensive guide by RaftLabs detailing the development of live and on-demand video streaming applications, including business use cases, essential features, and technology stacks.

Tags: live streaming, on-demand, app development, technology stack

A short history of video coding

https://www.slideshare.net/vcodex/a-short-history-of-video-coding?from_m_app=ios

Video coding is an essential component of video streaming, digital TV, video chat and many other technologies. This presentation, an invited lecture to the US ...

AWS Media Services Training

<https://aws.amazon.com/blogs/training-and-certification/three-free-resources-to-build-skills-in-media-services-workflows/>

Free digital courses and resources designed to build skills in cloud-based media services workflows. The offerings include the AWS Ramp-Up Guide for Media Services, courses on video processing and delivery, and introductions to AWS Media Services by use case.

Tags: AWS Media Services, cloud-based media, video processing, media workflows, training

About Frame Rates or Why 29.97?

<https://theautomaticfilmmaker.com/blog/2009/2/23/about-frame-rates-or-why-2997.html>

I recently remembered this popular post from my old blog. Since that blog no longer exists, I thought I would repost it here. Since I wrote this post about a decade ago, many others have done a much better job describing this in detail. I highly recommend the following two videos by Alec Watson from

An Overview of Emerging Video Coding Standards

<https://dl.acm.org/doi/abs/10.1145/3325867.3325873>

This article from GetMobile: Mobile Computing and Communications provides an overview of existing and emerging video coding standards, including the H.26X family, AV1, VP9, and VVC. It discusses the timeline of their development and future trends in video coding, offering valuable insights for those interested in the evolution of video compression technologies.

Tags: video coding standards, AV1, VP9, VVC, video compression

Ant Media Server Tutorials

<https://mirror.antmedia.io/tutorial/>

A collection of tutorials from Ant Media that cover various aspects of setting up and managing a video streaming server. Topics include WebRTC integration, RTMP, HLS, and CMAF protocols, providing practical guidance for developers building live streaming platforms.

Tags: video streaming, WebRTC, RTMP, HLS, tutorial

Avid Media Composer | First

<https://my.avid.com/get/media-composer-first>

Avid's free tier of Media Composer, offering a subset of the professional editing tools for beginners. It's widely used in the film & TV industry, making this entry-level version a resource for learning industry-standard workflows.

Tags: video editing, learning, professional tools, beginner, media composer

Awesome Live Stream

<https://github.com/liwf616/awesome-live-stream>

WebRTC & Nginx & DASH & Quick [fN`De™e6—Æÿ c ~íføe°N- .

BBC R&D Blog (Research & Development)

<https://www.bbc.co.uk/rd/blog>

The BBC Research & Development blog where broadcast technology advancements are discussed. Includes articles on new video compression techniques, streaming trials, and standards.

Tags: research, technology, broadcasting, video compression, streaming, standards

Back to Basics: Encoding Definition and Adaptive Bitrate

https://bitmovin.com/encoding-definition-bitrates/?utm_campaign=Newsletter&utm_medium=email&_hsenc=p2ANqtz-8MPFxbR7snQrxPYM7BI3UTEMgOh5ZXoDQCHjLl9lkskqE0lfBhEuz3us39Br-lvA_CnyNmQl6L5wqO6iKOFAJ8HznenQ&_hsmi=79678208&utm_content=79677632&utm_source=hs_email&hsCtaTracking=b8eb0e0a-f292-435e-8b99-719b75d81412%7C367afa65-d810-4c2e-aa2c-c87e897a8942

Guide explaining encoding fundamentals and adaptive bitrate streaming concepts, including how to deliver multiple quality versions optimized for different network conditions.

Background on the BAVC Model | Video History Project

<https://www.videohistoryproject.org/background-bavc-model>

An overview of the Bay Area Video Coalition's efforts in video preservation and remastering, highlighting the challenges and solutions in maintaining video art.

Tags: video preservation, BAVC, video art, remastering, media conservation

Best Practices for End-to-End Workflow Monitoring | Streaming Video Alliance

<https://www.streamingvideoalliance.org/project/best-practices-for-end-to-end-workflow-monitoring/>

Streaming Video Alliance guidelines for monitoring video workflows from ingest through delivery to end users.

Tags: workflow, monitoring, best practices, streaming, alliance

BigBlueButton

<https://bigbluebutton.org/>

An open source web conferencing system designed for online learning. BigBlueButton supports video, audio, slides, chat, and whiteboard, and is often integrated with learning platforms.

Tags: web conferencing, online learning, video streaming, education, open source

Bitmovin Tech Blog

<https://bitmovin.com/blog/>

Bitmovin's blog features articles on video streaming, codec comparisons, encoding techniques, and industry trends in OTT delivery (by a provider of streaming encoding/player solutions).

Tags: video streaming, encoding, codec, OTT, tutorial, technology

BoxCast Live Streaming Resources

<https://www.boxcast.com/learn>

An extensive learning hub offering support articles, step-by-step guides, courses, blogs, tech tips, and podcasts. Topics range from setting up encoders and simulcasting to advanced production features like geo-blocking and API keys.

Tags: live streaming, video production, encoder setup, simulcasting, geo-blocking

Build Your Own Video Compression Service with Rust WASM

<https://beendless.com/2024/08/02/Build%20Your%20Own%20Video%20Compression%20Service%20with%20Rust%20WASM/>

This tutorial guides you through creating a Rust application that compiles to WebAssembly and utilizes FFmpeg to build a video compression service. It covers setting up the environment, integrating FFmpeg.wasm, and handling JavaScript interoperability.

Tags: WebAssembly, Rust, FFmpeg, video compression, tutorial

Build Your Own Video Streaming Platform - Linode Tutorial

<https://www.classcentral.com/course/youtube-build-your-own-video-streaming-platform-it-s-easier-than-you-might-think-147570>

A free video tutorial that walks through the process of creating a video streaming website using HTML, CSS, and Next.js. It covers setting up a Next.js project, integrating Tailwind CSS, developing key components, and leveraging Linode Object Storage for video streaming.

Tags: Next.js, Tailwind CSS, Linode, video streaming, tutorial

Building Scalable Video Encoding Pipelines Essentials - Tonex Training

<https://www.tonex.com/training-courses/building-scalable-video-encoding-pipelines-essentials/>

A training course covering the fundamentals of building scalable video encoding pipelines, including topics like video codecs, cloud-based encoding, and workflow optimization.

Tags: video encoding, scalable pipelines, video codecs, cloud encoding, workflow optimization

Building Scalable Video Encoding Pipelines Training

<https://www.tonex.com/training-courses/building-scalable-video-encoding-pipelines-training/>

An in-depth training course by Tonex that focuses on designing and implementing robust video encoding systems. It covers principles, tools, and techniques to create efficient, scalable pipelines for high-quality video delivery, making it ideal for professionals aiming to enhance their video processing systems.

Tags: video encoding, scalable pipelines, video processing, training course, video delivery

CRF Guide (Constant Rate Factor in x264, x265 and libvpx)

<https://slhck.info/video/2017/02/24/crf-guide.html>

What is the Constant Rate Factor?

Can I use... Support tables for HTML5, CSS3, etc

<https://caniuse.com/#search=drm>

Browser compatibility data for DRM technologies, helping developers understand EME and content protection support across platforms.

Circles of Confusion

<https://tech.ebu.ch/publications/circles-of-confusion>

EBU technical paper on circles of confusion in imaging. Covers optical and resolution considerations for video.

Cloudinary's Guide on Live Streaming

<https://cloudinary.com/guides/live-streaming-video/a-guide-on-live-streaming-and-tips-for-making-it-a-success>

An in-depth guide explaining the live streaming process, including capturing video and audio, compressing and encoding video, and delivering content to audiences. It also offers tips for successful live streaming.

Tags: live streaming, video encoding, streaming tips, video compression, content delivery

Coding Video: A Practical Guide to HEVC and Beyond

<https://www.oreilly.com/library/view/coding-video/9781118711781/c03.xhtml>

A comprehensive book that provides a practical guide to High Efficiency Video Coding (HEVC) and other video coding standards. It includes a chapter on the history of video coding and video coding standards, offering insights into the

development and evolution of video compression technologies.

Tags: video coding, HEVC, standards, compression, history

Communicating Pictures

<https://www.elsevier.com/books/communicating-pictures/bull/978-0-12-405906-1>

2014-06-20 (1st Edition). Bull, David.

Coursera: AR & Video Streaming Services Emerging Technologies

<https://www.coursera.org/learn/ar-technologies-video-streaming>

An online course that explores augmented reality applications, technologies, products, and businesses, providing insights into video streaming methodologies and services.

Tags: augmented reality, video streaming, emerging technologies, online course, Coursera

Create a Video Streaming Platform with React

<https://www.geeksforgeeks.org/create-a-video-streaming-platform-with-react/>

This article from GeeksforGeeks guides readers through the process of creating a video streaming platform using React. It covers the setup of the React application, installation of required modules, and the development of components for video streaming. This tutorial is beneficial for developers interested in building front-end video streaming applications with React.

Tags: React, video streaming, tutorial, frontend development, web application

Create a Video Trimming App Using ffmpeg.wasm

https://cloudinary.com/blog/guest_post/create-a-video-trimming-app-using-ffmpeg-wasm

This tutorial guides you through building a simple video trimming application using ffmpeg.wasm. It provides a practical example of implementing video processing capabilities directly in the browser, making it a valuable resource for developers looking to integrate similar features into their web applications.

Tags: WebAssembly, FFmpeg, video trimming, tutorial, web development

Datavideo's 'Getting Started with Streaming' Tutorials

<https://www.datavideo.com/ap/page/getting-started-with-streaming/>

Datavideo offers a series of how-to videos and tutorials guiding users through the setup and operation of their streaming equipment, including encoders and mobile studios, with step-by-step instructions for streaming to platforms like Facebook Live and YouTube.

Tags: video streaming, tutorials, how-to videos, streaming setup, Datavideo

Decoding the Video Codec Wars: H.264, HEVC, and AV1 Compared for Streaming

<https://flussonic.com/cn/blog/news/evolution-of-video-codecs/>

This article explores the history and differences between leading video codecs: H.264, HEVC, and AV1. It provides insights into their development, benefits, and implications for TV/IPTV providers, OTT services, and professional production companies.

Tags: video codecs comparison, H.264, HEVC, AV1, video streaming

Demuxed 2016

<https://www.youtube.com/watch?v=kEo2TrXm7F4&list=PLkyaYNWEKcOekC2m9Na77G40Lmhb1bnsK>

2016 Demuxed talks & presentations

Tags: conference, talks, video technology, learning, presentations

Demuxed 2020

<https://www.youtube.com/playlist?list=PLkyaYNWEKcOcDIGjEbpxBe4woCJGHrarN>

2020 Demuxed talks & presentations

Tags: conference, talks, video technology, learning, presentations

Demuxed Conference Talks (YouTube Channel)

https://www.youtube.com/channel/UCFQoJk-PKfXI4M7qLrH_cCg

The YouTube channel for Demuxed, an annual conference for video engineers. Features recorded talks covering streaming protocols, codecs, player development, and industry case studies.

Tags: conference, video engineering, streaming, tutorials, technical talks

Demystifying HTML5 Video Player

<https://medium.com/@eyevinntechnology/demystifying-html5-video-player-e480846328f0>

In this post we will go under the hood of a HTML video player for video streaming. With the exception of Apple and their browser Safari, no...

Tags: HTML5, video player, web streaming, tutorial, browser technology

Digital Video

https://en.wikipedia.org/wiki/Digital_video

The Wikipedia page on Digital Video provides an overview of digital video technologies, including the history of video compression standards from H.261 to HEVC. It discusses the evolution of video coding formats and their applications, offering a broad perspective on digital video development.

Tags: digital video, video compression, standards, history, technology

Digital Video Introduction

https://github.com/leandromoreira/digital_video_introduction

A hands-on introduction to video technology: image, video, codec (av1, vp9, h265) and more (ffmpeg encoding).

Digital Video: How it Works, History, and Formats

<https://www.vodpod.com/video/digital/>

This resource explains the fundamentals of digital video, including how it works, its history, and various formats. It covers topics such as compression methods, encoding standards, and the differences between digital and analog video.

Tags: digital video, history, formats, compression, encoding

Elephants Dream (Open Movie)

<https://orange.blender.org/>

The first open movie from the Blender Foundation (2006). It's an animated short film widely used in the past as a standard test video for new compression codecs and media players.

Tags: open source, animation, codec testing, educational video, blender foundation

Encoding-for-HEVC

<https://streaminglearningcenter.com/wp-content/uploads/2018/05/Encoding-for-HEVC-SME-2018-Jan.pdf>

Guide covering encoding techniques and workflows specific to HEVC video codec implementation.

Encoding.com Resource Library

<https://www.encoding.com/resources/>

A collection of documents and solution briefs covering various aspects of video encoding, including automated quality control, adaptive bitrate streaming, and cloud-based processing.

Tags: video encoding, adaptive bitrate streaming, cloud processing, quality control, resources

Evolution of AVS Video Coding Standards: Twenty Years of Innovation and Development

<https://link.springer.com/article/10.1007/s11432-021-3461-9>

This research paper published in Science China Information Sciences provides a comprehensive survey of the three generations of AVS video coding standards developed over two decades. It discusses the advancements in coding performance, algorithm optimization, and hardware implementation, making it a valuable resource for understanding the progression of video coding technologies.

Tags: AVS standards, video coding evolution, video compression, coding performance, hardware implementation

Extending the Browser with WebAssembly

<https://web.dev/articles/wasm-av1>

An article that guides developers through the process of porting the AV1 video decoder to WebAssembly, enabling AV1 video playback in any modern browser. This resource is valuable for understanding how to extend browser capabilities using WebAssembly for advanced video decoding.

Tags: WebAssembly, AV1, video decoding, browser extension, tutorial

FFmpeg Encoding Course

<https://github.com/slhck/ffmpeg-encoding-course>

An introduction to FFmpeg and its tools.

FFmpeg Libav Tutorial

<https://github.com/leandromoreira/ffmpeg-libav-tutorial>

FFmpeg libav tutorial - learn how media works from basic to transmuxing, transcoding and more

Tags: ffmpeg, tutorial, media-processing, encoding, learning

FFmpeg Threads Command: How it Affects Quality and Performance

<https://streaminglearningcenter.com/blogs/ffmpeg-command-threads-how-it-affects-quality-and-performance.html>

So, I received an email from an acquaintance that read, "I was curious if there is actually any benefit to a "threads=" type custom command in x264. Specifically many streamers are buying 8 core/16 thread CPUs to encode as a standalone client capturing information from a video capture device." I had an article on FFmpeg...

FFmpeg Wiki: Streaming Guide

<https://trac.ffmpeg.org/wiki/StreamingGuide>

An official Wiki guide from FFmpeg explaining how to set up streaming using FFmpeg, including examples for streaming via RTMP to servers or using FFmpeg as a server for HLS/DASH.

Tags: ffmpeg, streaming, tutorial, rtmp, hls, dash, server configuration

FFmpeg and how to use it wrong

<https://videoblerg.wordpress.com/2017/11/10/ffmpeg-and-how-to-use-it-wrong/>

I've been in the streaming media industry since 2008 and have seen a lot of misinformation regarding both FFmpeg and libx264. In this post I hope to help shed some light on what does and does...

Foundry Nuke (Non-commercial)

<https://www.foundry.com/products/nuke/non-commercial>

Nuke is an industry-standard compositing tool. The Non-commercial edition provides free access for learning and personal projects, allowing users to explore Nuke's powerful node-based compositing workflows (with some limitations).

Tags: compositing, video editing, learning, tutorial, non-commercial

FrameFlow

<https://github.com/carsonDB/frameflow>

FrameFlow is a high-performance video processing library for web browsers, leveraging WebCodecs and FFmpeg compiled to WebAssembly. It supports hardware acceleration in Chromium-based browsers and offers features like stream processing, metadata extraction, and various video filters.

Tags: WebAssembly, video processing, FFmpeg, WebCodecs, browser

From Sys Admin to Netflix SRE

<https://www.youtube.com/watch?v=IZI51YzIgVE>

Talk by Jonah Horowitz, Albert Tobey What does it take to be a Netflix SRE? With tens of thousands of Linux instances in a distributed system architecture, a...

Fundamentals of Multimedia

<https://www.amazon.com/Fundamentals-Multimedia-Texts-Computer-Science/dp/303062126X/>

2022-02-17 (3rd Edition). Ze-Nian Li (Author), Mark S. Drew (Author), Jiangchuan Liu.

GStreamer Multimedia Framework

<https://gstreamer.freedesktop.org>

GStreamer is a pipeline-based multimedia framework that links together a wide variety of media processing systems to complete complex workflows. It supports various media-handling components, including audio and video playback, recording, streaming, and editing.

Tags: multimedia framework, video streaming, audio processing, open-source, GStreamer

Galicaster

<https://github.com/teltek/Galicaster>

An open source lecture capture system often used with the Opencast platform. It provides an interface for recording multiple video sources (camera, slides) in classrooms or conferences.

Tags: lecture capture, open source, education, multi-source recording, video capture

GstWASM

<https://fluendo.com/zh-cn/blog/gstwasm-the-open-source-project-for-gstreamer-development-in-the-web/>

GstWASM is an open-source project that brings GStreamer development to the web using WebAssembly. It provides features like SDL2 for video rendering, OpenAL for audio, and integration with WebCodecs, enabling developers to build multimedia applications that run efficiently in web browsers.

Tags: WebAssembly, GStreamer, multimedia, audio, video

H.261 Video Compression Standard

<https://en.wikipedia.org/wiki/H.261>

The Wikipedia page on H.261 provides detailed information about the first practical digital video coding standard developed by the ITU-T. It covers the history, technical details, and significance of H.261 in the evolution of video compression technologies. This resource is essential for understanding the foundations of modern video coding.

Tags: H.261, video compression, ITU-T, standards, history

HLS adaptive streaming tutorial with CloudFront & JW Player | Miracle Tutorials

<https://www.miracletutorials.com/hls-adaptive-streaming-tutorial-with-cloudfront-jw-player/>

A step-by-step HLS adaptive streaming tutorial with CloudFront & JW Player in two parts. It is easier than you think. This

tutorial presumes you have

HOW TO: View an HLS Stream in QuickTime or VLC – Softron Support Desk

https://softron.zendesk.com/hc/en-us/articles/207694617-HOW-TO-View-an-HLS-Stream-in-QuickTime-or-VLC?mobile_site=true

Tutorial demonstrating how to playback HLS streams using QuickTime and VLC media players on desktop systems.

HTTP Video Streaming Troubleshooting

<https://github.com/leandromoreira/http-video-streaming-troubleshooting>

A collection of fixes / problem solutions to HTTP video streaming

Tags: troubleshooting, http, streaming, solutions, guide

Historical Timeline of Video Coding Standards and Formats

<https://www.vcodex.com/historical-timeline-of-video-coding-standards-and-formats/>

An in-depth white paper by Vcodex BV that provides a comprehensive timeline of the development of video coding standards and formats. It covers the evolution of major standards like MPEG-1, MPEG-2, MPEG-4, H.261, H.263, H.264, H.265/HEVC, and H.266/VVC, offering insights into their key features and historical context.

Tags: video coding, standards, history, MPEG, H.26x

History of Video Technology

<https://public.websites.umich.edu/~evideo/1940.html>

A comprehensive timeline detailing the evolution of video technology from the 1940s to the 1990s, covering significant milestones in video recording and playback.

Tags: video technology, history, video recording, video playback, timeline

How To Stream Video: A Beginner's Guide

<https://web.archive.org/web/20260313011439/https://www.inetsolutions.org/how-to-stream-video-a-beginners-guide/>

A beginner's guide covering the essentials of setting up audio and video sources, choosing streaming destinations, and understanding the basics of live streaming.

Tags: video streaming, beginner guide, Inet Solutions

How to Build a Video Editor with Wasm in React

<https://img.ly/blog/how-to-build-a-video-editor-with-wasm-in-react/>

This tutorial walks you through building a WebAssembly-based video editor in React using ffmpeg.wasm. The application allows users to upload, trim, and convert videos to GIFs directly in the browser without backend services.

Tags: WebAssembly, React, FFmpeg, video editor, tutorial

How to Build a Video Streaming App? - Development Guide 2024

<https://theninehertz.com/blog/how-to/build-a-video-streaming-app>

A development guide by The NineHertz that provides insights into building a video streaming app, covering features, technology stack, and top video streaming apps.

Tags: video streaming app, development guide, features, technology stack

How to Develop a Video Streaming Service in 2024

<https://toxigon.com/developing-a-video-streaming-service>

A detailed guide by Toxigon that outlines the steps to develop a video streaming service, including planning, choosing the right technology stack, and ensuring video quality.

Tags: video streaming service, development, technology stack, video quality

How to Live Stream: Beginners Guide to Get Started

<https://academy.wedio.com/how-to-live-stream/>

A guide providing five simple steps to start live streaming, including choosing the right platform, setting up equipment, and engaging with your audience.

Tags: live streaming, beginner guide, Wedio

How video streaming works on the web: An introduction

<https://medium.com/canal-tech/how-video-streaming-works-on-the-web-an-introduction-7919739f7e1>

Note: this article is an introduction to video streaming in JavaScript and is mostly targeted to web developers. A large part of the...

HowVideo.works

<https://howvideo.works>

Educational resource explaining how video technology works from codec fundamentals to streaming delivery.

IBM Video Streaming: Basics of Streaming Video Production

<https://support.video.ibm.com/hc/en-us/articles/207852167-Basics-of-Streaming-Video-Production>

An article detailing the five essential elements of any streaming production setup, providing foundational knowledge for setting up and managing video streaming workflows.

Tags: video production, streaming setup, video signal, audio signal, encoding

In-browser Transcoding of Video Files with FFmpeg and WebAssembly

<https://blog.scottlogic.com/2020/11/23/ffmpeg-webassembly.html>

This article explores using the WebAssembly build of FFmpeg to perform video transcoding directly within the browser. It provides a step-by-step guide to setting up FFmpeg.wasm and demonstrates how to transcode video files client-side.

Tags: WebAssembly, FFmpeg, video transcoding, browser, tutorial

Integrating FFmpeg with React.js via WebAssembly

<https://web.archive.org/web/20251005160314/https://www.editframe.com/guides/integrating-ffmpeg-with-react.js-via-webassembly>

This guide explores utilizing FFmpeg within a React.js environment, leveraging WebAssembly for browser-based video processing tasks. It provides a step-by-step approach to setting up a React project with FFmpeg.wasm, demonstrating client-side video manipulations such as format conversion and adding text overlays.

Tags: WebAssembly, FFmpeg, React.js, video processing, client-side

Internet Video Streaming — ABR part 1

https://medium.com/@eyevinntechnology/internet-video-streaming-abr-part-1-b10964849e19?source=userActivityShare-94bcb50d11-1559723768&_branch_match_id=664736558865703297

First part of Eyevinn's introduction to adaptive bitrate streaming concepts and fundamentals for video delivery.

Internet Video Streaming — ABR part 2

https://medium.com/@eyevinntechnology/internet-video-streaming-abr-part-2-dbce136b0d7c?source=userActivityShare-94bcb50d11-1559723862&_branch_match_id=664736952377004405

Second part of Eyevinn's ABR series covering advanced adaptive bitrate streaming techniques and implementation.

Introduction to Digital Video Technology

<https://wangwei1237.github.io/2020/02/28/Introduction-to-digital-video-technology/>

An article providing an overview of how video codecs work, including a brief history and explanation of key concepts in digital video technology.

Tags: video codecs, digital video, technology, history, introduction

Introduction to H.264: (1) NAL Unit | Yumi Chan's Blog

<https://yumichan.net/video-processing/video-compression/introduction-to-h264-nal-unit/>

This article is to introduce H.264 and provide a overview of NAL Unit. It also explains NALU header byte which contains the basic information of a NALU.

Introduction to HTTP Live Streaming: HLS on Android and More

<https://www.toptal.com/apple/introduction-to-http-live-streaming-hls>

This article explains how HTTP Live Streaming works and demonstrates how to create an HLS player in Android.

Introduction to the WebCodec API - Real Time Video Encoding and Display

<https://dev.to/ethand91/introduction-to-the-webcodec-api-real-time-video-encoding-and-display-1b54>

A tutorial introducing the WebCodec API, demonstrating how to capture video from a user's camera, encode it in real time using VP8 encoding, and display both the raw video feed and the encoded frames on the screen.

Tags: WebCodec API, video encoding, real-time processing, VP8, video display

Learned Scalable Video Coding For Humans and Machines

<https://github.com/hadipardis/svc>

An end-to-end learnable video codec that supports machine vision tasks in its base layer and input reconstruction for human viewing in its enhancement layer. It introduces a new paradigm for video coding that efficiently represents and compresses video for both machine and human use.

Tags: video coding, machine vision, learnable codec, scalable video coding

Live Streaming for Dummies: How to Get Started

<https://umatechnology.org/live-streaming-for-dummies-how-to-get-started/>

A comprehensive guide that breaks down everything you need to know about getting started with live streaming, ensuring that even a complete novice can dive right in.

Tags: live streaming, beginner guide, video production

Matrox Video Learning Resources

<https://video.matrox.com/en/learning/resources>

A collection of technology guides, webinars, and on-site training seminars provided by Matrox Video experts. Topics include multi-channel encoding, AV over IP, streaming protocols, and KVM solutions, offering in-depth knowledge for both beginners and professionals.

Tags: multi-channel encoding, AV over IP, streaming protocols, KVM solutions, video technology

Muvi's Guide to Setting Up a Video Streaming Server

<https://www.muvi.com/blogs/video-streaming-server/>

A detailed guide that walks users through the process of setting up their own video streaming server in five simple steps. It covers topics such as video streaming basics, types of video streaming, and the technical architecture involved.

Tags: video streaming server, live streaming, on-demand streaming, streaming setup, video technology

Netflix

<https://github.com/nickdesaulniers/netfix>

Educational project walking through building a Netflix-like streaming platform. Demonstrates VOD architecture basics.

Netflix TechBlog – Official Blog

<https://netflixtechblog.com/>

Netflix's engineering blog often covers innovations in video encoding, quality metrics, and streaming architecture that Netflix has developed and open-sourced.

Tags: engineering, video-technology, streaming, encoding, open-source

Netflix Video Quality at Scale with Cosmos Microservices

<https://netflixtechblog.com/netflix-video-quality-at-scale-with-cosmos-microservices-35cf6047ed83>

A deep dive into Netflix's Cosmos platform and how its microservice architecture enables large-scale video quality improvements (like per-title and per-shot encoding) across Netflix's catalog.

Tags: microservices, netflix, video encoding, case study, scaling, video quality

Node.js Tutorial: How to Build a Video Streaming Application in 10 Minutes

<https://hackernoon.com/nodejs-tutorial-how-to-build-a-video-streaming-application-in-10-minutes>

This tutorial on HackerNoon provides a quick and practical guide to building a video streaming application using Node.js. It includes step-by-step instructions on setting up the server, serving HTML pages, and handling video streaming. This resource is particularly useful for developers looking to implement video streaming capabilities in their Node.js applications.

Tags: Node.js, video streaming, tutorial, web development, backend

OTT Content Delivery

<https://medium.com/@eyevinntechnology/ott-content-delivery-b43a35ef24f6>

Overview article on OTT content delivery architecture and strategies. Covers video distribution fundamentals.

OpenTelemetry Video QoE Metrics (Article)

<https://medium.com/@opentelemetry/video-streaming-qoe-monitoring-with-opentelemetry-1d9f3c2e0715>

An article on using OpenTelemetry (an open source observability framework) to collect and analyze Quality of Experience (QoE) metrics from video players. It demonstrates how to instrument players to send standardized metrics for buffering, startup time, etc., into a monitoring system.

Tags: observability, metrics, QoE, instrumentation, tutorial, performance

OpenVVC: A Lightweight Software Decoder for the Versatile Video Coding Standard

<https://github.com/OpenVVC/OpenVVC>

An open-source software decoder developed by researchers that supports a broad range of VVC functionalities. OpenVVC is designed to achieve real-time decoding of UHD video content with low memory requirements, making it suitable for embedded platforms. The source code is available on GitHub.

Tags: VVC, decoder, open-source, real-time, UHD

PTZOptics Hive Tutorial Series

<https://web.archive.org/web/20250905130722/https://remoteproduction.com/the-best-resources-for-learning-remote-production-technology/>

A YouTube tutorial series by PTZOptics focusing on remote camera control and cloud-based video production. The series offers step-by-step guides on effectively using Hive for managing remote productions, covering topics like camera linking and dynamic zoom.

Tags: remote production, video streaming, PTZOptics, tutorial, Hive

Per-Title Encoding: 5 Implementation Tips – Bitmovin

<https://bitmovin.com/per-title-encoding-5-tips/>

A Bitmovin guide that offers practical tips for implementing Per-Title encoding (content adaptive encoding) in a video workflow. It builds on Netflix's concept but provides advice gleaned from Bitmovin's own encoder, such as deciding on complexity metrics and fallback conditions.

Tags: encoding, tutorial, adaptive-bitrate, video-optimization

Qt Multimedia on WebAssembly

<https://doc.qt.io/qt-6/qtmultimedia-wasm.html>

This documentation provides insights into the availability and limitations of Qt Multimedia features on WebAssembly. It covers aspects such as device access, video playback, and performance considerations, serving as a valuable resource for developers looking to port multimedia applications to the web using Qt and WebAssembly.

Tags: WebAssembly, Qt, multimedia, documentation, video playback

Quantifying packaging overhead

<https://mux.com/blog/quantifying-packaging-overhead-2>

Mux makes adding video to your app or website as easy as making a single API call. But behind the scenes is a large multistep process to analyze and transform the video into something that can be easily consumed by a device. This process is commonly called a media "pipeline"

Tags: video processing, api, media pipeline, technical tutorial

Quick Tutorial: How to Increase Volume in Audacity [2019 Update]

<https://www.iskysoft.com/video-editing/how-to-increase-volume-in-audacity.html>

How to increase volume in Audacity? This article will guide you to change volume in Audacity and its alternative tool. You can pick up one of them to edit volume in Audacity as you like.

Tags: audio, tutorial, editing, volume, audacity

RaftLabs Guide to Building Video Streaming Apps

<https://www.raftlabs.com/blog/build-live-streaming-app-best-approach-tech-stack/>

A comprehensive guide detailing the development of live and on-demand video streaming applications. It covers types of video streaming, business use cases, essential and advanced features, tech stacks, and insights from RaftLabs' experience as a video app development company.

Tags: live streaming app, on-demand video, video app development, tech stack, business use cases

Rebuilding Netflix Video Processing Pipeline with Microservices

<https://netflixtechblog.com/rebuilding-netflix-video-processing-pipeline-with-microservices-7ed3d6f805bf>

Netflix Tech Blog post (part of a series) describing how Netflix migrated their video processing pipeline to a microservices architecture to increase agility and scalability.

Tags: microservices, netflix, video processing, architecture, case study

SDK Development Guide

<https://github.com/rokudev/SDK-Development-Guide>

Contribute to rokudev/SDK-Development-Guide development by creating an account on GitHub.

Saving on Encoding and Streaming: Deploy Capped CRF – Streaming Learning Center

<https://streaminglearningcenter.com/blogs/saving-encoding-streaming-deploy-capped-crf.html>

This is the second in a five-part series on how to cut your encoding and streaming costs. The first article was Saving on Encoding: Adjust Encoding Configuration to Increase Capacity. Article summary: Capped CRF encoding is a single-pass encoding method that can save encoding costs compared to two-pass VBR. Capped CRF is also a simple per-title...

Smithsonian Video Format Resources

<https://www.si.edu/tbma/video-format-resources>

A curated list of resources by the Smithsonian Institution, offering guidance on video format specifications, digitizing analog videotapes, and tools for video format identification.

Tags: video formats, digitization, video preservation, media tools, Smithsonian

Stack Exchange – Video Production

<https://video.stackexchange.com/>

A Q&A community for video production and post-production. Topics often include editing, codecs, formats, and troubleshooting video workflows (knowledge base format).

Tags: education, community, video production, knowledge base, troubleshooting, editing, codecs, formats

Streaming Learning Center's Beginner's Guide to Live Streaming

<https://streaminglearningcenter.com/learning/beginners-guide-to-live-streaming.html>

A beginner-friendly guide that introduces the basics of live streaming, including equipment setup, encoding, and streaming protocols. This resource is ideal for those new to live streaming and looking to understand the foundational concepts.

Tags: live streaming, beginner guide, video encoding, streaming protocols, equipment setup

Streaming Live From the Battlefield: Military Video in 2019

<https://www.streamingmedia.com/Articles/ReadArticle.aspx?ArticleID=135141>

Metadata and low-latency video create a tactical advantage in intelligence-gathering and decision making. Discover why HEVC is gaining momentum in the armed forces, and Android is preferred over iOS.

Streaming Media 101: Technical Onboarding for Streaming Media Professionals

<https://streaminglearningcenter.com/courses-streaming101>

An in-depth course designed to provide technical onboarding for streaming media professionals, covering essential topics in video streaming technology.

Tags: video streaming, technical onboarding, media professionals

Streaming Onboarding

<https://github.com/Eyevinn/streaming-onboarding>

New to streaming and don't know where to start? This is the place for you!

Tears of Steel (Open Movie)

<https://mango.blender.org/>

A Blender Foundation live-action/CGI short film (2012) released as open content. Often used as test material for visual effects, 4K streaming, and encoding performance evaluations.

Tags: open source, blender, 4K, visual effects, encoding, test material

The Complete Guide to Live Streaming

<https://www.wowza.com/blog/complete-guide-to-live-streaming>

An extensive guide by Wowza Media Systems that delves into the live streaming workflow, including capture, encoding, packaging, protocols, and delivery.

Tags: live streaming, workflow, encoding, protocols, delivery

The Good Parts of LibVLC

<https://mfkl.gumroad.com//libvlc-good-parts>

Guide focusing on practical LibVLC usage patterns. Helps developers integrate VLC media framework effectively.

The History Of Video File Formats [INFOGRAPHIC]

<https://umatechnology.org/the-history-of-video-file-formats-infographic/>

An informative article accompanied by an infographic that traces the evolution of video file formats from early analog formats to the latest digital formats, highlighting how technological advancements have revolutionized video storage and playback.

Tags: video file formats, history, analog to digital, technological advancements, video storage

The History of Video Compression Standards

<https://api.video/blog/video-trends/the-history-of-video-compression-starts-in-1929/>

This article from api.video traces the history of video compression standards from 1929 to the present. It highlights major milestones and developments in video compression, providing a clear understanding of how current standards evolved.

Tags: video compression, standards, history, video technology, learning resource

The Making of VES: the Cosmos Microservice for Netflix Video Encoding

<https://netflixtechblog.com/the-making-of-ves-the-cosmos-microservice-for-netflix-video-encoding-3e6177bfbeab>

A Netflix Tech Blog article detailing the design and development of Netflix's Video Encoding Service (VES), a microservice in their Cosmos platform to improve video encoding processes.

Tags: netflix, microservice, video encoding, case study, infrastructure

The State of Video Codecs: Evolution of Compression in a Video-First Economy

https://www.interdigital.com/white_papers/the-state-of-video-codecs-evolution-of-compression-in-a-video-first-economy-

This white paper by InterDigital explores the landscape of video compression technology, detailing the history of video coding standards and examining the advantages of modern codecs. It provides a comprehensive overview of the evolution of video compression, making it a valuable resource for understanding the current state and future of video coding.

Tags: video codecs, video compression, video technology evolution, learning resource, video streaming

The Ultimate Live Streaming Setup Guide for Beginners

<https://umatechnology.org/the-ultimate-live-streaming-setup-guide-for-beginners/>

An in-depth guide exploring every facet of creating a professional-grade live streaming setup for beginners, covering equipment, software, platform choices, and best practices.

Tags: live streaming setup, equipment guide, video production

The Video Insiders

<https://thevideoinsiders.simplecast.com/episodes>

Podcast covering video technology trends, streaming industry insights, and engineering best practices.

Tizen WASM Player

<https://developer.samsung.com/smarttv/develop/extension-libraries/webassembly/tizen-wasm-player/overview.html>

Tizen WASM Player is a Samsung TV extension API that provides WebAssembly applications with low-level access to the platform's media player. It accepts encoded audio and video frames, decoding and rendering them using the TV's multimedia pipeline. This is particularly useful for developing media applications on Tizen-based devices.

Tags: WebAssembly, Tizen, media player, video streaming, Samsung TV

Tizen WASM Player Sample Application

<https://developer.samsung.com/smarttv/develop/extension-libraries/webassembly/tizen-wasm-player/sample-wasm-player-application.html>

A sample application demonstrating how to use Tizen WASM Player to play media on Tizen TV devices using HTMLMediaElement with ElementaryMediaStreamSource as a data source in a WebAssembly module. It serves as a practical example for developers interested in building media applications for Tizen platforms.

Tags: WebAssembly, Tizen, media player, sample application, Samsung

Tizen WASM Video Decoder - Sample Application

https://github.com/SamsungDForum/WebAssemblyDemos/blob/master/video_decoder_sample/README.md

A sample application demonstrating the use of Tizen WASM Video Decoder to play media on Samsung Smart TV devices using a WebAssembly module. It showcases the integration of WebAssembly for media playback in smart TV applications.

Tags: WebAssembly, video decoding, Tizen, Samsung Smart TV, media playback

Tizen WASM Video Decoder Sample

<https://developer.samsung.com/smarttv/develop/extension-libraries/webassembly/sample-based-tutorials/tizen-wasm-video-decoder-sample.html>

This sample application demonstrates how to use the Tizen WASM Video Decoder to play media on Samsung devices using a WebAssembly module. It extends the Tizen WASM Player Sample Application, illustrating how to decode video frames to GL textures. This resource is beneficial for developers aiming to integrate video playback functionalities into Tizen-based applications.

Tags: Tizen, WebAssembly, video decoding, Samsung, GL texture

Tools | Video History Project

<https://www.videohistoryproject.org/tools>

A collection of resources detailing the technological challenges faced by media makers and the development of video tools from the 1960s onwards.

Tags: video tools, history, media technology, video art, resources

Understanding Rate Control Modes (x264, x265, vpx)

<https://slhck.info/video/2017/03/01/rate-control.html>

What is "rate control"? It's what a video encoder does when it decides how many bits to spend for a given frame. The goal of (lossy) video encoding is to sav...

Using Rust and WebAssembly to Process Pixels from a Video Feed

<https://dev.to/fallenstedt/using-rust-and-webassembly-to-process-pixels-from-a-video-feed-4hhg>

An article that explores how to use Rust and WebAssembly to process raw pixel data from a video feed. It provides insights into accessing and manipulating video frame data efficiently, which is beneficial for developers interested in video processing and analysis using WebAssembly.

Tags: WebAssembly, Rust, video processing, pixel data, tutorial

Video Annotator: Building Video Classifiers using Vision-Language

<https://netflixtechblog.com/video-annotator-building-video-classifiers-using-vision-language-techniques-8609e457d9c>

Netflix Tech Blog on a framework for training ML models for video understanding. Describes an internal system to create video classifiers using combined video and text (subtitle/metadata) analysis.

Tags: machine learning, video classification, netflix, vision-language model, ml framework

Video Codec

https://en.wikipedia.org/wiki/Video_codec

This resource offers an in-depth look at video codecs, explaining their function, history, and the various standards that have

been developed over time. It covers the progression from early codecs like H.261 to modern standards such as H.265, providing a solid foundation for understanding video compression.

Tags: video codec, video compression, video standards, H.261, H.265

Video Coding Basics - How is this so efficient?

<https://youtu.be/LDeL7-49qm4>

An introduction to the basics of video coding

Video Coding Experts Group (VCEG)

<https://www.itu.int/en/ITU-T/studygroups/2022-2024/16/video/Pages/vceg.aspx>

The Video Coding Experts Group (VCEG) is a working group of the ITU-T Study Group 16 responsible for developing international video coding standards. Their website provides access to various video coding standards, including H.120, H.261, and H.263, along with archives and resources related to the development of these standards.

Tags: VCEG, video coding standards, ITU-T, H.261, H.263

Video Coding History

<https://www.vcodex.com/video-coding-history>

An article by Vcodex BV that delves into the history of video coding, discussing the development of compression techniques and their impact on digital video. It highlights key milestones, including the invention of the Discrete Cosine Transform (DCT) and the evolution of video coding standards.

Tags: video coding, history, compression, DCT, standards

Video Coding Standards and Algorithms - Evolution & Patent Analysis

<https://www.copperpodip.com/post/different-types-of-video-coding-standards-and-algorithms-used-over-the-time>

An analysis of various video coding standards and algorithms over time, including their evolution and associated patents.

Tags: video coding, standards, algorithms, patents, evolution

Video Coding Standards and Algorithms — Evolution & Patent Analysis

<https://copperpod.medium.com/video-coding-standards-and-algorithms-evolution-patent-analysis-aaf8cd5994e3>

An article by Copperpod IP that explores the evolution of video coding standards and algorithms. It provides an overview of various video coding formats, their classifications, and the role of patents in the development of these technologies.

Tags: video coding, standards, algorithms, patents, evolution

Video Coding: An Introduction to Standard Codecs

<https://www.globalspec.com/reference/34024/203279/video-coding-an-introduction-to-standard-codecs>

This comprehensive guide explains video coding algorithms and compression systems, covering the history of video coding, principles of video compression, and standard codecs such as JPEG, H.261, MPEG-1, MPEG-2, H.263, and MPEG-4. It serves as an in-depth resource for understanding the development and functionality of standard video codecs.

Tags: video coding, standard codecs, compression, history, MPEG

Video Compression Handbook

<https://www.amazon.com/dp/0134866215/>

2018-07-03 (2nd Edition). Beach, Andy; Owen, Aaron.

Video Encoding - OTTVerse

<https://ottverse.com/video-encoding/>

A comprehensive resource covering various aspects of video encoding, including tools and libraries like FFmpeg, advanced topics such as content adaptive encoding, and discussions on codecs like HEVC and AV1. This resource is beneficial for developers looking to deepen their understanding of video encoding technologies.

Tags: video encoding, FFmpeg, HEVC, AV1, content adaptive encoding

Video Encoding 101: A Beginner's Guide to Key Concepts

<https://www.forasoft.com/blog/article/video-encoding-streaming-quality>

An introductory guide that explains the fundamental concepts of video encoding, including compression techniques, color encoding, and the differences between various encoding approaches. It's a valuable resource for beginners seeking to understand the basics of video encoding.

Tags: video encoding, compression techniques, color encoding, beginner's guide, video quality

Video Encoding and Transcoding Guide by OTTVerse

<https://ottverse.com/what-is-video-encoding-transcoding-codecs-compression/>

An in-depth guide explaining video encoding and transcoding concepts, including codecs, compression techniques, and the differences between encoding and transcoding.

Tags: video encoding, video transcoding, codecs, compression, video technology

Video Encoding for Beginners

<https://www.mindluster.com/certificate/16034/Video-encoding-video>

An introductory course that explores essential concepts and techniques for compressing and converting videos efficiently. Topics include codecs like H.264 and H.265, containers such as MP4 and MKV, resolution settings, bitrate adjustment, and practical sessions using tools like FFmpeg and HandBrake.

Tags: video encoding, H.264, H.265, FFmpeg, HandBrake

Video Encoding — Compression and Resolutions

<https://medium.com/@eyevinntechnology/chessboard-for-beginners-video-encoding-compression-and-resolutions-bcefe04fa639>

Written by: Boris Asadanin, Streaming Media Consultant at Eyevinn Technology

Video Encoding: The Big Streaming Technology Guide [2023]

<https://bitmovin.com/video-encoding-guide>

A comprehensive guide that delves into video encoding, covering topics such as video compression, encoding formats, hardware vs. software encoding, and adaptive bitrate streaming. This resource is essential for developers aiming to understand the intricacies of video encoding in streaming technology.

Tags: video encoding, video compression, streaming technology, adaptive bitrate, developer guide

Video Encoding: The Definitive Guide

<https://www.dacast.com/blog/what-is-video-encoding/>

An updated guide by Dacast that explains video encoding, compression, formats, codecs, and transcoding, providing a solid foundation for developers in the streaming industry.

Tags: video encoding, compression, codecs, transcoding

Video Frame Processing on the Web – WebAssembly, WebGPU, WebGL, WebCodecs, WebNN, and WebTransport

<https://webtrchacks.com/video-frame-processing-on-the-web-webassembly-webgpu-webgl-webcodecs-webnn-and-webtransport/>

This article discusses various technologies available for video frame processing on the web, including WebAssembly. It provides an overview of how these technologies can be utilized for real-time video processing, offering valuable insights for developers interested in advanced video processing techniques.

Tags: WebAssembly, video processing, WebGPU, WebGL, WebCodecs

Video History Project by the Experimental Television Center

<https://www.videohistoryproject.org/introduction>

The Video History Project is a dynamic platform that compiles an inclusive media history, featuring databases on people, tools, groups, and an extensive bibliography. It serves as a valuable resource for exploring the development and preservation of video technology.

Tags: video history, media preservation, experimental television, bibliography, media arts

Video Resource Center by TV Tech

<https://www.tvtechnology.com/miscellaneous/video-resource-center>

A curated collection of training and informational videos covering various aspects of video technology, including tutorials on MPEG compression and high-efficiency video coding.

Tags: video technology, resources, tutorials, compression

Video Streaming App Development: A Step-by-Step Guide

<https://www.wowza.com/blog/video-streaming-app-development-a-step-by-step-guide>

A comprehensive guide by Wowza that walks through the process of developing a video streaming application, covering aspects from defining goals to selecting the right technology stack.

Tags: video streaming, app development, technology stack, guide

Video Streaming Cheatsheet

<https://video-streaming-cheatsheet.s3.eu-west-2.amazonaws.com/artifacts/video-streaming-cheatsheet.pdf>

The Video Streaming Cheatsheet is a double sided page, suitable for printing, that contains common terminology used in the video streaming industry in a quick reference format

Video Streaming Server Guide by Ant Media

<https://antmedia.io/how-to-make-a-video-streaming-server/>

A comprehensive guide on building a video streaming server, covering topics from understanding video streaming platforms to step-by-step instructions on creating your own server. It's beneficial for developers looking to set up custom streaming solutions.

Tags: video streaming, server setup, guide, tutorial, Ant Media

Video Technology History (1960s-1990s)

<https://public.websites.umich.edu/~evideo/1960.html>

A series of pages detailing the history of video technology from the 1960s through the 1990s, covering significant developments such as the invention of helical scanning, the release of home VTRs, and the introduction of digital video formats.

Tags: video technology, history, VTR, digital video, developments

Video and containers

<https://neurocline.github.io/dev/2016/07/28/video-and-containers.html>

NALU, Annex B, and Start Codes

Video in the War Zone: The Current State of Military Streaming

<https://www.streamingmedia.com/Articles/ReadArticle.aspx?ArticleID=101310>

For the armed forces, streaming is a matter of national security. Here's an exclusive look at how the military, from analysts to ground troops, is using streaming video.

Video: HLS and DASH Multi-Codec Encoding & Packaging

<https://thebroadcastknowledge.com/2020/07/10/video-hls-and-dash-multi-codec-encoding-packaging/>

Free educational webinars, videos and other resources focused on the Broadcast Industry

VideoHelp

<https://www.videohelp.com/>

A comprehensive website offering tools, guides, and forums for video conversion, editing, and troubleshooting, popular among video enthusiasts for its vast information on formats and software.

Tags: video conversion, tutorials, software guides, community learning, video formats

Vittorio Giovara - Color Me Intrigued: A Jaunt Through Color Technology in Video

<https://www.youtube.com/watch?v=XMnvY7a4-As&feature=share>

This talk aims to shed light on colorspaces - what they are, how and why they work, why we should care about handling edge cases properly. Starting with hist...

WASM Video Filters

https://ontouchstart.pages.dev/wasm_video_filters

WASM Video Filters is a demonstration of applying video filters using WebAssembly. Based on the MDN tutorial on manipulating video using canvas, this project illustrates how to implement video processing functionalities in web applications through WebAssembly.

Tags: WebAssembly, video filters, video processing, web development, canvas

WASM-video-editor

<https://github.com/metal-face/WASM-video-editor>

A video editor built using WebAssembly and FFmpeg, this project provides a practical example of utilizing WebAssembly for client-side video editing tasks. It demonstrates the potential of combining WebAssembly with powerful multimedia libraries like FFmpeg to create efficient and responsive web applications.

Tags: WebAssembly, video editor, FFmpeg, client-side, multimedia

WasmVideoEncoder

<https://github.com/AlexVestin/WasmVideoEncoder>

WasmVideoEncoder utilizes FFmpeg compiled to WebAssembly to provide video encoding capabilities directly in the browser. It encodes raw image and audio data into MP4 format, making it suitable for applications like music video creation.

Tags: WebAssembly, FFmpeg, video encoding, browser-based, MP4

WasmVideoPlayer

<https://github.com/sonysuqin/WasmVideoPlayer>

WasmVideoPlayer is a web-based video player that utilizes WebAssembly, WebGL, and Web Audio API, with FFmpeg for multi-codec support, including H.265. It supports various streaming protocols such as HTTP, WebSocket, and HTTP-FLV. This project is valuable for developers aiming to implement versatile video playback solutions in web environments.

Tags: WebAssembly, video player, FFmpeg, WebGL, Web Audio API

Web-Demuxer

<https://foreversc.github.io/web-demuxer/>

Web-Demuxer is designed to demux various media formats directly in the browser using WebAssembly. It supports formats like mov, mp4, mkv, webm, flv, m4v, wmv, and avi, providing a friendly API for WebCodecs development.

Tags: WebAssembly, demuxer, media formats, WebCodecs, browser-based

WebASM HEVC Decoder SDK

<https://www.mainconcept.com/webasm>

The WebASM HEVC Decoder SDK by MainConcept provides a browser-based solution for HEVC/H.265 video decoding using WebAssembly. It enables high-quality, low-bandwidth video playback directly in browsers without the need for specialized hardware or software, supporting resolutions up to 8K and various decoding profiles.

Tags: WebAssembly, HEVC, H.265, video decoding, browser-based

WebAssembly (Wasm)

https://docs.google.com/presentation/d/1sonEk2neNVBcy8EzieUjWCNzj5SXN7dk-unkR_lpl8k/edit?usp=sharing

WebAssembly (Wasm) On the Edge Kevin Staunton-Lambert Solutions Architect R&D @kevleyski www.switch.tv Wasm)

WebAssembly Samples by Samsung Developer

<https://developer.samsung.com/smarttv/develop/samples/webassembly-samples.html>

A collection of sample applications demonstrating the use of WebAssembly for video playback and decoding on Tizen devices. These samples illustrate how to develop WASM Player and WASM Video Decoder in web applications, providing practical examples for integrating WebAssembly into video technology.

Tags: WebAssembly, video playback, Tizen, sample applications, video decoding

WebAssembly Video Filters

<https://github.com/HiDoYa/webassembly-video-filters>

This project implements performant video filters in the browser using WebAssembly. It includes various video scopes such as lumascope, RGB parade, and vectorscope, written in C++ and translated to WebAssembly for efficient in-browser video processing.

Tags: WebAssembly, video filters, C++, browser, video scopes

WebDSP

<https://github.com/shamadee/web-dsp>

WebDSP is a client-side signal processing library utilizing WebAssembly to perform high-performance operations on media data. It offers a collection of algorithms for video and image processing, including filters like grayscale, brighten, invert, and convolutional filters, designed to be building blocks for web applications.

Tags: WebAssembly, signal processing, video processing, image processing, browser

WebRTC Samples

<https://webrtc.github.io/samples/>

A collection of WebRTC demos and sample applications illustrating real-time peer-to-peer video, audio, and data streaming in web browsers using the WebRTC API.

Tags: WebRTC, tutorial, peer-to-peer, web, streaming, demo, javascript

WebRTC Tutorials - 36 Essential Learning Resources

<https://getstream.io/blog/webrtc-tutorials/>

A curated list of 36 essential WebRTC tutorials and learning resources, covering various aspects of WebRTC development and implementation.

Tags: WebRTC, tutorials, learning resources, real-time communication

WebRTC-Experiment

<https://github.com/muaz-khan/WebRTC-Experiment>

A repository of WebRTC experimental demos by Muaz Khan, including source code for various real-time communication use cases (screen sharing, peer connections, recording, etc.).

Tags: WebRTC, peer-to-peer, real-time communication, demos, source code

What Is Video Encoding? | Encoding Formats and How-To

<https://www.wowza.com/blog/what-is-video-encoding>

An informative article that explains video encoding, detailing various encoding formats suitable for live streaming, including H.264/AVC, H.265/HEVC, AV1, VP9, and H.266/VVC. This resource is beneficial for developers seeking to understand the different encoding formats and their applications in live streaming.

Tags: video encoding, encoding formats, live streaming, H.264, H.265

WildFires

<https://docs.google.com/presentation/d/1yiVEOq2rvtFynP1tLdJj7pBWkAEiE9g8BmaoryxRVrk/edit?usp=sharing>

VOD2Live Kevin Staunton-Lambert Solutions Architect R&D @kevleyski Wild Fire! How video engineers can help save lives www.switch.tv

Wowza Developer Portal

<https://www.wowza.com/developer>

A hub for developers offering tools, APIs, and documentation for building live streaming applications. It includes tutorials on live streaming basics, asset management, analytics, and more.

Tags: live streaming, APIs, developer tools, video analytics, streaming protocols

Wowza Streaming Engine: Play Your First Live Stream

<https://info.wowza.com/trial-getting-started-play-a-live-stream>

A step-by-step guide on setting up a live streaming application using Wowza Streaming Engine and Wowza GoCoder. The tutorial covers creating a live application, connecting to a live encoder, and testing incoming streams.

Tags: Wowza Streaming Engine, live streaming, GoCoder, streaming setup, video encoding

YouTube Creator Academy – Captions

<https://creatoracademy.youtube.com/page/lesson/captions>

A tutorial from YouTube's Creator Academy about adding captions and subtitles to videos. It explains how to create, upload, and auto-sync captions on YouTube, highlighting the importance of accessibility and offering tips for quality subtitle contributions.

Tags: tutorial, accessibility, captions, subtitles, youtube, education

ffmpeg tutorial

<http://www.dranger.com/ffmpeg/>

Tutorial series teaching FFmpeg usage and video processing fundamentals for developers.

Tags: ffmpeg, tutorial, video processing, encoding, learning

ffmpeg-wasm

<https://github.com/wide-video/ffmpeg-wasm>

ffmpeg-wasm is a WebAssembly port of FFmpeg, providing scripts and commands to build FFmpeg for WebAssembly. It powers applications like wide.video, a free online video editor, and ffmpeg.wide.video, an online FFmpeg tool.

Tags: WebAssembly, FFmpeg, video processing, online editor, browser

ffmpeg.wasm webcam Example

<https://gist.github.com/phgoodfriend/e356648e000658fa84ce50865fe3f189>

This example demonstrates how to use ffmpeg.wasm to record video from a webcam, transcode it to MP4 (x264), and play it back in the browser. It serves as a practical guide for integrating ffmpeg.wasm into web applications for real-time video processing.

Tags: WebAssembly, FFmpeg, webcam, video recording, browser

hls.js Low-Latency Playlist Simulator

https://hls-js.netlify.app/demo/?src=LL-HLS_URL

An online demo environment for hls.js where one can input a Low-Latency HLS playlist URL to test playback. Useful for testing LL-HLS streams in a real player implementation and observing buffering and timing behaviors.

Tags: hls, low-latency, streaming, demo, testing, playback

libass-wasm

<https://www.npmjs.com/package/libass-wasm>

libass-wasm is a WebAssembly port of libass, a library for rendering subtitles in the ASS format. It integrates easily with HTML5 videos, supporting most SSA/ASS features and various font formats, ensuring consistent subtitle rendering across platforms.

Tags: WebAssembly, subtitles, libass, HTML5 video, browser

libmpv-wasm

<https://github.com/brianhvo02/libmpv-wasm>

libmpv-wasm is a port of the mpv media player to WebAssembly, allowing it to run entirely in the browser. It aims to expand the range of audio and video codecs supported in web applications and provides a TypeScript API along with a React demo. This project is valuable for developers looking to integrate a comprehensive media player into web applications with broad codec support.

Tags: WebAssembly, mpv, media player, TypeScript, React

mp4-wasm-encoder

<https://github.com/mattdesl/mp4-wasm-encoder>

mp4-wasm-encoder is a client-side MP4 encoding demo that leverages WebAssembly and Web Workers to create H.264-encoded MP4 videos in the browser. It focuses on performance optimization and minimal dependency size, making it suitable for web applications requiring efficient video encoding.

Tags: WebAssembly, MP4, video encoding, client-side, performance

react-wasm-video-editor

<https://github.com/mike-at-redspace/react-wasm-video-editor>

This project integrates WebAssembly with React to create a video editor using FFmpeg. It serves as an example of how to build interactive video editing applications in the browser, leveraging the performance benefits of WebAssembly combined with the flexibility of React.

Tags: WebAssembly, React, video editing, FFmpeg, browser-based

server-side-wasm-video-editing

<https://github.com/second-state/server-side-wasm-video-editing>

This project demonstrates how to build server-side applications that utilize SecondState's Wasm VM for video editing. It showcases frame-by-frame, pixel-by-pixel processing using native Wasm data types, providing a foundation for developers interested in leveraging WebAssembly for server-side video processing tasks.

Tags: WebAssembly, video editing, server-side, SecondState, Wasm VM

video-editor-wasm-react

<https://github.com/imgly/video-editor-wasm-react>

A web-based video editor built with React and WebAssembly, featuring functionalities like video trimming and conversion to GIF. It leverages ffmpeg.wasm to perform video processing directly in the browser, offering a seamless user experience without server-side dependencies.

Tags: WebAssembly, React, video editing, ffmpeg.wasm, browser-based

wasm-av1

<https://github.com/GoogleChromeLabs/wasm-av1>

wasm-av1 is a project that ports the AV1 video codec to WebAssembly, allowing in-browser decoding of AV1 videos. This enables developers to experiment with and implement AV1 video playback directly within web applications.

Tags: WebAssembly, AV1, video codec, in-browser, video decoding

wasm-codecs

<https://github.com/adminky/wasm-codecs>

wasm-codecs combines libde265 and libopus into a single WebAssembly file, providing H.265 (HEVC) and Opus decoding capabilities for web applications. This project enables efficient decoding of high-quality video and audio streams directly in the browser.

Tags: WebAssembly, libde265, libopus, video decoding, audio decoding

wasm-media-encoders

<https://www.npmjs.com/package/wasm-media-encoders>

wasm-media-encoders provides LAME MP3 and Ogg Vorbis audio encoders compiled to minimal WebAssembly, ready for use in the browser or Node.js. It addresses the limitations of the MediaRecorder API by offering additional codec support, enabling efficient client-side audio encoding.

Tags: WebAssembly, audio encoding, MP3, Ogg Vorbis, browser

wasm-resources

<https://github.com/byrneml/wasm-resources>

A curated list of projects, companies, and videos related to WebAssembly, documenting various interesting aspects of the Wasm ecosystem. This resource is valuable for developers seeking to explore and learn about different applications of WebAssembly.

Tags: WebAssembly, resources, projects, companies, videos

wasmVision

<https://github.com/wasmvision/wasmvision>

wasmVision provides a high-performance computer vision processing engine designed to be customized and extended using WebAssembly. It captures video from various sources, processes frames with WebAssembly modules, and outputs results to streams or files.

Tags: WebAssembly, computer vision, video processing, GoCV, Wazero

wasmbooth: Video Filter Booth App Written in Rust

<https://github.com/mtharrison/wasmbooth>

wasmbooth is a video effect booth application written in Rust and WebAssembly. It streams webcam input into an offscreen video element, processes each frame with selected filters, and displays the output on a visible canvas. The project serves as a learning resource for integrating Rust with WebAssembly for video processing.

Tags: WebAssembly, Rust, video processing, filters, learning resource

wasmplyer

<https://github.com/dolphin-run/wasmplyer>

A WebAssembly-based player capable of playing H.264 videos using FFmpeg. This project demonstrates the potential of running complex video decoding tasks directly in the browser, reducing the need for native plugins or server-side processing.

Tags: WebAssembly, FFmpeg, video player, H.264, browser-based

webassembly-pixel-processing

<https://github.com/Fallenstedt/webassembly-pixel-processing>

A project that showcases how to access raw pixel data from a video feed using Rust and WebAssembly. It's particularly useful for developers interested in real-time video processing and analysis directly in the browser.

Tags: WebAssembly, Rust, video processing, pixel data, real-time

webm-wasm

<https://github.com/GoogleChromeLabs/webm-wasm>

webm-wasm enables creating WebM videos directly in JavaScript via WebAssembly. It consumes raw RGBA32 buffers, making it compatible with ImageData from a canvas, and converts them into WebM video.

Tags: WebAssembly, video encoding, JavaScript, WebM, canvas

x264 FFmpeg Options Guide - Linux Encoding

<https://sites.google.com/site/linuxencoding/x264-ffmpeg-mapping>

Reference guide mapping x264 encoder options to FFmpeg parameters for video encoding on Linux systems.

youtube-upload

<https://github.com/tokland/youtube-upload>

A command-line Python script that uploads videos to YouTube using the YouTube Data API, facilitating automated or batch uploads.

Tags: python, youtube, api, upload, automation, tutorial

Media Tools

3DLUT Creator

<https://3dlutcreator.com/>

A stand-alone software to create and edit 3D LUTs (Look Up Tables) for color grading. It provides a unique interface for manipulating the color response of images and videos, which can then be applied via LUT in other grading software or on monitors.

Tags: color grading, lut, video processing, color correction, post-production

4K Media | Free Ultra-HD / HDR / HLG / Dolby Vision 4K Video Demos

<https://4kmedia.org/>

Uncompressed 4K demos, samples, and trailers, to show off your new ultra-HD (2160p) HDR/HLG/Dolby Vision television or monitor.

Tags: 4K, HDR, demo, video samples, ultra-HD, display testing

AAC Audio Coding Standard - ISO/IEC 14496-3

<https://www.iso.org/standard/76383.html>

Defines the compressed audio format AAC

AM Viewer

<https://github.com/DolbyLaboratories/AM-Viewer>

Audio Metadata Viewer. Contribute to DolbyLaboratories/AM-Viewer development by creating an account on GitHub.

AVRate

<https://github.com/Telecommunication-Telemedia-Assessment/AVRate>

An open source modular Audio/Visual subjective evaluation test interface

Ad Insertion Sample

<https://github.com/OpenVisualCloud/Ad-Insertion-Sample>

The ad-insertion reference pipeline shows how to integrate various media building blocks, with analytics powered by the OpenVINO™ Toolkit, for intelligent server-side ad insertion. - OpenVisualClou...

Aegisub

<https://www.aegisub.org/>

A free, cross-platform open source tool for creating and modifying subtitles (advanced subtitle editor supporting ASS/SSA formats).

Tags: subtitle, editor, open-source, cross-platform, text-processing

Aeneas

<https://github.com/readbeyond/aeneas>

Aeneas is a Python library and command-line tool for automating the synchronization of audio and text (forced alignment). It can be used to align subtitles with audio tracks, making it useful for developers working on video streaming and encoding projects that require precise subtitle timing.

Tags: audio-text alignment, Python, forced alignment, subtitles, synchronization

Akamai Stream Validator

<https://players.akamai.com/stream-validator>

Akamai's Stream Validator is a tool designed to verify the integrity and compliance of streaming media, ensuring optimal playback across devices.

Amara

<https://amara.org/>

Amara is a web-based platform that enables users to create, edit, and translate subtitles collaboratively. It supports a wide range of video formats and integrates with platforms like YouTube and Vimeo, making it a versatile tool for developers working on video streaming and encoding projects.

Tags: subtitle, collaboration, web-based, translation, video

Anime4K

<https://github.com/bloc97/Anime4K>

A High-Quality Real Time Upscaler for Anime Video.

Applicaster Video Ads Integration

<https://docs.applicaster.com/using-zapp/monetization/video-ads>

Documentation and tools for integrating video ads into applications using Applicaster's platform, supporting VAST and VMAP tags with the IMA SDK.

Tags: Applicaster, video ads, integration, VAST, VMAP

Artillery Plugin HLS

<https://github.com/artilleryio/artillery-plugin-hls>

Load test HTTP Live Streaming (HLS) servers with Artillery Ø<ß¥

Audio Loudness | Conversational Actions | Google Developers

<https://developers.google.com/assistant/tools/audio-loudness>

Google developer tool for analyzing and measuring audio loudness levels in media content to meet broadcast and platform-specific audio standards.

Audio Offset Finder

<https://github.com/bbc/audio-offset-finder>

Find the offset of an audio file within another audio file

Audio Transcriber

https://github.com/shinobizero/audio_transcriber

Transcodes audio & video files to text, supports MP3, M4A, WAV, MP4, MKV, AVI, MPG & MPEG. No Online API's. Python 3

Audio normalization with ffmpeg using loudnorm (ebur128) filter

<https://bytesandbones.wordpress.com/2017/03/16/audio-normalization-with-ffmpeg-using-loudnorm-ebur128-filter/>

Tutorial on using FFmpeg's loudnorm filter based on EBU R128 standards to normalize and adjust audio levels in media files.

AudioTools Server by Telos Alliance

<https://success.telosalliance.com/file-based-audio-processing>

AudioTools Server is a collection of enterprise-ready audio solutions designed to add file-based expertise to automated environments. It automates sophisticated audio tasks and offers a wide variety of processing options, specifically created for use in cable, satellite, OTT, terrestrial and IPTV, radio, and post-production facilities.

Tags: audio processing, automation, broadcast, post-production, file-based

AudioTrack

<https://developer.android.com/reference/android/media/AudioTrack>

AudioTrack is an Android class that streams PCM audio buffers to audio hardware for playback. It provides low-level access to

audio hardware, allowing developers to implement custom audio playback solutions in their applications.

Tags: Android, audio playback, PCM, audio hardware, low-level API

Auto-Subs

<https://github.com/tmoroney/auto-subs>

Auto-Subs is a tool designed to automatically transcribe editing timelines using OpenAI Whisper and Stable-TS for extreme accuracy. It generates subtitles in a custom style, is completely free, and runs locally within DaVinci Resolve. It works on Mac, Linux, and Windows, supporting both Free and Studio versions of Resolve. Users can jump to positions on the timeline using the Subtitle Navigator and translate from any language to English. The tool provides a user-friendly interface for creating and customizing subtitles for video content.

Tags: automatic transcription, subtitle generation, DaVinci Resolve, OpenAI Whisper, video editing

AutoSub (abhirooptalasila)

<https://github.com/abhirooptalasila/AutoSub>

AutoSub is a CLI application to generate subtitle file (.srt) for any video file using Mozilla DeepSpeech

Automated Channel Testing

<https://github.com/rokudev/automated-channel-testing>

Roku Automated Channel Testing: Selenium-based WebDriver + Robot Framework + Samples

Automated Testing on Devices

<https://medium.com/netflix-techblog/automated-testing-on-devices-fc5a39f47e24>

key concepts and infrastructure

Autosub

<https://github.com/BingLingGroup/autosub>

Command-line utility to transcribe/translate from video/audio/subtitles to subtitles

Avatarify

<https://github.com/alievk/avatarify>

A real-time photorealistic avatar substitution tool. Avatarify uses neural networks to map your facial movements onto a target portrait (e.g., on Zoom or Skype) for fun impersonation or puppeteering.

BackgroundRemover

<https://github.com/nadermx/backgroundremover>

An AI-powered command-line tool to remove the background from images and videos. It uses deep learning to perform background removal with a simple interface.

Big Buck Bunny (4K UHD Demo)

<https://peach.blender.org/>

A short animated film by the Blender Foundation (2008). It's open content and commonly used as demo video footage for testing 4K, HDR, and high-bitrate video setups.

Tags: demo video, 4K, HDR, test footage, resolution testing, video quality

Bitmovin Advertising

<https://bitmovin.com/docs/player/articles/advertising>

Bitmovin's player supports client-side and server-side ad insertion, integrating with major ad servers and networks. It offers support for VAST, VMAP, and VPAID standards, providing a flexible solution for video ad integration.

Tags: video ads, Bitmovin, ad integration, VAST, VPAID

Blackmagic Fusion

<https://www.blackmagicdesign.com/products/fusion/>

A node-based compositing and visual effects software (with a free version integrated into DaVinci Resolve, and a Studio version). Fusion is used for 2D/3D compositing, motion graphics, and effects in films and TV.

Tags: compositing, visual effects, post-production, motion graphics, software

Blender (Video Sequence Editor)

<https://www.blender.org/>

The video sequence editor within Blender, an open source 3D creation suite. It provides basic cutting, transitions, color grading, and masking on top of Blender's powerful compositing nodes.

Tags: video editing, open source, compositing, post-production

Brave

<https://github.com/bbc/brave>

Basic Real-time AV Editor - allowing you to preview, mix, and route live audio and video streams on the cloud

Tags: live streaming, audio routing, video mixing, cloud media, broadcast tools

Brightcove Player Advertising Integration

<https://player.support.brightcove.com/getting-started/learning-guide-video-advertising.html>

Brightcove Player offers comprehensive support for integrating video advertisements using plugins for Google's IMA3 and FreeWheel. It provides detailed documentation and quick-start guides to help developers implement client-side ad insertion seamlessly.

Tags: video ads, ad integration, video player, IMA3, FreeWheel

Building In-Video Search at Netflix

<https://netflixtechblog.com/building-in-video-search-c2f93429d18b>

Netflix Tech Blog article about developing an internal system for searching within video content (e.g., finding scenes across the catalog). It covers the indexing of visual and audio data to enable search.

CALM Act (FCC Audio Loudness Rules)

<https://www.fcc.gov/document/calm-act-guide-industry>

The FCC's Commercial Advertisement Loudness Mitigation (CALM) Act requires broadcasters and streaming providers to ensure ads are not louder than the surrounding programming. This industry guide explains compliance for audio loudness, which influences how audio is processed in video streams.

CAMBI: A Banding Artifact Detector

<https://netflixtechblog.com/cambi-a-banding-artifact-detector-ea0f291a4dd8>

Netflix Tech Blog article introducing CAMBI (Contrast-Aware Multi-scale Banding Index), a metric Netflix developed to automatically detect banding artifacts in video content.

Tags: video quality, artifact detection, netflix, encoding, video processing

CEA-608/708 Closed Captioning Standards

<https://en.wikipedia.org/wiki/CEA-708>

The EIA-608 and EIA-708 standards define how closed captions are encoded in analog and digital television signals in the US. They are crucial for broadcast TV and cable, and understanding these can help when dealing with caption extraction or embedding in video files (e.g., in MPEG-2 or ATSC streams).

Tags: closed captions, EIA-608, EIA-708, broadcast standards, accessibility

Caption Inspector

<https://github.com/Comcast/caption-inspector>

Caption Inspector is a reference decoder for Closed Captions (CEA-608 and CEA-708).

Captions App

<https://topai.tools/s/video-subtitling-tool>

Captions App is an AI tool that simplifies adding subtitles and captions to videos with auto-generation, translation, and customization options. It also offers AI dubbing in over 100 languages, enabling creators to reach a global audience. The app provides a user-friendly interface for enhancing video content with accurate and customizable subtitles.

Tags: AI, subtitles, captioning, translation, video editing

Ccextractor

<https://github.com/CCExtractor/ccextractor>

CCExtractor - Official version maintained by the core team

Cea Extractor

<https://github.com/Comcast/cea-extractor>

Parsing and display logic for CEA-608 caption data in fragmented MP4 files.

Cea608 JS

<https://github.com/Dash-Industry-Forum/cea608.js>

A JavaScript project designed to extract CEA-608 captions.

Closed Captioning and Subtitling Products - MacCaption and CaptionMaker Overview - Telestream

https://www.telestream.net/captioning/overview.htm?utm_campaign=partners&utm_source=itunespartner.apple.com&utm_medium=text_link

Telestream Closed Captioning: MacCaption and CaptionMaker allow you to easily author, edit, create subtitles, and encode and repurpose video captions for television, web and mobile delivery

Tags: closed captioning, subtitles, video accessibility, encoding, media tools

Cobalt

<https://cobalt.tools/>

A simple and efficient multi-platform media downloader. Cobalt provides an easy way to download videos and audio from sites like YouTube, Instagram, and TikTok with a clean GUI.

Tags: downloader, media-utility, cross-platform, video-download, audio-download

Collection of VMAF Resources

<https://streaminglearningcenter.com/blogs/collection-of-vmaf-resources.html>

A colleague asked for some resources relating to VMAF. Rather than answer in an email I thought I would create a post around it. Some of these are from Netflix, most from me (Jan Ozer). I've broken the items into three groups; Computing VMAF, Using VMAF, and About VMAF. I hope you find this collection useful....

Comskip

<https://github.com/erikkaashoek/Comskip>

Tool for automatically detecting and marking commercial breaks in recorded video content.

Content Delivery Video Library (CDVL)

<https://www.cdvl.org/>

An online library of test video content and sequences provided for researchers and developers to evaluate video processing and streaming techniques (various resolutions, formats, and scenarios).

Tags: video testing, research, evaluation, video sequences, resolution testing

Conviva Streaming Performance Index (Report)

<https://web.archive.org/web/20230907033441/https://www.conviva.com/state-of-streaming/>

Conviva regularly publishes 'State of Streaming' reports which provide analytics on streaming quality (startup times, rebuffer rates, bitrate) across the industry. These reports are valuable to understand trends and the importance of monitoring QoE at scale.

Tags: analytics, performance, QoE, streaming metrics, industry report

DVB Inspector

<https://sourceforge.net/projects/dvbinspector/>

Download DVB Inspector for free. DVB transport stream analyzer written in java. DVB Inspector is a java program to analyze captured DVB-streams. It is not a real time tool that can interact with PC-TV DVB hardware directly.

DVB captions in media convert

<https://docs.aws.amazon.com/mediacore/latest/ug/dvb-sub-output-captions.html>

If your output captions are DVB-Sub, set them up in your outputs according to the following information.

DVBSnoop

<https://dvbsnoop.sourceforge.net/>

An analyzer and debugging tool for MPEG transport streams and DVB broadcasts. It allows inspection of packet structure, tables (PAT, PMT, etc.), and payload of digital video streams.

Tags: analysis, debugging, transport-stream, dvb, mpeg, inspection

DeepMedia Camera SDK

<https://deepmedia.io/>

DeepMedia's Camera SDK is a state-of-the-art toolkit designed to enhance mobile applications with rich media experiences and augmented reality, powered by AI. It supports features like AVC/HEVC video recording, high-quality image capture, AI portrait mode, concurrent cameras, and more. The SDK is available for both Android and iOS platforms.

Tags: camera SDK, AI, augmented reality, media enhancement, mobile development

Deepgram Captioning Packages

<https://deepgram.com/learn/subtitles-made-easy-deepgram-s-new-open-source-captioning-packages>

Deepgram has released two open-source captioning packages: deepgram-captions for Python and @deepgram/captions for JavaScript. These packages empower developers to effortlessly generate SRT and WebVTT captions, providing a flexible solution for various speech-to-text APIs, including Deepgram. Key features include SRT and WebVTT caption generation, API-agnostic design, speaker labeling and diarization compatibility, no Deepgram dependency, Whisper compatibility, and user-friendly caption creation.

Tags: captioning, speech-to-text, SRT, WebVTT, open-source

Def Video Ads

<https://defold.com/assets/defvideoads/>

An asset for the Defold game engine that enables the integration of UnityAds, facilitating the display of video advertisements within Defold projects.

Tags: Defold, UnityAds, ads, monetization, game development

DisplayCAL

<https://displaycal.net/>

An open source display calibration and profiling tool. While not directly a video editing tool, calibrating your monitor using DisplayCAL ensures accurate color representation, which is critical for color correction and grading tasks in video production.

Tags: color-calibration, color-correction, monitor-profiling, video-production-tools

DivFix++

<https://sourceforge.net/projects/divfixpp/>

An open source utility to repair and preview partially downloaded AVI files by rebuilding the index chunk, allowing playback of corrupted or incomplete AVIs.

Tags: file repair, video recovery, AVI, media utility, file indexing

Docker Serve

<https://github.com/Eyevinn/docker-serve>

A simple Python based HTTP server that sets CORS allow headers. Useful for streaming from files on local computer

Tags: http-server, cors, python, streaming-utility, docker

Dolby Stream Validator

https://ott.dolby.com/OnDel_tools/validator/Start_Here.html

Dolby's Stream Validator assesses streaming content for Dolby audio and video standards, ensuring high-quality media delivery.

EBU-TT Live Interoperability Toolkit

<https://ebu.github.io/ebu-tt-live-toolkit/>

Toolkit for implementing and testing EBU-TT Live, a subtitle standard used for live broadcast streaming across multiple platforms.

EU AVMSD Accessibility Requirements

<https://digital-strategy.ec.europa.eu/en/policies/accessibility-avmsd>

The European Union's Audiovisual Media Services Directive (AVMSD) includes requirements for accessibility (subtitles, audio description, sign language) on TV and on-demand services. Resources here outline how video service providers must comply to ensure content is accessible to people with disabilities.

Extract Subtitle

<https://github.com/xinnjie/extract-subtitle>

Tool to extract embedded subtitle tracks from video files for separate processing or storage.

Extract Subtitles

<https://github.com/shawnsky/extract-subtitles>

Utility to extract subtitle streams from video containers into separate text-based subtitle files.

Extracting contextual information from video assets

<https://medium.com/netflix-techblog/extracting-contextual-information-from-video-assets-ee9da25b6008>

for an improved Netflix user experience

Eyevinn Stream Corruptor

<https://github.com/Eyevinn/streaming-onboarding/blob/master/Stream-Corruptor.md>

Eyevinn's Stream Corruptor is a tool designed to intentionally corrupt streaming media, useful for testing error resilience in players.

FF Multi Converter

<https://github.com/ilstam/FF-Multi-Converter>

GUI File Format Converter. Contribute to ilstam/FF-Multi-Converter development by creating an account on GitHub.

FFmpeg Normalize

<https://github.com/slhck/ffmpeg-normalize#examples>

Audio Normalization for Python/ffmpeg.

FFmpeg Patch

<https://github.com/VCDP/FFmpeg-patch>

This repository contains a collection of FFmpeg* patches and samples to enable CNN model based video analytics capabilities (such as object detection, classification, recognition) in FFmpeg* framew...

FFmpeg Quality Metrics

<https://github.com/slhck/ffmpeg-quality-metrics>

Calculate quality metrics with FFmpeg (SSIM, PSNR, VMAF)

FGVC

<https://github.com/vt-vl-lab/FGVC>

[ECCV 2020] Flow-edge Guided Video Completion.

First Order Motion Model

<https://github.com/AliaksandrSiarohin/first-order-model>

A famous deep learning model for image animation (e.g., making a single portrait image speak or move) by modeling motion as a first order approximation. Often used in one-shot talking head video generation.

Fluster

<https://github.com/fluendo/fluster>

Testing framework for decoders conformance.

FreeWheel Ad Management Platform

<https://www.freewheel.com/>

FreeWheel provides a comprehensive ad management platform that enables publishers to manage and monetize their video content across various devices and platforms. It offers tools for ad decisioning, serving, and analytics, making it a valuable resource for integrating video ads into streaming services.

Tags: ad management, video ads, monetization, analytics, cross-platform

Frei0r Plugins

<https://github.com/dyne/frei0r>

A collection of free and simple video effect plugins. Frei0r plugins provide dozens of filters, mixers, and generators that can be used in compatible video editors (like LiVES, Flowblade, Kdenlive).

GNU MediaGoblin

<https://mediagoblin.org/>

A decentralized, federated media hosting platform supporting video, images, and audio. MediaGoblin allows users to run their own instances for sharing media content.

Tags: media hosting, decentralized, federation, open source, content sharing

Gaupol

<https://otsaloma.io/gaupol/>

An open source subtitle editor for text-based subtitle files, featuring timing adjustment, translation support, and built-in video preview (Linux).

Tags: subtitle, editor, text-processing, video-tools, linux

Gnome Subtitles

<https://gnomesubtitles.org/>

Gnome Subtitles is an open-source subtitle editor for the GNOME desktop environment. It supports popular subtitle formats and offers features like video previewing, timing synchronization, and subtitle translation, providing a comprehensive solution for developers focused on video streaming and encoding.

Tags: subtitle editing, open-source, GNOME, video synchronization, translation

Google IMA SDK

<https://developers.google.com/interactive-media-ads>

The Interactive Media Ads (IMA) SDK by Google enables developers to integrate video ads into their applications seamlessly. It supports various ad formats, including linear, non-linear, and companion ads, and is compatible with multiple platforms such as HTML5, Android, and iOS. This SDK is valuable for developers looking to monetize their video content effectively.

Tags: video ads, SDK, Google, monetization, cross-platform

HLS Stream Health Monitoring Tool

<https://github.com/imsanthosh/HLS-Stream-health-monitoring-tool>

HLS stream health monitoring utility tool provides an report of live HLS stream. This utility tool checks the all available bitrate streams and generates the report in html file format. HTML file i...

Tags: HLS, stream monitoring, health check, reporting, performance

Harmonic VOS360 Ad

<https://web.archive.org/web/20260508115505/https://www.harmonicinc.com/insights/blog/targeted-ads-innovations/>

Harmonic's VOS360 Ad is a cloud-based platform that leverages server-side ad insertion (SSAI) technology to seamlessly stitch personalized ads into video streams. This solution enhances the viewing experience by ensuring consistency and frame accuracy in streaming, while also supporting targeted advertising strategies.

Tags: server-side ad insertion, SSAI, targeted ads, video streaming, ad integration

Hippo Video SDK

<https://www.hippovideo.io/products/video-sdk.html>

Hippo Video SDK provides a complete video infrastructure that can be integrated into platforms with minimal code. It supports rapid deployment of video experiences, enabling developers to enhance their applications with video capabilities efficiently.

Tags: video infrastructure, SDK, Hippo Video, integration, developer tools

How to Set Audio Levels for Video

<https://www.premiumbeat.com/blog/how-to-set-audio-levels-for-video/>

Bad sound can easily ruin good footage. Use these tips when it comes time to set audio levels for video and film projects.

How to decode a video (memory file / byte string) and step through it frame by frame in Python?

<https://stackoverflow.com/questions/60558412/how-to-decode-a-video-memory-file-byte-string-and-step-through-it-frame-by-f>

I am using python to do some basic image processing, and want to extend it to process a video frame by frame. I get the video as a blob from a server - .webm encoded - and have it in python as a b...

HyBid SDK

<https://verve.com/blog/open-source-modular-sdk/>

An open-source and modular SDK designed for app developers to integrate ads, offering modules for various ad formats such as banners, interstitials, native ads, and rewarded video.

Tags: SDK, ads, modular, open-source, integration

IMSC 1.0.1 Text test content

https://github.com/w3c/IMSC-1.0.1_Text_TestContent/

GitHub repository containing test content files for validating IMSC 1.0.1 text-based subtitle implementations.

IMSC 1.1 Image test content

https://github.com/w3c/IMSC-1.1_Image_TestContent/

GitHub repository containing test content files for validating IMSC 1.1 image-based subtitle implementations.

IMSC 1.1 Text test content

https://github.com/w3c/IMSC-1.1_Text_TestContent/

GitHub repository containing test content files for validating IMSC 1.1 text-based subtitle implementations.

IMSC HRM validator

<https://github.com/sandflow/imscHRM>

Validator for the IMSC Hypothetical Render Model (HRM), which constrains TTML document complexity

IMSC Specification

<https://www.w3.org/TR/ttml-imsc1.1/>

W3C technical specification defining the IMSC 1.1 standard for timed text markup language used in subtitle and caption formatting.

IMSC validator

<https://imscv.sandflow.com/>

Online validation tool for checking compliance of subtitle files with the IMSC (International Movie Standardization Committee) specification.

InfraRecorder

<http://infrarecorder.org/>

An open source DVD burning tool that, among other features, can be used to author basic video DVDs (it includes support for creating video DVD file systems from video files).

Tags: dvd, burning, authoring, video-tools, file-system

Interactive Video Ads Integration Techniques

<https://fastercapital.com/content/Interactive-video-ads--Multi-Platform-Integration--Seamless-Multi-Platform-Integration-for-Interactive-Video-Ads.html>

This resource discusses various techniques for integrating interactive video ads across multiple platforms, enhancing user engagement and providing seamless ad experiences. It includes case studies and best practices for implementing interactive ad formats.

Tags: interactive ads, multi-platform integration, video advertising, user engagement, ad formats

Isoviewer

<https://github.com/sannies/isoviewer>

GUI application to have closer look ISO 14496-12 and other MP4 files.

Jubler

<https://www.jubler.org/>

A cross-platform open source subtitle editor written in Java. It supports multiple subtitle formats, style editing, timing synchronization, and real-time preview of subtitles on video.

Tags: subtitles, editor, java, cross-platform, media-processing

Jukebox

<https://github.com/openai/jukebox>

Code for the paper "Jukebox: A Generative Model for Music"

Kaltura Video Player

<https://github.com/kaltura/kaltura-player-js>

Kaltura's Video Player is an open-source, customizable player that supports advanced video advertising features. It integrates with various ad servers and networks, supporting standards like VAST and VPAID, and offers both client-side and server-side ad insertion.

Tags: video ads, Kaltura, VAST, VPAID, open-source

Kapwing

<https://web.archive.org/web/20250126100102/https://thinkbrandedmedia.com/the-best-ai-tools-for-adding-subtitles-improve-your-videos-and-retain-customers/>

Kapwing is a user-friendly AI tool that allows users to easily add subtitles, captions, and annotations to videos. It's ideal for non-profit video production, where making video content accessible to a diverse audience is essential. Kapwing offers automatic captioning, helping to increase the accessibility of videos. The platform supports various video formats and provides options for optimizing video titles and adding metadata, which helps improve search rankings on video platforms.

Tags: video editing, subtitles, captions, AI tools, video accessibility

LEADTOOLS Audio and Video Processing Filters

<https://www.leadtools.com/sdk/multimedia/filters/audio-video-processing-filters>

LEADTOOLS offers a comprehensive collection of DirectShow filters and Media Foundation transforms for processing and enhancing audio and video content within .NET and C/C++ applications. These filters can be applied in real-time during playback or live capture, or during conversion for post-processing and editing.

Tags: audio processing, video processing, DirectShow, Media Foundation, SDK

LEADTOOLS Multimedia SDK

<https://www.leadtools.com/sdk/multimedia>

LEADTOOLS Multimedia SDK provides a comprehensive set of DirectShow and Media Foundation filters for audio and video processing. It offers developers tools to implement features such as playback, capture, conversion, and processing of multimedia content within .NET and C/C++ applications.

Tags: multimedia, SDK, DirectShow, Media Foundation, processing

LEXI AI-Powered Captioning Toolkit

<https://www.ai-media.tv/knowledge-hub/insights/free-captioning-tools/>

The LEXI AI-Powered Captioning Toolkit leverages AI advancements to provide a comprehensive suite covering live and recorded captioning, translation, display, archiving, and search. It includes LEXI Automatic Captioning, delivering AI-powered automatic captions with quality rivaling human captions at a fraction of the cost. LEXI Recorded makes captioning recorded content quick and easy with fully automated, fast turnaround times. LEXI Translate offers efficient and cost-effective solutions for global reach through automatic translation. LEXI Local provides on-premises solutions for live, automatic captions, ensuring elevated security and greater control. LEXI Library offers a caption archive and search tool for easy viewing, searching, editing, and archiving of live captions.

Tags: AI, captioning, translation, archiving, search

LiVES Video Editing System

<http://lives-video.com/>

LiVES is an open-source video editing system designed for both professional and amateur video editors. It supports real-time

and non-linear editing, a wide range of video formats, and includes features like frame accurate cutting, real-time effects, and support for various audio and video codecs.

Tags: video editing, open-source, real-time effects, non-linear editing, multimedia

Libcaption

<https://github.com/szatmary/libcaption>

Free open-source CEA608 / CEA708 closed-caption encoder/decoder

Libgosubs

<https://github.com/wargarblgarbl/libgosubs>

A library for parsing and manipulating subtitle files in various formats for processing and conversion.

Tags: subtitles, captions, library, tools, text processing

Librosa

<https://librosa.org/>

Librosa is a Python library for analyzing and processing audio signals. It provides functionalities such as feature extraction, visualization, and audio effects, which are essential for audio analysis in video applications.

Tags: audio processing, Python, signal analysis, feature extraction, visualization

Livestreamer

<https://github.com/chrippa/livestreamer>

The precursor to Streamlink, a command-line utility that extracted streams from various services to feed into a media player. (No longer maintained, replaced by Streamlink.).

Low Latency Android iOS Linux Windows tvOS macOS Interactive Audio Platform

<https://github.com/superpoweredSDK/Low-Latency-Android-iOS-Linux-Windows-tvOS-macOS-Interactive-Audio-Platform>

Superpowered Audio, Networking and Cryptographics SDKs. High performance and low latency for Android, iOS, macOS, tvOS, Linux, Windows and modern web browsers.

Lux (Media Downloader)

<https://github.com/iawia002/lux>

A command-line media downloader (formerly known as "annie") that supports downloading videos and audio from various websites in high quality, similar to youtube-dl but in Go.

M2amedia SCTE35 Dump for MPEGTS

<https://github.com/m2amedia/scte35dump>

M2A Media's tool for extracting SCTE-35 markers from MPEG-TS streams.

MKVToolNix

<https://mkvtoolnix.download/>

A set of tools to create, alter, and inspect Matroska (MKV) files. Includes mkvmerge for muxing streams into MKV and mkvinfo/mkvextract for inspecting and extracting content.

Tags: container, muxing, media-manipulation, mkv, video-tools

MMAction2

<https://github.com/open-mmlab/mmaction2>

An open source toolkit for video understanding by OpenMMLab. It supports state-of-the-art models for action recognition, temporal action detection, and spatio-temporal detection in videos.

MPEG Ts Inspector

<https://github.com/brendanlong/mpeg-ts-inspector>

A command-line tool for inspecting MPEG-TS files. Contribute to brendanlong/mpeg-ts-inspector development by creating an account on GitHub.

MediaArea Tools (MediaInfo, etc.)

<https://mediaarea.net/>

MediaArea produces open source software for media analysis, including MediaInfo (displays technical and tag data for files), MediaConch (policy compliance for archival video), and BWF MetaEdit (for WAV metadata).

Tags: metadata, analysis, file-inspection, archival, WAV, media-info

MediaConch

<https://mediaarea.net/MediaConch>

An open source policy compliance checker for video files (especially in archival). MediaConch can batch-scan a collection of files to ensure they meet format specifications or institutional policies (like checking if files conform to MKV/FFV1 or

MOV/ProRes standards).

MedialInfo (GitHub)

<https://github.com/MediaArea/MedialInfo>

Convenient unified display of the most relevant technical and tag data for video and audio files.

Meteorite

<https://sourceforge.net/projects/meteorite/>

The first (experimental) open source MKV/Matroska repair tool. Meteorite can attempt to fix corrupted.mkv files or those still downloading, by reconstructing missing or damaged segments.

Tags: repair, mkv, file-recovery, video-repair, open-source

Minyami

<https://github.com/Last-Order/Minyami>

A lovely video downloader for HLS videos.

Tags: downloader, HLS, video-utility, streaming-tool

Mp4analyser

<https://github.com/essential61/mp4analyser>

mp4 file analyser written in Python.

Mutagen

<https://github.com/quodlibet/mutagen>

Python module for handling audio metadata.

Natron

<https://natron.fr/>

A free and open source node-based compositing software, similar to After Effects or Nuke, for visual effects and motion graphics.

Tags: compositing, visual-effects, motion-graphics, post-production

Netflix Timed Text Style Guide (English)

<https://partnerhelp.netflixstudios.com/hc/en-us/articles/215758617-Timed-Text-Style-Guide-General-Requirements>

Netflix's style guide for subtitles and captions outlines the editorial and formatting standards for timed text on their platform. It's a comprehensive resource for best practices in captioning (e.g., line length, timing, speaker labels) that is often referenced industry-wide.

Tags: subtitles, captions, style guide, text formatting, accessibility

Node Thumbnails Webvtt

<https://github.com/estliberitas/node-thumbnails-webvtt>

Video thumbnail generator generating WebVTT spec file

Node Webvtt

<https://github.com/osk/node-webvtt>

Parse WebVTT files, segments and generates HLS playlists for them.

OOONA Tools

<https://web.archive.org/web/20251115022642/https://theiabm.org/bamproducts/online-caption-subtitle-toolkit/>

OOONA Tools offers professional online tools for subtitling and captioning, accessible from any computer at any time. The platform provides a simple and intuitive web interface for creating and editing captions and subtitles in any language. Features include frame-accurate text timing with an advanced timeline, video grid for precise caption positioning, audio waveform, and scene-change detection for accurate subtitle spotting. It supports import and export in various caption and subtitle formats, including TTML/DFXP, VTT, SCC, CAP, EBU-STL, SRT, IMSC1, PAC, 890, and more. Additionally, it supports generating image-based subtitles for DVD/Blu-ray/DCP authoring. Powerful Pro apps allow users to run automated QA scripts to check and fix files, as well as customize hotkeys and project settings.

Tags: subtitling, captioning, video editing, localization, media production

Oboe

<https://github.com/google/oboe>

Oboe is a C++ library that wraps OpenSL ES and AAudio, offering high-performance audio operations for Android applications. It simplifies the development of real-time audio apps by providing a consistent API across different Android versions.

Tags: Android, audio processing, real-time audio, C++, library

Open Video Downloader (YouTube-dl GUI)

<https://github.com/jely2002/youtube-dl-gui>

A cross-platform GUI for youtube-dl/yt-dlp that makes it easy to download videos from YouTube and other platforms without using the command line.

OpenColorIO

<https://opencolorio.org/>

An open source color management solution geared towards motion picture production. Developed by the Academy Software Foundation, OpenColorIO handles color space conversions and is a core part of VFX/animation workflows.

Tags: color management, vfx, post-production, color grading, academy software foundation

OpenFX (OFX)

<https://github.com/AcademySoftwareFoundation/openfx>

An open standard for visual effects plugins, widely supported in compositing software. It defines a C API for writing plugins (e.g., for Nuke, Natron) that perform image and video processing tasks.

OpenIMAJ Audio Processing Libraries

<https://openimaj.org/openimaj-audio/>

OpenIMAJ provides a suite of audio processing and analysis tools, including implementations of various audio processors such as multichannel conversion and volume change. It is designed for developers working on audio-related functionalities.

Tags: audio processing, analysis, multichannel, volume control, open-source

OpenPose

<https://github.com/CMU-Perceptual-Computing-Lab/openpose>

A real-time multi-person keypoint detection library (for body, face, and hands) by CMU. Often used on video to extract pose information frame-by-frame for animation or analysis.

Opencast

<https://opencast.org/>

An open source lecture capture and video management system. Opencast allows universities and organizations to schedule recordings of lectures, process the videos (including multi-source), and distribute them to students.

Tags: lecture capture, video management, education, content distribution, multi-source recording

Otter.ai

<https://otter.ai/>

Otter.ai is an AI-powered transcription service that automatically generates subtitles for videos. It integrates with platforms like YouTube and Vimeo, offering features like real-time transcription and collaboration tools, streamlining the subtitle integration process.

Tags: AI transcription, automatic subtitles, platform integration, real-time transcription, collaboration tools

Plex

<https://www.plex.tv/>

A popular media server and client platform for organizing and streaming personal video (and other media) collections across devices.

Tags: media server, streaming, personal media, cross-platform, media organization

Pmd Tool

https://github.com/DolbyLaboratories/pmd_tool

A pmd_tool is a command line utility that converts between different representations of SMPTE RDD49 metadata

Processing Video Library

<https://processing.org/reference/libraries/video/>

The Processing Video library enables playback of movie files and capture of video data from cameras. It supports a wide range of media formats and is based on the GStreamer multimedia framework, facilitating video processing tasks in Java.

Tags: video playback, video capture, GStreamer, Java, multimedia

Promising Initial Results with AV1 Testing - Streaming Learning Center

<https://streaminglearningcenter.com/blogs/promising-initial-results-with-av1-testing.html>

I began testing AV1 early this week. Briefly, my tests involve 16 ten-second clips in four genres (movies, sports, animations, gaming) and an "other" category (music video, nature video). I've completed the first set of tests with FFmpeg 4.3, benchmarking x264, x265, and the latest version of the Alliance for Open Media AV1 codec,...

PySceneDetect

<https://github.com/Breakthrough/PySceneDetect>

An open source Python tool for automatic scene cut detection in videos. It can split video content into scenes by detecting fades, cuts, and other transitions.

Pycaption

<https://github.com/pbs/pycaption>

Python module to read/write popular video caption formats

Pycaption CLI

<https://github.com/jnorton001/pycaption-cli>

A command line interface for the pycaption module.

Pymediainfo

<https://github.com/sbraz/pymediainfo>

A Python wrapper around the MediaInfo library. Contribute to sbraz/pymediainfo development by creating an account on GitHub.

Pysub Parser

<https://github.com/federicocalendino/pysub-parser>

Utility to extract the text and timestamps of a subtitle file (.srt,.ssa,.sub,.txt).

QCTools

<https://mediaarea.net/QCTools>

Quality Control Tools for Video Preservation – an open source toolset (by BAVC) to analyze digitized video files, with visualizations for signal quality issues.

QCTools Documentation

<https://bavc.github.io/qctools/>

QCTools (Quality Control Tools for Video Preservation) is a free and open source software tool that helps users analyze and understand their digitized video files through use of audiovisual analytics and filtering. QCTools is funded by the National Endowment for the Humanities and the Knight Foundation, and is developed by the Bay Area Video Coalition.

Qctools (GitHub)

<https://github.com/bavc/qctools>

A quality control tool for analyzing video files, detecting defects, and evaluating technical quality metrics for archival and broadcast purposes.

Tags: quality control, video analysis, metrics, testing, media tools

Qtff Parameter Editor

<https://github.com/bbc/qtff-parameter-editor>

QuickTime file parameter editor for modifying transfer function, colour primary and matrix characteristics.

Quality of Experience in Streaming

https://medium.com/@eyevinntech/quality-of-experience-in-streaming-5c25355a4111?source=userActivityShare-94bccb50d11-1559724940&_branch_match_id=664741478927428385

In Eyevinn Technology's ambition to broaden our sharing of knowledge we now expand this with addressing quality. In today's landscape of...

R4WG20 QoE Metrics

<https://github.com/cta-wave/R4WG20-QoE-Metrics>

Issue tracking repository for the R4-Wg20 QoE Initiative

Radiant Media Player

<https://www.radiantmediaplayer.com/docs/v9/video-ads-documentation.html>

A comprehensive media player that offers built-in support for video ads using the Google IMA HTML5 SDK, facilitating seamless ad integration.

Tags: media player, video ads, Google IMA, HTML5, integration

RawCooked

<https://mediaarea.net/RawCooked>

A tool that wraps video, audio, and data files (like DPX sequences or WAV) into a Matroska container without loss. Often used in digital archiving to store large raw content more efficiently while preserving all data for exact restoration.

Remotion

<https://www.remotion.dev/>

A framework for creating videos programmatically using React. Developers can use web technologies to render dynamic video content (with animations, data, etc.) and export it as MP4 videos.

Tags: video-generation, react, programmatic-video, web-technology, mp4-export

Renee Video Editor Pro

<https://www.reneelab.com/add-subtitles-to-video.html>

Renee Video Editor Pro is a comprehensive video editing software that allows users to add subtitles and text to videos effortlessly. It supports various video formats and provides multiple editing tools, including cutting, merging, cropping, rotating, and transforming videos. Users can change and add background music, subtitles, and remove watermarks from videos. The software also supports format conversion, enabling users to convert videos and audios into different formats. Additionally, it offers screen recording capabilities, allowing users to record screen and webcam activities on Windows computers. The intuitive interface and diverse functionalities make it suitable for both beginners and professionals.

Tags: video editing, subtitles, format conversion, screen recording, media production

Rev.com

<https://www.rev.com/>

Rev.com offers professional captioning services that combine AI and human editing to provide accurate subtitles. It supports various video formats and platforms, making it a reliable tool for integrating high-quality subtitles into video content.

Tags: captioning services, AI, human editing, video formats, platform support

SIMID

<https://github.com/InteractiveAdvertisingBureau/SIMID>

Secure Interactive Media Interface Definition (SIMID) is a project by the Interactive Advertising Bureau (IAB) that defines a standard for interactive video ads. It aims to provide a secure and consistent framework for interactive media ads across different platforms, enhancing user engagement and ad performance.

Tags: video-ads, interactive-ads, iab, standard, ad-integration

SRT Metadata Extractor

<https://github.com/Eyevinn/srt-metadata-extractor>

Contribute to Eyevinn/srt-metadata-extractor development by creating an account on GitHub.

STRCleaner

<https://github.com/Marcos-A/STRCleaner>

Script that extracts all the text from a subtitles file ignoring time indications, HTML tags and other alien info.

Scenecut Extractor

<https://github.com/slhck/scenecut-extractor>

Extract scenecuts from video files using ffmpeg

ScreenToGif

<https://www.screentogif.com/>

An open source tool to record a selected screen area, webcam, or whiteboard and export it as an animated GIF or video. It includes basic editing like frame-by-frame editing and text/brush overlays.

Tags: screen recording, gif creation, video editing, screen capture, animation

Seafile

<https://github.com/haiwen/seafile>

High performance file syncing and sharing, with also Markdown WYSIWYG editing, Wiki, file label and other knowledge management features.

Tags: file-sharing, collaboration, markdown, wiki, knowledge-management

Server-Side Ad Insertion Techniques

<https://www.tothenew.com/blog/techniques-for-online-video-ad-integration/>

An in-depth look at server-side ad insertion (SSAI) methods, discussing their advantages and challenges. This resource is valuable for developers seeking to implement SSAI for a more consistent viewing experience across devices.

Tags: server-side ad insertion, SSAI, video ads, ad integration, video streaming

Serverless Subtitles

<https://github.com/awslabs/serverless-subtitles>

Serverless Subtitles can handle a video input, extract the sound, transcript it and generate different subtitle files for your video.

Shaka Packager

<https://github.com/google/shaka-packager>

An open source tool by Google for packaging media content (e.g., creating MPD/manifest and segments) with support for encryption (Widevine, PlayReady, FairPlay). It's often used to prepare content for DASH and HLS with CMAF and DRM, complementing encoding pipelines.

ShortGPT

<https://github.com/RayVentura/ShortGPT>

An experimental AI framework to automate the creation of YouTube Shorts or TikTok videos. ShortGPT can generate video content (scenes, subtitles, etc.) using generative AI models.

Shot Detection Benchmarks

<https://github.com/albanie/shot-detection-benchmarks>

A comparison of FFmpeg, Shotdetect and PySceneDetect for shot transition detection

SimpleVideoCutter

<https://github.com/bartekmotyl/simple-video-cutter>

A straightforward Windows tool for browsing and cutting segments from video files without re-encoding. It provides a preview and easy selection of in/out points to extract clips quickly.

Sintel (Open Movie)

<https://durian.blender.org/>

A 15-minute open source animated film (2010) by Blender Institute. It's frequently used for benchmarking video encoders and playback devices (available in 4K and various formats).

Tags: benchmarking, 4K, encoder testing, video quality, open source

Smart City Sample

<https://github.com/OpenVisualCloud/Smart-City-Sample>

The smart city reference pipeline shows how to integrate various media building blocks, with analytics powered by the OpenVINO™ Toolkit, for traffic or stadium sensing, analytics and management tasks.

Snowmix - The Swiss Army Knife of Open Source Live Video Mixing.

<https://snowmix.sourceforge.io>

Open source live video mixing software for combining multiple video sources and creating real-time composites.

Tags: video mixer, open source, live video, video processing

SoX (Sound eXchange)

<https://sox.sourceforge.net/>

SoX is a command-line utility for audio manipulation and processing. It supports various audio formats and provides a range of effects, making it useful for developers needing to process audio within video projects.

Tags: audio processing, command-line, audio manipulation, effects, multiformat

Software Tools

<https://github.com/VQEG/software-tools>

VQEG's Software and Tools Website.

SponsorBlock

<https://github.com/ajayyy/SponsorBlock>

An open source browser extension to skip sponsor segments in YouTube videos.

SpotLink

<https://www.tvtechnology.com/news/free-opensource-software-introduced-for-dynamic-ad-insertion>

An open-source software for dynamic ad insertion in VOD, IPTV, broadband video, and network DVRs, connecting SCTE 130-compliant dynamic ad insertion systems to broadband ad servers using the IAB's VAST standard.

Tags: ad-insertion, VOD, IPTV, broadband, VAST

SpotX Video Ad Serving Platform

<https://www.spotx.tv/>

SpotX is a video ad serving platform that offers solutions for publishers and advertisers to manage and optimize their video ad inventory. It provides tools for programmatic ad buying, real-time bidding, and analytics, facilitating efficient video ad integration and monetization.

Tags: video ads, ad serving, programmatic, monetization, analytics

Stb-tester

<https://github.com/stb-tester/stb-tester>

An automation tool for testing set-top boxes and video devices. stb-tester uses a Raspberry Pi with a capture card to run automated tests (via video analysis and remote control input) on HDMI outputs.

Tags: testing, automation, video-analysis, quality-assurance, set-top-box, raspberry-pi

Stream Analyzer - ts analyzer, stream validation, ETSI TR 101 290 | Elecard: Video Compression Guru

<https://www.elecard.com/products/video-analysis/stream-analyzer>

Professional Video Analysis Tool For Syntax Analysis Of Encoded Media Streams. Operates With MPEG-2 PS/TS, VES and MP4 Files. Automation, batch execution via Command Line Interface. Check your multiplexer

Streaming Video Experimentation at Netflix

<https://netflixtechblog.com/streaming-video-experimentation-at-netflix-visualizing-practical-optimization-834152da60a5>

Netflix Tech Blog article explaining how Netflix conducts A/B testing and experimentation on streaming video quality and features to optimize QoE for its users at scale.

Tags: QoE, A/B testing, streaming optimization, performance, netflix

Streamsim

<https://github.com/VQEG/streamsim>

VQEG/streamsim - A GitHub repository for video/multimedia development.

SubEasy

<https://topai.tools/s/automated-video-subtitling-tool>

SubEasy is an AI-driven subtitles and transcriptions generator supporting over 100 languages. It provides high accuracy with subtitle reflow technology and offers seamless video creation with high-precision video AI transcription and translation.

SubEasy simplifies the process of adding subtitles to videos, enhancing accessibility and engagement for a global audience.

Tags: AI, subtitles, transcription, translation, video editing

SubRip

<https://zuggy.wz.cz>

SubRip is a free software program for Windows that extracts subtitles and their timings from various video formats to a text file. Its subtitle format, .srt, is widely supported, making it a valuable tool for developers needing to extract and integrate subtitles into video streaming and encoding projects.

Tags: subtitle extraction, Windows, srt format, video integration, encoding

Subs Extract

https://github.com/cessen/subs_extract

Extracts per-sentence subtitles + audio from a subtitle file + video file.

Subsync

<https://github.com/smacke/subsync>

Automagically synchronize subtitles with video.

Tags: subtitles, synchronization, media-processing, video-tools

Subtitld

<https://subtitld.org/en>

Subtitld is an open-source software designed to edit, transcribe, and create subtitles. It supports various formats like SRT, SSA, TTML, and more. Features include timeline adjustments, playback speed control, and the ability to export videos with burned-in subtitles, enhancing the workflow for video developers.

Tags: subtitle editing, transcription, open-source, multi-format support, video processing

Subtitle Composer

<https://github.com/maxrd2/subtitlecomposer>

A Qt-based subtitle editor for KDE/Linux that supports multiple subtitle formats and provides tools for timing, text editing, and translation.

Tags: subtitle, editor, KDE, Linux, text-editing, translation

Subtitle Edit

<https://github.com/SubtitleEdit/subtitleedit>

A free, open source subtitle editor that supports over 200 subtitle formats, with tools for sync, translation, and conversion.

Tags: subtitle, editor, translation, conversion, open-source

Subtitle Horse

<https://www.subtitle-horse.com/>

Subtitle Horse is a browser-based captions editor for subtitling videos. This free online editor features real-time validation, an interactive timeline, shortcuts, and more. It can be used within or from a CMS, or on-premise with a paid version, making it versatile for various development needs.

Tags: browser-based, captions editor, real-time validation, interactive timeline, free tool

Subtitle Workshop

<https://www.uruworks.net/>

Subtitle Workshop is a free subtitle editor with a user-friendly interface and support for multiple subtitle formats. It offers features like spell checking, translation mode, and subtitle synchronization tools, making it suitable for both beginners and experienced users.

Tags: subtitle editing, free, multi-format support, spell checking, translation mode

Subtitle Workshop XE

<https://subworkshop.sourceforge.io/>

One of the oldest and popular freeware tools for subtitle editing. Subtitle Workshop XE is a modern update that supports a variety of subtitle formats, offering spell check, timing adjustment, and format conversion in a user-friendly environment.

Tags: subtitle, editing, format-conversion, text-processing, media-tool

Subtitles

<https://github.com/active-video/subtitles>

AV Platform MPEG DASH subtitles.

Syncplay

<https://syncplay.pl/>

A tool that synchronizes media playback across multiple computers over the internet. Syncplay allows friends to watch the same video file or stream in lockstep with a shared experience (works with MPC-HC, mpv, VLC).

Tags: synchronization, media-playback, collaborative-viewing, video-sharing

TTML/WebVTT Coding of audio-visual objects - ISO/IEC 14496-30

<https://www.iso.org/standard/75394.html>

Defines formatting for text tracks webvtt or ttml

TV Subtitle Extraction

<https://github.com/statsbiblioteket/tv-subtitle-extraction>

System for extraction of subtitles from TV broadcasts.

Tartube

<https://github.com/axcore/tartube>

A GUI front-end for youtube-dl/yt-dlp on Linux and Windows that helps manage video downloads, subscriptions to channels/playlists, and metadata for offline viewing.

Tero Subtitler

<https://github.com/URUWorks/TeroSubtitler>

Tero Subtitler is an open-source, cross-platform, and free subtitle editing software with a user-friendly interface. It offers fully-fledged editing with SMPTE and MEDIA modes, support for various subtitle formats, multi-level undo/redo, search and replace, auto-backup, source and transcription modes, translation memory, audiovisual preview, timeline with waveform visualizer, manipulation tools, formatting options, quality control features, translation and transcription capabilities, validation tools, automation for correcting errors, and more. The tool utilizes powerful multimedia playback engines like mpv, advanced audio/video manipulation tools like FFmpeg, tools for automatic transcription like whisper.cpp/Faster-Whisper, auto-translation API like Google Translate, and ElevenLabs TTS for video dubbing.

Tags: subtitle editing, cross-platform, open-source, video processing, translation

Test Engine Live Tools

<https://github.com/ebu/test-engine-live-tools>

Small tools and scripts for the EBU test engine platform.

Test Engine Ondemand

<https://github.com/ebu/test-engine-ondemand>

EBU's on-demand test engine for validating broadcast and streaming workflows against technical specifications.

Text2Video-Zero

<https://github.com/Picsart-AI-Research/Text2Video-Zero>

An AI model (ICCV 2023) for zero-shot text-to-video generation using image diffusion models. It allows generating short video clips from text prompts without training on video data.

The Challenge to Maintain and Translate Creative Visual Ideas to Everyone's Viewing Devices

<https://medium.com/@eyevinntechnology/the-challenge-to-maintain-and-translate-creative-visual-ideas-to-everyones-viewing-devices-a88e1a841439>

Many articles have already been posted drawing conclusions on what went wrong with the visual quality of a very popular TV show that was...

The ultimate guide to CCs

<https://www.3playmedia.com/resources/popular-topics/closed-captioning/>

Comprehensive guide covering closed captioning practices, standards, and implementation methods for video accessibility.

TimeSide

<https://timeside.readthedocs.io/en/stable/intro.html>

TimeSide is a Python framework for scalable audio processing, analysis, imaging, transcoding, streaming, and labeling. It provides a high-level API designed to enable complex processing on large datasets of audio or video assets with a plug-in architecture and an extensible web frontend.

Tags: audio processing, Python, transcoding, streaming, analysis

TorchAudio

<https://github.com/pytorch/audio>

TorchAudio provides a set of tools and building blocks for audio and speech processing, designed to accelerate the development and deployment of machine learning applications in these domains. It offers GPU-compatible, differentiable, and production-ready components, making it valuable for integrating audio processing into video streaming and encoding workflows.

Tags: audio processing, machine learning, speech processing, PyTorch, deep learning

Toward A Practical Perceptual Video Quality Metric

<https://medium.com/netflix-techblog/toward-a-practical-perceptual-video-quality-metric-653f208b9652>

measuring video quality accurately at scale

Toward a Better Quality Metric for the Video Community

<https://netflixtechblog.com/toward-a-better-quality-metric-for-the-video-community-7f94ba0d008>

Netflix Tech Blog article discussing the need for improved video quality metrics beyond existing ones like PSNR or SSIM, and how metrics like VMAF were developed to better reflect human perception.

Tags: quality metrics, video perception, VMAF, PSNR, SSIM, Netflix

Transcode (bmhayward)

<https://github.com/bmhayward/Transcode>

Tools to batch transcode and process videos.

Tube Archivist

<https://www.tubearchivist.com/>

A self-hosted YouTube archival and library management application. It lets you subscribe to channels, automatically download videos, and index them with search and tags for offline access.

Tags: archival, library, video-management, offline-access, content-indexing

Type Studio

<https://topai.tools/s/automated-video-subtitling->

Type Studio is an online video text converter and editor tool that can transcribe video text automatically, support multiple languages, and export in various formats. It simplifies the process of adding subtitles to videos by providing automatic transcription and editing features, making it valuable for content creators aiming to enhance accessibility and engagement.

Tags: video editing, transcription, multilingual support, online tool, automatic subtitles

Unified Streaming Validator

<https://validator.unified-streaming.com/>

Unified Streaming's Validator checks the compliance and quality of streaming content, supporting various streaming protocols.

Unison

<https://github.com/bcpierce00/unison>

File synchronization tool useful for keeping video assets and media files in sync across multiple servers or locations.

Tags: file sync, file transfer, media management, utility

Untrunc

<https://github.com/ponchio/untrunc>

A tool to recover and repair corrupted video files (e.g., missing headers) by reconstructing them using a reference file of the same format.

VAST 4.0 Sample Tags

https://github.com/InteractiveAdvertisingBureau/VAST_Samples

The Interactive Advertising Bureau (IAB) provides sample VAST 4.0 tags to help developers understand and implement the latest standards in video ad serving. These samples are valuable resources for testing and ensuring compliance with VAST 4.0 specifications.

Tags: VAST 4.0, IAB, video ads, sample tags, ad serving

VAST 4.0 Specification

<https://iabtechlab.com/standards/vast/>

The Video Ad Serving Template (VAST) 4.0 is a standard developed by the Interactive Advertising Bureau (IAB) that provides a common protocol for serving in-stream video ads. It defines the structure for ad responses and supports various ad formats, including linear and non-linear ads. Implementing VAST 4.0 ensures compatibility with a wide range of video players and ad servers.

Tags: VAST, video ads, ad serving, IAB standard, ad integration

VAST Inspector

<https://developers.google.com/interactive-media-ads/docs/sdks/html5/vastinspector>

The VAST Inspector is a tool provided by Google to validate VAST (Video Ad Serving Template) ad tags. It helps developers ensure that their VAST tags are correctly formatted and functioning as intended, which is crucial for delivering video ads without errors. This tool is essential for developers working with VAST-compliant ad servers.

Tags: VAST, ad validation, Google, video ads, developer tool

VHS (Terminal Recorder)

<https://github.com/charmbracelet/vhs>

A tool for turning terminal sessions into animated GIFs or videos. VHS lets you script terminal interactions and outputs them as recordings, useful for demos and documentation.

VMAF

<https://github.com/Netflix/vmaf/>

Perceptual video quality assessment based on multi-method fusion.

VMAF: The Journey Continues

<https://medium.com/netflix-techblog/vmaf-the-journey-continues-44b51ee9ed12>

by Zhi Li, Christos Bampis, Julie Novak, Anne Aaron, Kyle Swanson, Anush Moorthy and Jan De Cock

VOCAL Technologies Video Software Library

<https://vocal.com/software-modules/video-software/>

VOCAL Technologies provides a comprehensive video software library that includes industry-standard video codecs, multi-channel video frameworks, voice and video over IP solutions, VoIP vocoders, and voice enhancement tools. These resources support a range of video communications applications from surveillance to teleconferencing.

Tags: video codecs, VoIP, voice enhancement, video processing, communications

VQEG Tools and Subjective Labs Setup

<https://vqeg.github.io/software-tools/>

Providing the video quality research community with a wide variety of software tools and guidance in order to facilitate research.

VQMT

<https://github.com/Rolinh/VQMT>

VQMT: Video Quality Measurement Tool. Fast implementations of the following objective image quality metrics: PSNR, SSIM, MS-SSIM, VIFp, PSNR-HVS and PSNR-HVS-M.

VTCLab Media Analyzer

<https://media-analyzer.pro>

In-browser tool that helps to analyze the internal structure of MPEG-TS and MP4/MOV files.

Vaapi Fits

<https://github.com/intel/vaapi-fits>

Intel/vaapi-fits - A GitHub repository for video/multimedia development.

Video Bench — How measure your video quality easily

<https://medium.com/@jnoduq/video-bench-how-measure-your-video-quality-easily-85a0feb8f6e2>

Guide to using automated tools for measuring and benchmarking video quality metrics in encoding workflows.

Video Hub App

<https://github.com/whyboris/Video-Hub-App>

An electron-based video library organizer that scans folders for video files and presents an interface with thumbnails and search for browsing a video collection.

Video Quality Assessment

https://medium.com/@eyevinntechnology/video-quality-assessment-34abd35f96c0?source=userActivityShare-94bccb50d11-1560983815&_branch_match_id=670021582869771680

In Eyevinn's initiative to share our knowledge around quality we continue with addressing video quality assessment; from both a subjective...

Video Tensorflow

<https://docs.google.com/presentation/d/1NAqYWmFOWxJEacZCuPLdX0mRNRFPFgeRbsm22EaxerU/edit?usp=sharing>

Using Tensorflow For Audience Measurement Kevin Staunton-Lambert Solutions Architect R&D @keveyski www.switch.tv

Video Thumbnail Generator

<https://github.com/flavioribeiro/video-thumbnail-generator>

Generate thumbnail sprites from videos.

Video.js Ads Plugin

<https://github.com/dmlap/videojs-ads>

A plugin for Video.js that provides common functionality needed by video ad integrations, simplifying the process of incorporating ads into video content.

Tags: video.js, ads, plugin, video, integration

Video.js IMA Plugin

<https://github.com/googleads/videojs-ima>

The Video.js IMA Plugin integrates the Video.js player with Google's IMA SDK, enabling seamless ad integration in HTML5 video players. It supports various ad formats and provides a consistent ad experience across different platforms.

Tags: video ads, IMA SDK, Video.js, ad integration, HTML5

VideoCaptioner

<https://github.com/WEIFENG2333/VideoCaptioner>

VideoCaptioner is a video subtitle processing assistant based on large language models (LLM). It supports speech recognition, subtitle segmentation, optimization, and translation, providing a comprehensive solution for subtitle generation and integration. The tool offers features such as accurate subtitle generation without GPU, intelligent segmentation and sentence splitting based on LLM, AI subtitle optimization and translation, and batch video subtitle synthesis.

Tags: subtitle-processing, speech-recognition, LLM, video-subtitles, AI

VideoLingo

<https://github.com/Huanshere/VideoLingo>

VideoLingo is an all-in-one video translation, localization, and dubbing tool designed to generate Netflix-quality subtitles. It automates processes such as subtitle cutting, translation, alignment, and dubbing, enabling seamless video localization across language barriers. Key features include YouTube video download via yt-dlp, word-level subtitle recognition with WhisperX, NLP and GPT-based subtitle segmentation, and dubbing alignment with GPT-SoVITS.

Tags: video-translation, subtitle-integration, dubbing, localization, AI

VideoReTalking

<https://github.com/OpenTalker/video-retalking>

A research project (SIGGRAPH Asia 2022) providing code for audio-driven lip synchronization in talking head videos. It enables realistic re-sync of lip movements to new audio on existing footage.

Videobench

<https://github.com/JNoDuq/videobench>

Command-line tool for analyzing video quality using VMAF, PSNR, and bitrate measurements.

Videocr

<https://github.com/apm1467/videocr>

Extract hardcoded subtitles from videos using machine learning

Videogrep

<https://github.com/antiboredom/videogrep>

automatic video supercuts with python.

Voice Data Extract

<https://github.com/abinashmeher999/voice-data-extract>

A command line interface to combine text information from subtitles with voice data in the video. Provides a convenient way to generate training data for speech-recognition purposes. - abinashmeher...

Vtt JS

<https://github.com/video-dev/vtt.js>

A JavaScript implementation of the WebVTT specification

Wav2Lip

<https://github.com/Rudrabha/Wav2Lip>

An AI model that achieves accurate lip-syncing in videos. Given an input video of a person and a target speech audio, Wav2Lip generates a video where the person's lip movements match the audio perfectly.

Web Video Text Tracks Format (WebVTT)

https://developer.mozilla.org/en-US/docs/Web/API/WebVTT_API

Web Video Text Tracks Format (WebVTT) is a format for displaying timed text tracks (such as subtitles or captions) using the track element.

Web Video Text Tracks Format (WebVTT) (W3C)

<https://www.w3.org/TR/webvtt1/>

The W3C recommendation for WebVTT, a text format for captioning and subtitling web video. It defines the syntax for caption files used in HTML5 video.

Tags: subtitles, captions, HTML5, web video, text tracks

Webvtt Py

<https://github.com/glut23/webvtt-py>

Read, write and segment WebVTT caption files in Python.

Webvtt Ruby

<https://github.com/opencoconut/webvtt-ruby>

WebVTT Ruby parser and segmenter.

Wisual

<https://github.com/MarcAntoine-Arnaud/wisual>

Web for Visual Quality Assessment.

YUView

<https://github.com/IENT/YUView>

YUView is a QT based, cross-platform YUV player with an advanced analytic toolset.

YouTube-DL

<https://yt-dl-org.github.io/youtube-dl/>

The classic command-line program for downloading videos from YouTube and hundreds of other sites. youtube-dl is open source and supports extracting audio, selecting formats, and many customization options.

Tags: download, video, extraction, command-line, open-source

Zubtitle

<https://zubtitle.com/>

Zubtitle automates the subtitling process, making it easy for beginners to add accurate captions. This AI-powered tool is designed to create professional subtitles quickly.

Tags: subtitle integration, video editing, AI tools, automation

api.video In-stream Ads

<https://docs.api.video/delivery/ads>

Documentation on implementing in-stream ads in videos and live streams using api.video's platform, supporting VAST and VPAID tags for ad integration.

Tags: api.video, in-stream ads, VAST, VPAID, video streaming

create-subtitles

<https://github.com/geomedialab/create-subtitles>

The 'create-subtitles' tool is designed to generate subtitles from a video and transcript using Python and the YouTube Data API. It assists in turning verbatim transcripts for video recordings into subtitle files, facilitating the integration of subtitles into videos. The tool can also be used for concatenating standalone subtitle files, making it versatile for various subtitle integration tasks.

Tags: subtitle-generation, Python, YouTube-API, video-transcripts, automation

easyVmaf

<https://github.com/gdavila/easyVmaf>

Python script to easily compute VMAF using FFmpeg. It allows to deinterlace, scale and sync Ref and Distorted video automatically

ffsubsync

<https://github.com/smacke/ffsubsync>

A tool to automatically synchronize subtitles with video by analyzing audio tracks. It uses speech detection to align subtitle timing via FFmpeg and machine learning.

flvmeta

<https://github.com/noirotm/flvmeta>

An open source tool to manipulate Adobe FLV files. flvmeta can inject onMetaData tags, fix timestamps, and generally prepare FLV files for optimal playback (seeking). Useful for legacy content or repackaging streams from RTMP to file.

go-astisub

<https://github.com/asticode/go-astisub>

A Golang library for manipulating subtitles (supports formats like SRT, WebVTT, SSA/ASS, etc.), useful for automated subtitle processing.

Tags: subtitle, library, golang, processing, SRT, WebVTT, SSA, ASS

imscJS: IMSC/TTML/SMPTE-TT/EBU-TT-D renderer

<https://github.com/sandflow/imscJS/>

Renders IMSC/TTML/SMPTE-TT/EBU-TT-D subtitles and captions to HTML

ironSource Mobile Ad SDK

<https://www.is.com/mobile-advertising-sdk/>

ironSource provides a comprehensive mobile ad SDK for Android and iOS, enabling developers to integrate various ad formats, including rewarded videos, interstitials, and banners. The SDK focuses on maximizing user engagement and revenue through effective ad monetization strategies.

Tags: mobile ads, SDK, monetization, ironSource, ad integration

libebur128

<https://github.com/jiixyj/libebur128>

An open source C library implementing the EBU R128 loudness standard. It can be used to measure and normalize audio loudness (LUFS) for broadcast compliance.

Tags: audio, loudness, broadcast, standards, normalization, LUFS

madman-android

<https://github.com/flipkart-incubator/madman-android>

Madman (Media Ads Manager) is a high-performance alternative to Google's standard IMA Android SDK, allowing developers to render video ads with full control over the UI using their own VAST server.

Tags: Android, ads, VAST, IMA SDK, media

normalizing Audio

<https://www.learndigitalaudio.com/normalize-audio>

Educational resource explaining audio normalization techniques for adjusting audio levels to consistent volumes across digital media.

product_video_ads

https://github.com/google/product_video_ads

A solution developed by Google to build video ads at scale, enabling the creation of product video ads efficiently.

Tags: video ads, Google, product ads, video, advertising

pysrt

<https://github.com/byroot/pysrt>

pysrt is a Python library for parsing, modifying, and composing SubRip (SRT) files. It provides developers with tools to handle subtitle files programmatically, enabling tasks such as reading, writing, and editing subtitles within Python applications.

Tags: subtitle-parsing, SRT, Python, library, subtitles

pysubs2

<https://github.com/tkarabela/pysubs2>

pysubs2 is a Python library for editing subtitle files, supporting various formats including SubRip (SRT), SubStation Alpha (SSA), and Advanced SubStation Alpha (ASS). It allows for reading, writing, and modifying subtitle files, making it a versatile tool for developers working with subtitles in Python.

Tags: subtitle-editing, Python, library, SSA, ASS

srt

<https://github.com/cdown/srt>

srt is a simple library and set of tools for parsing, modifying, and composing SRT files. It offers a straightforward approach to handling subtitle files, making it suitable for developers who need to work with subtitles in their applications.

Tags: subtitle-parsing, SRT, Python, library, subtitles

swiftsubtitles

<https://github.com/dagronf/swiftsubtitles>

A Swift package for reading and writing various subtitle formats, including SRT, SBV, SUB, VTT, CSV, LRC, and Podcast Index. It enables iOS and macOS developers to integrate subtitle functionalities into their applications.

Tags: swift, subtitles, subtitles-parsing

threefive

<https://github.com/futzu/threefive>

A Python library for parsing SCTE-35 in various formats, including MPEGTS, HLS, and DASH, facilitating ad insertion and management in video streams.

Tags: SCTE-35, ad-insertion, MPEGTS, HLS, DASH

truex-ad-renderer-web-integration

<https://github.com/socialvibe/truex-ad-renderer-web-integration>

Documentation and resources for integrating true[X]'s CTV web ad renderer, providing guidelines for implementing interactive ad experiences in connected TV environments.

Tags: true[X], CTV, ad renderer, interactive ads, integration

ttconv: subtitle/caption format converter

<https://github.com/sandflow/ttconv>

Converts EBU STL, IMSC/TTML/SMPTE-TT/EBU-TT-D and 608/SCC into IMSC, WebVTT and SRT

v4l (Go Video4Linux)

<https://pkg.go.dev/github.com/korandiz/v4l>

A Go library providing bindings to Video4Linux2 (V4L2) APIs, enabling video capture and camera control in Linux from Go applications.

videojs-ads

<https://github.com/GLStephen/videojs-ads>

A plugin for Video.js that provides common functionality needed by video ad integrations, simplifying the process of incorporating advertisements into video content.

Tags: video.js, ads, plugin, video, integration

videojs-contrib-ads

<https://videojs.github.io/videojs-contrib-ads/integrator/getting-started.html>

A Video.js plugin that provides a framework for creating ad integrations, offering a set of events and methods to manage ad playback and content video interactions.

Tags: Video.js, ads, plugin, integration, framework

vigenair

<https://github.com/google-marketing-solutions/vigenair>

Recrafting Video Ads with Generative AI, this project leverages machine learning to enhance video advertisements, utilizing technologies like Google Cloud Vision and Vertex AI.

Tags: machine-learning, video-ads, AI, Google-Cloud, generative-ai

web-monetisation-video-ads

<https://libraries.io/npm/web-monetisation-video-ads>

A utility that enables developers to monetize videos by utilizing web monetization when available and loading ads as a fallback through the IMA SDK.

Tags: web-monetization, IMA SDK, advertising, monetization, video

x9k3

<https://github.com/futzu/x9k3>

An HLS segmenter with SCTE-35 support, enabling live streaming from non-live sources and looping, useful for dynamic ad insertion scenarios.

Tags: HLS, SCTE-35, live-streaming, ad-insertion, segmenter

xk media library

<https://github.com/chapmanjacobd/library>

Scan millions of files with FFmpeg and access the metadata as a SQLite database. Also, a CLI alternative to media browsers like Plex or Jellyfin.

yt-dlp

<https://github.com/yt-dlp/yt-dlp>

A command-line program to download videos from YouTube and many other video platforms. It's a fork of youtube-dl with additional features and fixes, widely used for video retrieval.

zveloDB

<https://zvelo.com/case-study/digital-video-ad-platform-roi-brand-safety/>

zveloDB is a URL database that enhances brand safety in digital video advertising platforms. By integrating zveloDB, platforms can ensure that video ads are served in appropriate contexts, protecting brand reputation and providing greater transparency in ad placements.

Tags: brand safety, URL database, ad placement, transparency, video ads

Infrastructure & Delivery

9 Best Home Server Apps to Automate Media Management

<https://www.smarthomebeginner.com/best-home-server-apps/>

These are top 9 best home server apps to automate media management, so you get the latest Movies, Music and TV Shows in the best quality available.

AVideo CDN Storage

<https://github.com/WWBN/AVideo/wiki/CDN-Storage>

AVideo offers a CDN Storage solution that automatically replicates video files globally, ensuring low latency and high throughput for video streaming. It provides infinite scalability, strategic positioning of data, and direct CDN integration, making it valuable for developers seeking efficient content delivery.

Tags: CDN, video streaming, content delivery, scalability, low latency

AWS CDN Video Streaming Service

<https://github.com/Shereefo/AWS-CDN-Video-Streaming-Service>

This project leverages Amazon S3, Amazon CloudFront, and React to build a scalable and efficient video streaming service. It allows users to view a variety of streaming content through a web-based interface, ensuring optimized security, performance, and reliability through AWS's global network of data centers.

Tags: AWS, CDN, video streaming, React, scalable

AWS CloudFront Live failover

https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/high_availability_origin_failover.html

You can set up CloudFront with origin failover for scenarios that require high availability. To get started, create an origin group in which you designate a primary origin for CloudFront plus a second origin that CloudFront automatically switches to when the primary origin returns specific HTTP status code failure responses.

AWS CloudFront for Live Streaming

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/live-streaming.html>

To use AWS Media Services with CloudFront to deliver live content to a global audience, follow the guidance included in this section.

AWS Elemental Media Services

<https://aws.amazon.com/media/>

AWS Elemental Media Services offer a suite of cloud-based solutions for video processing and delivery, including live and on-demand streaming. These services enable scalable, secure, and cost-effective video workflows, making them valuable for developers building video streaming and encoding applications.

Tags: video streaming, cloud services, AWS, media processing, live streaming

AWS Elemental MediaConnect

<https://aws.amazon.com/mediaconnect/features/>

AWS Elemental MediaConnect is a reliable, secure, and flexible transport service for live video that enables broadcasters and content owners to build live video workflows and securely share live content with partners and customers. It supports a range of protocols for video delivery, including Zixi, RIST, SRT, RTP, and RTP with FEC, facilitating high-quality video sharing and real-time monitoring.

Tags: live video transport, video streaming, AWS, media services, video protocols

AWS Elemental MediaConvert

<https://aws.amazon.com/mediaconvert/>

AWS Elemental MediaConvert is a file-based video transcoding service that allows for the conversion of content into different formats to ensure compatibility across multiple playback devices. It provides comprehensive cloud-based video processing solutions, including encoding, transcoding, packaging, and live streaming services.

Tags: video transcoding, file-based processing, cloud service, AWS, media processing

AWS Elemental MediaLive

<https://aws.amazon.com/medialive/>

AWS Elemental MediaLive is a cloud-based live video processing service that enables users to encode live video streams for broadcast and streaming to various devices. It integrates seamlessly with other AWS services to create a comprehensive video processing pipeline in the cloud.

Tags: live streaming, video encoding, AWS, cloud, broadcast

AWS Elemental MediaPackage

<https://aws.amazon.com/mediapackage/>

AWS Elemental MediaPackage is a video origination and just-in-time packaging service that allows users to securely and reliably deliver live and on-demand video content to a wide range of devices. It supports multiple delivery formats, content protection mechanisms, and integrates with other AWS services to build scalable video workflows.

Tags: video packaging, live streaming, on-demand video, AWS, content delivery

AWS Elemental MediaTailor

<https://aws.amazon.com/mediatailor/>

AWS Elemental MediaTailor is a content personalization and monetization service that allows users to serve targeted ads to viewers while maintaining broadcast-quality streaming. It supports server-side ad insertion, ensuring a seamless viewing experience across devices, and integrates with other AWS services for comprehensive video workflows.

Tags: ad insertion, content personalization, monetization, AWS, video streaming

AWS Media & Entertainment Cloud Solutions

<https://aws.amazon.com/media/resources/>

AWS offers a range of cloud solutions tailored for media and entertainment, including OTT video workflows, live sports workflows, machine learning and analytics for media, and monetization of video workflows. These solutions enable developers to build secure, scalable, and integrated video streaming infrastructures, focusing on innovation rather than infrastructure management.

Tags: AWS, media solutions, cloud video, OTT workflows, video monetization

AWS Media Services

<https://aws.amazon.com/media-services/>

AWS Media Services offer a suite of cloud-based services that enable the creation, transformation, and delivery of digital content. These services include AWS Elemental MediaConnect for live video transport, AWS Elemental MediaConvert for file-based video transcoding, and AWS Elemental MediaLive for live video processing. They provide scalable and flexible solutions for video streaming and encoding, allowing developers to build and adapt video workflows quickly without the overhead of managing physical infrastructure.

Tags: cloud video services, video streaming, video encoding, AWS, media processing

Adxchange Engine

<https://github.com/Eyevinn/adxchange-engine>

Eyevinn Adxchange Engine is a microservice placed between the server-side ad-insertion component and the adserver or SSP

Amazon CloudFront Streaming Tutorials

<https://aws.amazon.com/cloudfront/streaming/>

Amazon CloudFront offers detailed tutorials and best practices for implementing video streaming workflows using their CDN services. These resources guide developers through setting up scalable, distributed architectures for ingesting, storing, processing, and delivering video content efficiently.

Tags: video streaming, CDN, Amazon CloudFront, video delivery, AWS

Amazon S3 | Fastly Help Guides

<https://docs.fastly.com/en/guides/amazon-s3>

Documentation for integrating Amazon S3 object storage with Fastly's CDN platform to optimize content delivery and distribution.

Ant Media Server

<https://antmedia.io/>

Ant Media Server is a highly scalable real-time video streaming platform that supports live and on-demand streaming with ultra-low latency. It offers compatibility with various cloud networks, including AWS, Azure, GCP, and Oracle Cloud, and provides SDKs for multiple platforms to facilitate broad audience reach.

Tags: video streaming, real-time, low latency, cloud, SDK

Ant Media Server (GitHub)

<https://github.com/ant-media/Ant-Media-Server>

Ant Media Server supports RTMP, RTSP, WebRTC and Adaptive Bitrate. It can also record videos in MP4, HLS and FLV

Tags: streaming, server, RTMP, RTSP, WebRTC, recording, media server

Apache Traffic Control

<https://trafficcontrol.apache.org/>

An open source CDN control plane originally contributed by Comcast. Apache Traffic Control can create a CDN with edge cache servers (like Apache Traffic Server) and provides tooling to manage content delivery, making it useful for building an edge caching infrastructure for video.

Tags: cdn, content delivery, edge caching, infrastructure, open source

Apache Traffic Server

<https://trafficserver.apache.org/>

An open source HTTP/1.1 and HTTP/2 caching proxy server that can be used as an edge cache in video streaming content delivery networks (CDNs) to improve scalability.

Tags: caching, proxy, CDN, edge cache, HTTP, content delivery

Ateme

<https://www.ateme.com/>

Ateme is a multinational company specializing in video compression, CDN/streaming, cloud recording, and advertising. They develop software for video compression based on standards like MPEG2, H.264/AVC, H.265/HEVC, AV1, and H.266/VVC. Ateme's solutions are used by content providers, broadcasters, and streaming providers for delivery over various networks, offering tools for efficient video processing and delivery.

Tags: video compression, CDN, streaming, cloud recording, video processing

Ateme KYRION

<https://www.ateme.com/products/kyrion/>

Ateme's KYRION is a high-quality, ultra-low-latency encoding and decoding appliance. It is designed for live video delivery, providing broadcasters and service providers with reliable and efficient solutions for real-time content distribution.

Tags: encoding, decoding, live streaming, low latency, broadcast

Ateme NEA

<https://www.ateme.com/products/nea/>

Ateme's NEA is a software suite that provides origin, packaging, storage, analytics, and CDN capabilities. It enables efficient content delivery over various networks, including cable, satellite, IPTV, terrestrial, and OTT, supporting formats like MPEG2-TS, HLS, and DASH.

Tags: content delivery, CDN, packaging, analytics, OTT

Ateme PILOT

<https://www.ateme.com/products/pilot/>

Ateme's PILOT is a management and operations software suite that facilitates the deployment and monitoring of video delivery services. It offers tools for resource management, service orchestration, and operational analytics, ensuring optimal performance and reliability.

Tags: management, operations, orchestration, analytics, video delivery

Avbroadcast

<https://github.com/media-toolbox/avbroadcast>

Avbroadcast - republish media streams for mass consumption

BLSS

<https://github.com/gnolizuh/BLSS>

NGINX-based Live Media Streaming Server.

Tags: nginx, live streaming, media server, server-side, streaming infrastructure

BemTV

<https://github.com/streamroot/bemtv>

BemTV is an open-source P2P library tailored for the HTTP Live Streaming (HLS) protocol. It introduces a hybrid CDN/P2P architecture for live video broadcasts over the internet, leveraging WebRTC to decentralize delivery. This approach reduces the number of requests to CDN servers, thereby decreasing transmission costs and enhancing system scalability. BemTV is valuable for developers exploring peer-to-peer solutions to scale live video streaming without external plugins.

Tags: P2P streaming, HLS, WebRTC, CDN integration, open source

Bitmovin

<https://bitmovin.com/>

Bitmovin provides cloud-based video encoding and player solutions, offering high-quality streaming experiences with support for various codecs and formats. It enables efficient video workflows with features like adaptive streaming and analytics.

Tags: video encoding, adaptive streaming, cloud, analytics, player

Bitmovin CDN Output

<https://developer.bitmovin.com/encoding/docs/bitmovin-cdn-output>

Bitmovin's Content Delivery Network (CDN) simplifies content distribution by automatically distributing your encoded content globally. It removes the need to set up your own output storage, enabling fast and efficient delivery to viewers without additional effort.

Tags: CDN, video encoding, content distribution, Bitmovin, cloud service

Bitmovin Encoding API

<https://bitmovin.com/encoding/>

Bitmovin's Encoding API offers cloud-based video encoding services that support a wide range of codecs and formats. It provides high-speed encoding with features like per-title encoding, multi-codec streaming, and DRM integration. This API is designed for developers seeking efficient and scalable video encoding solutions in the cloud.

Tags: video encoding, cloud encoding, API, multi-codec, DRM

Bitmovin Streams

<https://bitmovin.com/video-streaming-apis-sdks-ovps-explained/>

Bitmovin Streams is a cloud-based video streaming solution that simplifies the process of uploading, encoding, and delivering on-demand and live video content. It features a customizable player, in-depth analytics, and content protection mechanisms, providing developers with an end-to-end streaming infrastructure.

Tags: video streaming, live streaming, video encoding, analytics, content protection

Bitmovin Video Encoding API

<https://bitmovin.com/video-encoding/>

Bitmovin offers a cloud-based video encoding solution that supports HEVC and VP9 live streaming with up to 60FPS and 4K resolution. It provides containerized video encoding using Docker and Kubernetes, ensuring scalability and flexibility for developers. The platform is known for its in-house development, ease of integration, and high customizability, making it suitable for enterprise-level applications.

Tags: video encoding, cloud video, HEVC, VP9, Docker

Bitmovin Video Streaming APIs and SDKs

<https://bitmovin.com/blog/video-streaming-apis-sdks-ovps-explained/>

Bitmovin provides video streaming APIs and SDKs that offer comprehensive functionality across encoding, playback, and analytics. Their solutions support per-title encoding, allowing customized settings for each video to ensure high-quality viewing experiences and efficient data usage. Bitmovin's VOD and Live encoders can be deployed on various cloud infrastructures, providing flexibility and cost efficiency. They also offer customizable HTML5 players and SDKs for different platforms.

Tags: video streaming APIs, SDKs, per-title encoding, HTML5 player, cloud deployment

BlazingCDN

<https://blazingcdn.com/>

BlazingCDN offers a cloud-based content delivery network optimized for video streaming. It provides low-latency, high-speed delivery of video content to a global audience, ensuring a smooth streaming experience. The platform supports various video formats and integrates with existing video platforms, making it a valuable resource for developers seeking efficient content distribution solutions.

Tags: CDN, video streaming, content delivery, cloud services, BlazingCDN

BlazingCDN Integration Best Practices

<https://blog.blazingcdn.com/en-us/best-practices-for-cdn-integration-with-existing-it-systems>

BlazingCDN offers a detailed blog post outlining best practices for integrating CDNs with existing IT systems. The article emphasizes seamless integration and configuration, including DNS setup, cache rules, load balancing, and security measures. This resource is beneficial for developers aiming to enhance video streaming services through effective CDN integration.

Tags: CDN integration, BlazingCDN, best practices, load balancing, security measures

Brightcove

<https://www.brightcove.com/>

Brightcove is a cloud-based video platform that provides on-demand and live video streaming, with an emphasis on monetization and marketing. It offers tools for generating followers, driving brand awareness, producing pipeline, increasing income, and interacting with employees.

Tags: video streaming, cloud video, monetization, marketing, enterprise video

Brightcove Open Source Projects

<https://general.support.brightcove.com/general/brightcove-open-source-projects.html>

A collection of open-source tools and libraries developed by Brightcove to enhance video streaming capabilities. These projects include API wrappers, troubleshooting scripts, and sample applications, providing developers with resources to integrate and manage video content effectively.

Tags: video streaming, open source, API, Brightcove, developer tools

Byte Down: Making Netflix's Data Infrastructure Cost-Effective

<https://netflixtechblog.com/byte-down-making-netflixs-data-infrastructure-cost-effective-fee7b3235032>

Netflix engineering blog post describing strategies for reducing data infrastructure costs in large-scale video operations.

BytePlus Live

<https://www.byteplus.com/en/topic/86429?title=top-10-tools-for-video-streaming-api-in-2024-byteplus-live-leads-the-pack>

BytePlus Live offers a comprehensive toolkit for creating seamless, scalable video streaming applications. Key features include low-latency video transmission, cross-platform compatibility, adaptive bitrate streaming, advanced analytics and monitoring, and robust security features. BytePlus Live is designed to help developers and SMBs deliver high-quality, real-time video experiences efficiently.

Tags: video streaming API, low-latency streaming, adaptive bitrate, video analytics, security

CDN Transcode Sample

<https://www.intel.com/content/www/us/en/developer/articles/technical/cdn-transcode-sample-powered-by-open-visual-cloud.html>

The CDN Transcode Sample, powered by Open Visual Cloud, demonstrates how to integrate media transcoding and streaming services optimized for Intel® Xeon® processors. It utilizes an FFmpeg-based media transcoding stack and an NGINX-based streaming and web hosting stack. This sample is beneficial for developers aiming to build efficient and scalable video streaming solutions with CDN integration.

Tags: media transcoding, CDN integration, FFmpeg, NGINX, Intel Xeon

CDN Video Streaming

<https://github.com/shaowenwei/cdn-video-streaming>

This project delves into video streaming via CDN, focusing on implementing adaptive bitrate selection, DNS load balancing, and an HTTP proxy server to stream video efficiently. It provides insights into building a video CDN, making it a valuable resource for developers aiming to enhance their understanding of video content distribution and CDN integration.

Tags: video streaming, CDN, adaptive bitrate, DNS load balancing, HTTP proxy

CDN Video Uploader and Transcoder

<https://github.com/SoftUni/CDN-Video-Uploader>

A tool designed for transcoding and uploading videos to a CDN network for HLS video streaming. It supports various input formats and resolutions, utilizes hardware-accelerated video transcoding, and automates the upload process to the target CDN.

Tags: video transcoding, CDN integration, HLS streaming, hardware acceleration, video uploader

CasparCG

<https://casparcg.com/>

An open source broadcast graphics and video playout server. CasparCG is used to deliver layered videos, images, and templates to professional broadcast environments in real-time.

Tags: broadcast, graphics, playout, real-time, server

CasparCG (GitHub)

<https://github.com/CasparCG/server>

A professional open source graphics and video playback server used in TV broadcasting for overlaying graphics, playing videos, and mixing live inputs on multiple outputs.

Channel Engine

<https://github.com/Eyevinn/channel-engine>

Engine for generating linear OTT TV channels from VOD content. Useful for creating 24/7 streaming channels.

ClipBucket

<https://github.com/arslancb/clipbucket>

An open source video sharing platform (YouTube-like) allowing users to upload, manage, and stream video content on their own server.

Cloud-Based Video Streaming

<https://github.com/spatil36/Cloud-Based-Video-Streaming>

This project demonstrates the development of a secure, on-demand video streaming platform using AWS services. It includes a React-based frontend, AWS Cognito for user management, and AWS Media Package for video processing, offering a comprehensive solution for building scalable video streaming applications.

Tags: video streaming, AWS, serverless, React, cloud computing

Cloudinary

<https://cloudinary.com/>

Cloudinary is an AI-powered cloud platform that automates the creation, management, and delivery of video with AI and machine learning, accelerating video workflows at scale. It offers features such as multiple-source fetches, uploads, auto-tagging, advanced search, transcription, format and codec selection, content-aware cropping, effects, previews, and thumbnails.

Tags: video management, AI-powered, cloud platform, media processing, automation

Cloudinary Cloud Video Streaming

<https://cloudinary.com/glossary/cloud-video-streaming>

Cloudinary provides cloud-based video streaming services that enable efficient storage and delivery of video content over the internet. It offers features like API access, live streaming, transcoding, and video analytics, allowing developers to integrate and manage video streaming capabilities seamlessly.

Tags: video, cloud, streaming, API, transcoding

Cloudinary Video API

https://cloudinary.com/video_api

Cloudinary provides a cloud-based video content management solution with APIs for uploading, storing, transcoding, and delivering videos. It offers real-time video manipulation, customizable video players, live streaming capabilities, and AI-based video tagging and transcription, simplifying video workflows for developers.

Tags: video API, video management, live streaming, AI tagging, video transcoding

Cloudinary Video CDN

<https://cloudinary.com/guides/web-performance/video-cdn-why-you-need-it-and-top-5-video-cdns>

Cloudinary provides a comprehensive Video CDN solution that enhances video delivery through features like adaptive bitrate streaming, video prefetching for live events, and optimized video formats for various devices. It also offers advanced capabilities such as dynamic overlays, allowing developers to add text, images, or other videos over the main video stream, facilitating personalized and interactive content delivery.

Tags: CDN, video streaming, adaptive bitrate, dynamic overlays, content delivery

Cloudinary Video Management Platform

https://cloudinary.com/blog/introducing_the_complete_video_solution_for_web_and_mobile_developers

Cloudinary offers a comprehensive cloud-based video content management solution for developers. It provides APIs and UI widgets for video upload, real-time video transcoding and manipulation via CDN delivery URLs, a customizable video player with user engagement analytics, live video streaming directly from web and mobile apps, and AI-based video tagging and transcription. This platform simplifies the video workflow for web and mobile developers while enhancing the end-user experience.

Tags: video management, cloud-based, video transcoding, live streaming, AI video tagging

Coconut Video Encoding API

<https://www.coconut.co/features/coconut-api>

A scalable and flexible video encoding API that integrates seamlessly into any existing platform. Coconut harnesses cloud technology to provide a powerful solution for video encoding, supporting multiple regions to ensure low latency.

Tags: video encoding, API, cloud technology, scalable, multi-region

Create your own video streaming server with Linux

<https://opensource.com/article/19/1/basic-live-video-streaming-server>

Using Nginx to create a streaming server using RTMP and nginx.

Creating a secure video-on-demand (VOD) platform using AWS

<https://aws.amazon.com/blogs/media/creating-a-secure-video-on-demand-vod-platform-using-aws/>

Authored by Chirag Oswal, Solution Architect, AWS, and Vikas Tiwari, Solution Architect Manager, AWS Video has become the primary means of Information sharing and learning. Customers are investing in innovative solutions to tap into the e-learning and video space. Video content is their IP and needs to be protected and securely delivered. Online video is a [...]

DTube

<https://d.tube/>

A decentralized video platform built on blockchain (Steem/LBRY), offering an alternative to YouTube where videos are distributed via a peer-to-peer network and blockchain stores metadata and rewards.

Tags: blockchain, decentralized, peer-to-peer, video platform, distributed infrastructure

Dacast Video Streaming Platform

<https://www.dacast.com/>

Dacast is a comprehensive video streaming platform that integrates with top-tier CDNs to provide reliable and high-quality video delivery. It offers features like video hosting, monetization options, and robust security protocols. Dacast's user-friendly interface and API access make it a valuable tool for developers looking to implement video streaming solutions with CDN support.

Tags: Dacast, video streaming, CDN integration, monetization, security

Dionysia

<https://github.com/dionysia-dev/dionysia>

Dionysia is a live streaming platform based on open-source software. It integrates various open-source projects to create a seamless experience for video developers, supporting multiple ingest protocols, customizable transcoding profiles, and adaptive bitrate streaming.

Tags: live streaming, open-source, video platform, transcoding, adaptive bitrate

Djmediastreamer

<https://github.com/davidbt/djmediastreamer>

A Django project that allows you to catalog and stream your videos (using FFmpeg to add subtitles and transcode).

Docker NGINX RTMP

<https://github.com/alfg/docker-nginx-rtmp>

Ø=Ü A Dockerfile for nginx-rtmp-module + FFmpeg from source with basic settings for

Tags: nginx, rtmp, docker, streaming, hls, server

Dtt2ip

<https://github.com/ebu/dtt2ip>

Broadcast to IP conversion for Wifi indoor coverage

Dvbshout

<https://github.com/njh/dvbshout>

Dvbshout takes an MPEG transport stream from a DVB card, extracts audio channels from stream, and sends the audio to an Icecast / Shoutcast server and/or a RTP multicast packet stream.

Dynamic Adaptive Streaming over HTTP (DASH)

https://en.wikipedia.org/wiki/Dynamic_Adaptive_Streaming_over_HTTP

Dynamic Adaptive Streaming over HTTP (DASH), also known as MPEG-DASH, is an international standard for adaptive bitrate streaming over HTTP. It enables high-quality streaming of media content by breaking the content into small segments and delivering them over HTTP. DASH is codec-agnostic and supports various container formats, making it versatile for different streaming scenarios. It allows for adaptive bitrate streaming, adjusting the quality of the media stream in real-time based on network conditions and device capabilities, ensuring a smooth viewing experience for users.

Tags: adaptive bitrate, HTTP streaming, MPEG-DASH, streaming standard, codec-agnostic

Edge Computing with Fastly CDN and Varnish VCL for Authenticated Requests - Endertech

<https://endertech.com/blog/edge-computing-fastly-cdn-varnish-vcl-authenticated-requests>

A brief explanation of how to use Fastly CDN and Varnish with authenticated requests to offload static file serving from the origin and edge caching.

EdgeCDN-X Open Source CDN

<https://github.com/EdgeCDN-X>

EdgeCDN-X is an open source CDN built on Kubernetes with ingress-nginx and CoreDNS for routing, and ArgoCD for fast config distribution to the edges. Supports static upstreams, S3 origins, URL signatures, static IP prefix routing, and geo routing.

Example Project: Video Streaming Platform

<https://docs.osaas.io/osaas.wiki/Example-Project%3A-Video-Streaming-Platform.html>

An example of a Video on Demand (VOD) streaming platform built with open web services in Eyevinn Open Source Cloud. It includes a VOD preparation pipeline, orchestrator, database, and a simple web application.

Tags: VOD, streaming, open source, cloud, Eyevinn

Eyevinn Live Encoding

<https://github.com/Eyevinn/live-encoding>

Eyevinn Live Encoding is an open-source live encoder based on ffmpeg and Shaka packager. It supports live transcoding to HLS and optionally MPEG-DASH, providing an origin for CDN shield to pull streams as well as push to CDN origin. It's a valuable tool for setting up live streaming infrastructures.

Tags: live encoding, ffmpeg, Shaka packager, HLS, MPEG-DASH

Fastly CDN

<https://www.fastly.com/>

Fastly is a real-time content delivery network that offers advanced features for video streaming, including instant purging, real-time analytics, and edge computing capabilities. Its integration with video platforms allows for low-latency streaming and efficient content delivery. Fastly's programmable edge platform enables developers to customize content delivery and optimize performance for their specific needs.

Tags: Fastly, CDN, video streaming, edge computing, real-time analytics

Flussonic Media Server

<https://flussonic.com/flussonic-media-server/>

Flussonic Media Server is a multi-purpose software solution for launching high-load video streaming services of any scale. It delivers video to millions of viewers worldwide and offers features like live streaming, recording, transcoding, and access authorization.

Tags: media server, streaming, transcoding, recording, high-load

Galene

<https://galene.org/>

A simple self-hosted SFU (Selective Forwarding Unit) for videoconferencing. Galene is lightweight and supports multiple rooms and groups with direct browser clients (using WebRTC).

Tags: webrtc, videoconferencing, server, self-hosted, real-time

Gohls

<https://github.com/shimberger/gohls>

A server that exposes a directory for video streaming via web interface

Tags: go, hls, video-server, web-streaming, directory-streaming

Google Cloud Media Supply Chain Solutions

<https://cloud.google.com/blog/products/media-entertainment/how-cloud-enables-the-media-supply-chain>

Google Cloud offers comprehensive solutions for building cloud-native broadcast media supply chains. These solutions enable media companies to streamline operations, reduce costs, and deliver engaging content to global audiences. Key features include AI integration, scalability, and a focus on openness and efficiency.

Tags: media supply chain, cloud-native, Google Cloud, AI integration, media operations

Google Cloud Video Services C++ Client Library

<https://googleapis.github.io/google-cloud-cpp/google/cloud/video/>

An idiomatic C++ client library for Google Cloud's video services, including Live Stream API, Transcoder API, and Video Stitcher API. This library enables developers to integrate live video streaming, video file transcoding, and dynamic content generation into their applications.

Tags: C++, Google Cloud, video services, API, client library

Google Media Transcoder

<https://cloud.google.com/transcoder>

Google Media Transcoder is a cloud-based service that provides scalable and efficient video transcoding capabilities. It supports a wide range of input and output formats, enabling developers to deliver high-quality video content across various devices and platforms.

Tags: video transcoding, Google Cloud, video processing, cloud services, developer tools

HLS.js P2P Engine

<https://github.com/cdnbye/hlsjs-p2p-engine>

A hls.js plugin to offload bandwidth from expensive traditional CDNs while also maximizing

Tags: p2p, cdn, hls, streaming, bandwidth-optimization

HTTP Live Streaming (HLS)

https://en.wikipedia.org/wiki/HTTP_Live_Streaming

HTTP Live Streaming (HLS) is an HTTP-based adaptive bitrate streaming protocol developed by Apple Inc. It allows for the delivery of live and on-demand content by breaking the media into small segments and serving them over HTTP. HLS supports encryption and secure key distribution, providing a simple DRM system. It is widely supported across various devices and platforms, making it a popular choice for streaming media content. HLS enables adaptive bitrate streaming, adjusting the quality of the stream based on network conditions and device capabilities to provide a seamless viewing experience.

Tags: adaptive bitrate, HTTP streaming, HLS, Apple, streaming protocol

How To Setup NGINX For HLS Video Streaming On CentOS 7

<https://dev.to/samuyi/how-to-setup-nginx-for-hls-video-streaming-on-centos-7-3jb8>

How to live stream videos with Nginx.

HunyuanVideo: A Systematic Framework For Large Video Generative Models

<https://github.com/Tencent/HunyuanVideo>

HunyuanVideo is an open-source video foundation model that demonstrates performance in video generation comparable to, or even surpassing, leading closed-source models. It encompasses a comprehensive framework integrating data curation, advanced architectural design, progressive model scaling and training, and efficient infrastructure tailored for large-scale model training and inference. HunyuanVideo aims to bridge the gap between closed-source and open-source communities, empowering individuals to experiment with their ideas and fostering a more dynamic video generation ecosystem.

Tags: video generation, open-source, large-scale model, deep learning, video technology

IBM Cloud Video

<https://www.ibm.com/cloud/video>

IBM Cloud Video is an end-to-end platform providing seamless live and on-demand video content delivery across multiple platforms, including employee communications and virtual conferences. With advanced AI capabilities, it enhances viewer engagement and optimizes video performance in real time.

Tags: video streaming, cloud video, enterprise video, AI, video platform

Icecast

<https://icecast.org/>

An open source streaming media server primarily for audio streaming (Ogg/Opus), which can also deliver video in formats like Ogg/Theora. Commonly used for internet radio and basic video streams.

Tags: streaming, server, open-source, audio, video, internet radio

ImageKit Video CDN

<https://imagekit.io/guides/video-cdn>

ImageKit offers a Video CDN solution that includes features like dynamic overlays, allowing developers to add text, images, or other videos over the main video stream. This capability enables personalized and interactive content delivery, enhancing user engagement and experience.

Tags: CDN, video streaming, dynamic overlays, content delivery, user engagement

Inca — Message Tracing and Loss Detection For Streaming Data @ Netflix

<https://medium.com/@NetflixTechBlog/inca-message-tracing-and-loss-detection-for-streaming-data-netflix-de4836fc38c9>

At Netflix, our real-time data infrastructure have embraced the multi-cluster Kafka architecture and Flink powered stream processing...

Intel Cloud Streaming

<https://github.com/intel/cloud-streaming>

A comprehensive suite of software, drivers, and utilities to set up streaming and video encoding services in the cloud. It enables remote access to Android and Windows cloud instances, providing tools for screen capture, encoding, and streaming.

Tags: Intel, cloud streaming, video encoding, remote access, Android, Windows

Janus WebRTC Server

<https://janus.conf.meetecho.com/>

An open source, general-purpose WebRTC server (SFU and gateway). Janus is modular and can support video conferencing, streaming, and SIP/RTSP/WebRTC interop through its plugin architecture.

Tags: WebRTC, server, SFU, gateway, video conferencing, streaming

Janus WebRTC Server (GitHub)

<https://github.com/meetecho/janus-gateway>

An open source WebRTC server (SFU and gateway) that allows the creation of videoconferencing, streaming, and other real-time multimedia applications.

Jellyfin

<https://jellyfin.org/>

An open source media server and client solution, providing streaming of video (and other media) to a variety of devices as a self-hosted alternative to Plex.

Tags: media server, self-hosted, streaming, open source, video hosting

Kaltura

<https://corp.kaltura.com/>

Kaltura is a comprehensive video platform offering solutions for webinars, virtual events, enterprise video content management, educational technology, and Cloud TV. It provides tools such as video portals, LMS and CMS extensions, virtual event and webinar platforms, and TV streaming apps, making it valuable for developers seeking versatile video streaming and encoding solutions.

Tags: video platform, enterprise video, webinars, Cloud TV, educational technology

Kaltura Community Edition

<https://github.com/kaltura/server>

The open source edition of the Kaltura video platform, which offers video content management, transcoding, and streaming services that can be self-hosted.

Kaltura Platform Install Packages

<https://github.com/kaltura/platform-install-packages>

Official deployment packages to install the Kaltura platform on server or cluster environments using native OS package managers. Kaltura provides a comprehensive video platform for managing, publishing, and distributing video content.

Tags: video platform, Kaltura, deployment, server, cluster

Kaltura nginx-vod-module

<https://github.com/kaltura/nginx-vod-module>

An NGINX module for Video on Demand streaming that enables MP4 repackaging and dynamic adaptive streaming (HLS/DASH) support on Nginx servers.

Kurento Media Server

<https://doc-kurento.readthedocs.io/en/stable/>

An open source WebRTC media server with advanced capabilities like mixing, transcoding, and computer vision integration. Kurento allows building videoconferencing, broadcasting, and augmented video applications by handling streams on the server side.

Tags: WebRTC, media server, transcoding, video processing, server-side streaming

Lambda Edge Tutorial

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/lambda-edge-how-it-works-tutorial.html>

This tutorial shows you how to get started with Lambda@Edge by helping you create and add a sample Node.js function that runs in CloudFront. The example that we walk through adds HTTP security headers to a response, which can improve security and privacy for a website. (That said, you don't need a website for this walkthrough; we simply add security headers to a response when CloudFront retrieves a file.)

Lambda@Edge Design Best Practices | Amazon Web Services

<https://aws.amazon.com/blogs/networking-and-content-delivery/lambdaedge-design-best-practices/>

Lambda@Edge transforms CloudFront into a highly programmable CDN with serverless compute capabilities closer to your viewers around the world. This blog is the first in a series that explains best practices associated with using Lambda@Edge functions to customize your content delivery.

Live Low-Latency Cloud-based Adaptive Video Streaming Evaluation Framework

<https://github.com/cd-athena/LLL-CAdViSE>

This open-source framework allows for the evaluation of adaptive video streaming in cloud environments. It supports testing multiple media players under different network conditions, providing insights into performance and quality of experience metrics.

Tags: adaptive streaming, low latency, cloud evaluation, video quality, open source

LiveAPI

<https://liveapi.com/>

LiveAPI offers a robust API for building live and on-demand video streaming applications. It provides features such as video upload, encoding, adaptive bitrate streaming, and global CDN delivery. LiveAPI simplifies the complexities of video streaming by handling encoding, storage, and delivery, making it a valuable resource for developers aiming to integrate video streaming capabilities into their applications.

Tags: video API, live streaming, on-demand streaming, CDN integration, adaptive bitrate streaming

LiveKit

<https://github.com/livekit/livekit>

End-to-end stack for WebRTC with a powerful SFU server, client SDKs, and cloud infrastructure. Supports scalable video conferencing, live streaming, and real-time communication with features like adaptive streaming, recording, ingress/egress, and comprehensive APIs for multiple platforms.

Livepeer

<https://livepeer.org/>

A decentralized live video streaming platform and network built on blockchain. Livepeer provides an open source protocol for scalable, affordable live streaming.

Tags: blockchain, decentralized, streaming infrastructure, live streaming, protocol

Marsha

<https://github.com/openfun/marsha>

A self-hosted opensource LTI video provider.

Media Server

<https://github.com/ireader/media-server>

RTSP/RTP/RTMP/FLV/HLS/MPEG-TS/MPEG-PS/MPEG-DASH/MP4/fMP4

MediaCMS

<https://mediacms.io/>

An open source video and media CMS for building a YouTube-like platform. It supports uploading, transcoding, and managing video content with a modern UI.

Tags: cms, video-platform, content-management, transcoding, video-hosting

MediaMTX

<https://github.com/bluenvirom/mediamtx>

MediaMTX is a free, open-source media server supporting real-time video streaming, RTSP, RTMP, HLS, and WebRTC. It enables the management and streaming of video from various sources, including RTSP cameras, with low-latency performance.

Tags: media server, streaming, RTSP, RTMP, open-source

MediaPlatform Video Delivery Services

<https://www.mediaplatform.com/platforms/video-delivery-services/>

MediaPlatform offers enterprise video delivery solutions that optimize live and on-demand video streaming across corporate networks. Their services include eCDN (Enterprise Content Delivery Network) for efficient video caching and delivery, Smartpath for rules-based distribution, and integration with various peering and WAN optimization technologies. These solutions ensure high-quality video experiences with minimal network impact, making them valuable for organizations looking to enhance their internal video communications.

Tags: enterprise video delivery, eCDN, video streaming, network optimization, corporate communications

Microsoft Rocket: Hybrid Edge + Cloud Video Analytics Platform

<https://www.microsoft.com/en-us/research/video/microsoft-rocket-hybrid-edge-cloud-video-analytics-platform/>

An extensible software stack developed by Microsoft Research that leverages both edge and cloud computing for scalable video analytics. Rocket is designed to facilitate real-time analysis of live camera streams, making it suitable for applications like security surveillance and traffic planning.

Tags: video analytics, edge computing, cloud computing, Microsoft, real-time analysis

Minio

<https://github.com/minio/minio>

MinIO is a high performance object storage server compatible with Amazon S3 APIs

MistServer

<https://mistserver.org/>

An open source streaming media server that supports multiple protocols (HLS, RTMP, WebRTC, etc.). It focuses on easy setup and compatibility with any device.

Tags: open source, media server, streaming, multi-protocol, HLS, RTMP, WebRTC

NGINX S3 Proxy

<https://github.com/coopernurse/nginx-s3-proxy>

nginx compiled with aws-auth support, suitable for S3 reverse proxy usage

NGINX with nginx-rtmp-module

<https://nginx.org/>

Nginx, combined with the nginx-rtmp-module, provides a powerful solution for live video streaming. This setup supports HTTP Live Streaming (HLS) in live mode, enabling efficient and scalable live video delivery. It's widely used for setting up live streaming servers due to its performance and flexibility.

Tags: live streaming, HLS, Nginx, RTMP, video delivery

Netflix Titus (Container orchestration for video)

<https://netflix.github.io/titus/>

Netflix's open source container management platform that runs video processing workloads among others. While not video-specific, Titus is key to how Netflix deploys and scales encoding and other batch jobs in a containerized fashion, ensuring efficient resource usage.

Tags: container, orchestration, netflix, cloud, kubernetes, infrastructure

Nimble Streamer

<https://wmspanel.com/nimble>

A lightweight, free live streaming server supporting HLS, DASH, RTMP, SRT, and more, often used for low-latency streaming and transmuxing in production workflows.

NodeTube

<https://github.com/mayeaux/nodetube>

An open-source self-hosted online video platform (YouTube alternative) supporting video/audio upload, livestreaming, and monetization.

Nvidia Docker

<https://github.com/NVIDIA/nvidia-docker>

Build and run Docker containers leveraging NVIDIA GPUs

OBS Studio

<https://github.com/obsproject/obs-studio>

OBS Studio - Free and open source software for live streaming and screen recording

OTT Content Delivery– CDN Alternatives

https://medium.com/@eyevinntechnology/ott-content-delivery-cdn-alternatives-cafe75dab71d?source=userActivityShare-94bcb50d11-1560983135&_branch_match_id=670018733519578135

Article exploring CDN alternatives for OTT content delivery. Covers infrastructure options for video streaming.

OTT Content Delivery– Multi CDN

https://medium.com/@eyevinntechnology/ott-content-delivery-multi-cdn-8cd90ad2628a?source=userActivityShare-94bcb50d11-1560983307&_branch_match_id=670019455010399744

Eyevin article explaining multi-CDN strategies and architectures for OTT video content distribution.

OblivCDN: A Practical Privacy-preserving CDN with Oblivious Content Access

<https://arxiv.org/abs/2501.07262>

OblivCDN is a privacy-preserving system designed for seamless integration with existing real-world Internet-CDN infrastructure. It adapts Range ORAM primitives to optimize memory and disk seeks when accessing contiguous blocks of CDN content, preserving both content confidentiality and user access pattern hiding features. OblivCDN eliminates the need for trusted hardware on edge servers, enhancing compatibility with networks containing legacy devices. In real-world streaming evaluations, it demonstrated remarkable performance, downloading a 256 MB video in just 5.6 seconds.

Tags: privacy-preserving, CDN, ORAM, content confidentiality, user privacy

Open Broadcaster Software | OBS

<https://obsproject.com/>

OBS (Open Broadcaster Software) is free and open source software for video recording and live streaming. Stream to Twitch, YouTube and many other providers or record your own videos with high quality H264 / AAC encoding.

Open Source Cloud by Eyevinn Technology

<https://www.osaas.io/>

A service that allows users to launch open-source video streaming and encoding tools in the cloud without the need to build, deploy, maintain, or host them. Open Source Cloud provides full code transparency and contributes back to the open-source community, offering a sustainable model for utilizing open-source software.

Tags: open source, cloud services, video streaming, encoding, Eyevinn Technology

OpenVisualCloud

<https://github.com/OpenVisualCloud>

OpenVisualCloud is an open-source project that provides optimized media, analytics, and graphics software stacks. It offers Dockerfiles for various media processing tasks, including transcoding and streaming, making it valuable for developers building cloud-based video solutions.

Tags: media processing, cloud computing, video streaming, open source, Docker

Opencast (GitHub)

<https://github.com/opencast/opencast>

An open source enterprise video capture and distribution system, commonly used by educational institutions to manage lecture recordings, process videos, and distribute them to students.

OvenMediaEngine

<https://github.com/AirenSoft/OvenMediaEngine>

A real-time streaming server with sub-second latency support. OvenMediaEngine offers WebRTC and Low Latency DASH/HLS for interactive live streaming.

Overcast

<https://www.overcasthq.com/>

Overcast is a Video Content-as-a-Service platform that automates complex video tasks using AI and ML. It offers customizable workflows using cloud applications, enabling efficient video content management and distribution.

Tags: video content management, AI, ML, cloud, automation

Owncast

<https://owncast.online/>

A self-hosted live video streaming server with a built-in chat. Owncast enables anyone to set up a live stream akin to Twitch, but on their own infrastructure.

Tags: self-hosted, live streaming, server, open-source, chat integration

P2P CDN SDK for JavaScript

<https://github.com/vadootvpeer/p2p-cdn-sdk-javascript>

A free JavaScript SDK that reduces video streaming costs by up to 90% and improves scalability by 6x using WebRTC-based peer-to-peer technology. It supports major players like Hls.js, Shaka Player, and Video.js, and both HLS and DASH streaming protocols.

Tags: P2P CDN, WebRTC, video streaming, JavaScript SDK, HLS, DASH

P2P Media Loader

<https://github.com/Novage/p2p-media-loader>

P2P Media Loader is an open-source engine designed for peer-to-peer streaming of live and on-demand video directly within web browsers. By leveraging WebRTC, it reduces the load on traditional CDNs and enhances scalability, making it a valuable tool for developers aiming to optimize video delivery and reduce infrastructure costs.

Tags: P2P streaming, WebRTC, live streaming, on-demand video, CDN optimization

Panopto Video Cloud Platform

<https://www.panopto.com/features/video-cms/video-cloud/>

Panopto's Video Cloud Platform provides secure storage and management of video content, enabling organizations to host private, internal videos with ease. It offers features like elastic scaling, regional hosting options, and intelligent video content delivery, ensuring optimal playback across devices. This platform is particularly beneficial for enterprises looking to manage and deliver video content efficiently.

Tags: video cloud, secure storage, elastic scaling, content delivery, enterprise video

PeerTube

<https://joinpeertube.org/>

A decentralized video sharing platform powered by ActivityPub and peer-to-peer (BitTorrent) directly in the browser, as a federated alternative to centralized sites.

Tags: decentralized, peer-to-peer, federation, video sharing, distributed infrastructure

PlayoutAutomation

<https://github.com/jaskie/PlayoutAutomation>

Television broadcast automation system.

Pseudo Live Playout

<https://github.com/Eyevinn/pseudo-live-playout>

Contribute to Eyevinn/pseudo-live-playout development by creating an account on GitHub.

PyLivestream

<https://github.com/scivision/PyLivestream>

Pure Python FFmpeg-based live video / audio streaming to YouTube, Facebook, Periscope, Twitch, and more

QUANTEEC

<https://quanteec.com/>

QUANTEEC provides a production-ready plugin that enables hybrid P2P CDN delivery to reduce delivery costs and improve QoE during peaks. The QUANTEEC SDK connects users to share data directly with each other, relieving dependence on CDN, rendering delivery more robust and suppressing over-deployments at peak levels. Every user becomes a delivery node for others. Quanteec chases resources where they are available to avoid deploying more infrastructures.

Tags: DRM, video-encryption, secure-streaming, video-player, content-protection

Qencode Cloud Video Services

<https://cloud.qencode.com/>

Qencode offers a suite of cloud-based video services, including video transcoding, live streaming, media storage, and content delivery. Their APIs allow developers to transcode videos into various formats and codecs, launch high-quality live streams globally, and manage media storage efficiently. The platform also provides a customizable HTML5 video player and features like per-title encoding to optimize file sizes using AI-powered video optimization.

Tags: video transcoding, live streaming, media storage, content delivery, video player

RTSP Simple Server

<https://github.com/aler9/rtsp-simple-server>

A simple, ready-to-use RTSP streaming server written in Go. It supports publishing and reading streams via RTSP and can also re-distribute via other protocols.

Rclone

<https://github.com/rclone/rclone>

rsync for cloud storage - Google Drive, Amazon Drive, S3, Dropbox, Backblaze B2, One Drive, Swift, Hubic, Cloudfiles, Google Cloud Storage, Yandex Files

Tags: cloud storage, sync, backup, file transfer, cloud services

Red5 Media Server

<https://www.red5.net/open-source-live-streaming/>

Red5 is a free, open-source media streaming server implemented in Java. It provides services similar to Adobe Flash Media Server, including streaming video and audio, recording client streams, and support for various protocols like RTMP, RTMPT, RTMPS, and RTMPE.

Tags: media server, open-source, streaming, Java, RTMP

Red5 Pro

<https://www.red5.net/>

Red5 Pro is a streaming server that leverages cloud infrastructure to dynamically meet scaling needs, avoiding the massive infrastructure costs associated with traditional CDNs. It offers sub-500 millisecond latency for live broadcasts and is suitable for developers requiring scalable and low-latency streaming solutions.

Tags: streaming server, cloud infrastructure, low latency, scalability, live streaming

Red5 Server

<https://github.com/Red5/red5-server>

Open source media server supporting RTMP, HLS, and WebRTC streaming protocols in Java.

Riot Games Keeps League of Legends Esports Rolling With Fully Cloud-Based Virtualized Workflow

<https://www.sportsvideo.org/2020/03/27/riot-games-keeps-league-of-legends-esports-rolling-with-fully-cloud-based-virtualized-production-workflow/>

Although the traditional sports world has come to a standstill due to the coronavirus pandemic, many major esports organizations are still going strong, hosting competitions remotely. Riot Games keeps League of Legends esports rolling with a fully cloud-based, virtualized production workflow.

SCTE-104/35 and Beyond: A Look at Ad Insertion in an OTT World

<https://www.tvtechnology.com/opinions/scte10435-and-beyond-a-look-at-ad-insertion-in-an-ott-world>

Ad Insertion is a very important part of many video delivery systems because of the monetization aspect—it generates revenue!

SRT Open Source project

<https://github.com/Haivision/srt>

Open source implementation of the Secure Reliable Transport protocol for low-latency video streaming over unreliable networks.

Secure Video Streaming with Signed URLs

<https://github.com/KHAOUITI-Apps/CDN>

A tutorial-based project that guides you through building a YouTube-like video streaming system using Spring Boot, securing video access with signed URLs. It implements token-based authentication to ensure secure and time-limited access to video streams.

Tags: secure video streaming, Spring Boot, signed URLs, token-based authentication, video content delivery

Srs

<https://github.com/ossrs/srs>

SRS is a simple live streaming cluster, a simple joy.

Tags: live streaming, rtmp, server, cluster, open-source

Storm Streaming Server

<https://www.stormstreaming.com/>

Storm Streaming Server is a fully scalable live video streaming software available for on-premise deployment. It is designed to manage mass concurrent streams with thousands of online viewers, offering ultra-low latency and easy integration with existing websites or applications. Storm Streaming Server is ideal for organizations with dedicated hardware and network infrastructure.

Tags: streaming server, ultra-low latency, scalable, live streaming, on-premise

Stream New

<https://github.com/muxinc/stream.new>

The repo for <https://stream.new>.

Streama

<https://github.com/streamaserver/streama>

Self hosted streaming media server. [HTTPS://docs.streama-project.com/](https://docs.streama-project.com/)

Streaming With SRT Protocol in OBS

<https://obsproject.com/wiki/Streaming-With-SRT-Protocol>

Guide for configuring and using the SRT protocol within OBS Studio for reliable streaming to various destinations.

Streamline: End-to-End Live Streaming System

<https://github.com/streamlinevideo/streamline>

A reference system for end-to-end live streaming video, covering capture, encoding, packaging, uplink, origin, CDN, and player. It utilizes commodity hardware and free software to provide a high-quality live streaming solution.

Tags: live streaming, end-to-end system, video capture, CDN integration, video player

SuperStreamer

<https://github.com/matvp91/superstreamer>

An open, scalable, online streaming setup that provides an all-in-one toolkit from ingest to adaptive video playback. Built for developers in need of video tooling, SuperStreamer offers a comprehensive solution for setting up and managing video streaming infrastructure.

Tags: video streaming, scalable, adaptive playback, developer tools, open source

Swift

<https://github.com/openstack/swift>

OpenStack object storage system commonly used for storing and serving large video files at scale.

Tags: cloud storage, object storage, distributed systems, infrastructure

Telestream Cloud Services

<https://www.telestream.net/telestream-cloud/>

Telestream offers cloud-based media processing services, including Vantage Gateway, Vantage Cloud, Qualify, and Timed Text Speech. These services provide unmatched format support, dynamic job orchestration, automated quality control, and auto-transcription capabilities. Developers can leverage these tools to ingest and output content in various broadcast and OTT streaming formats, scale workflows dynamically, and ensure content quality and compliance.

Tags: media processing, video transcoding, quality control, auto-transcription, cloud services

Terraform AWS Media Services Modules

<https://registry.terraform.io/modules/terraform-aws-modules/mediaConvert/aws/latest>

Terraform modules and examples for deploying AWS MediaConvert and MediaLive as infrastructure-as-code. This helps automate the provisioning of cloud encoding and live streaming resources, demonstrating best practices for treating video pipelines as code.

This is How I Reduced My CloudFront Bills by 80%

<https://medium.com/faun/this-is-how-i-reduced-my-cloudfront-bills-by-80-a7b0dfb24128>

If you are using S3 and CloudFront to host your content and noticed that your bills are increasing, read this!

ThreeFive SCTE35

https://github.com/futzu/scte-35_threefive

An SCTE-35 parser and decoder for analyzing ad markers in MPEG-TS streams, designed for broadcast workflows.

Tube

<https://github.com/prologic/tube>

Ø=Üú a Youtube-like (without censorship and features you don't need!) Video Sharing automatic transcoding to MP4 H.265 AAC, multiple collections and R...

Understanding Real-time Bidding for AVOD Services

<https://medium.com/@eyevinntechnology/understanding-real-time-bidding-for-avod-services-861ebfa8bd13>

We have in previous blog articles described the principles behind server-side ad-insertion and described some of the challenges with it as...

Understanding Server-Side Dynamic Ad Insertion

<https://medium.com/@eyevinntechnology/understanding-server-side-dynamic-ad-insertion-d7ed90e34aa2>

In this post we're explaining the principles behind Server-Side Dynamic Ad Insertion technology. If you are already familiar with video...

Unified Origin

<https://www.unified-streaming.com/products/unified-origin>

Unified Origin is a software solution that enables streaming of live and on-demand content using various protocols, including MPEG-DASH. It supports integration with CDNs and offers features like DRM support and adaptive bitrate streaming. This project is beneficial for developers aiming to deliver high-quality streaming experiences across multiple devices.

Tags: streaming server, MPEG-DASH, DRM, adaptive bitrate streaming, CDN integration

Unified Streaming Demos

<https://demo.unified-streaming.com/>

A showcase of Unified Streaming's capabilities, including adaptive streaming formats and DRM-protected playback.

Unified Streaming Features Page

<https://demo.unified-streaming.com/k8s/features/stable/#!/mpd>

A demonstration of Unified Streaming's feature set, including on-the-fly packaging and multi-format support.

VOD on AWS

<https://s3.amazonaws.com/solutions-reference/video-on-demand-on-aws/latest/video-on-demand-on-aws.pdf>

AWS reference architecture document for implementing video-on-demand streaming solutions using Amazon Web Services infrastructure and tools.

VOD2Live

https://docs.google.com/presentation/d/1Ua76BBaZKtTmaZrlfM_eG0vwz0ZAqPljreCSfB4qFQQ/edit?usp=sharing

VStream

<https://github.com/Benj-Qu/VStream>

VStream is a project focused on video streaming via CDN, implementing adaptive bitrate selection, DNS load balancing, and an HTTP proxy server. It provides a comprehensive setup for streaming video at high bitrates from the closest server to a client, including a round-robin load balancer and geographic distance load balancer. This project is beneficial for developers looking to understand and implement efficient video streaming solutions using CDN technologies.

Tags: video streaming, CDN integration, adaptive bitrate, DNS load balancing, HTTP proxy

VdoCipher

<https://www.vdocipher.com/>

VdoCipher offers a secure video streaming solution with DRM encryption to prevent unauthorized downloads and piracy. It provides a cloud-based platform for hosting, streaming, and protecting video content, ensuring secure delivery to viewers.

Tags: video streaming, DRM, video security, cloud video platform, video hosting

Video Streaming Studio

<https://github.com/TianYao12/Video-Streaming-Studio>

A video streaming platform similar to Netflix, allowing users to upload and watch videos. It utilizes React.js, Express.js, MongoDB, Amazon S3, and CloudFront CDN for efficient video delivery and storage.

Tags: video streaming, AWS S3, CloudFront, React.js, Express.js

Video Streaming via CDN

<https://github.com/andrewwong97/video-streaming-cdn>

This project implements adaptive bitrate selection, DNS load balancing, and an HTTP proxy server to stream video at high bit rates from the closest server to a given client. It includes components like miProxy (an HTTP proxy), a nameserver for DNS load balancing, and utilizes Mininet for network simulation.

Tags: adaptive bitrate, DNS load balancing, HTTP proxy, video streaming, CDN

Video Streaming via CDN by heaventourist

<https://github.com/heaventourist/Video-Streaming-via-CDN>

This project focuses on implementing adaptive bitrate selection, DNS load balancing, and an HTTP proxy server to stream video at high bitrates from the closest server to a given client. The HTTP proxy accepts connections from web browsers, modifies video chunk requests, resolves the web server's DNS name, opens a connection with the resulting IP address, forwards the modified request to the server, and returns the unmodified video chunks to the browser. The DNS server implements load balancing using round-robin and geographic distance methods.

Tags: adaptive bitrate, DNS load balancing, HTTP proxy, video streaming, CDN

Video on Demand on AWS Foundation

<https://docs.aws.amazon.com/solutions/latest/video-on-demand-on-aws-foundation/solution-overview.html>

This AWS solution provides a customizable architecture to build a video-on-demand workflow. It automatically provisions the necessary AWS services to create a scalable, distributed video-on-demand system, facilitating efficient video processing and delivery.

Tags: video on demand, AWS, cloud architecture, media processing, scalable workflows

Video-Streaming-Service

<https://github.com/hetjagani/Video-Streaming-Service>

This project presents a cloud computing-based video streaming service with a REST API implemented using the Spring framework. It features a load balancer to distribute requests across multiple servers and utilizes SQLite for database management, providing a scalable and efficient solution for video delivery.

Tags: video streaming, cloud computing, REST API, Spring framework, load balancing

Vimeo

<https://vimeo.com/>

Vimeo is an American video hosting, sharing, and services provider focusing on the delivery of high-definition video across a range of devices. It provides tools for video creation, editing, and broadcasting, as well as enterprise software solutions for video professionals.

Tags: video hosting, video sharing, video creation, enterprise video, video editing

Visual Cloud

https://en.wikipedia.org/wiki/Visual_Cloud

Visual Cloud refers to the implementation of visual computing applications that rely on cloud computing architectures, cloud-scale processing and storage, and ubiquitous broadband connectivity. It encompasses services such as media

processing, enhancement, and delivery, leveraging cloud infrastructure to provide scalable and efficient solutions for video streaming and encoding. This approach allows for optimized infrastructure tailored to visual computing application requirements.

Tags: visual computing, cloud computing, media processing, video streaming, cloud infrastructure

Vmapproxy

<https://github.com/Eyevinn/vmapproxy>

Proxy service from Eyevinn for handling VMAP and VAST ad insertion metadata in video streams.

WebRTC

<https://webrtc.org/>

WebRTC (Web Real-Time Communication) is a free and open-source project that provides web browsers and mobile applications with real-time communication capabilities via simple APIs. It enables audio and video communication to work inside web pages without the need for plugins, facilitating peer-to-peer data sharing.

Tags: real-time communication, peer-to-peer, audio, video, open-source

Wifibroadcast – Analog-like transmission of live video data

<https://befinitiv.wordpress.com/wifibroadcast-analog-like-transmission-of-live-video-data/>

Wifibroadcast is a project aimed at the live transmission of HD video (and other) data using wifi radios. One prominent use case is to transmit camera images for a first person view (FPV) of remote...

Wowza

<https://www.wowza.com/>

Wowza is a robust cloud video streaming platform built for businesses of all sizes. It offers a range of features that allow you to deliver high-quality video content without interruptions. With patented software designed specifically for live streaming, Wowza ensures flawless performance even during peak traffic conditions.

Tags: live streaming, video hosting, cloud video, video platform, enterprise video

Wowza CDN Integration Overview

<https://www.wowza.com/blog/how-do-content-delivery-networks-cdns-work>

Wowza offers a comprehensive overview of how Content Delivery Networks (CDNs) work, focusing on their integration with video streaming services. The resource explains the benefits of using CDNs, such as reduced latency and improved scalability, and provides insights into load balancing and real-time analytics. This guide is valuable for developers aiming to enhance their video streaming infrastructure through CDN integration.

Tags: CDN integration, Wowza, video streaming, load balancing, real-time analytics

Wowza Streaming Cloud

<https://www.wowza.com/products/streaming-cloud>

Wowza Streaming Cloud is a robust cloud-based platform designed for live and on-demand video streaming. It offers features like adaptive bitrate streaming, cloud transcoding, and low-latency delivery, making it suitable for businesses aiming to reach global audiences with high-quality video content.

Tags: live streaming, video on demand, cloud transcoding, low-latency streaming, adaptive bitrate

Wowza Streaming Engine

<https://www.wowza.com/products/streaming-engine>

Wowza Streaming Engine is a robust and customizable media server software that supports live and on-demand streaming. It offers support for various streaming protocols, including HLS, and provides features like real-time transcoding, adaptive bitrate streaming, and content security. It's suitable for developers looking to build scalable and reliable streaming solutions.

Tags: media server, live streaming, on-demand streaming, HLS, adaptive bitrate

YouPHPTube (AVideo)

<https://github.com/WWBN/AVideo>

An open source, self-hosted video sharing website platform (formerly YouPHPTube). It supports user accounts, video uploads, streaming, and plugins for features like live broadcasting and advertisement.

YuJa Enterprise Video Platform

<https://www.yuja.com/video-cms/video-cloud/>

YuJa offers an Enterprise Video Platform that serves as a comprehensive Video Content Management System (VCMS). It enables organizations to securely store, organize, and access media across platforms. Key features include automatic transcoding, deep branding capabilities, search-inside-video functionality, and secure content protection, making it a valuable resource for developers focusing on video infrastructure and delivery.

Tags: video content management, VCMS, enterprise video, video repository, secure video storage

ZLMediaKit

<https://github.com/ZLMediaKit/ZLMediaKit>

A high-performance C++ media server framework supporting WebRTC, RTSP, RTMP, HTTP-FLV, HLS, WebSocket streaming, and SRT, for building streaming server/client applications.

api.video

<https://api.video/>

api.video provides a cloud-based video platform that allows developers to integrate video functionalities into their applications seamlessly. It offers services for video uploading, hosting, transcoding, and delivery, enabling on-demand and live-streaming capabilities. The platform is designed for scalability and ease of use, making it suitable for developers looking to incorporate video features without managing complex infrastructure.

Tags: video API, cloud video platform, video streaming, video hosting, developer tools

cdnjs

<https://cdnjs.com/>

cdnjs is a free and open-source content delivery network (CDN) hosted by Cloudflare. It serves over 4,000 JavaScript and CSS libraries, which are stored publicly on GitHub. cdnjs is included in millions of websites, making it the second most popular CDN for JavaScript. It is maintained by the community and sponsored by companies like DigitalOcean, Algolia, Heroku, Atlassian, Sentry, and Lean20.

Tags: CDN, JavaScript, CSS, open source, Cloudflare

datarhei Restreamer

<https://datarhei.com/>

datarhei Restreamer is an open-source video streaming server that provides a simple yet powerful solution for live streaming. It features a fully documented API, making it easy to integrate into various projects. Restreamer is suitable for both personal and commercial applications.

Tags: streaming server, live streaming, open-source, API, self-hosted

go2rtc

<https://github.com/AlexxIT/go2rtc>

A flexible high-performance streaming server written in Go that can ingest RTSP, WebRTC, RTMP, HTTP-FLV, and LL-HLS and output streams to multiple protocols (commonly used for home camera streaming).

hlsjs-p2p-engine

<https://cdnbye.github.io/hlsjs-p2p-engine/>

hlsjs-p2p-engine enhances video streaming scalability by implementing WebRTC data channels to create a peer-to-peer (P2P) network overlaying existing CDNs or origin servers. Compatible with hls.js, it supports both live and VOD streams over the HLS protocol and integrates seamlessly with popular HTML5 players like video.js, JWPlayer, and Flowplayer. This approach reduces server load and bandwidth costs while improving streaming performance, offering developers an innovative solution for efficient and scalable video delivery.

Tags: P2P streaming, WebRTC, HLS, CDN, video.js

jsDelivr

<https://www.jsdelivr.com/>

jsDelivr is a free, open-source CDN that allows developers to host and load code and other resources from repositories on GitHub, npm, and the WordPress theme and plugin directories. It supports loading specific versions of software packages or the latest available versions. jsDelivr can also minify JavaScript, CSS, or SVG files to reduce loading times. It is powered by multiple CDN providers, including Cloudflare, Fastly, and BunnyCDN, ensuring high availability and performance.

Tags: CDN, JavaScript, CSS, open source, multi-CDN

lhls-simple-live-platform

<https://slides.com/jordicenzano/deck-973aed>

You can build your own live platform just wiring up some open source tools, this is a demo video of [HTTPS://GitHub.com/jordicenzano/lhls-simple-live-platform](https://github.com/jordicenzano/lhls-simple-live-platform).

librist

<https://code.videolan.org/rist/librist>

A library that can be used to easily add the RIST protocol to your application.

olaris / olaris-server

<https://gitlab.com/olaris/olaris-server>

Self-hosted media server with transcoding and streaming capabilities for personal video libraries.

simple-peer - Simple WebRTC video, voice, and data channels

<https://github.com/feross/simple-peer>

Simple WebRTC library for video, voice, and data channels with a concise API. Supports dynamic stream addition, Node.js usage via wrtc, and works in browsers without configuration. Powers many P2P projects including file sharing and video chat applications.

switch media adease

<https://www.switch.tv/mediahq/adease/>

Switch Media's analytics and monitoring tool for tracking streaming performance and ad delivery metrics.

switch media live2vod

<https://www.switch.tv/mediahq/live2vod/>

Switch Media's live-to-VOD conversion service for transforming live streams into on-demand content.

switch media mediahq

<https://www.switch.tv/mediahq/>

Switch Media's VOD streaming server platform for delivering video-on-demand content.

General Tools

A Docker container with the video streaming tools you need

<https://medium.com/@eyevinntechology/a-docker-container-with-the-video-streaming-tools-you-need-b8319e98f36a>

As a video streaming technician there are a number of tools that you find yourself using on a daily basis. Wouldn't it be handy if all...

AWS SPEKE Reference Server

<https://github.com/aws-labs/speke-reference-server>

The Secure Packager and Encoder Key Exchange (SPEKE) reference server by AWS provides a framework for communication between media services and DRM system key servers. It helps developers implement content encryption protection strategies effectively.

Tags: DRM, content-encryption, AWS, SPEKE, media-services

Advanced Encryption Techniques: ContentProtection Tags for MPDs and PSSH Boxes for DASH.

https://go.buysdr.com/thedrmblog/advanced_encryption_techniques

In this installment of TheDRMBlog we take an in-depth look at Advanced Encryption Techniques.

Aes Decrypter

<https://github.com/videojs/aes-decrypter>

Contribute to videojs/aes-decrypter development by creating an account on GitHub.

AutoConfigShellScript

<https://github.com/sunhailin-Leo/AutoConfigShellScript>

Automatically compile and configure FFmpeg, Python 3.7.2(default), PyAV, OpenCV, Keras, TensorFlow(CPU Mode) and other relative environment.

AutoVideo

<https://github.com/datamllab/autovideo>

AutoVideo is a Python system designed for automated video action recognition. It features a modular infrastructure, an extensive list of primitives for pipeline construction, data-driven tuners for pipeline optimization, and an easy-to-use graphical user interface. This tool is valuable for developers working on video analysis and understanding tasks.

Tags: video-processing, action-recognition, automation, Python, machine-learning

AviSynth

http://avisynth.nl/index.php/Main_Page

AviSynth is a powerful scripting-based tool for video post-production. It allows users to process videos through scripts without a GUI, enabling tasks like cutting, trimming, cropping, and applying various filters. AviSynth acts as a non-linear video editor controlled entirely by scripting, making it highly flexible for complex video processing workflows.

Tags: scripting, video-editing, video-processing, non-linear, tool

Axinom DRM Decoders

<https://tools.axinom.com/>

Axinom offers DRM tools for encoding and decoding, ensuring secure streaming media delivery.

Axinom Public Test Vectors - Multi-DRM Test Content

<https://github.com/Axinom/public-test-vectors>

Public test vectors for adaptive streaming playback and multi-DRM scenarios. Includes working license servers for Widevine, FairPlay, PlayReady, and Clear Key testing with pre-generated license tokens and various test configurations.

Azure Media Services

<https://azure.microsoft.com/en-us/products/media-services/>

Azure Media Services is a cloud-based platform by Microsoft that offers scalable media processing and delivery solutions. It provides built-in DRM support, enabling secure streaming of on-demand and live video content to a global audience.

Tags: DRM, video streaming, cloud services, media processing, Microsoft

BONES: Near-Optimal Neural-Enhanced Video Streaming

<https://github.com/UMass-LIDS/bones>

BONES is a Neural-Enhanced Streaming (NES) control algorithm that jointly manages network and computational resources to maximize the quality of experience (QoE) for users. It formulates NES as a Lyapunov optimization problem and solves it online with near-optimal performance. BONES increases QoE by 5% to 20% over state-of-the-art algorithms with minimal overhead, making it a significant advancement in video streaming technology.

Tags: video-streaming, neural-enhancement, QoE, Lyapunov-optimization, NES

Binary to base64: Convert between bytes and base64 — Cryptii

<https://cryptii.com/pipes/binary-to-base64>

Base64 encoding schemes are used when binary data needs to be stored or transferred as textual data. Therefore 64 characters are chosen that are both members of a subset common to most encodings (ASCII), and also printable.

Bitmovin Player Web SDK

<https://developer.bitmovin.com/playback/docs/streaming-drm-protected-content-with-bitmovin-player-web-sdk>

Bitmovin Player Web SDK is a powerful video player that supports streaming DRM-protected content. It provides comprehensive documentation and examples for integrating DRM systems like Widevine, PlayReady, and FairPlay, facilitating secure and seamless playback across various devices and platforms.

Tags: DRM, video player, streaming, content protection, web SDK

Bitmovin Test Stream

<https://bitmovin.com/demos/stream-test>

Bitmovin's Test Player allows users to experience their HTML5 video player demo, showcasing adaptive streaming capabilities.

BrightWebSocket

<https://github.com/rolandoislas/BrightWebSocket>

RFC 6455 WebSocket Library for the Roku.

BuyDRM_KeyOS_PlatformOverview_FUUGO_062118

https://go.buydrm.com/hubfs/BuyDRM_KeyOS_Content_Protection_in_DASH_DASHIF_Workshop_Comcast_August2018-1.pdf

BuyDRM presentation covering KeyOS content protection implementation in DASH workflows and architecture.

Chrome Media Internals

<https://developer.chrome.com/docs/devtools/media-panel/>

A built-in Chrome tool that provides detailed logs and diagnostics for media playback and streaming issues.

Coconut DRM and AES Encryption

<https://www.coconut.co/features/drm>

Coconut provides a video transcoding API that includes DRM and AES encryption features, supporting Google Widevine, Apple FairPlay, and Microsoft PlayReady to protect premium video content.

Tags: DRM, video transcoding, content protection, Coconut, encryption

Common Encryption (CENC) Standard

<https://www.iso.org/standard/68042.html>

The ISO/IEC 23001-7 standard for Common Encryption in ISO base media file format files. CENC allows a single video file to be encrypted once and used with multiple DRM systems (Widevine, PlayReady, etc.) by defining a common encryption scheme and signaling, forming the basis of DRM interoperability in DASH and HLS fMP4.

Community events of the VideoLAN non-profit organization

<https://www.videolan.org/videolan/events>

Event listings for the VideoLAN non-profit organization. Community calendar for VLC and open video development.

Content Manager

<https://github.com/ebu/content-manager>

The Content Manager is a visual production tool which is able to generate on the fly visualisation for DAB slideshow and RadioVIS.NET framework based, it is developed in C# and is distributed und...

Cpix Validator

<https://github.com/Axinom/cpix-validator>

Web app for validating CPIX documents. Contribute to Axinom/cpix-validator development by creating an account on GitHub.

DASH & HLS Sample Streams

<https://bitmovin.com/mpeg-dash-hls-examples-sample-streams/>

Collection of sample streams in DASH and HLS formats provided by Bitmovin for testing adaptive bitrate streaming implementations and player functionality.

DRM Web3 Player Demo

<https://github.com/gsofter/drm-player-demo>

This project offers a demonstration of integrating Axinom DRM with web-based video players. It provides a quick start guide for protecting and playing back premium video content using DRM technologies such as Google Widevine, Microsoft PlayReady, and Apple FairPlay Streaming. The demo includes a sample application showcasing the usage of Axinom DRM, making it a valuable resource for developers looking to implement DRM in web-based video players.

Tags: DRM, video player, Axinom, Widevine, PlayReady

DeepFaceLab

<https://github.com/iperov/DeepFaceLab>

The leading software for creating deepfakes. It's an open source toolkit that allows users to swap faces in videos using machine learning, with support for multiple models and GPU acceleration.

Deploying KeyOS with AWS Elemental Media Services

<https://go.buysdr.com/thedrmblog/deploying-keyos-with-aws-elemental-media-services>

In this post we provide an in-depth first party overview of the complex integration between the KeyOS Platform and the Elemental Media Services via the SPEKE API.

Devine

<https://github.com/devine-dl/devine>

Devine is a modular software designed for archiving movies, TV shows, and music. It features built-in DASH and HLS parsers, Widevine DRM integration via pywidevine, and supports local and remote DRM key vaults. Devine is valuable for developers looking to implement DRM-protected media archival and playback solutions.

Tags: DRM, media archival, Widevine, DASH, HLS

Digital Rights Management (DRM) – Everything you need to know

<https://bitmovin.com/digital-rights-management-everything-to-know/>

Bitmovin's comprehensive guide covering digital rights management concepts, technologies, and implementation approaches for protecting video content.

Digital Rights Management (multi - drm) – aameer.GitHub.io

<https://aameer.github.io/articles/digital-rights-management-multi-drm>

Article explaining multi-DRM concepts and implementation strategies for cross-platform protected video delivery.

Digital Rights Management Project

<https://github.com/davidgmorais/digital-rights-management>

A project involving cryptography for DRM, developed during a course on Security in Informatics and Organizations. It explores concepts associated with a secure media player, enabling clients to consume media content while enforcing strong restrictions through cryptographic primitives.

Tags: DRM, cryptography, secure media player, content protection, video security

Docker Bento4

<https://github.com/alfg/docker-bento4>

A dockerized Bento4 from source. Built on Alpine Linux.

Dolby Audio Sample

<https://github.com/rokudev/dolby-audio-sample>

A collection of Dolby test content available in different streaming protocols.

EZDRM + Bento4 Open Source DRM Guide

<https://www.ezdrm.com/Documentation/EZDRM%20Bento%204%20Open%20Source%20v1.0.pdf>

Guide for implementing DRM workflows using EZDRM service with Bento4 open-source packaging tools.

Editly

<https://github.com/mifi/editly>

Editly is a command-line tool for declarative video editing, allowing users to create videos programmatically using JSON configurations. It supports features like text overlays, transitions, and audio tracks, making it a valuable resource for developers seeking to automate video creation processes.

Tags: video-editing, command-line, automation, JSON, Node.js

Eel

<https://github.com/Comcast/eel>

A simple proxy service to forward JSON events and transform or filter them along the way.

Encryption & DRM with Multiple Keys — Unified Streaming

<https://docs.unified-streaming.com/documentation/package/multiple-keys.html>

DRM with multiple keys for Unified Packager.

Eyevinn Streaming Samples

<https://github.com/eyevinn/http-streaming-samples>

A collection of HTTP streaming samples curated by Eyevinn, available for testing and development.

FFCreator

<https://github.com/tnfe/FFCreator>

FFCreator is a fast video processing library based on Node.js. It allows developers to create videos with various effects, transitions, and animations programmatically, making it suitable for applications like video generation and editing.

Tags: video-creation, nodejs, ffmpeg, video-editing, video-processing

FFMediaToolkit

<https://github.com/radek-k/FFMediaToolkit>

FFMediaToolkit is a cross-platform video decoder/encoder library for .NET that utilizes FFmpeg native libraries. It supports video frame extraction, reading stream metadata, and creating videos from bitmaps in any format supported by FFmpeg. This toolkit is particularly useful for developers working with multimedia applications in .NET environments.

Tags: FFmpeg, video processing, decoder, encoder, dotnet

FFmpeg General Documentation

<https://ffmpeg.org/general.html>

The General Documentation for FFmpeg offers comprehensive guidance on various aspects of FFmpeg, including compilation guides, encoding instructions, hardware acceleration, streaming, filtering, and more. It serves as a crucial resource for developers aiming to leverage FFmpeg's full potential in their projects.

Tags: documentation, ffmpeg, compilation, encoding, streaming

FFmpeg Utilities Documentation

<https://ffmpeg.org/ffmpeg-utils.html>

The FFmpeg Utilities Documentation provides detailed information on various utilities and features offered by FFmpeg, including syntax, quoting and escaping mechanisms, date and time duration formats, video size and rate specifications, and color definitions. This resource is invaluable for developers seeking to understand and utilize FFmpeg's capabilities effectively.

Tags: documentation, ffmpeg, utilities, syntax, video-processing

FFmpeg Wiki

<https://trac.ffmpeg.org/wiki>

The FFmpeg Wiki is a community-contributed resource that supplements the official FFmpeg documentation. It includes guides on compiling FFmpeg, encoding, hardware acceleration, streaming, filtering, and more. This wiki is particularly useful for developers seeking practical examples and user-generated insights.

Tags: wiki, ffmpeg, community, guides, video-processing

FFmpeg Windows Build Helpers

<https://github.com/rdp/ffmpeg-windows-build-helpers>

Helper script for cross compiling some media tools for windows, like customizable ffmpeg.exe (with or without non-free components, etc), and some other bonuses like mplayer, mp4box, mxf, etc. - rdp...

FFmpeg-For-Beginners-Ebook

<https://github.com/jdriselvato/FFmpeg-For-Beginners-Ebook>

An open-source ebook titled 'FFmpeg For Beginners - Edit Audio & Video Like a Pro for YouTube and Social Media'. It serves as a comprehensive guide for newcomers to FFmpeg, covering a wide range of topics from basic commands to advanced editing techniques.

Tags: FFmpeg, ebook, tutorial, audio editing, video editing

FaceSwap

<https://github.com/deepfakes/faceswap>

An open source deepfake application that provides tools to extract, train, and swap faces in videos. FaceSwap has an active community and supports plugins for different neural network models.

Fast Video Cataloger Scripting

<https://videocataloger.com/feature1/scripting-in-fast-video-cataloger/>

Fast Video Cataloger offers scripting support for managing video databases, allowing users to create custom user interfaces and extend functionalities using C#. It provides access to the .NET library and supports external libraries, making it suitable for developers looking to customize video cataloging solutions.

Tags: video-cataloging, scripting, C#, customization, video-management

GPAC Project on Advanced Content

<https://gpac.io/>

GPAC is an open-source multimedia framework focused on modularity and standards compliance. It provides tools for processing, inspecting, packaging, streaming, and playing back media content. GPAC is known for its wide MP4/ISOBMFF capabilities and is popular among video enthusiasts, academic researchers, and professional broadcasters. It is valuable for developers working on multimedia applications requiring advanced content packaging and streaming functionalities.

Tags: multimedia, streaming, packaging, open-source, MP4

High Dynamic Range Television and Hybrid Log-Gamma - BBC R&D

<https://www.bbc.co.uk/rd/projects/high-dynamic-range>

BBC R&D's project page on HDR television and Hybrid Log-Gamma. Resource for HDR workflow development.

How to Protect Your Content With DRM

<https://www.streamingmedia.com/Articles/ReadArticle.aspx?ArticleID=132289&pageNum=2>

Lock it down. If you're streaming proprietary or premium online video, it's time to take the step up to true digital rights management protection. Here's how to get started.

Hybrik API Reference

<https://docs.hybrik.com/api/v1/HybrikAPI.html?#getting-started>

API reference documentation for Hybrik, a cloud-based media processing platform that provides endpoints for managing video encoding, transcoding, and delivery workflows.

Integrate BuyDRM for a Multi-DRM workflow

<https://bitmovin.com/integrate-buydrm-multi-drm-system>

Integration guide for connecting Bitmovin encoding and playback services with BuyDRM multi-DRM platform.

JRE FFmpeg Apline

<https://github.com/silencecorner/jre-ffmpeg-apline>

Dockerfile combining the fabric8 JRE 8 base image with jrottenberg's FFmpeg image to produce an Alpine-based Java runtime with FFmpeg available.

JW Player

<https://github.com/jwplayer/jwplayer>

A proprietary JavaScript video player that supports MPEG-DASH, DRM integration (in collaboration with Vualto), interactive advertisements, and customization through CSS. JW Player is widely used for embedding videos into web pages and offers a free version under a Creative Commons license.

Tags: video player, DRM, MPEG-DASH, JavaScript, JW Player

Jaffree

<https://github.com/kokorin/Jaffree>

Java ffmpeg and ffprobe command-line wrapper.

Kaltura uDRM Module

<https://knowledge.kaltura.com/help/drm-support>

Kaltura's uDRM module offers a comprehensive multi-DRM solution supporting major DRM schemas including Microsoft

PlayReady, Google Widevine, and Apple FairPlay. It integrates seamlessly with Kaltura's business modules, enabling content owners to define complex business scenarios and ensure content protection across various devices and platforms.

Tags: DRM, video streaming, content protection, multi-DRM, Kaltura

List Of Streams

<https://github.com/bengarney/list-of-streams>

Community list of public test streams for HLS and DASH.

Man in the Middle Proxy

<https://github.com/mitmproxy/mitmproxy>

An interactive HTTPS proxy that allows inspection, modification, and debugging of network traffic, useful for video streaming analysis.

Manim

<https://www.manim.community/>

Manim is an animation engine for explanatory math videos. It allows for the creation of precise animations programmatically, making it a powerful tool for educators and developers interested in producing mathematical visualizations.

Tags: animation, math, visualization, python

Media-Transport-Library

<https://github.com/OpenVisualCloud/Media-Transport-Library>

Developed by OpenVisualCloud, this library provides a real-time media transport stack for both raw and compressed video. It supports various transport methods, including DPDK, AF_XDP, and RDMA, facilitating efficient video streaming and processing.

Tags: video-streaming, media-transport, real-time

Microsoft PlayReady

<https://www.microsoft.com/playready>

Microsoft PlayReady is a comprehensive DRM technology designed to protect digital media content across various devices and platforms, ensuring secure distribution and playback.

Tags: DRM, content protection, video streaming, Microsoft, PlayReady

Microsoft PlayReady Documentation

<https://learn.microsoft.com/playready/>

Microsoft's documentation for PlayReady DRM, including whitepapers and implementation guidance for integrating PlayReady content protection.

Tags: microsoft, content protection, digital rights management, playready, documentation

MoviePy

<https://zulko.github.io/moviepy/>

MoviePy is a Python library that provides a user-friendly interface for video editing and creation. It allows for tasks such as cutting, concatenating, and applying effects to video clips, as well as adding audio tracks. MoviePy simplifies complex video processing tasks with its intuitive API.

Tags: video-editing, python, video-processing, audio, effects

Mp

<https://github.com/sitkevij/mpi>

A media inspection and analysis tool that provides information and metadata about media files.

Tags: cli, utility, command-line, wrapper

NVIDIA Video Processing Technologies

<https://developer.nvidia.com/video-processing>

NVIDIA offers a suite of GPU-accelerated video processing technologies, including SDKs and libraries for tasks such as video encoding, decoding, and streaming. These tools leverage the power of NVIDIA GPUs to enhance performance and efficiency in video processing applications.

Tags: GPU-acceleration, video-processing, encoding, decoding, streaming

Nimble Streamer DRM

<https://softvelum.com/nimble/drm/>

Nimble Streamer is a media server that supports Digital Rights Management (DRM) for various encryption providers, including Widevine, PlayReady, and FairPlay. It offers features like live streaming encryption, DVR encryption, and VOD content encryption, ensuring secure content delivery across multiple platforms.

Tags: DRM, video streaming, media server, content protection, encryption

OpenDRM

<https://github.com/willkk/opendrm>

OpenDRM is an open-source implementation of Digital Rights Management (DRM) systems, including support for PlayReady, Widevine, and FairPlay. It aims to provide a common key system implementation compatible with ISO Common Encryption (CENC) and W3C Encrypted Media Extensions (EME), offering developers a foundation for building DRM solutions.

Tags: DRM, open-source, content protection, encryption, video streaming

OpenDVC: Open Source Implementation of Deep Video Compression

<https://github.com/RenYang-home/OpenDVC>

OpenDVC is an open-source TensorFlow implementation of the Deep Video Compression (DVC) method. It provides both PSNR-optimized and MS-SSIM-optimized models, achieving better MS-SSIM performance than the Low-Delay P (LDP) very fast setting of x265 and comparable PSNR performance. OpenDVC serves as a valuable resource for researchers and developers interested in learned video compression methods, offering pre-trained models and source codes for further development.

Tags: video-compression, deep-learning, TensorFlow, PSNR, MS-SSIM

PallyCon DRM CLI Packager

<https://pallycon.com/docs/en/multidrm/getting-started/downloads/>

A command-line tool that packages original content into encrypted streaming content with Multi-DRM support. It facilitates the integration of DRM services into your platform, ensuring secure content distribution.

Tags: DRM, video packaging, content protection, CLI tool, streaming

PallyCon Multi-DRM Cloud

<https://aws.amazon.com/marketplace/solutions/media-entertainment/drm/>

PallyCon Multi-DRM Cloud by INKA Entworks is a SaaS solution that simplifies DRM licensing services for online video providers. It integrates with AWS Media Services, offering DRM license management, API integration for HTML5 and mobile players, and cloud-based forensic watermarking for enhanced content security.

Tags: DRM, video streaming, content protection, multi-DRM, PallyCon

PallyCon Multi-DRM Integration Samples

<https://github.com/inka-pallycon/html5-player-drm-samples>

A collection of sample projects demonstrating the integration of PallyCon Multi-DRM services with various HTML5 players. These samples are valuable for developers looking to implement multi-DRM solutions in their web-based video applications.

Tags: DRM, video player, PallyCon, HTML5, multi-DRM

Play your own DRM content on ExoPlayer - Taku Semba - Medium

<https://medium.com/@takusemba/play-your-own-drm-content-on-exoplayer-e8ed73d5864c>

Digital rights management, or you could just refer to it as DRM, is a way of controlling what users can do with some sort of digital..

Process Home Videos

<https://github.com/mtlynch/process-home-videos>

A collection of helper scripts designed to process home videos and upload them to MediaGoblin. The scripts include functionalities for converting CSV metadata to YAML, rendering scenes by extracting clips based on metadata, and publishing processed videos to a MediaGoblin instance. This project is valuable for automating the organization and sharing of personal video collections.

Tags: video-processing, automation, MediaGoblin, home-videos, metadata

Pssh box

<https://raw.githubusercontent.com/google/shaka-packager/master/packager/tools/pssh/pssh-box.py>

Python utility for parsing and generating PSSH boxes used in content protection and multi-DRM workflows.

PyTorchVideo

<https://pytorchvideo.org/>

PyTorchVideo is a deep learning library designed for video understanding tasks. It offers modular and efficient components for video classification, detection, and self-supervised learning, facilitating the development of advanced video analysis models.

Tags: deep-learning, video-understanding, classification, detection

QUANTEEC Digital Rights Management

<https://web.archive.org/web/20260122131732/https://doc.quanteec.com/advanced-topics/drm/>

QUANTEEC provides a plugin that enables DRM-protected playback by encrypting video data on the server side and decrypting it on the player side. It is compatible with various players and DRM systems, facilitating secure video streaming.

Qencode Digital Rights Management

<https://cloud.qencode.com/drm>

Qencode offers a comprehensive DRM solution integrated with their video transcoding and live streaming services. It supports multiple DRM systems, providing developers with tools to protect video content effectively.

Tags: DRM, video-transcoding, live-streaming, content-protection, video-security

RAPIQUE

<https://github.com/vztu/RAPIQUE>

RAPIQUE (Rapid and Accurate Video Quality Evaluator) is a model for blind video quality assessment of user-generated content. It combines quality-aware scene statistics features with semantics-aware deep convolutional features, offering efficient and accurate video quality predictions. This tool is beneficial for developers focusing on video quality analysis and enhancement.

Tags: video-quality, assessment, machine-learning, Python, UGC

Rokumote

<https://github.com/nod/rokumote>

osx app for controlling your roku because sometimes your kids lose the remote

Secure HLS streaming using DRM encryption

<https://www.wowza.com/docs/how-to-secure-apple-hls-streaming-using-drm-encryption>

Protect live and on-demand HLS streaming using DRM encryption in Wowza Streaming Engine.

Securing OTT Content — DRM

https://medium.com/@eyevinntechnology/securing-ott-content-drm-1af2c08fdd31?source=userActivityShare-94bccb50d11-1560983518&_branch_match_id=670020366479331042

Written by: Boris Asadanin, Streaming Media Consultant and Partner at Eyevinn Technology.

Seriously.js

<https://github.com/brianchirls/Seriously.js>

Seriously.js is a real-time, node-based video compositor for the web. Inspired by professional software like After Effects, it allows dynamic and interactive rendering of high-quality video effects directly in the browser.

Tags: video-compositing, javascript, real-time, video-effects, web

Sffmpeg

<https://github.com/pyke369/sffmpeg>

Full-featured static FFmpeg build helper.

Springer Free Books

https://github.com/alexgand/springer_free_books

Python script to download all Springer books released for free during the 2020 COVID-19 quarantine

Streamlink

<https://github.com/streamlink/streamlink>

Streamlink is a CLI utility which pipes video streams from various services into a video player

Streams

<https://github.com/video-dev/streams>

A repository of shared streams - no media uploads.

THEOplayer DRM Integration Guide

<https://github.com/THEOplayer/documentation/blob/main/theoplayer/how-to-guides/04-drm/00-introduction.md>

THEOplayer provides a detailed guide for integrating various DRM systems, including FairPlay, Widevine, and PlayReady, into their video player, facilitating secure content playback across platforms.

Tags: DRM, video player, content protection, THEOplayer, integration guide

Text To Speech Nodejs

<https://github.com/watson-developer-cloud/text-to-speech-nodejs>

Sample Node.js Application for the IBM Watson Text to Speech Service.

The Hitchhiker's Guide to DRM

<https://ottverse.com/hitchhikers-guide-to-drm-2/>

A gentle guide to the world of Digital Rights Management. It includes a technology overview of AES, EME, CDM, CENC, Keys, and then explores popular DRM technologies such as Google Widevine, Apple FairPlay, Microsoft PlayReady, and finally, Multi-DRM.

The Top 656 Video Open Source Projects

<https://awesomeopensource.com/projects/video>

Browse The Most Popular 656 Video Open Source Projects.

TheDRMBlog | CENC

<https://go.buydrm.com/thedrmblog/topic/cenc>

CENC | The DRM Blog - Your New Official Source for Everything DRM.

This gist will generate an Intel QSV-enabled FFmpeg build using the open source Intel Media SDK. Testbed used: Ubuntu 18.04LTS. A fallback is also provided for the intel vaapi driver where needed.

<https://gist.github.com/SeanMollet/0eed16e80630ab67532890a9d42132af>

This gist will generate an Intel QSV-enabled FFmpeg build using the open source Intel Media SDK. Testbed used: Ubuntu 18.04LTS. A fallback is also provided for the intel vaapi driver where needed....

Toolbox

<https://github.com/Eyevinn/toolbox>

A set of Docker containers with Streaming tools.

VPF: Hardware-Accelerated Video Processing Framework in Python

<https://github.com/NVIDIA/VideoProcessingFramework>

VPF is a Python library that provides a simple yet powerful tool for fully hardware-accelerated video encoding, decoding, and processing. Built on top of NVIDIA's Video Codec SDK, VPF allows developers to achieve high GPU utilization with minimal code. It exposes decoded video frames as NumPy arrays or CUDA device pointers, facilitating seamless interaction and extension. VPF is ideal for developers seeking to leverage GPU acceleration for video processing tasks.

Tags: video-processing, GPU-acceleration, python, NVIDIA, CUDA

VPF: Video Processing Framework

<https://developer.nvidia.com/blog/vpf-hardware-accelerated-video-processing-framework-in-python/>

VPF is a hardware-accelerated video processing framework in Python developed by NVIDIA. It provides developers with tools for fully hardware-accelerated video encoding, decoding, and processing, leveraging NVIDIA GPUs for high-performance video applications.

Tags: hardware-acceleration, video-processing, nvidia, gpu

VPlayed DRM Platform

<https://www.vplayed.com/video-security-drm-platform.php>

VPlayed offers a comprehensive DRM platform designed to protect video content against piracy. It features AES256 encryption, dynamic watermarks, and access management, ensuring secure video and live streaming across various devices and platforms.

Tags: DRM, video security, content protection, encryption, video streaming

VapourSynth

<https://www.vapoursynth.com/>

VapourSynth is a video processing framework with a scripting interface. It is designed to be a modern replacement for AviSynth, offering support for multithreading and various video formats. VapourSynth allows developers to write scripts for video processing tasks, making it suitable for complex workflows and automation.

Tags: video-processing, scripting, framework, multithreading, tool

VdoCipher Developer Platform

<https://www.vdocipher.com/developer/>

VdoCipher offers a developer-centric video hosting platform with advanced player features and technical support, focusing on secure video streaming with DRM protection.

Tags: DRM, video hosting, content protection, VdoCipher, developer tools

Verimatrix DRM

<https://www.verimatrix.com/products/multi-drm>

Verimatrix DRM is a comprehensive content protection solution that supports multiple DRM systems, including Widevine, PlayReady, and FairPlay. It offers cloud-based services for secure video streaming and provides real-time analytics to help optimize content delivery and user engagement.

Tags: multi-DRM, content protection, video streaming, analytics, cloud-based

VidGear

<https://github.com/abhiTronix/vidgear>

VidGear is a high-performance cross-platform video processing Python framework that provides an easy-to-use, highly extensible, and thoroughly optimized multi-threaded API. It integrates various libraries like OpenCV, FFmpeg, and ZeroMQ, enabling developers to perform complex video processing tasks with minimal code.

Tags: video-processing, python, ffmpeg, opencv, multithreading

Video Containers Debugging Tools

<https://github.com/leandromoreira/video-containers-debugging-tools>

A set of command lines to debug video streaming files like mp4 (MPEG-4 Part 14), ts (MPEG-2 Part 1), fmp4 in Dash, HLS, or MSS, with or without DRM.

Video Decrypter

https://github.com/CrackerCat/video_decrypter

Decrypt video from a streaming site with MPEG-DASH Widevine DRM encryption.

Video-Super-Resolution-Library

<https://github.com/OpenVisualCloud/Video-Super-Resolution-Library>

This library offers a native C and C++ implementation of RAISR (Rapid and Accurate Image Super-Resolution). It's designed to enhance video quality through super-resolution techniques, making it valuable for developers working on video enhancement projects.

Tags: video-enhancement, super-resolution, image-processing

VideoProcessingTools

<https://github.com/Otis-Laboratory/VideoProcessingTools>

A collection of video processing scripts designed to curate pre-training and fine-tuning data. It includes tools to split video files into frames and deduplicate them based on hash similarity, facilitating efficient data preparation for machine learning models.

Tags: video-processing, data-curation, machine-learning, frame-extraction, deduplication

Vitrium Security

<https://www.vitrium.com/drm-for/technology-and-manufacturing>

Vitrium Security provides DRM solutions for protecting digital content, including videos, documents, images, and audio files. It offers secure web formats accessible on any device, integration with existing business systems, and features like annotation tools for enhanced user experience.

Tags: DRM, content protection, video security, multi-format, Vitrium

WebGazer.js

<https://webgazer.cs.brown.edu/>

WebGazer.js is an eye-tracking library that uses common webcams to infer the eye-gaze locations of web visitors in real-time. It self-calibrates by observing user interactions, making it valuable for creating interactive web applications.

Tags: eye-tracking, javascript, webcam, real-time, video-processing

Widevine

<https://widevine.com>

Widevine is a proprietary digital rights management (DRM) system developed by Google. It is integrated into most major web browsers and operating systems, including Android and iOS. Widevine is utilized by streaming services such as Netflix, Amazon Prime, and Hulu to securely deliver content to authorized users while preventing unauthorized copying.

Tags: DRM, video streaming, content protection, Google, Widevine

Widevine DRM

<https://developer.widevine.com/solutions/widevine-drm/>

Widevine DRM is Google's content protection system for premium media, utilized by major partners globally. It provides secure content protection, supports multiple DRM systems, and offers a fully featured HTML5 video player with adaptive streaming. Widevine is essential for developers needing robust DRM solutions for their video streaming services.

Tags: DRM, content protection, video streaming, adaptive streaming, Google

Widevine DRM Documentation (Google)

<https://developers.google.com/widevine>

Official documentation for Google Widevine DRM, detailing integration of content protection on various platforms and browsers using Widevine Modular DRM.

Wowza Streaming Engine DRM

<https://www.wowza.com/docs/wowza-drm>

Wowza Streaming Engine offers integration with various third-party DRM systems, including BuyDRM KeyOS, EZDRM, and Verimatrix VCAS. It provides comprehensive documentation and support for configuring DRM to protect live and on-demand

streams, making it a robust solution for secure video delivery.

Tags: DRM, video streaming, content protection, Wowza, streaming engine

Wowza Streaming Engine DRM Integration

<https://www.wowza.com/docs/about-digital-rights-management-and-wowza-streaming-engine>

Wowza Streaming Engine provides integration with third-party DRM key management systems to add on-the-fly encryption to live and video-on-demand workflows. It supports key rotation and integration with systems like BuyDRM KeyOS, offering a comprehensive solution for secure streaming. This is valuable for developers seeking robust DRM integration in their streaming infrastructure.

Tags: DRM, streaming engine, encryption, BuyDRM, key management

Wsgidav

<https://github.com/mar10/wsgidav>

A generic and extendable WebDAV server based on WSGI

XML + Code + Good times = RSG Application

<https://medium.com/plexlabs/xml-code-good-times-rsg-application-b963f0cec01b>

Written by John Zolezzi - April 6th 2018.

Xiph.org :: Test Media

<https://media.xiph.org/>

Repository of royalty-free test media files and sample streams for video development and testing.

Zype DRM

<https://www.zype.com/product/zype-streaming-platform/videometa-cms/drm>

Zype provides a Digital Rights Management (DRM) solution that integrates with standards like FairPlay, Widevine, and PlayReady. It offers turnkey implementation, cross-device security support, and API-driven management, enabling content creators to safeguard their revenue by preventing unauthorized access and piracy.

Tags: DRM, video streaming, content protection, encryption, API

bento4.mp4encrypt

<https://www.bento4.com/documentation/mp4encrypt/>

Part of the Bento4 suite, mp4encrypt is a command-line utility to apply DRM encryption (e.g., Common Encryption) to MP4 files. It's used to encrypt media files for DASH (cenc) or HLS (sample AES) and is essential for preparing content protection in adaptive streaming workflows.

Tags: drm, encryption, mp4, content-protection, streaming-security

cinelerra-cv

<https://github.com/cinelerra-cv-team/cinelerra-cv>

Cinelerra-CV is an advanced non-linear video editing and compositing software for Linux. It provides a comprehensive suite of tools for video processing, making it suitable for developers seeking a robust video editing platform.

Tags: video-editing, video-compositing, non-linear-editing

compile and install latest FFmpeg source as pkg

<https://gist.github.com/krzemienski/e51a0b7a6ba77e616f954e516783270c#file-compile-and-install-latest-ffmpeg-source-sh-L2>

compile and install latest ffmpeg source as pkg. GitHub Gist: instantly share code, notes, and snippets.

drm_server

https://github.com/gyr66/drm_server

An open-source video upload and sharing platform utilizing DRM technology. It offers features like anti-screen recording, virtual machine playback restriction, and secure video encryption, making it suitable for developers seeking to implement DRM in video sharing applications.

Tags: DRM, video sharing, content protection, open-source, video platform

drmx-integration-learndash

<https://github.com/Haihaisoft/drmx-integration-learndash>

A WordPress plugin integrating Haihaisoft DRM-X 4.0 with LearnDash LMS. It offers encrypted video, audio, and PDF protection, preventing screen recording, displaying user information watermarks, and binding user devices to prevent account sharing.

Tags: DRM, WordPress, LearnDash, video protection, encryption

drmx-integration-tutorlms

<https://github.com/Haihaisoft/drmx-integration-tutorlms>

A WordPress plugin integrating Haihaisoft DRM-X 4.0 with Tutor LMS. It provides encrypted video, audio, and PDF protection,

preventing screen recording, displaying user information watermarks, and binding user devices to prevent account sharing.

Tags: DRM, WordPress, Tutor LMS, video protection, encryption

drmx-integration-woocommerce

<https://github.com/Haihaisoft/drmx-integration-woocommerce>

This WordPress plugin integrates Haihaisoft's DRM-X 4.0 digital copyright protection platform with WooCommerce. It supports encrypted video, audio, and PDF, preventing screen recording and unauthorized sharing, making it ideal for protecting digital courses and media.

Tags: DRM, WordPress, WooCommerce, video-protection, encryption

ffmpeg-concat

<https://github.com/transitive-bullshit/ffmpeg-concat>

ffmpeg-concat is a Node.js module that concatenates a list of videos together using FFmpeg with smooth OpenGL transitions. It provides an easy-to-use interface for merging multiple video files into a single output with customizable transitions, making it ideal for creating video montages or compilations.

Tags: nodejs, ffmpeg, video-concatenation, transitions, video-editing

ffmpeg-gl-transition

<https://github.com/transitive-bullshit/ffmpeg-gl-transition>

ffmpeg-gl-transition is an FFmpeg filter that applies GLSL transitions between video streams. It provides a collection of high-quality transitions that can be used to enhance video editing and production workflows.

Tags: ffmpeg, glsl, transitions, video-editing, filters

ffmpeg-static

<https://github.com/eugeneware/ffmpeg-static>

ffmpeg-static provides static FFmpeg binaries for macOS, Linux, and Windows. It simplifies the process of including FFmpeg in projects by offering precompiled binaries, eliminating the need for users to compile FFmpeg from source. This is particularly useful for CI/CD pipelines and cross-platform development.

Tags: ffmpeg, static-binaries, cross-platform, ci-cd, video-processing

ffmpeg-tools

<https://github.com/utility-toolbox/ffmpeg-tools>

ffmpeg-tools offers a collection of helpful FFmpeg commands designed to simplify common video processing tasks. It provides scripts for operations like splitting videos into chapters, concatenating multiple videos, and stacking videos horizontally or vertically. This tool is beneficial for developers and archivists who frequently work with FFmpeg for video manipulation.

Tags: FFmpeg, video processing, utilities, command-line, scripting

media-tools

<https://github.com/MartinCastroAlvarez/media-tools>

media-tools is a video processing pipeline that leverages the FFmpeg command-line interface to perform various video manipulations. It includes functionalities such as downloading videos from YouTube, compressing videos, cutting segments, applying effects like blur and gamma correction, and converting videos to different formats. This project is valuable for developers looking to automate complex video processing workflows using FFmpeg.

Tags: FFmpeg, video processing, pipeline, command-line, automation

sensorReporter

<https://github.com/rkoshak/sensorReporter>

A python based service that receives sensor inputs and publishes them over REST (should work with any API but mainly tested with openHAB) or MQTT. It can also receive commands and perform an action...

video_processing_utils

https://github.com/dmcken/video_processing_utils

A collection of general video processing utilities primarily using FFmpeg. It provides command-line tools for concatenating videos, converting formats, and merging files, facilitating efficient batch processing and automation of video tasks.

Tags: FFmpeg, video utilities, batch processing, automation, command-line

Encoding & Codecs

A Large-Scale Comparison of x264, x265, and libvpx

<https://medium.com/netflix-techblog/a-large-scale-comparison-of-x264-x265-and-libvpx-a-sneak-peek-2e81e88f8b0f>

Netflix's technical analysis comparing encoding efficiency and performance across x264, x265, and libvpx codecs.

ABR Broadcaster

<https://github.com/jkarthic-akamai/ABR-Broadcaster>

A real time encoder for Adaptive Bitrate Broadcast

AMD Advanced Media Acceleration (AMA) SDK

https://amd.github.io/ama-sdk/latest/git_hub.html

A comprehensive SDK from AMD that provides source codes for various elements of media acceleration, including sample codes, third-party integrations, and utilities. It is particularly useful for developers looking to leverage AMD hardware for efficient video encoding and decoding.

Tags: AMD, media acceleration, SDK, video encoding, hardware acceleration

AMD Advanced Media Framework (AMF SDK)

<https://github.com/GPUOpen-LibrariesAndSDKs/AMF>

AMD's GPUOpen Video SDK that provides developers access to GPU-accelerated encoding and decoding (H.264, HEVC, etc.) on AMD hardware.

AMF Encoder Developer Guide (Wiki)

<https://github.com/GPUOpen-LibrariesAndSDKs/AMF/wiki/Guide-for-Video-CODEC-Encoder-App-Developers>

Official AMF wiki guide for developers building video codec encoder applications with AMD's Advanced Media Framework SDK — covers encoder component setup, property configuration, and submission/polling workflow. Companion documentation to the AMF SDK repository entry.

Tags: video encoding, hardware acceleration, AMD, C++, developer guide

AV1 Codec

https://docs.google.com/presentation/d/12_Vewc0SDpB1FycflfT4us9eipRCy0HWJVSaDMDifRs/edit?usp=sharing

Working with the AV1 Codec Kevin Staunton-Lambert Solutions Architect R&D @keveleyski www.switch.tv

Tags: codec, video compression, AV1, open source, streaming technology

AV1 decoder model

https://norkin.org/research/av1_decoder_model/

Description of the AV1 decoder model.

AV1 vs HEVC – Bitmovin Compression Efficiency Results

<https://bitmovin.com/av1-showing-greater-compression-efficiency-than-hevc/>

A Bitmovin blog post that discusses internal testing results comparing AV1 and HEVC encoders. It provides charts and analysis of bitrate savings, encoding speed, and visual quality, contributing practical data to the conversation about next-gen codec adoption.

Tags: codec, compression, AV1, HEVC, bitrate, encoding, comparison

AWS Lambda FFmpeg

<https://github.com/binoculars/aws-lambda-ffmpeg>

An S3-triggered Amazon Web Services Lambda function that runs your choice of FFmpeg uploads the outputs to a bucket.

Tags: ffmpeg, aws, lambda, video-processing, s3, transcoding

Adjust and Normalize Your Music Files with FFMPEG - Make Tech Easier

<https://www.maketecheasier.com/normalize-music-files-with-ffmpeg/>

If your music files are too loud, too soft, or have obnoxious peaks and irregular volume, you can use FFmpeg to normalize your music files. Here's how.

Adobe Media Encoder

<https://www.adobe.com/products/media-encoder.html>

Adobe Media Encoder is a professional video encoding software that integrates seamlessly with Adobe's suite of creative tools. It supports a wide range of formats and offers features like batch processing, GPU acceleration, and preset templates, making it ideal for high-quality video encoding tasks.

Tags: video encoding, professional, batch processing, GPU acceleration, Adobe integration

Adobe Media Encoder (AME) Guide

<https://helpx.adobe.com/media-encoder/user-guide.html>

User guide for Adobe Media Encoder, a tool that automates batch transcoding and export from Adobe's video apps. Useful resource for understanding batch encoding settings for various formats and how to optimize output for different platforms.

Alliance for Open Media (AOMedia)

<https://aomedia.org/>

An industry consortium that develops open, royalty-free media formats. AOMedia created the AV1 video codec and works on related audio/image formats and tools.

Tags: codec, AV1, open source, video compression, royalty-free

Alpine Bento FFmpeg

<https://github.com/realeyes-media/alpine-bento-ffmpeg>

Alpine Linux with FFMPEG, Bento, and PM2.

Alpine Node Video Multitool

<https://github.com/realeyes-media/alpine-node-video-multitool>

Contribute to realeyes-media/alpine-node-video-multitool development by creating an account on GitHub.

Android FFmpeg Transcoder

<https://github.com/kibotu/Android-FFmpeg-Transcoder>

An Android library for performing video transcoding on-device, leveraging FFmpeg (or MediaCodec) to extract frames or encode videos on Android.

Android-Video-Trimmer

<https://github.com/iknow4x/Android-Video-Trimmer>

An Android project that implements video trimming functionality using MediaMetadataRetriever to obtain video frames and FFmpeg for video cropping and compression.

Tags: Android, video-trimming, FFmpeg, video-compression, video-editing

Arch FFmpeg GPAC

<https://github.com/majamee/arch-ffmpeg-gpac>

A ready-prepared video transcoding pipeline to create DASH/ HLS compatible video files & playlists

Tags: ffmpeg, transcoding, dash, hls, video-processing

Ateme TITAN

<https://www.ateme.com/products/titan/>

Ateme TITAN is a software suite designed for video origination, acquisition, and compression. It offers high-quality encoding solutions suitable for various applications, including broadcasting and OTT services.

Tags: video, compression, broadcasting, OTT, encoding

Audio Video Tools

<https://github.com/okorach/audio-video-tools>

Python based batch tools to transcode audio and video conveniently (leverages FFmpeg)

Auto-Editor

<https://github.com/WyattBlue/auto-editor>

An automatic video editor written in Python that can remove silence, jump cuts, and more without manual editing. It processes video/audio to create a faster-paced output automatically.

Av1an

<https://github.com/master-of-zen/Av1an>

Cross-platform command-line AV1 encode toolkit. Contribute to master-of-zen/Av1an development by creating an account on GitHub.

Av1transcoder

<https://github.com/luziferius/av1transcoder>

Transcode video files to the AV1 format using FFmpeg and libaom-av1.

AviSynth+

<https://avs-plus.net/>

An improved version of the classic AviSynth frameserver. AviSynth+ allows video processing and filtering through scripting, enabling complex edits and conversions on Windows with high performance.

Tags: frameserver, video processing, scripting, windows, filtering

Avidemux

<https://avidemux.sourceforge.net/>

A simple, free video editor for cutting, filtering, and encoding tasks, supporting many file types and codecs.

Tags: video editor, encoding, free tool, codec support, video processing

Avtk

<https://github.com/senko/avtk>

A Python library for audio and video processing that provides tools for encoding and manipulating media files.

Tags: ffmpeg, video-processing, tool, encoding

Awesome FFmpeg

<https://github.com/transitive-bullshit/awesome-ffmpeg>

Ø=Ü{ A curated list of awesome FFmpeg resources.

Tags: ffmpeg, resources, video-tools, encoding

Batch Py Remux

<https://github.com/ZaifSenpai/Batch-Py-Remux>

Convert mkv video to hevc (h.265).

Batch Python FFmpeg Tutorial

<https://github.com/Azure-Samples/batch-python-ffmpeg-tutorial>

A Python application that uses Batch to process media files in parallel with the FFmpeg open-source tool.

Beamr Content-Adaptive Bitrate Encoding

<https://www.encoding.com/media-services-overview>

Beamr's Content-Adaptive Bitrate (CABR) technology is integrated into Encoding.com's platform, allowing for streaming video with the highest possible picture quality while reducing bitrate by up to 50%. This technology is valuable for efficient video delivery and storage.

Tags: CABR, video streaming, bitrate reduction, video quality, encoding

Bento

<https://github.com/bento-video/bento>

Bento is an open-source, high-speed serverless video transcoding pipeline designed for deployment on Amazon Web Services (AWS). It offers a fast and simple solution for individuals and small businesses needing efficient video transcoding capabilities.

Tags: serverless, video, transcoding, AWS, pipeline

Bento4 | Fast, Modern Tools and C++ Class Library for all your MP4 and DASH media format needs

<https://www.bento4.com/>

Bento4 is a comprehensive toolkit and C++ library providing command-line tools and APIs for MP4 and DASH media format creation, manipulation, and packaging.

Bitrate Viewer

<https://www.winhoros.de/docs/bitrate-viewer/>

Bitrate Viewer is a free Windows application that allows users to analyze the bitrate of H.264 encoded video files. It provides a graphical representation of bitrate distribution, aiding in the assessment of video quality and encoding efficiency.

Tags: bitrate analysis, H.264, video quality, Windows, freeware

ButtleOFX

<https://github.com/ButtleOFX/ButtleOFX>

An open source nodal video compositing software built on top of OpenFX. It provides a user-friendly interface to apply and connect video effects (an experimental alternative to commercial compositors).

CDN Transcode Sample (GitHub)

<https://github.com/OpenVisualCloud/CDN-Transcode-Sample>

Media transcoding is a key function for live video broadcasting, streaming, and video on demand use cases in a CDN network. The CDN Transcode sample (not a finished product) provides a reference pi...

CUDA GPU Accelerated h264/h265/HEVC Video Encoding with ffmpeg

<https://ntown.at/de/knowledgebase/cuda-gpu-accelerated-h264-h265-hevc-video-encoding-with-ffmpeg/>

How to use CUDA GPU hardware encoding with ffmpeg to encode h264 and h264 HEVC movies in high quality and highspeed with our optimized parameter settings.

Can FFmpeg extract closed caption data

<https://stackoverflow.com/questions/3169910/can-ffmpeg-extract-closed-caption-data>

I am currently using ffmpeg to convert videos in various formats to flv files. One request has also come up and that is to get closed caption info out o the file as well. Does anyone have any exper...

CineForm SDK

<https://gopro.github.io/cineform-sdk/>

GoPro's CineForm codec SDK, open-sourced, providing tools and libraries for the CineForm HD wavelet-based video codec used for high-fidelity video compression.

Tags: codec, sdk, compression, video-encoding, gopro

Cinelerra GG Infinity

<https://cinelerra-gg.org/>

A robust open source non-linear editor for Linux, derived from the long-running Cinelerra project. Cinelerra GG Infinity offers professional-grade editing, compositing, and audio features for advanced users on Linux.

Tags: linux, video editing, open source, non-linear editor, professional editing

Cisco OpenH264

<https://github.com/cisco/openh264>

An open source H.264 (AVC) encoder and decoder library by Cisco, provided under a license that allows free use (Cisco covers MPEG LA licensing for binaries).

Cisco Thor Video Codec

<https://tools.ietf.org/html/draft-fuldseth-netvc-thor-00>

Thor was an experimental video codec developed by Cisco as a candidate for the NETVC (Next-Gen Video) initiative. While Thor itself is not used as-is, some of its technology contributed to the development of AV1. It's a piece of codec research history with source available for study.

Cloud Transcoder

<https://github.com/xyk2/cloud-transcoder>

Transcoding long (>1 hour) videos quickly and cost-effectively on GCP to adaptive HLS and MP4 mezzanine files. Up to 40x \$ savings.

CloudTranscode

<https://github.com/bfansports/CloudTranscode>

Distributed videos and images encoding/transcoding using Amazon SFN, FFMpeg and ImageMagic

Tags: transcoding, ffmpeg, amazon-sfn, distributed-encoding, cloud-encoding

Codem Transcode

<https://github.com/madebyhiro/codem-transcode>

Offline video transcoder written in node.js.

CompressAI

<https://arxiv.org/abs/2011.03029>

CompressAI is a PyTorch library and evaluation platform for end-to-end compression research. It provides custom operations, layers, models, and tools to research, develop, and evaluate image and video compression codecs. CompressAI includes pre-trained models and evaluation tools to compare learned methods with traditional codecs, facilitating advancements in compression technology.

Tags: compression research, PyTorch, video encoding, evaluation platform, open-source

Correcting for audio/video sync issues with the ffmpeg program's ITSOFFSET switch

<https://wjwoodrow.wordpress.com/2013/02/04/correcting-for-audiovideo-sync-issues-with-the-ffmpeg-programs-itsoffset-switch/>

The ffmpeg program has numerous "switches" that help to adjust and convert audio and video files. Some of them are not explained very well in the documentation, and many websites have c...

CrafyVideoJS

<https://github.com/chijete/CrafyVideoJS>

CrafyVideoJS is a lightweight, hardware-accelerated JavaScript library that enables in-browser manipulation, decoding, and encoding of MP4 H.264 (AVC) videos. It leverages the browser's VideoEncoder and VideoDecoder APIs along with mp4box.js to provide efficient video processing without server-side support. Features include video trimming, bitrate adjustment, resolution changes, and metadata extraction.

Tags: JavaScript, video processing, MP4, H.264, browser-based

DaVinci Resolve (Free Edition)

<https://www.blackmagicdesign.com/products/davinciresolve>

A professional cross-platform video editing, color grading, and post-production suite with a free version available.

Daala Codec (Xiph)

<https://xiph.org/daala/>

An experimental video codec from the Xiph.Org Foundation that aimed to achieve next-generation compression (the effort preceded and influenced AV1). Daala introduced novel techniques like lapped transforms and per-pixel entropy coding. Its codebase is open and served as a sandbox for ideas that made it into AV1.

Tags: codec, compression, open-source, experimental, video-encoding

Dav1d (videolan)

<https://code.videolan.org/videolan/dav1d>

dav1d is the fastest AV1 decoder on all platforms.

Demo Encoder

<https://github.com/realeyes-media/demo-encoder/>

A nodejs encoding system based on ffmpeg and configured to write HLS streaming files to S3

Distributed FFmpeg Transcoding Cluster

https://github.com/chn-lee-yumi/distributed_ffmpeg_transcoding_cluster

R ^ _ F F M p e g • l x - Æ • ¢ 0 A F F M p e g transcoding cluster runs in variable CPUs, including ARM. You can use it to run a RaspberryPi cluster.

DivX Converter

<https://www.divx.com/en/software/converter/>

DivX Converter is a video encoding software that allows users to compress and convert video files into the DivX digital format, known for its high-quality output and efficient compression.

Tags: video, compression, conversion, high-quality, efficient

Django Video Encoding

<https://github.com/escaped/django-video-encoding>

django-video-encoding helps to convert your videos into different formats and resolutions.

Tags: django, video-encoding, transcoding, python, conversion

Django Video Transcoding

<https://github.com/just-work/django-video-transcoding>

Simple video transcoding application for Django Framework

Dlb Mp4base

https://github.com/DolbyLaboratories/dlb_mp4base

The Dolby MP4 streaming muxer (dlb_mp4base) is a software implementation of a muxer of fragmented or unfragmented ISO base media file format (mp4). It supports muxing of Dolby Digital (AC-3), Dolby...

Dlb Mp4demux

https://github.com/DolbyLaboratories/dlb_mp4demux

Dolby Laboratories' official MP4 streaming demuxer (dlb_mp4demux) — a reference implementation for demuxing fragmented or unfragmented ISO base media files.

Dlb Mp4demux (ShaoWeiguo)

https://github.com/ShaoWeiguo/dlb_mp4demux

Community mirror of dlb_mp4demux, the Dolby MP4 streaming demuxer for fragmented and unfragmented ISO base media files.

Docker FFmpeg

<https://github.com/linuxserver/docker-ffmpeg>

Linuxserver/docker-FFmpeg - A GitHub repository for video/multimedia development.

Docker HEVC

<https://github.com/Eyevinn/docker-hevc>

Docker container to create HEVC streaming packages

Docker Java Media

<https://github.com/ElderByte-/docker-java-media>

JRE 10 (Java 10) and media tools (ffmpeg).

Tags: docker, java, ffmpeg, media-tools, encoding

Docker Jit Capture

<https://github.com/Eyevinn/docker-jit-capture>

A Docker container for an open source Just-In-Time Capture Origin

Dolby Vision Professional Decoder Plugin

https://github.com/DolbyLaboratories/dolby_vision_professional_decoder_plugin

DolbyLaboratories/dolby_vision_professional_decoder_plugin - A GitHub repository for video/multimedia development

Encode HDR with VP9

<https://developers.google.com/media/vp9/hdr-encoding>

Hands on tutorial of using FFmpeg to do hdr encoding.

Express FFmpeg

<https://github.com/bookyo/express-ffmpeg>

Node.js FFmpeg video transcode webui ȳ WúNŽ Node.js v„N'•lx |û~ß H T T P S : // w w w . e f v c m s . c o m

FFCommand Engine

https://github.com/ColorlabMD/FFCommand_Engine

Create and execute FFmpeg commands.

FFV1 Codec

<https://ffmpeg.org/~michael/ffv1.html>

FFmpeg's FFV1 is a lossless intra-frame video codec. It's open, extremely efficient for archiving video, and is defined in an ongoing standardization (supported by archival institutions).

FFmbc

<https://github.com/bcoudurier/FFmbc>

FFmpeg customized for broadcast and professional usage

FFmpeg

<https://ffmpeg.org>

FFmpeg is a free and open-source software project consisting of a suite of libraries and programs for handling video, audio, and other multimedia files and streams. It is widely used for format transcoding, basic editing, and standards compliance.

Tags: multimedia framework, video processing, audio processing, open-source, transcoding

FFmpeg (GitHub mirror)

<https://github.com/FFmpeg/FFmpeg>

Mirror of [git://source.ffmpeg.org/ffmpeg.git](https://source.ffmpeg.org/ffmpeg.git).

Tags: encoding, video processing, open-source, multimedia, codec

FFmpeg AWS Lambda Layer

<https://github.com/serverlesspub/ffmpeg-aws-lambda-layer>

FFmpeg/FFprobe AWS Lambda layer.

FFmpeg Android Java

<https://github.com/WritingMinds/ffmpeg-android-java>

Android java library for FFmpeg binary compiled using <https://github.com/writingminds/ffmpeg-android>

Tags: ffmpeg, android, encoding, library, mobile

FFmpeg AutoGen (C#)

<https://github.com/Ruslan-B/FFmpeg.AutoGen>

A set of auto-generated P/Invoke bindings for FFmpeg libraries. FFmpeg.AutoGen allows .NET developers to directly use FFmpeg's C API in C# for advanced media processing tasks.

FFmpeg Batch Converter (Windows)

<https://ffmpeg-batch.sourceforge.io/>

A Windows front-end that allows batch processing with FFmpeg. Users can queue up multiple files with presets for conversion or processing, automating repetitive FFmpeg tasks without writing a script.

Tags: batch conversion, windows, automation, video processing, ffmpeg-frontend

FFmpeg Bitrate Stats

<https://github.com/slhck/ffmpeg-bitrate-stats>

Calculate bitrate statistics using FFmpeg

FFmpeg Build Script

<https://github.com/markus-perl/ffmpeg-build-script>

The FFmpeg build script provides an easy way to build a static FFmpeg on OSX and Linux with non-free codecs included.

Tags: ffmpeg, build-script, codec, static-build, linux, macos

FFmpeg CLI Wrapper

<https://github.com/bramp/ffmpeg-cli-wrapper>

Java wrapper around the FFmpeg command line tool.

Tags: ffmpeg, java, encoding, wrapper, cli

FFmpeg Debug Qp

<https://github.com/slhck/ffmpeg-debug-qp>

FFmpeg Debug Script for QP Values

FFmpeg Docker Container

<https://github.com/tamara-schmitz/ffmpeg-docker-container>

This project provides a Docker container for FFmpeg, facilitating the use of FFmpeg in isolated environments. It simplifies the deployment and execution of FFmpeg commands, ensuring consistency across different systems and easing the setup process for developers.

Tags: Docker, FFmpeg, containerization, video processing, transcoding

FFmpeg Docker Images (jrottenberg)

<https://github.com/jrottenberg/ffmpeg>

Docker build for FFmpeg on Ubuntu / Alpine / CentOS 7 / Scratch

FFmpeg JS

<https://github.com/Kagami/ffmpeg.js>

Port of FFmpeg with Emscripten.

Tags: ffmpeg, emscripten, javascript, encoding, web, porting

FFmpeg Libvpx HDR Static

<https://github.com/id3as/ffmpeg-libvpx-HDR-static>

A script to build a static binary of FFmpeg optimised for libvpx (HDR 10bit) encoding.

FFmpeg Nvidia Adaptive VOD Transcoder

<https://github.com/SmurfManX/ffmpeg-nvidia-adaptive-vod-transcoder>

Bash script which will detect video new file in folder and transcode it to adaptive bitrate

FFmpeg Python (kkroening)

<https://github.com/kkroening/ffmpeg-python>

A Python wrapper for FFmpeg that simplifies interaction with the FFmpeg command-line tool for encoding, decoding, and processing video and audio.

Tags: ffmpeg, python, encoding, video-processing, library

FFmpeg Scale Benchmark

<https://github.com/Saurabh702/ffmpeg-scale-benchmark>

Saurabh702/FFmpeg-scale-benchmark - A GitHub repository for video/multimedia development.

FFmpeg Transcoder

https://github.com/sambios/ffmpeg_transcoder

Video transcode based on FFmpeg, support H264/HEVC and more.

FFmpeg Verbose Logging

<https://ffmpeg.org/ffmpeg.html#Generic-Options>

Guide to FFmpeg's -loglevel and debug options for troubleshooting encoding and decoding issues, covering the log levels from quiet to debug and flags like -report that write detailed logs to a file.

FFmpeg Wiki: Filtering Guide

<https://trac.ffmpeg.org/wiki/FilteringGuide>

Documentation on using FFmpeg's filtering system, demonstrating how to chain video filters (scale, crop, overlay, etc.) and audio filters for processing media with examples.

FFmpeg Wiki: H.264 Encoding Guide

<https://trac.ffmpeg.org/wiki/Encode/H.264>

A comprehensive guide on the FFmpeg Wiki detailing how to encode video to H.264 using FFmpeg. It covers recommended settings, presets, CRF usage, and quality considerations.

FFmpeg Wiki: H.265 (HEVC) Encoding Guide

<https://trac.ffmpeg.org/wiki/Encode/H.265>

An FFmpeg Wiki guide for encoding video to H.265/HEVC. It describes how to use x265 via FFmpeg, including tuning parameters, bitrate vs quality (CRF), and preset trade-offs.

FFmpeg iOS

<https://sourceforge.net/projects/ffmpeg-ios/>

FFmpeg iOS provides precompiled FFmpeg libraries tailored for iOS development. These libraries enable developers to incorporate robust video processing capabilities into iOS applications, supporting a wide range of codecs and formats essential for video streaming and encoding tasks.

Tags: FFmpeg, iOS, video processing, video encoding, video decoding

FFmpeg iOS Build Script

<https://github.com/kewlbear/FFmpeg-iOS-build-script>

Shell scripts to build FFmpeg for iOS and tvOS.

Tags: ffmpeg, ios, tvos, build-script, encoding

FFmpeg-go

<https://github.com/u2takey/ffmpeg-go>

A Golang binding for FFmpeg that enables developers to generate and run FFmpeg commands programmatically in Go.

FFmpeg.WebAssembly

<https://github.com/ffmpegwasm/ffmpeg.wasm>

A pure WebAssembly port of FFmpeg that enables video and audio processing directly in the browser. It provides an API to run FFmpeg commands for transcoding or editing media on the web.

FFmpegKit

<https://github.com/tanersener/ffmpeg-kit>

A complete, cross-platform wrapper for FFmpeg, providing ready-to-use binaries and libraries for Android, iOS, Linux, Windows, and more.

Fantastic Transcoder

<https://github.com/ClearSlide/Fantastic-Transcoder>

Fantastic transcoder is a video transcoder which utilizes massively parallel compute to achieve ludicrous conversion speeds.

FastPix Cloud Transcoding

<https://www.fastpix.io/blog/cloud-transcoding-choosing-the-right-solution-for-your-video-centric-product>

FastPix provides a cloud-based transcoding solution that converts video files into multiple resolutions, bitrates, and formats optimized for playback across devices and networks. Key benefits include scalability to handle peak traffic, cost-efficiency with pay-as-you-go models, and accessibility through powerful APIs for automation and dynamic workflows. FastPix supports modern codecs like AV1 and legacy ones like H.264, ensuring compatibility across various platforms.

Tags: cloud transcoding, video encoding, API integration, scalability, cost-efficiency

Fflock

<https://github.com/ericgriffin/fflock>

Distributed video transcoding.

Ffmediaelement

<https://github.com/unosquare/ffmediaelement>

FFME: The Advanced WPF MediaElement (based on FFmpeg)

Tags: ffmpeg, wpf, media-element, video-rendering, dotnet

Ffmpeg4java

<https://github.com/nextbreakpoint/ffmpeg4java>

FFmpeg4Java provides a JNI wrapper of FFmpeg library

FfmpegGUI (ffWorks)

<https://www.fworks.net/>

A macOS graphical front-end to FFmpeg (commercially known as ffWorks). It simplifies complex FFmpeg operations with a drag-and-drop interface and preset configurations, while harnessing FFmpeg's full power under the hood.

Tags: ffmpeg, gui, video encoding, macOS, tool

Ffmpy

<https://github.com/Ch00k/ffmpy>

A lightweight Python wrapper for FFmpeg that enables simple encoding and transcoding of video and audio files.

Tags: ffmpeg, encoding, video-tools, python

Flowblade

<https://github.com/jliljeb1/flowblade>

A free, open-source video editor for cutting, compositing, and editing video with a timeline-based interface.

Tags: video editing, non-linear editing, open source, video processing

GMF (Go Media Framework)

<https://github.com/3d0c/gmf>

A Go binding for FFmpeg that simplifies working with multimedia in Go, providing high-level access to FFmpeg's decoding and encoding functionality.

GPAC

<https://github.com/gpac/gpac>

Multimedia framework for MP4, DASH, and HLS packaging, streaming, and playback with MPEG-4 Systems support.

GPAC MP4Box

<https://github.com/gpac/gpac/wiki/mp4box>

MP4Box is a multimedia packager from the GPAC framework, used for encoding and packaging media files.

Gcloud Node Video Transcoding

<https://github.com/diego3g/gcloud-node-video-transcoding>

Transcode Google Cloud Storage video files with Node.js and FFmpeg

Generate MPEG-TS from file with FFmpeg

https://medium.com/@eyevinntechnology/generate-mpeg-ts-from-file-with-ffmpeg-7561181e6369?source=userActivityShare-94bccb50d11-1560983471&_branch_match_id=670020142756633081

In this post I will describe how an MPEG-TS multicast stream can be generated with FFmpeg by looping an MP4 file and a Docker container..

Gmmlib

<https://github.com/intel/gmmlib>

Intel/gmmlib - A GitHub repository for video/multimedia development.

Gots

<https://github.com/Comcast/gots>

MPEG Transport Stream handling in Go.

Gstreamill

<https://github.com/i4tv/gstreamill>

encoder with hls output based on gstreamer.

Guide to HEVC/H.265 Encoding and Playback

<https://www.techspot.com/article/1131-hevc-h256-enconding-playback/>

HEVC's main advantage over H.264 is that it offers roughly double the compression ratio for the same quality. This means that a video file encoded with HEVC can occupy half the space of its H.264 equivalent with no noticeable change in quality, or the same amount of space with improved quality.

H.264 profiles and levels | Inside & Outside MediaCoder

<https://blog.mediacoderhq.com/h264-profiles-and-levels>

Explains H.264 encoding profiles and levels, helping developers choose appropriate settings for compatibility and quality.

H265ize

<https://github.com/FallingSnow/h265ize>

A node utility utilizing ffmpeg to encode videos with the hevc codec.

Tags: ffmpeg, encoding, hevc, h265, node, video-processing, codec

HEVC DRM Market Update

<https://go.buysdrm.com/thedrmblog/hevc-drm-market-update>

Since time eternal, the streaming industry has toiled with and extolled the virtues of CODECs and their key enablement of the entire digital video experience. Now comes the latest candy in the increasingly large bowl, H.265 (MPEG-H Part 2) or as it's more commonly known. HEVC.

HEVC/H.265 Video Coding Standard: Part 1

<https://www.youtube.com/watch?v=TLNkK5C1KN8&feature=youtu.be>

Part one of the HEVC/H.265 video coding standard lecture series by Dr. Dan Grois, Benjamin Bross, Dr. Detlev Marpe and Karsten Sühling, introducing the standard's core coding tools and structure.

HEVC/H.265 Video Coding Standard: Part 2

<https://www.youtube.com/watch?v=V6a1AW5xyAw&feature=youtu.be>

Part two of the HEVC/H.265 video coding standard lecture series by Dr. Dan Grois, Benjamin Bross, Dr. Detlev Marpe and Karsten Sühling, covering the Range Extensions and the Scalable Extension.

HEVC/h265 Video Coding Standard - ISO/IEC 23008-2

<https://www.iso.org/standard/85457.html>

Defines the compressed video format HEVC/h265

HLS To Dash

<https://github.com/Eyevinn/hls-to-dash>

Open source packager and tools to rewrap live HLS to live MPEG DASH

HandBrake

<https://handbrake.fr/>

A popular open source video transcoder that converts video from nearly any format to a selection of modern, widely supported codecs.

Tags: transcoder, open source, video conversion, codec

HandBrake Web

<https://github.com/TheNickOfTime/handbrake-web>

A self-hosted platform to use HandBrake on headless devices via a bespoke web interface. It allows users to harness the processing power of multiple devices to work on a single queue, facilitating efficient video transcoding workflows.

Tags: web, HandBrake, transcoding, self-hosted, video

Hardware-Assisted Video Transcoding At Dailymotion

<https://medium.com/dailymotion/hardware-assisted-video-transcoding-at-dailymotion-66cd2db448ae>

What if you could save time, power consumption and therefore money, while still keeping a decent quality for your converted video ?

Hybrik Samples

<https://github.com/hybrik/hybrik-samples>

Sample job JSON files and API integration scripts for Dolby Hybrik, a cloud-based media transcoding and QC platform. The repo demonstrates transcode presets, analysis/QC jobs, and workflow automation against the Hybrik API.

IHMC Video Codecs

<https://github.com/ihmcrobotics/ihmc-video-codecs>

A Java library providing support for H.264 and MJPEG encoding and decoding, as well as MP4 support. It utilizes native code for efficient video processing and includes automatic downloading of the OpenH264 library, ensuring compliance with licensing requirements.

Tags: Java, H.264, MJPEG, MP4, video encoding

Imageio Ffmpeg

<https://github.com/imageio/imageio-ffmpeg>

Python wrapper around Ffmpeg for video processing. Simplifies Ffmpeg integration in Python applications.

Imf Conversion

<https://github.com/DSRCorporation/imf-conversion>

NF IMF media conversion utility allows to handle flat file creation from a specified CPL within the IMF package

Improving Our Video Encodes for Legacy Devices

<https://netflixtechblog.com/improving-our-video-encodes-for-legacy-devices-9f39f7774850>

A Netflix Tech Blog post describing techniques to ensure Netflix content remains compatible and efficient on older or less powerful devices, through specialized encoding profiles and optimizations.

Intel Vaapi Driver

<https://github.com/intel/intel-vaapi-driver>

VA-API user mode driver for Intel GEN Graphics family

Intel oneAPI Video Processing Library (oneVPL)

<https://software.intel.com/content/www/us/en/develop/tools/oneapi/components/onevpl.html>

Intel's oneAPI Video Processing Library, the successor to Intel Media SDK. It offers a unified API for GPU-accelerated video encode/decode and processing (Quick Sync Video and other accelerators).

Intel® Video Processing Library (Intel® VPL)

<https://www.intel.com/content/www/us/en/developer/tools/vpl/overview.html>

A library offering advanced access to specialized media hardware, including encode, decode, and video processing features on Intel GPUs. It supports hardware-accelerated codecs and programmable graphics, improving frame rates and reducing power usage compared to traditional CPUs.

Tags: Intel, video processing, hardware acceleration, video encoding, video decoding

Internet Friendly Media Encoder (IFME)

<https://github.com/Anime4000/IFME>

A cross-platform GUI front-end for x265, x264, and VP9 encoders. IFME makes it easy to batch encode videos with advanced compression (HEVC) settings, including support for plugins and multiple audio streams.

Introducing SVT-AV1: a scalable open-source AV1 framework

<https://medium.com/netflix-techblog/introducing-svt-av1-a-scalable-open-source-av1-framework-c726cce3103a>

By Andrey Norkin, Joel Sole, Kyle Swanson, Mariana Afonso, Anush Moorthy, Anne Aaron.

Is it possible to get FFmpeg to use hardware acceleration for HEVC transcoding on macOS?

<https://superuser.com/questions/1295957/ffmpeg-and-hardware-acceleration-of-hevc-transcoding-on-mac>

I have a MacBook Pro with a Kaby Lake processor running macOS High Sierra (10.12). Is it possible somehow to setup FFmpeg to utilize hardware encoding of HEVC with toolbox, instead of libx265?

JCodec

<https://github.com/jcodec/jcodec>

A pure Java library for video and audio codecs, including support for H.264 and AAC decoding, aimed at processing media without native dependencies.

JPEG XL Reference Implementation

<https://github.com/libjxl/libjxl>

The official reference implementation of the JPEG XL image format, offering both lossy and lossless compression. This project is valuable for developers working on image and video processing applications that require efficient compression techniques.

Tags: JPEG XL, image compression, lossy, lossless, reference implementation

JVET VVC Test Model (VTM) Reference Software

<https://jvet.hhi.fraunhofer.de/>

Official overview site for Versatile Video Coding (VVC) and the VVC Test Model (VTM) reference software from the Joint Video Experts Team. Hosts VTM encoder/decoder, experimental JEM software, and Benchmark Set (BMS) tools.

Kdenlive

<https://kdenlive.org/>

A free and open source non-linear video editor for GNU/Linux, Windows, and macOS. It offers multi-track video editing, a wide range of effects, and GPU acceleration.

Tags: video editor, open source, multi-track, gpu acceleration, cross-platform

Kvazaar HEVC Encoder

<https://github.com/ultravideo/kvazaar>

An open source HEVC encoder developed by Tampere University. Kvazaar is aimed at academic research and ultra HD content encoding, offering a modular codebase for experimentation with HEVC encoding algorithms while achieving competitive performance.

Tags: encoder, codec, research, academic, open-source, ultra-hd

LAV Filters

<https://github.com/Nevcairiel/LAVFilters>

An open source set of DirectShow filters based on FFmpeg that allow the playback of virtually any audio/video format on Windows (often used within media players like MPC-HC).

Lagarith Lossless Codec

<https://lags.leetcode.net/codec.html>

A lossless video codec for Windows that offers high compression ratios. Useful for archival and intermediate video editing workflows where quality must be preserved (open source).

Libavcodec

<https://ffmpeg.org/libavcodec.html>

Libavcodec is a free and open-source library of codecs for encoding and decoding video and audio data. It is an integral part of many open-source multimedia applications and frameworks, providing implementations for a wide range of codecs. Libavcodec is known for its portability and performance, offering optimized implementations of common multimedia processing tasks.

Tags: codec library, video encoding, audio encoding, open-source, multimedia processing

Libva

<https://github.com/intel/libva>

Libva is an implementation for VA-API (Video Acceleration API)

Lightworks

<https://lwks.com/>

A professional NLE (non-linear editor) with a free edition. Lightworks has been used in Hollywood and offers multi-track editing, effects, and support for various formats on Windows, Mac, and Linux.

Tags: NLE, video editing, professional software, multi-platform

Liquid DL

https://github.com/Kthulu120/liquid_dl

Liquid-dl is a simple tool for utilities such as FFmpeg, youtube-dl, and sctl. It provides a simple framework with simple point and click options allowing users to just click on what they need and u...

Tags: ffmpeg, download, utility, media-processing, cli-tool

Live Video Transmuxing/Transcoding: FFmpeg vs TwitchTranscoder, Part 2

https://blog.twitch.tv/live-video-transmuxing-transcoding-ffmpeg-vs-twitchtranscoder-part-ii-4973f475f8a3?source=userActivityShare-94bccb50d11-1561003748&_branch_match_id=670105191114382351&gi=fd8d504494f4

Technical comparison examining FFmpeg and Twitch's proprietary transcoder for live video transmuxing and transcoding in production environments, Part 2.

Live Video Transmuxing/Transcoding: FFmpeg vs TwitchTranscoder, Part I

<https://blog.twitch.tv/en/2017/10/10/live-video-transmuxing-transcoding-f-ffmpeg-vs-twitch-transcoder-part-i-489c1c125f28/>

Technical comparison examining FFmpeg and Twitch's proprietary transcoder for live video transmuxing and transcoding in production environments, Part 1.

Live Video Transmuxing/Transcoding: FFmpeg vs TwitchTranscoder, Part I (Medium)

<https://medium.com/twitch-news/live-video-transmuxing-transcoding-ffmpeg-vs-twitchtranscoder-part-i-489c1c125f28>

Part one of Twitch's engineering deep-dive comparing FFmpeg with their in-house TwitchTranscoder for live video transmuxing and transcoding, covering system architecture and design constraints at Twitch scale.

Live Video Transmuxing/Transcoding: FFmpeg vs TwitchTranscoder, Part II

<https://medium.com/twitch-news/live-video-transmuxing-transcoding-ffmpeg-vs-twitchtranscoder-part-ii-4973f475f8a3>

Part two of Twitch's engineering deep-dive comparing FFmpeg with TwitchTranscoder, covering benchmark results, performance trade-offs, and lessons from running live transcoding in production.

Loop file and generate multiple video bitrates muxed in MPEG-TS with FFmpeg

https://medium.com/@eyevinntechnology/loop-file-and-generate-multiple-video-bitrates-muxed-in-mpeg-ts-with-ffmpeg-85658d0b74bb?source=userActivityShare-94bccb50d11-1560983383&_branch_match_id=670019768959110835

In a previous post I described how an MPEG-TS multicast stream can be generated with FFmpeg by looping an MP4 file. In this post I will..

Lossless Cut

<https://github.com/mifi/lossless-cut>

A tool for trimming and cutting video files without re-encoding, preserving the original quality and codec data.

Tags: trimming, video-editing, cutting-tool, media-processing

Lossless H.264 Codec

<https://github.com/inno-v/losslessH264>

An open-source H.264 codec implementation focusing on lossless compression. This project is beneficial for developers requiring high-quality video encoding without loss of data, suitable for applications where preserving the original video quality is paramount.

Tags: H.264, lossless compression, video encoding, open-source, codec

Low-complexity Deep Video Compression with A Distributed Coding Architecture

<https://github.com/Xinjie-Q/Distributed-DVC>

This project introduces a deep video compression framework with a distributed coding architecture, aiming to reduce encoding complexity while maintaining competitive performance. It offers significant encoding speed improvements, making it suitable for deployment on resource-constrained devices.

Tags: video compression, deep learning, distributed coding, low complexity, resource-constrained devices

MLT Framework

<https://www.mltframework.org/>

An open source multimedia framework for video editing applications. MLT provides a toolkit for developers to build editors (used by Kdenlive, Shotcut) with support for a range of effects and transitions.

Tags: video editing, framework, multimedia, open source, development

MP4Box JS

<https://github.com/gpac/mp4box.js>

JavaScript version of GPAC's MP4Box tool.

Tags: mp4, javascript, encoding, media-processing, tool

MP4Joiner & MP4Splitter

<https://www.mp4joiner.org/en/>

Free open-source tools to losslessly join and split MP4 files without re-encoding. Useful for simple video file merging and cutting tasks.

Tags: mp4, video-editing, lossless, file-manipulation, open-source

MainConcept Codec SDK

<https://www.mainconcept.com/codecs>

MainConcept offers a comprehensive Codec SDK that provides developers with a suite of high-quality codecs for integrating into their applications. The SDK supports a wide range of formats, including H.264/AVC, H.265/HEVC, and MPEG-2, facilitating efficient video encoding and decoding.

Tags: codec SDK, video encoding, video decoding, H.264, H.265

Manifesto

<https://github.com/gitfu/manifesto>

Manifesto is an HLS tool for creating multiple variants, a master.m3u8 file, and converting 608 captions to segmented webvtt subtitles via ffmpeg.

Tags: ffmpeg, hls, encoding, captions, subtitle-conversion

Media Driver

<https://github.com/intel/media-driver>

Intel/media-driver - A GitHub repository for video/multimedia development.

Media Works

https://github.com/quarkscript/media_works

Transcode video by FFmpeg with nvenc; normalize the volume; force dynamic range compression to the volume

MediaArchiver

<https://github.com/zolinux/MediaArchiver>

Transcode video files using FFmpeg and python3.

MediaConversions

<https://github.com/thisguysouldworkforus/MediaConversions>

A project aimed at learning PowerShell 7+, HandBrakeCLI, and FFmpeg by creating scripts for media conversion. It focuses on optimizing media libraries for a wide range of mid-to-low bandwidth Roku devices by creating multi-stream videos.

Tags: media conversion, PowerShell, HandBrakeCLI, FFmpeg, Roku optimization

MediaInfo

<https://mediaarea.net/en/MediaInfo>

MediaInfo is a free, cross-platform tool that provides detailed technical and tag information about video and audio files. It supports a wide range of formats and is essential for video engineers who need to analyze media files.

Tags: media analysis, technical information, cross-platform, video, audio

MediaMolder - FFmpeg for the AI era

<https://github.com/MediaMolder/mediamolder>

A rewrite of the interface and orchestration layers of FFmpeg: a graphical interface to visualize, edit, validate and run processing graphs, jobs defined in JSON files, import of any FFmpeg command line, and a real-time encoding control system.

Tags: video, encoding, AI

Mixwave

<https://github.com/matvp91/mixwave>

Self hostable platform that aims to simplify the complexities of video delivery. Transcode and package your media for online streaming with simple API calls and sane defaults, or craft dynamic HLS playlists on the fly with bumpers, ads and filters.

Mlt

<https://github.com/mltframework/mlt>

Cross-platform multimedia framework for video editing, processing, and playback applications.

Tags: multimedia, framework, video processing, encoding, library

Mmf

<https://github.com/mrpdaemon/mmf>

Video transcoding ffmpeg frontend in Python.

Mobile FFmpeg

<https://github.com/tanersener/mobile-ffmpeg>

FFmpeg for Android, iOS and tvOS.

More Efficient Mobile Encodes for Netflix Downloads

<https://netflixtechblog.com/more-efficient-mobile-encodes-for-netflix-downloads-625d44b2daef>

Netflix Tech Blog post summarizing improvements in Netflix's encoding for mobile downloads, targeting low-bandwidth scenarios. It discusses techniques like per-title optimizations for mobile.

Moscow State University Video Quality Measurement Tool (VQMT)

<https://streaminglearningcenter.com/learning/free-must-have-apps-for-video-engineers.html>

VQMT is a tool that computes VMAF, SSIM, and PSNR, displaying the scores in a graph. It allows users to view the frames in the video, making it useful for measuring video quality metrics.

Tags: video quality measurement, VMAF, SSIM, PSNR, video analysis

Moviepy (GitHub)

<https://github.com/Zulko/moviepy>

A Python library for video editing and composition that uses FFmpeg backend to create, edit, and manipulate video files programmatically.

Tags: python, video-editing, effects, compositing, library

Mp4Composer-android

<https://github.com/MasayukiSuda/Mp4Composer-android>

An Android library to compose and edit MP4 videos using MediaCodec, supporting filters, scaling, trimming, transcoding, cropping, time scaling, and rotation.

Mp4parser

<https://github.com/sannies/mp4parser>

A Java API to read, write and create MP4 files. Contribute to sannies/mp4parser development by creating an account on GitHub.

Mugi FFmpeg

<https://github.com/mugiseyebrows/mugi-ffmpeg>

Gui for FFmpeg to simplify transcoding and embedding audio / subtitles in mkv videos

MystiQ Video Converter

<https://github.com/swl-x/MystiQ>

A cross-platform FFmpeg GUI for converting audio and video. MystiQ offers an intuitive interface, preset profiles for common formats (MP4, WEBM, MP3, etc.), and options for advanced users to tweak encoding parameters.

NVIDIA Video Codec SDK

<https://developer.nvidia.com/nvidia-video-codec-sdk>

NVIDIA's official SDK for hardware-accelerated video encoding and decoding (NVENC/NVDEC). Provides APIs to leverage GPU acceleration for H.264, HEVC, and AV1 codecs.

NextLevelSessionExporter

<https://github.com/NextLevel/NextLevelSessionExporter>

NextLevelSessionExporter is a Swift library that facilitates media export and transcoding on iOS. It provides a simple API for exporting and transcoding media sessions, making it a valuable tool for developers working on video applications for Apple

devices.

Tags: iOS, Swift, video, transcoding, export

Node FFmpeg Stream

<https://github.com/phaux/node-ffmpeg-stream>

Node.js bindings to FFmpeg command, exposing stream based API

Node Fluent FFmpeg

<https://github.com/fluent-ffmpeg/node-fluent-ffmpeg>

A fluent API to FFMPEG (<http://www.ffmpeg.org>).

Node Transcoding

<https://github.com/benvanik/node-transcoding>

node.js video transcoding library.

NotEnoughAV1Encodes

<https://github.com/Alkl58/NotEnoughAV1Encodes>

GUI Handler for AV1 Encoders (aomenc, rav1e & svt-av1)

OTT Packager

<https://github.com/cannonbeach/ott-packager>

OTT/ABR streaming encoder (H264/HEVC) and packager for DASH and HLS

Olive Video Editor

<https://olivevideoeditor.org/>

A next-generation open source non-linear video editor under development aiming for professional features and performance (currently in alpha).

Tags: video editor, open source, non-linear editing, professional editing

Open Source Media Framework

<https://www.intel.com/content/www/us/en/developer/topic-technology/open/opensource-media-framework/overview.html>

Intel's Open Source Media Framework provides tools and libraries to increase the speed and performance of video decoding, encoding, processing, and transcoding on compressed digital video and imaging. It supports workload offloading from the CPU to the GPU.

Tags: media framework, video processing, hardware acceleration, open source

Open Transcoder

<https://github.com/yksuzuki/open-transcoder>

Open Transcoder is a media transcoding tool that integrates FFmpeg and OpenH264. It offers a command-line interface for converting media files between various formats, providing a flexible solution for developers needing to transcode media efficiently.

Tags: transcoding, FFmpeg, OpenH264, media conversion, video processing

OpenCV

<https://opencv.org/>

An open source computer vision and machine learning library with extensive support for image and video analysis. Often used for video processing tasks like object detection, tracking, and filtering.

Tags: computer vision, video processing, object detection, machine learning, image analysis

OpenEncoder

<https://github.com/openencoder/openencoder>

OpenEncoder is a free audio and video transcoder based on FFmpeg. It offers an easy-to-use interface with features like drag-and-drop file addition, multi-file conversion queues, and support for professional formats such as DNxHD, ProRes, and CineForm. This tool is valuable for developers seeking a user-friendly transcoding solution with robust format support.

Tags: transcoder, audio, video, FFmpeg, open-source

OpenH264

<https://www.openh264.org/>

OpenH264 is an open-source H.264 codec library developed by Cisco. It supports H.264 encoding and decoding and is suitable for real-time applications such as WebRTC. Cisco provides this library under a BSD license, making it freely available for use in various applications.

Tags: H.264, codec, open-source, encoding, decoding

OpenHEVC Decoder

<https://github.com/OpenHEVC/openHEVC>

An open source high-performance HEVC decoder. OpenHEVC was developed to provide a fast HEVC/H.265 software decoding solution and has been integrated into projects like VLC. It supports real-time decoding of 1080p and higher resolutions on typical PC hardware.

Tags: codec, decoder, open-source, h265, video-decoding

OpenShot Library (libopenshot)

<https://github.com/OpenShot/libopenshot>

The C++ library behind OpenShot Video Editor that provides core video editing functionalities (timeline, effects, keyframes). libopenshot can be used independently to build custom video editing applications.

OpenShot Video Editor

<https://www.openshot.org/>

An easy-to-use, cross-platform open source video editor. It supports timeline editing, effects, and keyframes for creating videos on Linux, Mac, and Windows.

Tags: video editor, cross-platform, open source, editing, timeline

Openfaas Transcode

<https://github.com/cpitkin/openfaas-transcode>

Serverless video transcoding pipeline built on OpenFaaS for function-as-a-service deployments.

Optimized Shot-Based Encodes for 4K: Now Streaming!

<https://netflixtechblog.com/optimized-shot-based-encodes-for-4k-now-streaming-9e4236f07f0d>

Netflix Tech Blog article on Netflix's rollout of shot-based encoding for 4K titles. It explains how splitting content by scene (shot) and encoding with per-shot parameters improved quality.

Other Video Transcoding

https://github.com/donmelton/other_video_transcoding

Other tools to transcode videos.

PLVC

<https://github.com/RenYang-home/PLVC>

PLVC (Perceptual Learned Video Compression) is a deep learning-based video compression approach that employs a recurrent conditional GAN. It focuses on achieving high perceptual quality at low bitrates, outperforming traditional codecs like HEVC in user studies and perceptual quality metrics.

Tags: video compression, deep learning, GAN, perceptual quality, low bitrate

Per-Title Encode Optimization – Netflix TechBlog

<https://netflixtechblog.com/per-title-encode-optimization-7e99442b62a2>

Netflix's seminal whitepaper on Per-Title Encoding Optimization, describing how they determine optimal encoding settings for each content title to maximize quality and minimize bitrate.

Pitivi

<https://www.pitivi.org/>

A free, open source video editor for Linux built on the GStreamer framework, focusing on an intuitive user interface and integration with GNOME.

Tags: linux, open-source, video-editor, gstreamer, gnome

PyAV

<https://github.com/PyAV-Org/PyAV>

A Pythonic binding for FFmpeg's libraries (Libav). PyAV allows Python programs to read, write, and manipulate video and audio data by leveraging the powerful FFmpeg backend.

PyFFmpeg

<https://pypi.org/project/pyffmpeg/>

A Python wrapper for FFmpeg, allowing the composition and execution of FFmpeg commands through a fluent Python interface.

Tags: python, ffmpeg, video-processing, encoding, wrapper

Python FFmpeg

<https://github.com/jonghwanhyeon/python-ffmpeg>

A python interface for FFmpeg using asyncio.

Python FFmpeg Video Streaming (aminyazdanpanah)

<https://github.com/aminyazdanpanah/python-ffmpeg-video-streaming>

Package media content for online streaming(DASH and HLS) using FFmpeg

Pyup Transcoder

<https://github.com/sw360cab/pyup-transcoder>

A Python-based software to transcode videos and upload files to a remote server or S3-bucket

QADRA

<https://github.com/PhoenixVideo/QADRA>

QADRA (Quality-Aware Dynamic Resolution Adaptation) is an open-source Python-based framework designed for adaptive video streaming applications. It determines encoding resolutions and quantization parameters to maximize perceptual quality while adhering to encoding and decoding time constraints, enhancing streaming efficiency.

Tags: adaptive streaming, video encoding, quality optimization, Python, open-source

QCTools (Quality Control Tools for Video Preservation)

<https://bavc.org/programs/preservation/preservation-tools/>

QCTools is a free, open-source software developed by BAVC Media in collaboration with Dave Rice and the Dance Heritage Coalition. It enables conservators and archivists to inspect video signal characteristics for batches of digital media, facilitating quality control, error detection, and increased trust in video digitization efforts.

Tags: video preservation, quality control, open-source, video analysis, digitization

QWinFF

<https://sourceforge.net/projects/qwinff/>

A simple cross-platform FFmpeg GUI (using Qt) for converting audio and video, providing preset profiles and the ability to batch process files.

Tags: ffmpeg, conversion, gui, cross-platform, batch processing

RIFE (AI Frame Interpolation)

<https://github.com/hzwer/arXiv2020-RIFE>

An AI-based method (Real-Time Intermediate Flow Estimation) for frame interpolation to create slow-motion videos by generating intermediate frames between existing ones.

RTSP Stream

<https://github.com/Roverrr/tsp-stream>

Out of box solution for RTSP - HLS live stream transcoding. Makes RTSP easy to play in browsers.

Rasbora

<https://rasbora.openseawave.com/>

Rasbora is a distributed, scalable open-source video transcoding solution with hardware acceleration, providing an alternative to mainstream cloud video transcoding services. It offers components for video transcoding, task management, system monitoring, and more.

Tags: video transcoding, distributed system, hardware acceleration, open source, scalable

Rate Control Tests

<https://github.com/slhck/rate-control-tests>

Tests for different rate control modes in x264. Contribute to slhck/rate-control-tests development by creating an account on GitHub.

Rav1e

<https://github.com/xiph/rav1e>

The fastest and safest AV1 encoder.

React Native Transcode

<https://github.com/selsamman/react-native-transcode>

Video Transcoder for React Native.

Restreamer

<https://github.com/datarhei/restreamer>

A user-friendly solution to re-stream video sources (like local RTSP webcams) to streaming platforms. It wraps FFmpeg to publish streams to YouTube, Twitch, or custom servers with a simple UI.

Running FFmpeg on AWS Lambda for 1.9% the cost of AWS Elastic Transcoder

<https://intoli.com/blog/transcoding-on-aws-lambda/>

A guide to building a transcoder using Exodus, FFmpeg, and AWS Lambda.

RxFFmpeg

<https://github.com/microshow/RxFFmpeg>

Ø=Ý% R x F F m p e g f/WúNŽ (F F m p e g 4.0 + X 2 6 4 + m p 3 l a m e + f d k - a a c)

Tags: ffmpeg, encoding, x264, mp3, aac, mobile

SVT-AV1

<https://github.com/AOMediaCodec/SVT-AV1>

An open-source AV1 encoder and decoder from the Scalable Video Technology project. SVT-AV1 is optimized for performance and multi-threading, enabling faster AV1 encoding.

Tags: codec, open-source, encoding, performance, multi-threading

SVT-AV1: an open-source AV1 encoder and decoder

<https://netflixtechblog.com/svt-av1-an-open-source-av1-encoder-and-decoder-ad295d9b5ca2>

By Andrey Norkin, Joel Sole, Mariana Afonso, Kyle Swanson, Agata Opalach, Anush Moorthy, Anne Aaron.

SVT-HEVC

<https://github.com/OpenVisualCloud/SVT-HEVC>

An open source high-performance HEVC encoder from Intel (Scalable Video Technology for HEVC). SVT-HEVC is optimized for speed and latency, allowing real-time encoding of HD/UHD content on multi-core CPUs.

Tags: codec, encoder, intel, high-performance, real-time, multi-core

SVT-VP9

<https://github.com/OpenVisualCloud/SVT-VP9>

Intel's Scalable Video Technology implementation of the VP9 codec. SVT-VP9 is an open source VP9 encoder/decoder designed to leverage multi-core processors for faster encoding of VP9 videos.

Tags: codec, encoder, decoder, open source, intel, multi-core, video compression

Scalable, Parallel Video Transcoding on Ubuntu

<https://blog.dustinkirkland.com/2014/07/scalable-parallel-video-transcoding-on.html>

Transcoding video is a very resource intensive process. It can take many minutes to process a small, 30-second clip, or even hours to p...

Selene Media Encoder

<https://github.com/teejee2008/selene>

A minimalistic audio/video converter for Linux. Selene provides a simple GUI to convert media files to formats like MP4, MKV, OGG, with support for latest codecs (H.264, H.265, VP9) and basic editing (trim, crop).

Server-less Just-in-Time Packaging with AWS Fargate and Unified Origin by Unified Streaming

https://medium.com/@eyevinntechnology/server-less-just-in-time-packaging-with-aws-fargate-and-unified-origin-by-unified-streaming-c1682dc051ca?source=userActivityShare-94bccb50d11-1559724204&_branch_match_id=664738392430917730

In this blog article Jonas Rydholm Birmé describes how he created a server-less just-in-time packaging origin, using AWS ECS Fargate tasks..

Server-less Video Backend

<https://medium.com/@eyevinntechnology/server-less-video-backend-1a142d1d2ba>

In this blog post by Jonas Rydholm Birmé he describes how a completely server-less video backend on AWS would look like.

Shaka Packager – opensource.google

<https://opensource.google/projects/shaka-packager>

Google's open-source media packaging tool for creating DASH and HLS content with encryption support.

Shotcut

<https://shotcut.org/>

A free, open source, cross-platform video editor. It offers a wide format support via FFmpeg, a customizable UI, and a range of video and audio filters.

Tags: video editor, open source, cross-platform, ffmpeg, video processing

Shutter Encoder

<https://www.shutterencoder.com/>

A free, cross-platform video and audio converter that uses FFmpeg behind the scenes. Shutter Encoder offers a user-friendly GUI with presets and advanced options.

Tags: ffmpeg, converter, video-encoding, cross-platform, gui

Simple Binary Encoding (SBE)

<https://github.com/real-logic/simple-binary-encoding>

SBE is a high-performance message codec designed for low-latency applications. It provides efficient encoding and decoding of messages, making it suitable for real-time video streaming and encoding scenarios.

Tags: message codec, low-latency, high-performance, encoding, decoding

Snickers

<https://github.com/snickers/snickers>

An open source alternative to the video cloud encoding services.

Standardization of High Efficiency Video Coding (HEVC)

<https://youtu.be/p6dLZfs0jTY>

Standardization of High Efficiency Video Coding (HEVC) Jens-Rainer Ohm, Institute of Communication Engineering, RWTH Aachen University, DE

StaxRip

<https://github.com/staxrip/staxrip>

A versatile video encoding GUI for Windows, utilizing a variety of open source tools (AviSynth/VapourSynth, x264/x265, etc.) for advanced encoding workflows.

StreamPack - Multiprotocol live streaming broadcaster libraries for Android

<https://github.com/ThibaultBee/StreamPack>

StreamPack is a lightweight, high-performance toolkit for building streaming video processing pipelines on Android. It provides reusable components for frame sources, transformations, encoding, and output, making it easy to develop efficient real-time video applications with a modular and composable architecture.

Tags: android, srt, rtmp, streaming, camera

Swift Encoder

<https://github.com/ptrandev/swift-encoder>

A fire-and-forget shell script that encodes multiple video and audio files with ffmpeg.

Swift Video Generator

<https://github.com/dev-labs-bg/swift-video-generator>

Dev-labs-bg/swift-video-generator - A GitHub repository for video/multimedia development.

Telestream Vantage

<https://www.telestream.net/telestream-solutions/digital-asset-management.htm>

Telestream's Vantage is a comprehensive video transcoding and workflow automation solution that integrates with various digital asset management systems. It provides multiformat digital video and audio ingest, metadata extraction, keyframe extraction for storyboarding, automated proxy generation, transcoding between formats, and content repurposing for web, mobile, and more. Vantage is designed to streamline media workflows, ensuring efficient and high-quality video processing.

Tags: video transcoding, workflow automation, digital asset management, media processing, content repurposing

The Market Significance of Apple's Adopting HEVC: Here's What I Think

<https://www.linkedin.com/pulse/market-significance-apples-adopting-hevc-heres-what-i-jan-ozar>

At the recent World Wide Developer's Conference (WWDC), Apple announced that the next versions of Safari, iOS, and tvOS will support HLS with HEVC encoded video. This puts Apple firmly in the HEVC camp, with the Alliance for Open Media camp (Amazon, Cisco, Intel, Google, Microsoft, Mozilla, Netflix,

The Netflix IMF Workflow

https://medium.com/netflix-techblog/the-netflix-imf-workflow-f45dd72ed700?source=userActivityShare-94bccb50d11-1568773157&_branch_match_id=702692448596112473

Interesting architectural implications.

Theora

<https://www.theora.org/>

Theora is a free and open video compression format developed by the Xiph.Org Foundation. Derived from the VP3 codec, Theora offers a royalty-free alternative for video compression, making it suitable for developers seeking open standards for video streaming and encoding.

Tags: video compression, open standard, royalty-free, streaming, open source

Transcode

<https://github.com/bbxnet/transcode>

Bbxnet/transcode - A GitHub repository for video/multimedia development.

Transcode (cwinging)

<https://github.com/cwinging/transcode>

Python-based transcoding server by cwinging for handling video conversion tasks.

Transcode (kees)

<https://github.com/kees/transcode>

Collection of command-line tools for video format conversion and transcoding operations.

Transcode Web Video

<https://github.com/monking/transcode-web-video>

Quickly transcode a source video to MP4, OGV, and WebM, with scale, bitrate, and screengrab options.

TranscodeVideo

<https://github.com/Ponyboy47/TranscodeVideo>

A Swift wrapper around the transcode-video command

Transcoder

<https://opensource.deepmedia.io/transcoder>

Transcoder is a hardware-accelerated video transcoding library for Android, utilizing the MediaCodec APIs. It supports features such as cropping, concatenation, clipping, audio processing, and video speed adjustments. Designed for fast and efficient video processing on Android devices, Transcoder offers a fluent API and is suitable for applications requiring on-device video manipulation.

Tags: Android, video transcoding, hardware acceleration, video processing, open-source

Transcoder (GitHub)

<https://github.com/Vilsol/Transcoder>

Docker container to transcode videos in mounted volume to H265 using FFmpeg

Transcoding

<https://github.com/AlvianPrasetya/transcoding>

FFmpeg transcoders benchmark. Contribute to AlvianPrasetya/transcoding development by creating an account on GitHub.

Transcoding FFmpeg

https://github.com/LordCrainer/transcoding_ffmpeg

Transcoding video usando ffmpeg.

Transcoding Performance Trial

<https://github.com/2501world/transcoding-performance-trial>

Runs FFmpeg transcoding processes simultaneously and measures CPU performance

Ts2mpa

<https://github.com/njh/ts2mpa>

Ts2mpa is a simple tool to extract MPEG Audio from a MPEG-2 Transport Stream.

UT Video Codec Suite

http://omezawa.dyndns.info/wordpress/?page_id=67

An open source lossless video codec from Umezawa that provides fast encoding/decoding and is often used for lossless video compression on Windows (supports RGB and YUV formats).

Use of Look-Up Tables (LUTs) in FFmpeg

https://downloads.bbc.co.uk/rd/pubs/papers/HDR/BBC_HDRTV_Use_of_LUTs_FFmpeg.pdf

How to use luts with FFmpeg for converting between different HDR encodes.

Useful FFmpeg Commands

<https://github.com/mitio/useful-ffmpeg-commands>

A collection of FFmpeg commands taken from practice

VCT (Video Converter & Transcoder)

<https://github.com/zbabac/VCT>

VCT is a frontend for FFmpeg that allows users to set options, drag multiple files, transcode between formats (e.g., MKV to MP4), manually edit FFmpeg commands, record network streams, and manage batch processing. It's particularly useful for developers needing a versatile tool for video conversion and transcoding tasks.

Tags: video conversion, transcoding, FFmpeg frontend, batch processing, network stream recording

VCT - Video Converter & Transcoder

<https://sourceforge.net/projects/videoconvertertranscoder/>

VCT is an open-source video/audio converter GUI based on FFmpeg. It works natively for Windows and Linux, allowing batch conversion of audio/video collections to H.264 or H.265 HEVC formats, with features like video rotation, resizing, and audio extraction.

Tags: video conversion, transcoding, FFmpeg GUI, open source, cross-platform

VCodec Interface C++ Library

<https://github.com/ConstantRobotics-Ltd/VCodec>

VCodec is a C++ library that provides a standardized interface for various video codecs, facilitating video encoding and decoding processes. It defines data structures and rules for different video codec classes, supporting multiple raw pixel formats and compressed formats like JPEG, H264, and HEVC. This library is valuable for developers seeking a consistent and extensible framework for integrating video codec functionalities into their applications.

Tags: video encoding, video decoding, C++ library, video codecs, VCodec

VLC Live Transcoding Guide

<https://gist.github.com/teocci/a7eae1b60ec018b64bdd13bb7a3499e0>

A comprehensive guide detailing various methods for live transcoding using VLC and FFmpeg. It includes examples for creating TCP video streaming connections, streaming recorded video files, and sending VLC output to stdout.

Tags: VLC, live transcoding, FFmpeg, video streaming, TCP connections

VLC Media Player

<https://www.videolan.org/vlc/>

VLC is a free and open-source multimedia player that also offers video conversion tools. It supports a wide range of formats and provides basic video encoding functionalities, making it a versatile tool for both media playback and simple encoding tasks.

Tags: media player, video conversion, open-source, multiformat support, cross-platform

VLMC (VideoLAN Movie Creator)

<https://www.videolan.org/vlmc/>

An experimental non-linear video editing software from the VideoLAN project. Aims to offer basic editing (cutting, transitions, etc.) with a simple interface, built on libVLC.

Tags: video editing, libVLC, non-linear editing, open source

VP9 Dash

<https://github.com/Eyevinn/vp9-dash>

FFmpeg wrapper script to create VP9 MPEG-DASH packages

VVCEasy

<https://github.com/MartinEesmaa/VVCEasy>

Ready-to-use prebuilt VVC (H.266) binaries and players for Windows, macOS, Linux, FreeBSD, Haiku, Android, and WebAssembly. Includes vvdec, vvenc, uvg266 encoders/decoders, and FFmpeg builds with VVC support.

VVdeC (Fraunhofer VVC Decoder)

<https://github.com/fraunhoferhhi/vvdec>

An open source Versatile Video Coding (H.266/VVC) software decoder by Fraunhofer HHI. VVdeC supports Main10 profile and can decode 4K/8K content in real-time with optimizations.

VVenC (Fraunhofer Versatile Video Encoder)

<https://github.com/fraunhoferhhi/vvenc>

An open source H.266/VVC encoder implementation by Fraunhofer HHI, designed to be fast and efficient while supporting all features of the VVC standard.

VVenC and VVdeC

<https://arxiv.org/abs/2107.13385>

VVenC is a fast and efficient VVC encoder developed by Fraunhofer HHI, while VVdeC is a live decoder for VVC. Together, they provide an open-source end-to-end toolchain for the Versatile Video Coding (VVC) standard, enabling significant bit-rate reductions for the same subjective video quality compared to previous standards.

Tags: VVC, video encoding, video decoding, open-source, video compression

Vaapi Video Converter

<https://github.com/git-developer/vaapi-video-converter>

A docker-based video converter that uses VAAPI-compatible hardware for transcoding

Vantage Gateway

<https://www.encoding.com/>

Vantage Gateway, formerly known as Encoding.com, is a cloud-based media processing platform that offers comprehensive transcoding, packaging, quality control, and content protection services. It supports a wide range of video formats and provides an API for seamless integration into existing workflows, making it a robust solution for video streaming and encoding needs.

Tags: transcoding, cloud-based, video processing, API, content protection

Vantage Transcode

<https://www.telestream.com/vantage/transcode/>

Vantage Transcode by Telestream provides deep and broad support for all major professional video and audio formats, containers, and packages. It offers faster-than-real-time support, automation of conversion workflows, support for the latest UHD+ and HDR standards, and the ability to work in Vantage Cloud. Vantage Transcode simplifies media processing with dynamic workflows, making it a reliable solution for video streaming and encoding.

Tags: transcoding, video processing, automation, UHD, HDR

Vantage Transcode by Telestream

<https://www.telestream.net/vantage/transcode.htm>

Vantage Transcode is a powerful transcoding solution that automates the conversion of media files into various formats. It supports a broad array of codecs and containers, including MPEG-2, H.264, HEVC, and more, making it suitable for professional media workflows.

Tags: transcoding, media conversion, professional workflows, codec support, automation

Vcodes

<https://www.vcodes.tv/>

Vcodes offers high-performance media processing solutions, supporting major formats and standards used by TV, OTT, and post-production houses. It enables automated media processing workflows, eliminating the need for expensive in-house video expertise. Vcodes provides features like content-aware encoding, automated quality checks, and easily scalable resources, making it a valuable tool for video streaming and encoding developers.

Tags: transcoding, media processing, automation, content-aware encoding, scalability

VidCoder

<https://github.com/RandomEngy/VidCoder>

An open source Windows application for converting and ripping video files and DVDs/Blu-rays, built on the HandBrake engine with an enhanced user interface.

VidCutter

<https://github.com/ozmartian/vidcutter>

An open source cross-platform tool for fast video trimming and joining. VidCutter focuses on doing simple cuts without re-encoding, using intelligent scene detection and supporting many formats via FFmpeg.

Video Compression Motion Estimation Block Video Encoder

<https://github.com/AK1194/Video-Compression-motion-estimation-block-video-encoder>

This repository is about video compression, and more specifically about the motion estimation block (ME block) of a video encoder. It is a research project for developing an efficient motion estima...

Video Processing Benchmarks and Research

<https://videoprocessing.github.io/>

Comprehensive video codec benchmarking and research platform with ML-based optimization methods, stereoscopic quality assessment, and codec comparison studies. Includes research on x264/x265 preset optimization and video quality metrics analysis.

Video Processing Project

<https://sourceforge.net/projects/vpp/>

The Video Processing Project offers a collection of C++ source code for various video processing algorithms and tools used in software media pipelines. It includes DirectShow Filters such as H.264 encoder/decoder, RTSP sink/source filters, and more, providing essential components for developers working on video streaming and encoding solutions.

Tags: video processing, C++, DirectShow Filters, H.264, RTSP

Video Transcode

<https://github.com/GeoHaber/Video-Transcode>

FFmpeg H264 H265 HEVC MPEG Video Trans-code Image-Matrix Collage

Video Transcoder

<https://github.com/andressspinetti/video-transcoder>

Serverless transcoding solution using AWS S3 for storage, Lambda for compute, and MediaConvert or Elastic Transcoder.

Video Transcoder (Mozilla-Open-Lab-Etwas)

<https://github.com/Mozilla-Open-Lab-Etwas/Video-Transcoder>

FFmpeg WebAssembly Video Transcoder.

Video Transcoding

https://github.com/donmelton/video_transcoding

Tools to transcode, inspect and convert videos.

Video Transcoding API

<https://github.com/video-dev/video-transcoding-api>

Agnostic API to transcode media assets across different cloud services.

Video Transcoding Scripts

<https://github.com/lisamelton/video-transcoding-scripts>

A collection of Bash scripts designed to transcode, inspect, and convert videos using tools like HandBrake, MKVToolNix, MPlayer, FFmpeg, and MP4v2. These scripts serve as intelligent wrappers around these tools, facilitating efficient video processing.

Tags: video transcoding, Bash scripts, FFmpeg, HandBrake, video processing

Video Utils

https://github.com/kwodzicki/video_utils

Python package containing many tools useful for converting video files to h264/h265 encoded MP4 or MKV files

VideoTranscoding Backend

<https://github.com/cash2one/VideoTranscoding-Backend>

This application transcode a video that you send on all formats what you want and diferent resolutions.

Videomorph

<https://github.com/videomorph-dev/videomorph>

A user-friendly Video Converter based on FFmpeg and writen in Python/PyQt5.

Vireo

<https://github.com/twitter/vireo>

Vireo is a lightweight and versatile video processing library written in C++11

VirtualDub

<https://www.virtualdub.org/>

A classic open source video capture and processing utility for Windows, useful for quick linear operations, filtering, and format conversions on AVI files.

Tags: video processing, windows, avi, open source, encoding

Voctomix

<https://github.com/voc/voctomix>

Full-HD Software Live-Video-Mixer in python.

Tags: python, live-mixer, video-processing, software-tool, encoding

Vrecord

<https://github.com/amiaopensource/vrecord>

An open source tool for capturing analog video (VHS, Beta, etc.) to digital formats using Blackmagic or AJA capture cards.

Vrecord provides a scriptable FFmpeg-based workflow for archivists with features like live histogram and QA metrics.

WinFF

<https://github.com/WinFF/winff>

A graphical user interface for FFmpeg that runs on Windows and Linux. WinFF makes it easy to convert multiple media files by providing presets and a batch-processing interface.

Tags: ffmpeg, media conversion, batch processing, cross-platform

Wirecast

<https://www.telestream.net/wirecast/>

Wirecast is a professional live video streaming production software that enables users to create high-quality live-streamed events. It supports multiple cameras, graphics, and media sources, making it suitable for broadcasting and live event

production.

Tags: live-streaming, production, broadcasting, multicamera, graphics

XMedia Recode

<https://www.xmedia-recode.de/en/>

A free Windows tool for converting media files to various formats. It supports a broad array of codecs (built on FFmpeg) and provides fine control over encoding settings.

Tags: ffmpeg, media conversion, encoding, transcoding, windows

XVC Codec (Divideon)

<https://github.com/divideon/xvc>

XVC is a proprietary but openly documented video codec by Divideon that claims high compression efficiency. An open source evaluation implementation was released for research purposes. XVC serves as a modern codec design to compare against standards like HEVC and AV1 in compression efficiency research.

Xabe.FFmpeg

<https://github.com/tomaszzmuda/Xabe.FFmpeg>

A .NET Standard wrapper for FFmpeg, enabling developers to compose and execute FFmpeg commands in C# with a fluent interface.

Xiph.Org Foundation Projects

<https://xiph.org/>

The Xiph.Org Foundation develops free and open protocols and software for multimedia, including well-known projects like the Vorbis audio codec and the Theora video codec. Their work focuses on providing open-source alternatives for multimedia processing, making them valuable resources for developers seeking royalty-free codecs.

Tags: open source, audio codec, video codec, multimedia, royalty-free

Xuggler

<https://github.com/artclarke/xuggle-xuggler>

An open source Java library (now discontinued) that wrapped FFmpeg functionality, enabling Java developers to decode, encode, and stream video.

Yamdi (Yet Another Metadata Injector)

<https://yamdi.sourceforge.net/>

A simple, fast tool to inject metadata (onMetaData) into FLV files. Yamdi can also insert an onLastSecond event and is often used to prepare live-recorded FLV files for on-demand playback by adding missing metadata for durations and keyframes.

Tags: metadata, FLV, encoding, video processing, file manipulation

avTranscoder

<https://github.com/avTranscoder/avTranscoder>

C++ wrapper providing a high-level API around FFmpeg and LibAV for video transcoding tasks.

Tags: ffmpeg, encoding, c++, libav, transcoding, video processing

cuda/ubuntu16.04/FFmpeg-gpu/Dockerfile · master · nvidia / container-images / samples

<https://gitlab.com/nvidia/container-images/samples/blob/master/cuda/ubuntu16.04/ffmpeg-gpu/Dockerfile>

Sample Dockerfiles for Docker Hub images.

dav1d

<https://www.videolan.org/projects/dav1d.html>

dav1d is an open-source AV1 decoder developed by the VideoLAN and FFmpeg communities. It is designed for speed and efficiency, making it suitable for a wide range of devices and applications.

Tags: AV1, video decoding, open-source, high performance

dolby-encoding-engine/plugins at master · DolbyLaboratories/dolby-encoding-engine

<https://github.com/DolbyLaboratories/dolby-encoding-engine/tree/master/plugins>

Contribute to DolbyLaboratories/dolby-encoding-engine development by creating an account on GitHub.

ffmpeg-apple-arm64-build

<https://github.com/Vargol/ffmpeg-apple-arm64-build>

A build script for FFmpeg targeting the latest open-source video codecs running on macOS using Apple's M1 processor.

Tags: FFmpeg, macOS, Apple-M1, video-codecs, build-script

ffmprovisr

<https://amiaopensource.github.io/ffmprovisr>

Cookbook of commonly used FFmpeg recipes with descriptions of how each command works and how to modify it to fit your needs.

go-astlav

<https://github.com/asticode/go-astlav>

A comprehensive Go binding for FFmpeg's libav* libraries, facilitating the building of media applications in Go with FFmpeg's capabilities.

goav

<https://github.com/giorgisio/goav>

Go bindings for FFmpeg's libav* libraries. It allows developers to leverage FFmpeg's powerful audio and video processing features in Go applications.

gstreamer-rs

<https://github.com/sdroege/gstreamer-rs>

Rust language bindings for the GStreamer multimedia framework, allowing developers to create GStreamer pipelines and plugins in Rust.

jrottenberg/FFmpeg Docker Images

<https://hub.docker.com/r/jrottenberg/ffmpeg>

A popular Docker image repository for FFmpeg maintained by the community. These images package FFmpeg (with various codecs enabled) for easy use in CI pipelines or cloud deployments, simplifying environment setup for video processing tasks.

libVLC

<https://www.videolan.org/vlc/libvlc.html>

libVLC is the core engine and interface of the VLC media player, providing a modular framework for multimedia applications. It supports a wide range of media formats and streaming protocols, enabling developers to integrate powerful video playback and streaming capabilities into their applications.

Tags: multimedia, streaming, playback, open-source, library

libaom

<https://aomedia.org/oss/>

libaom is the reference implementation for the AV1 video coding format developed by the Alliance for Open Media. It provides tools for encoding and decoding AV1 video streams, offering improved compression efficiency over previous formats.

Tags: AV1, video codec, open-source, video encoding, compression efficiency

libaom: AV1 Codec Library

<https://aomedia.googlesource.com/aom>

libaom is the official AOMedia video codec library for AV1. It provides the tools necessary for encoding and decoding AV1 video streams, offering a royalty-free alternative to other codecs with improved compression efficiency.

Tags: AV1, video codec, encoding, decoding, compression

libde265

<https://github.com/strukturag/libde265>

An open source H.265/HEVC video decoder written in C. libde265 can be used to decode HEVC bitstreams in software and is often integrated into players or frameworks needing HEVC support.

libmpeg2

<https://libmpeg2.sourceforge.net/>

A free library for decoding MPEG-1 and MPEG-2 video streams. It's lightweight and portable, used in several open source media players for MPEG-2 playback.

Tags: codec, mpeg, decoding, library, open-source

libvpx

<https://www.webmproject.org/code/>

libvpx is an open-source project that provides libraries for encoding and decoding VP8 and VP9 video codecs. Developed by Google, it is widely used for web video applications and offers high-quality compression.

Tags: VP8, VP9, video encoding, decoding, open-source

libvpx (VP8/VP9 Codec Library)

<https://github.com/webmproject/libvpx>

The reference implementation of VP8 and VP9 video codecs (WebM Project). libvpx provides encoding and decoding for VP8/VP9, which are open and royalty-free codecs.

libvpx: VP8/VP9 Codec SDK

<https://chromium.googlesource.com/webm/libvpx/>

libvpx is the official WebM project library for VP8 and VP9 video codec implementations. It provides developers with the tools to encode and decode VP8 and VP9 video streams, facilitating high-quality video compression and playback.

Tags: VP8, VP9, video codec, encoding, decoding

libwebm: WebM File Parser

<https://chromium.googlesource.com/webm/libwebm/>

libwebm is a library for parsing WebM files, enabling developers to read and write WebM container files. It supports the WebM multimedia format, which is designed for web video streaming.

Tags: WebM, file parser, video container, multimedia, streaming

mediaEngine by nablet GmbH

<https://nablet.com/media-engine>

mediaEngine is a professional video transcoding software that unites multiple components into a single solution, reducing integration time and accelerating time-to-market. It offers high-speed encoding and superior video quality, making it suitable for developers seeking efficient transcoding solutions.

Tags: video transcoding, professional software, high-speed encoding, video quality, integration

mofo7777/Stack Overflow

<https://github.com/mofo7777/Stackoverflow>

All source codes I've provided on Stack Overflow as an answer, usually under tag ms-media-foundation. Mediafoundation, audio, video, 3D, decoder, encoder. - mofo7777/Stack Overflow.

mp4ff

<https://github.com/edgeware/mp4ff>

A Golang library and set of tools for parsing and writing MP4 (ISO BMFF) files, including support for fragmented MP4 and progressive MP4.

multicoreware / x265 / wiki / Home — Bitbucket

https://bitbucket.org/multicoreware/x265_git/wiki/Home

Documentation and development wiki for x265, an open-source H.265/HEVC video encoder project hosted on Bitbucket.

oneVPL (oneAPI Video Processing Library)

<https://github.com/oneapi-src/oneVPL>

Intel's oneAPI Video Processing Library, providing a unified API to utilize hardware accelerators (CPU/GPU) for video encoding, decoding, and processing.

rffmpeg

<https://github.com/joshuaboniface/rffmpeg>

rffmpeg is a remote SSH FFmpeg wrapper tool that allows users to execute FFmpeg commands on remote servers via SSH. This is particularly useful for offloading transcoding tasks to more powerful machines, enabling efficient media processing and load balancing.

Tags: remote processing, FFmpeg, SSH, transcoding, media server

scikit-video

<https://github.com/aizvorski/scikit-video>

A set of video processing routines for use with SciPy, providing basic video reading and writing capabilities for scientific computing workflows in Python.

slowmoVideo

<http://slowmovideo.granjow.net/>

An open source tool that creates slow-motion videos from existing footage by interpolating intermediate frames, allowing extreme slow motion effects.

Tags: frame interpolation, slow motion, video processing, open source, video enhancement

transcode2H265

<https://github.com/amaurypm/transcode2H265>

Unattended video transcoder to H265 and ACC codecs, in MKV containers.

tsMuxer

<https://github.com/justdan96/tsMuxer>

tsMuxer is a transport stream muxer for remuxing/muxing elementary streams, EVO/VOB/MPG, MKV/MKA, MP4/MOV, TS, M2TS to TS to M2TS. Supported video codecs H.264/AVC, H.265/HEVC, VC-1, MPEG2. Support...

uv266 - Open-source VVC/H.266 Encoder

<https://github.com/ultravideo/uv266>

Open-source VVC/H.266 encoder licensed under 3-clause BSD, developed by University of Helsinki's Ultra Video Group. Still under active development with ongoing RD-quality and speed improvements.

vid.stab

<https://github.com/georgmartius/vid.stab>

An open source video stabilization library, used with FFmpeg to stabilize shaky footage by analyzing and smoothing camera motion.

video-transcoding-api

<https://pkg.go.dev/github.com/video-dev/video-transcoding-api/v2>

The video-transcoding-api provides an HTTP API for transcoding media files into different formats using pluggable providers. It supports multiple transcoding services, offering flexibility and scalability for developers integrating video transcoding into their applications.

Tags: API, transcoding, media files, Go, pluggable providers

video4j

<https://github.com/metaloorm/video4j>

A Java video processing library providing a set of tools and APIs to handle video operations in Java applications.

videojs-record ffmpeg.wasm Plugin

<https://github.com/collab-project/videojs-record/blob/master/docs/plugins/ffmpeg.wasm.md>

The ffmpeg.wasm plugin for videojs-record allows developers to run FFmpeg directly in the browser using WebAssembly. This enables on-the-fly transcoding of recorded media without the need for server-side processing. It's particularly useful for applications requiring real-time video manipulation and supports various FFmpeg options for customized encoding.

Tags: FFmpeg, WebAssembly, video.js, transcoding, browser-based

webm-tools: WebM Utilities

<https://chromium.googlesource.com/webm/webm-tools/>

webm-tools is a collection of utilities for working with WebM files, including tools for handling alpha channels, encryption, and streaming-related tasks. These tools assist developers in managing and processing WebM video content.

Tags: WebM, utilities, alpha channel, encryption, streaming

webmdshow: DirectShow Filters for WebM

<https://chromium.googlesource.com/webm/webmdshow/>

webmdshow provides DirectShow filters for WebM, allowing developers to integrate WebM playback and encoding capabilities into Windows applications using the DirectShow framework.

Tags: WebM, DirectShow, filters, Windows, multimedia

webmquicktime: QuickTime Components for WebM

<https://chromium.googlesource.com/webm/webmquicktime/>

webmquicktime offers QuickTime components for WebM, enabling QuickTime-based applications to support WebM video playback and encoding, thus expanding compatibility with the WebM format.

Tags: WebM, QuickTime, components, video playback, encoding

x264 (H.264/AVC Encoder)

<https://code.videolan.org/videolan/x264>

The most widely used open source H.264/AVC video encoder, known for its high quality and efficiency across many profiles and presets.

x264: H.264 Video Encoder

<https://github.com/mirror/x264>

x264 is a free software library for encoding video streams into the H.264/MPEG-4 AVC format. It is widely used due to its high performance and quality, making it a staple in video encoding applications.

Tags: H.264, video encoding, MPEG-4, AVC, compression

x265 (H.265/HEVC Encoder)

<https://x265.org/>

A high-performance open source H.265/HEVC encoder. x265 is used to create HEVC video streams and is known for being very configurable to balance quality/speed.

Tags: encoder, HEVC, H.265, video compression, open source

x265 Documentation — x265 documentation

<https://x265.readthedocs.io/en/master/>

Complete documentation for x265, an open-source HEVC video encoder library and application.

x266 - Next Generation VVC/H.266 Encoder by MulticoreWare

<https://multicorewareinc.com/what-we-do/audio-video-solutions/x266-vvc-encoder/>

Open-source VVC/H.266 compliant encoder built on top of x265, featuring HDR/WCG support, 360° video encoding, and SDR video. Developed for achieving twice the coding efficiency of HEVC.

xvc

<https://xvc.io/>

xvc is a modern and flexible open-source video codec that offers high compression performance at reasonable encoding and decoding complexity. It is based on technologies from standards like AVC/H.264 and HEVC, and includes additional advancements to enhance video compression efficiency.

Tags: video codec, compression, encoding, decoding, open source

Players & Clients

ABR Player Chrome

<https://github.com/Eyevinn/abr-player-chrome>

Chrome extension that uses Eyevinn HTML player to be able to play HLS and MPEG-DASH natively

AV1 Player

<https://github.com/Eyevinn/av1-player>

Video player specifically for AV1 codec playback. Helps developers test and deploy AV1 streaming content.

AVPlayer

<https://developer.apple.com/documentation/avfoundation/avplayer>

AVPlayer is Apple's framework for playing and streaming audio and video content on iOS and macOS. It supports a variety of formats and provides developers with a powerful API to control playback, manage buffering, and handle interruptions, making it a fundamental tool for media playback in iOS applications.

Tags: video player, iOS, macOS, media playback, streaming

Able Player

<https://ableplayer.github.io/ableplayer>

A fully accessible, cross-browser HTML5 media player library that supports captions, audio descriptions, and sign language tracks, designed to be inclusive for users with disabilities.

Adaptive Video Player

<https://github.com/IvanoBilenchi/Adaptive-Video-Player>

HLS player for iOS that supports manual selection for the quality of adaptive streams

Tags: HLS, iOS, adaptive streaming, video player, mobile

Advantech Video Solutions

<https://www.advantech.com/en-us/solutions/video-solution>

Advantech offers a range of industrial-grade video solutions, including high-quality, real-time video transmission systems with UHD resolutions and low latency. Their VEGA series supports various protocols, aiding in broadcasting, mobile, gaming, and medical applications, accelerating the deployment of next-generation media solutions.

Tags: video transmission, UHD, low latency, VEGA series, industrial-grade

Agora Video SDK for Android

<https://docs.agora.io/en/interactive-live-streaming/get-started/compile-run-sample-project>

Agora provides open-source sample projects demonstrating the implementation of basic and advanced Video SDK features for Android. These samples help developers integrate video calling and live streaming functionalities into their applications, leveraging Agora's global real-time communication network.

Tags: Android, Video SDK, Live Streaming, Video Calling, Agora

Agora.io Live Streaming SDK

<https://www.agora.io/en/products/interactive-live-streaming/>

Agora.io provides a live streaming SDK that enables developers to build interactive live streaming applications on mobile platforms. It offers features like dual video streaming, co-hosting capabilities, and adaptive bitrate streaming up to 1080p at 30fps.

Tags: live streaming, SDK, iOS, Android, interactive

Akamai Players

<https://players.akamai.com/players>

Akamai provides a suite of media players optimized for high-quality streaming across various devices.

AllPlayer

<https://www.allplayer.org/>

A free media player for Windows with support for a wide range of video formats and subtitles.

Tags: media player, windows, video formats, subtitles, desktop application

Amazon Kinesis Video Streams WebRTC SDKs

<https://docs.aws.amazon.com/kinesisvideostreams-webrtc-dg/latest/devguide/webrtc-sdks.html>

Amazon Kinesis Video Streams offers WebRTC SDKs for C, JavaScript, Android, and iOS, enabling real-time, two-way audio and video streaming. These SDKs facilitate low-latency live streaming and data exchange between web, Android, iOS applications, and embedded devices.

Tags: WebRTC, SDK, real-time streaming, mobile, Amazon Kinesis

Amplitude Brightscript

<https://github.com/gabek/Amplitude-Brightscript>

A Brightscript (Roku) library for submitting analytics to Amplitude

Tags: roku, analytics, brightscript, sdk, platform-specific

Android Oembed Video

<https://github.com/TalbotGooday/Android-Oembed-Video>

An Android library for parsing and playing video links (YouTube, Vimeo, etc.) within a WebView without requiring official APIs, using oEmbed.

Aniview Android Video Player SDK

<https://aniview.com/mobile-sdk/android-sdk/>

Aniview's Android Video Player SDK allows developers to integrate video experiences into Android apps, supporting multiple protocols like HLS and MPEG-DASH. It offers features such as subtitle support, customizable player UI, and monetization options through video ads, including support for VAST, VPAID, and VMAP standards.

Tags: Android, video player, streaming, SDK, monetization

Aniview Mobile Video Player SDKs

<https://aniview.com/mobile-sdk/>

Aniview provides native video player SDKs for Android and iOS, allowing developers to integrate high-quality video playback with support for VAST and VPAID ads. The SDKs offer features like pre-roll, mid-roll, and post-roll ad slots, customizable player interfaces, and support for various streaming protocols, enhancing the video experience in mobile applications.

Tags: video player, SDK, Android, iOS, video ads

Aniview iOS Video Player SDK

<https://aniview.com/mobile-sdk/ios-sdk/>

Aniview's iOS Video Player SDK enables developers to incorporate video playback into iOS applications, supporting protocols like HLS and MPEG-DASH. It provides features such as subtitle support, customizable player UI, and monetization capabilities through video ads, with support for VAST, VPAID, and VMAP standards.

Tags: iOS, video player, streaming, SDK, monetization

Ant Media Server Mobile Streaming SDKs

<https://antmedia.io/solutions/mobile-streaming-application/>

Ant Media Server provides WebRTC-based streaming SDKs for mobile applications, enabling developers to build live streaming apps with low latency and adaptive bitrate streaming. The SDKs support both iOS and Android platforms, facilitating quick integration and deployment of mobile streaming solutions.

Tags: WebRTC, mobile streaming, SDK, iOS, Android

Apprentice Video

<https://sourceforge.net/projects/apprenticevideo/>

An open source cross-platform media player focusing on high quality playback and unique features like alpha channel preview and timeline looping.

Tags: media player, cross-platform, playback, open source, timeline, alpha channel

Awesome Smart TV

<https://github.com/vitalets/awesome-smart-tv>

A curated list of awesome resources for building Smart TV apps.

Awesome Video

<https://awesome.video/>

Awesome Video is a curated list of video players and related resources for various platforms, including set-top boxes and Smart TVs. It provides links to useful utilities, frameworks, and sample channels for devices like Roku, Samsung HbbTV, and more. This resource is valuable for developers looking to build or enhance video player applications across different hardware platforms.

Tags: video players, set-top box, Smart TV, frameworks, resources

BMPlayer

<https://github.com/BrikerMan/BMPlayer>

A video player for iOS, based on AVPlayer, support the horizontal, vertical screen. support adjust volume, brightness and seek by slide, support subtitles.

Tags: ios, video player, avplayer, mobile, subtitles

Banshee

<https://bansheemedioplayer.github.io/>

An open source media player for Linux (and other platforms) that can play and organize video and audio, with support for podcasts and device sync.

Tags: media player, linux, open source, audio, video, desktop application

BestMediaPlayer

<https://github.com/bestmedioplayer/medioplayer>

BestMediaPlayer is a free, open-source, no-download HTML5 media player that supports a wide range of video and audio formats, including FLV, MP4, WEBM, and more. It aims to provide a comprehensive media playback solution directly in the browser without the need for additional downloads.

Tags: HTML5, media player, open-source, browser-based, multi-format

Bifserver

<https://github.com/anachirino/bifserver>

Server which creates and serves up BIF files for Roku players

Tags: roku, server, bif, video-metadata, streaming-support

Bigscreen Player

<https://github.com/bbc/bigscreen-player>

Simplified media playback for bigscreen devices.

Tags: bigscreen, media playback, device-specific, hardware player

Bitmovin Native Player SDKs

<https://bitmovin.com/demos/native-sdks>

Bitmovin provides native player SDKs for Android, iOS, Android TV, tvOS, and Fire OS, enabling developers to deliver high-quality video experiences across a wide range of devices. These SDKs support multiple streaming protocols such as MPEG-DASH, HLS, and Smooth Streaming, and offer a flexible API with extensive event handling capabilities.

Tags: video player, SDK, mobile, adaptive streaming, cross-platform

Bitmovin Player SDK

<https://bitmovin.com/video-player/android-sdk>

Bitmovin's Player SDKs for Android and iOS offer a unified API and UI for seamless integration of live and on-demand video streaming. They support various features including DRM, offline playback, subtitles, and advertising, enabling developers to build robust video applications.

Tags: video player SDK, Android, iOS, DRM, offline playback

Bitmovin Player SDKs

<https://bitmovin.com/delivering-high-quality-video-ios-android-sdks/>

Bitmovin offers Player SDKs for HTML5, Android, iOS, and Smart TVs, enabling high-quality video playback across various devices. These SDKs support DRM and content protection, ensuring secure and reliable video delivery.

Tags: video player, SDK, DRM, mobile, Bitmovin

Bitmovin Video Player SDKs

<https://bitmovin.com/video-player/>

Bitmovin provides video player SDKs designed for flawless playback across a wide range of devices. Their SDKs support platforms such as HTML5, Android, iOS, Smart TVs, Roku, gaming consoles, React Native, and Flutter, ensuring high-quality

streaming experiences with features like adaptive streaming, DRM, and low latency.

Tags: video player SDK, adaptive streaming, cross-platform, DRM, low latency

Brighterscript Formatter

<https://github.com/RokuCommunity/brighterscript-formatter>

A code formatter for BrighterScript (and BrightScript)

Brightscript JSON

<https://github.com/chrishoffman/brightscript-json>

JSON parser for Roku's proprietary Brightscript language

Tags: roku, json, parsing, brightscript, development

Brightscript State Machine

<https://github.com/schtanislau/brightscript-state-machine>

State management for Roku channel..

Brs Testbed

<https://github.com/sjbarag/brs-testbed>

A simple, buildable Roku channel that executes arbitrary BrightScript files.

Build a basic Cast Receiver

<https://codelabs.developers.google.com/codelabs/cast-receiver/#0>

Google Codelab tutorial for building a basic Cast Receiver application to enable video playback on Google Cast-compatible smart TVs.

Building native video Pins

<https://medium.com/pinterest-engineering/building-native-video-pins-7ff89ad3ec33>

Billions of videos are viewed across the internet every day, but video on Pinterest is unique. On Pinterest, you've always been able to save videos from around the web, and in 2013, we made it..

CastReceiver

<https://github.com/googlecast/CastReceiver>

Reference Receiver: CastReceiver shows how to develop a fully Cast Design Checklist compliant receiver with additional features.

Celluloid (GNOME MPV)

<https://github.com/celluloid-player/celluloid>

A simple GTK+ frontend for the mpv player (formerly GNOME MPV). Celluloid provides a lightweight graphical interface to mpv on Linux, integrating features like playlists and settings into a GNOME-friendly UI.

Channel Engine Multiview

<https://github.com/Eyevinn/channel-engine-multiview>

A multiview frontend for Eyevinn Channel Engine.

Chimee

<https://github.com/Chimeejs/chimee>

a video player framework aims to bring wonderful experience on browser

Tags: javascript, video player, browser, web framework, HTML5

Clapper

<https://github.com/Rafostar/clapper>

Clapper is a media player designed for GNOME that utilizes GStreamer for media playback. It supports hardware-accelerated video playback, providing smoother performance and improved efficiency on compatible hardware. Clapper offers a simple and modern user interface, making it a suitable choice for users seeking a native GNOME media player.

Tags: GNOME, media player, GStreamer, hardware acceleration, Linux

Clappr

<https://github.com/clappr/clappr>

An open source HTML5 video player that supports custom plugins and playback of various formats.

Tags: html5, video player, javascript, web, open source, plugin support

ComQi Digital Display Players

<https://comqi.com/display-hardware/>

ComQi provides a range of digital media players, including the EMP-200, EMP-300, EMP-500, and OPS 400, designed for

various content playback needs. These players support resolutions up to 4K and are suitable for applications ranging from simple multi-zone layouts to complex HTML5 applications and interactive touch screen content. ComQi's solutions cater to diverse digital signage requirements.

Tags: digital media player, 4K content, HTML5 applications, interactive content, digital signage

CyTube

<https://github.com/calzoneman/sync>

A self-hosted web application for synchronized media playback and chat. CyTube allows multiple users to watch videos together in real-time in various channels.

DKVideoPlayer

<https://github.com/Doikki/DKVideoPlayer>

An Android video player library wrapping MediaPlayer, ExoPlayer, and IJKPlayer with features like preload, list playback, floating window, and more.

Tags: android, video-player, mobile, library, playback

DLGPlayer

<https://github.com/DeviLeo/DLGPlayer>

A media player for iOS based on FFmpeg 4.0.

Tags: ios, media player, ffmpeg, mobile, video playback

DPlayer

<https://dplayer.diygod.dev/>

DPlayer is a free, open-source, lightweight, and simple HTML5 video player that supports a wide range of features, including clarity switching, bullet screen (danmaku) support, double-speed playback, and streaming formats like HLS, DASH, and FLV. It is designed for easy integration into web applications and offers a rich API for customization.

Tags: HTML5, video player, open-source, danmaku, streaming

DPlayer (GitHub)

<https://github.com/MoePlayer/DPlayer>

Wow, such a lovely HTML5 danmaku video player.

Tags: html5, video player, web, javascript, interactive, danmaku

Daily Video SDKs

<https://www.daily.co/products/video-sdk/>

Daily provides cross-platform video SDKs for real-time video, live streaming, and recording. With support for JavaScript, React Native, iOS, Android, Python, and Flutter, developers can build scalable and flexible video applications with features like adaptive video codecs, RTMP, media ingest, and real-time transcription.

Tags: video SDK, real-time video, live streaming, cross-platform, recording

Deskreen

<https://github.com/pavlobu/deskreen>

An open source application that turns any device with a web browser into a secondary screen for your computer by streaming your desktop over WebRTC in real-time.

Distributed Codec Engine (DCE)

https://en.wikipedia.org/wiki/Distributed_Codec_Engine

An open-source project that provides a framework for offloading multimedia processing tasks to dedicated hardware accelerators. DCE is particularly useful for developers working with Texas Instruments OMAP platforms, offering support for hardware-accelerated video decoding and encoding.

Tags: codec, hardware acceleration, video processing, OMAP, open-source

Docker Html5player

<https://github.com/Eyevinn/docker-html5player>

A Docker containerized HTML5 player based on Shaka Player

Dolby.io Client SDKs

<https://optiview.dolby.com/docs/millicast/client-sdks/>

Dolby.io offers Client SDKs for developers aiming to create applications with streaming capabilities. These SDKs support various platforms, including JavaScript for web applications, React Native, Flutter, and native SDKs for Android, iOS, Mac, Windows, and Linux. They assist in connecting, capturing, publishing, and viewing streams from the Dolby.io Streaming Platform, providing a comprehensive solution for building streaming applications.

Tags: JavaScript, React Native, Flutter, Android, iOS

Emby

<https://emby.media/>

A media server designed to organize, play, and stream audio and video to a variety of devices. Emby uses a client-server model, with Emby Server available for multiple operating systems and clients for platforms including mobile devices, streaming devices, smart TVs, and gaming consoles. It offers features like live TV, DVR, and cloud sync.

Tags: media server, streaming, cross-platform, live TV, DVR

Emby Roku

<https://github.com/MediaBrowser/Emby.Roku>

Official Roku client application for the Emby media server platform for streaming personal video libraries.

Tags: roku, media browser, streaming client, hardware player

Enhancer for YouTube

<https://www.mrfdev.com/enhancer-for-youtube>

A browser extension that adds a variety of features to enhance the YouTube watching experience (ad control, playback speed, themes, etc.).

Epic Video Comparator

<https://github.com/epiclabs-io/epic-video-comparator>

Javascript library which implements a video comparator component: two overlaped and synchronized video players each one playing an independent source.

Tags: javascript, video player, web, comparison, synchronization

ExoPlayer

<https://exoplayer.dev/>

An open source, application-level media player for Android by Google. ExoPlayer provides an easy way to play adaptive streaming content (DASH, HLS, SmoothStreaming) with features like DRM support and custom renderers.

Tags: android, mobile, media-player, streaming, adaptive-streaming, DRM

ExoPlayer (GitHub)

<https://github.com/google/ExoPlayer>

ExoPlayer is an application level media player for Android.

Eyevinn Player

<https://github.com/Eyevinn/eyevinn-player>

Throttled video player to test video streams.

FFEngine Framework

<https://github.com/xiewei-wayne/FFEngine.framework>

FFEngine framework is a high performance player sdk for iOS based on ffmpeg.

FFmpeg Avplayer For iOS tvOS

<https://github.com/imoreapps/ffmpeg-avplayer-for-ios-tvos>

A tiny but powerful iOS and Apple TV OS av player framework that's based on the FFmpeg library.

FOSDEM Video Hardware

<https://github.com/FOSDEM/video-hardware>

An open-source project aimed at developing a fully free and open hardware video streaming platform. It serves as a successor to existing FOSDEM video boxes, focusing on creating a compact, lightweight, and cost-effective device for video streaming. The project utilizes the Olimex A20-Olinuxino-Lime2 as the mainboard and is working on custom HDMI input solutions.

Tags: open-source, video streaming, hardware, FOSDEM, video technology

FPGA-Video-Player

<https://github.com/sicajc/FPGA-Video-Player>

FPGA-Video-Player is a fork of the FPGAmP project, offering similar capabilities as a 720p FPGA-based media player. It supports Motion JPEG encoded video playback over HDMI or VGA, hardware-accelerated JPEG decoding, SD/MMC card interface with FAT16/32 support, MP3 playback, JPEG stills display, and IR remote control. The project is designed for use with Digilent Arty A7 and associated peripherals.

Tags: FPGA, media player, hardware acceleration, RISC-V, open-source

FPGAmP

<https://github.com/ultraembedded/FPGAmP>

FPGAmp is a 720p FPGA-based media player capable of playing Motion JPEG encoded video over HDMI or VGA on commonly available FPGA boards. It features hardware-accelerated JPEG decoding, SD/MMC card interface with FAT16/32 support, MP3 playback, JPEG stills display, and IR remote control. The project serves as an interesting test case for various open-source digital IPs, including a RISC-V CPU and audio/video controllers.

Tags: FPGA, media player, hardware acceleration, RISC-V, open-source

FVV Live

<https://arxiv.org/abs/2007.00558>

A real-time free-viewpoint video system designed for low-cost and real-time operation using consumer electronics hardware. FVV Live enables high-quality free-viewpoint video using consumer-grade cameras and hardware, making it suitable for immersive event broadcasting or videoconferencing. The system supports an arbitrary number of cameras and capture servers, offering low motion-to-photon and end-to-end delays for seamless free-viewpoint navigation.

Tags: free-viewpoint video, real-time, consumer hardware, immersive broadcasting, videoconferencing

Flowplayer

<https://flowplayer.com/>

A lightweight HTML5 video player known for fast performance and plugin support, historically used as an open source web player.

Tags: html5, web player, open source, javascript, video playback

Fluid Player

<https://github.com/fluid-player/fluid-player>

Fluid Player is a free, open-source HTML5 video player that is lightweight and easy to integrate. It offers advanced VAST (Video Ad Serving Template) capabilities, making it suitable for developers looking to implement video advertising. Fluid Player supports various customization options and is licensed under the MIT License.

Tags: HTML5, video player, open-source, VAST, advertising

FreeTube

<https://freetubeapp.io/>

A standalone YouTube client for Windows, macOS, and Linux that is open source and privacy-friendly. FreeTube lets users watch YouTube content without ads or tracking.

Tags: desktop, open-source, privacy, video-player, cross-platform

GNOME Videos (Totem)

<https://apps.gnome.org/Totem/>

The default video player for the GNOME desktop (formerly Totem). It provides simple playback for local and network video streams on Linux.

Tags: desktop, linux, media-player, open-source

GOM Player

<https://www.gomlab.com/gomplayer-media-player/>

A popular free media player for Windows with built-in support for many codecs and a user-friendly interface.

Tags: media player, windows, codec support, free software, video playback

GPUVideo-android

<https://github.com/MasayukiSuda/GPUVideo-android>

An Android library that applies GPU-accelerated video filters to videos, integrates with ExoPlayer, and supports video recording with Camera2.

Tags: mobile, android, gpu, video-processing, exoplayer, camera2

GSPlayer

<https://github.com/wxxsw/GSPlayer>

#i Video player, support for caching, preload, fullscreen transition and custom contro

GSYVideoPlayer

<https://github.com/CarGuo/GSYVideoPlayer>

An open source video player library for Android, offering multiple playback engines (ExoPlayer, IjkPlayer etc.), gesture controls, and rich features for building video apps.

Tags: android, video player, mobile, playback, ExoPlayer, IjkPlayer

GetStream JavaScript Video SDK

<https://github.com/GetStream/stream-video-js>

The GetStream JavaScript Video SDK offers a comprehensive solution for integrating video streaming capabilities into web

applications. It includes a low-level client for managing call lifecycles, React utilities and hooks for state management, and sample applications to demonstrate platform capabilities, facilitating the development of robust video streaming features.

Tags: video SDK, JavaScript, React, video streaming, client

Google Media Framework iOS

<https://github.com/googleads/google-media-framework-ios>

The Google Media Framework (GMF) is a lightweight media player designed to make video playback and integration with the Google IMA SDK on iOS easier.

Tags: ios, media-player, google-ads, sdk, video-integration

HLS with Widevine for Android - Taku Semba - Medium

<https://medium.com/@takusemba/hls-with-widevine-for-android-de3f41027ed2>

As of the version of 2.6.0, ExoPlayer started supporting Widevine + HLS playback. WideVine is the one of the DRM schemes defined by Google..

HTY360Player

<https://github.com/hanton/HTY360Player>

Open Source iOS 360 Degree Panorama Video Player.

Tags: ios, video player, 360-degree, panorama, mobile video

HVD - Hardware Video Decoder

<https://github.com/bmegli/hardware-video-decoder>

HVD is a C library that provides a simple zero-copy interface to FFmpeg for hardware-accelerated video decoding. It supports multiple technologies and is designed for efficient video processing, making it valuable for developers working on performance-critical video applications.

Tags: C library, video decoder, hardware acceleration, FFmpeg, zero-copy

HWAcceleratedVP9Player

<https://github.com/rinsuki/HWAcceleratedVP9Player>

Hardware Accelerated VP9 Player in macOS 11.0 Big Sur beta 4+

Hardware Video Streaming

<https://github.com/bmegli/hardware-video-streaming>

Hardware Video Streaming is a meta repository encompassing several projects focused on hardware-accelerated video encoding and streaming. It includes components like HVE (Hardware Video Encoder), MLSP (Minimal Latency Streaming Protocol), NHVD (Network Hardware Video Decoder), and NHVE (Network Hardware Video Encoder). These tools facilitate efficient video streaming with low latency, leveraging hardware acceleration for improved performance.

Tags: hardware video streaming, video encoding, low latency, hardware acceleration, open-source

HbbPlayer

<https://github.com/Samsung/HbbPlayer>

HbbPlayer is HbbTV application which can playback media by using url of media as parameter. It conforms to W3C and HbbTV specification.

HybridMediaPlayer

<https://github.com/mkaflowski/HybridMediaPlayer>

Android music and video player. Uses ExoPlayer 2 and MediaPlayer for lower APIs and makes using ExoMediaPlayer easier. If you need advanced options such as handling Chromecast it is delivered by Ex...

IINA

<https://iina.io/>

A modern, open source macOS media player built on mpv. IINA features a sleek UI, gesture support, and wide format compatibility.

Tags: media player, macOS, open source, mpv, video playback

IVB - Interactive Video Builder

<https://github.com/kleftodimos/IVB-An-interactive-video-builder-that-supports-learning-analytics>

IVB is an open-source interactive video builder that supports learning analytics. It allows educators and developers to create interactive video-based learning environments using open-source technologies. IVB integrates with MediaElement, an open-source HTML5 video player, to provide a seamless video playback experience.

Tags: interactive video, learning analytics, open-source, education, HTML5

Iina (GitHub)

<https://github.com/iina/iina>

The modern video player for macOS.

Tags: macOS, video player, desktop application, media playback

Indigo Player

<https://github.com/matvp91/indigo-player>

Highly extensible, modern, JavaScript video player. Handles MPEG-Dash / HLS / MPEG-4 and is built on top of the HTML5 video element.

Tags: javascript, html5, video player, streaming, MPEG-DASH, HLS

Inputstream Adaptive

<https://github.com/peak3d/inputstream.adaptive>

kodi inputstream addon for several manifest types.

Invidious

<https://github.com/iv-org/invidious>

An open source, alternative front-end for YouTube that is privacy-respecting. It streams YouTube content without ads or tracking, and supports lightweight instances.

JRiver Media Center

<https://jriver.com/>

A comprehensive media center application with high-quality video and audio playback, library management, and home theater features.

Tags: media player, home theater, library management, video playback, audio playback

JW Player (Free Version)

<https://www.jwplayer.com/>

A popular embeddable web video player with a free tier, originally an open source project, offering HLS and DASH support and extensive customization.

Tags: web player, HTML5, streaming, HLS, DASH, embeddable

JW Player Mobile SDKs

<https://jwplayer.com/mobile-sdk/>

JW Player provides native SDKs for iOS and Android, enabling developers to build and launch cross-platform video experiences. These SDKs offer extensive ad insertion capabilities, support for analytics, and are regularly updated for platform compatibility. They also include a React Native Player SDK for cross-platform development.

Tags: video player, SDK, mobile, React Native, ad insertion

Jasmine Roku

<https://github.com/veeta-tv/jasmine-roku>

Example jasmine tests using node-roku-test for verifying Roku channel behavior

JetAudio

<https://www.jetaudio.com/>

An integrated multimedia player for Windows capable of playing video and audio with an array of sound effects and visualization features.

Tags: desktop player, multimedia, audio player, video player, Windows

Jitsi Meet

<https://jitsi.org/jitsi-meet/>

An open source video conferencing platform that you can self-host. Jitsi Meet supports multi-user video calls with screen sharing, encryption, and no account needed.

Tags: video conferencing, web player, open source, multi-user, screen sharing

KMPlayer

<https://www.kmplayer.com/>

A versatile media player for Windows and mobile that supports various video formats and provides advanced playback controls.

Tags: media player, video playback, cross-platform, multimedia, windows, mobile

Kaltura HTML5 Video Player

<https://player.kaltura.com/>

Kaltura Player is a free-to-use, open-source HTML5 video player that allows for the creation of custom inter-browser and inter-device skins to match your website's design. It offers robust performance, multi-platform support, and integrates with various video ad networks, supporting most ad formats including VAST v. 3.0.

Tags: HTML5, video player, open-source, customizable, advertising

Kaltura Mobile Video Player SDKs

<https://github.com/kaltura/DeveloperPortalDocs/blob/master/documentation/Mobile-Video-Player-SDKs/IntroductionV2.md>

Kaltura offers Mobile Video Player SDKs for iOS and Android, enabling developers to embed the Kaltura Video Player into native applications. These SDKs support features like DRM, IMA (DFP), Chromecast, AirPlay, offline mode (including DRM), and Picture-in-Picture (PIP). The architecture allows seamless integration across platforms, providing a consistent user experience.

Tags: Kaltura, mobile SDK, video player, iOS, Android

Kaltura Mobile Video SDK

https://site.kaltura.com/Kaltura_Mobile_Video_SDK_Page.html

Kaltura provides mobile video SDKs and reference applications for iOS and Android, enabling seamless, high-quality video delivery across mobile devices. These SDKs support live and on-demand video streaming, enhancing mobile video experiences.

Tags: video SDK, mobile video, iOS, Android, live streaming

Kaltura Player SDK for Android

<https://developer.kaltura.com/player/android/getting-started-android/>

Kaltura's Player SDK for Android provides a comprehensive solution for integrating video playback into Android applications. It supports various streaming protocols and offers features like DRM support, analytics, and customization options to enhance the video playback experience.

Tags: Android, Video Player, Streaming, DRM, Customization

Kodi

<https://kodi.tv/>

A free and open source media center software that plays videos, music, and more across multiple platforms (formerly XBMC).

Tags: media player, cross-platform, open source, video playback, media center

Kodi Add-ons (PVR IPTV Simple Client)

<https://kodi.tv/addons>

An example official Kodi add-on that allows playing live TV streams via IPTV. Kodi's rich add-on ecosystem includes clients for various streaming protocols, enabling Kodi to serve as a full-fledged PVR.

LFLiveKit

<https://github.com/LaiFengiOS/LFLiveKit>

An open source iOS live streaming framework that supports RTMP broadcasting with H.264 (GPU) and AAC encoding.

Larix Broadcaster SDK for Android

https://softvelum.com/larix/android_sdk/

Larix Broadcaster Android SDK provides source code and binary libraries to help developers create applications with live streaming capabilities. It includes sample applications demonstrating features like screen capture, background streaming, and video encoding, facilitating integration of streaming functionalities into Android apps.

Tags: Android, Live Streaming, Video Encoding, Screen Capture, SDK

Larix Player SDK for Android

https://softvelum.com/player/android_sdk/

Softvelum's Larix Player SDK for Android provides a rich set of protocols to play live streaming content on mobile devices and Android TV STBs. It supports SRT, SLDP, RTMP/RTMPS, Icecast, HLS, and MPEG-DASH protocols, offering a comprehensive solution for streaming playback.

Tags: video player SDK, Android, live streaming, protocol support, developer tools

Larix Player SDKs

<https://softvelum.com/mobile/>

Softvelum's Larix Player SDKs enable playback of various streaming protocols including SRT, SLDP, RTMP, Icecast, HLS, and MPEG-DASH on iOS and Android devices. These SDKs support features like low latency and adaptive bitrate streaming, providing developers with tools to build efficient streaming applications.

Tags: video player SDK, low latency, adaptive bitrate, SRT, HLS

Libvlc Go

<https://github.com/adrg/libvlc-go>

Go bindings for libVLC and high-level media player interface.

Libvlcsharp

<https://github.com/videolan/LibVLCSharp>

Cross-platform.NET/Mono bindings for LibVLC

Light Alloy

<https://light-alloy.com/>

A lightweight, freeware media player for Windows known for its fast startup and support of all common video formats with configurable keyboard shortcuts.

Tags: media player, windows, video playback, freeware

Lightspark

<https://github.com/lightspark/lightspark>

An open source flash player implementation.

Tags: flash, player, open source, video playback

MHVideoPhotoGallery

<https://github.com/mariohahn/MHVideoPhotoGallery>

An iOS Photo and Video Gallery component that supports playing videos and viewing photos in a gallery interface.

MPV Media Player

<https://www.open-source-alternative.to/project/mpv>

A free, open-source, and cross-platform media player designed for the command line. MPV supports a wide variety of media file formats, audio and video codecs, and subtitle types. It offers powerful scripting capabilities for extensive customization, high-quality video output with support for OpenGL, Vulkan, and D3D11, and GPU video decoding across platforms.

Tags: media player, open-source, cross-platform, video playback, customization

MPlayer

<http://www.mplayerhq.hu/>

One of the original open source media players for UNIX-like systems, known for its wide codec support and command-line interface. MPlayer can play virtually any format and was the basis for mpv and many other players.

Tags: open-source, media-player, unix, codec-support, command-line

Media Element Syncer

<https://github.com/tjenkinson/media-element-syncer>

Synchronise two or more HTML5 media elements.

Media Player Classic - Black Edition (MPC-BE)

<https://sourceforge.net/projects/mpcbe/>

A community fork of MPC-HC with additional features and improvements, offering enhanced audio/video playback and a dark-themed interface.

Tags: desktop player, media player, video playback, open source, community fork

Media Player Classic - Home Cinema (MPC-HC)

<https://github.com/clsid2/mpc-hc>

A lightweight, open source Windows media player modeled after the classic Windows Media Player 6.4. It supports a wide range of formats with GPU acceleration and subtitle support.

MediaPortal

<https://www.team-mediaportal.com/>

An open source media center for Windows that turns a PC into a complete home theater hub for playing TV, video, and music.

Tags: media center, home theater, windows, video player, media playback

Mediaelement

<https://github.com/mediaelement/mediaelement>

HTML5 <audio> or <video> player with support for MP4, WebM, and MP3 as well as HLS, Dash, YouTube, Facebook, SoundCloud and others with a common HTML5 MediaElement API,...

MiSTer FPGA

https://github.com/MiSTer-devel/Main_MiSTer

MiSTer FPGA is an open-source project that aims to recreate various classic computers, game consoles, and arcade machines using modern FPGA-based hardware. It allows software and video game images to run as they would on original hardware, using peripherals such as mice, keyboards, joysticks, and other game controllers. MiSTer provides a versatile platform for enthusiasts interested in retro computing and gaming.

Tags: FPGA, retro gaming, open source, hardware emulation, classic computers

Milkymist One and Flickernoise

<https://en.wikipedia.org/wiki/M-Labs>

An open-source video synthesizer and reconfigurable computer designed for live performances. Milkymist One features multi-standard video input, DMX512 ports, MIDI support, and SVGA output. It runs Flickernoise, a video synthesis software inspired by MilkDrop, allowing interactive visual effects programming.

Tags: video synthesizer, open-source, live performance, MIDI, DMX512

MobilePlayer

<https://github.com/mobileplayer/mobileplayer-ios>

A powerful and completely customizable media player for iOS, with playback controls and easy integration.

ModernAVPlayer

<https://github.com/noreasonprojects/ModernAVPlayer>

ModernAVPlayer is a persistence AVPlayer wrapper.

Tags: AVPlayer, iOS, video playback, persistence, mobile player

Mpv (GitHub)

<https://github.com/mpv-player/mpv>

Lightweight command-line video player supporting wide format compatibility and advanced playback features.

Tags: video player, command-line, open source, media playback

Muvi Flex Mobile SDKs

<https://www.muvi.com/flex/mobile-sdk/>

Muvi Flex offers Mobile SDKs for Android and iOS, providing features like DRM and non-DRM content streaming, subtitle support, multiple video resolutions, customizable subtitle fonts, colors, and sizes, watermarking, adaptive bitrate streaming, and minimal coding requirements. These SDKs enable developers to deliver seamless video playback experiences in their mobile applications.

Tags: Android, iOS, video player, streaming, SDK

Muvi Live SDK

<https://www.muvi.com/live/features/live-sdk/>

Muvi Live SDK enables developers to add live streaming capabilities to existing iOS and Android applications with minimal coding. It offers features like adaptive bitrate streaming, DRM support, and customizable subtitles, providing a comprehensive solution for seamless live video integration.

Tags: live streaming, video SDK, iOS, Android, adaptive bitrate

Muvi Live Streaming SDK

<https://www.muvi.com/live/>

Muvi offers a live streaming SDK that allows developers to build live streaming applications for web and mobile platforms. It supports HLS and RTMP feeds, provides options for recording live streams, and includes security features like screen recording protection and geo-blocking.

Tags: live streaming, SDK, HLS, RTMP, security

Muvi One SDK

<https://www.muvi.com/one/features/software-development-kit/>

Muvi One SDKs enable developers to integrate video streaming capabilities into applications across various platforms and devices, offering features like live streaming, video-on-demand, monetization options, and analytics.

Tags: Video Streaming, SDK, Live Streaming, VOD, Analytics

Muvi Player SDK

<https://www.muvi.com/player/android-sdk/>

Muvi Player SDK provides a unified solution for seamless video playback across Android devices. It addresses device fragmentation issues and offers features like adaptive bitrate streaming, DRM support, subtitle customization, and watermarking, ensuring a consistent and high-quality streaming experience.

Tags: video player, Android SDK, adaptive bitrate, DRM, subtitle support

Muvi Video Player SDK

<https://www.muvi.com/player/>

Muvi offers a powerful online video player SDK designed for seamless integration into mobile applications. It supports adaptive streaming technologies like HLS and DASH, cross-platform compatibility, detailed analytics, and monetization capabilities. The SDK is customizable to align with brand design guidelines.

Tags: video player, SDK, adaptive streaming, cross-platform, monetization

Mux Live Streaming API

<https://www.mux.com/live>

Mux provides a developer-friendly live streaming API that allows for easy integration of live streaming capabilities into mobile and web applications. It supports adaptive bitrate streaming, real-time video performance analytics, and scalable infrastructure, ensuring high-quality video delivery.

Tags: live streaming, API, adaptive bitrate, video analytics, scalable infrastructure

Mux Video API

<https://www.mux.com/>

Mux offers a developer-friendly streaming infrastructure with APIs for live and on-demand video. It provides adaptive bitrate streaming, real-time video performance analytics, and scalable infrastructure, suitable for integrating video streaming into mobile applications.

Tags: video streaming, API, adaptive bitrate, analytics, mobile

MythTV

<https://www.mythtv.org/>

An open source DVR and media center that lets you capture broadcast TV, schedule recordings, and play them back on various devices.

Tags: DVR, media center, playback, open source, recording

Network Benchmark

<https://github.com/exegersha/network-benchmark>

Proof of concept. Roku app implementing network layer using scene graph nodes.

Tags: roku, network, scene graph, proof of concept, app development

NymphCast

<https://github.com/MayaPosch/NymphCast>

An open source audio and video casting system (similar to Chromecast) that allows streaming media and running custom applications on a receiver device.

OMXPlayer

<https://github.com/popcornmix/omxplayer>

A hardware-accelerated command-line video player for the Raspberry Pi that utilizes the Pi's GPU decoding for smooth playback.

OTT Multiview

<https://github.com/Eyevinn/ott-multiview>

This is a web based multiview screen for HLS and MPEG-DASH streams based on hls.js and Shaka Player.

Open Source Embedded Video Player

<https://av.tib.eu/media/42094>

A project that demonstrates how to build a video playback system for embedded devices using solely open-source software. It utilizes hardware acceleration on i.MX6 platforms with components like GStreamer, Qt QML, the etnaviv GPU driver, and the coda video decoder driver. This project is valuable for developers interested in creating video playback systems for embedded devices without relying on proprietary drivers.

Tags: embedded systems, video playback, open-source, hardware acceleration, i.MX6

OpenMAX

<https://www.khronos.org/openmax/>

A non-proprietary, cross-platform set of C-language programming interfaces that provide abstractions for routines useful in processing audio, video, and still images. OpenMAX is designed for low power and embedded system devices, facilitating efficient multimedia data processing. It offers three layers of interfaces: application layer (AL), integration layer (IL), and development layer (DL), catering to various aspects of media processing.

Tags: API, multimedia, cross-platform, embedded systems, Khronos Group

OpenPlayer.js

<https://github.com/openplayerjs/openplayerjs>

OpenPlayer.js is a lightweight HTML5 video and audio player written in TypeScript. It offers smooth controls and the capability to play VAST, VPAID, and VMAP ads. The player supports modern browsers, provides a responsive design by default, and allows for easy customization, including adding custom buttons and controls.

Tags: HTML5, video player, audio player, ads, TypeScript

Panopto Certified Video Capture Appliances

<https://www.panopto.com/features/video-recording/video-capture-hardware/>

Panopto offers certified video capture appliances that seamlessly integrate with their video platform, providing a comprehensive solution for recording and streaming. These appliances are designed for high-quality video capture and are suitable for various applications, including education and corporate communications. Panopto's hardware solutions ensure reliability and ease of use for content creators.

Tags: video capture, streaming, certified appliances, education, corporate communications

Periscope VideoViewController

<https://github.com/gontovnik/Periscope-VideoViewController>

An iOS video view controller that mimics Periscope's fast rewind control for quick scrubbing through video playback.

Phils Players

<https://github.com/GeneticGenesis/phils-players>

A collection of video players with vaguely simple GUIs for video engineers.

Player

<https://github.com/piemonte/Player>

⌚ video player in Swift, simple way to play and stream media on iOS/tvOS

Plyr

<https://plyr.io/>

Plyr is a simple, lightweight, and accessible HTML5 media player that supports video, audio, and even YouTube and Vimeo embeds. It offers a clean and customizable interface, making it easy to integrate into various projects.

Tags: HTML5, media player, JavaScript, accessible, customizable

Plyr (GitHub)

<https://github.com/sampotts/plyr>

A simple HTML5, YouTube and Vimeo player.

Popcorn.js

<https://popcornjs.org/>

Popcorn.js is an open-source JavaScript library for HTML5 media developers. It uses native HTMLMediaElement properties, methods, and events, normalizing them into an API and providing a plugin system. Popcorn.js supports playing non-native media through wrappers and is part of a Mozilla program to promote web video creation via open standards.

Tags: HTML5, JavaScript, media library, open-source, plugins

Projekktor

<https://github.com/frankyghost/projekktor>

An open source HTML5 video player from the early 2010s. Projekktor provided a unified player with plugin support to handle both HTML5 and Flash fallback in an era of transition. While development has slowed, it remains a piece of the open source player history.

Tags: html5, video player, open source, javascript, web technology

PryntTrimmerView

<https://github.com/HHK1/PryntTrimmerView>

An iOS UI component that provides a video trimming interface, allowing users to select and cut segments of a video with a visual timeline.

RAF4RSG Sample

<https://github.com/rokudev/RAF4RSG-sample>

sample demonstrating the Roku Advertising Framework in SceneGraph

RTMP Video Player For iOS

<https://github.com/xiewei-wayne/rtmp-video-player-for-ios>

Based on FFEngine framework, a rtmp video player for apple iOS devices.

React Native Media Console

<https://github.com/lunatiquecoder/react-native-media-console>

React Native Media Console is a video player component for React Native, built with TypeScript. It provides customizable controls and supports various video formats, offering a flexible solution for integrating media playback into React Native applications.

Tags: React Native, video player, TypeScript, media playback, customizable controls

React Native VLC Media Player

<https://github.com/razorrun/react-native-vlc-media-player>

React Native VLC Media Player is a media player component for React Native applications, supporting video streaming and playback. It supports protocols like RTSP and RTMP, leveraging VLC's capabilities to provide a robust media player for both Android and iOS platforms.

Tags: React Native, VLC, media player, video streaming, cross-platform

React Native Video

<https://github.com/react-native-video/react-native-video>

A React Native component for video playback that supports both iOS and Android using native player APIs.

ReactPlayer

<https://github.com/cookpete/react-player>

A React component for playing a variety of URLs, including YouTube, Vimeo, SoundCloud, and many more.

Red5 Pro Mobile Live Video Streaming SDK

<https://www.red5.net/blog/mobile-live-video-streaming-sdk/>

Red5 Pro offers mobile SDKs for iOS and Android that enable developers to build live video streaming applications. These SDKs provide tools for real-time streaming, low-latency broadcasting, and adaptive bitrate streaming, facilitating the creation of interactive live video experiences on mobile devices.

Tags: Red5 Pro, Live Streaming, Mobile SDK, iOS, Android

Red5 Pro Mobile SDKs

<https://www.red5.net/live-video-streaming-native-sdks/>

Red5 Pro provides live video streaming SDKs for native applications across multiple platforms, including Android, iOS, Windows, macOS, and Linux. These SDKs are optimized to increase server capacity and reduce hosting costs, offering out-of-the-box functionality for live streaming.

Tags: live streaming, SDK, cross-platform, native applications

Redoku

<https://github.com/briandunnington/Redoku>

Roku application or development tool by Brian Dunnington for video streaming on Roku devices.

Tags: roku, development, redux, streaming, hardware-player

Replay

<https://github.com/vimond/replay>

A React video player facilitating adaptive stream playback with custom UI and a React-friendly API.

Tags: react, video player, adaptive streaming, web, javascript

Rise Vision Media Player Hardware as a Service

<https://www.risevision.com/rise-vision-media-player-hardware-as-a-service>

Rise Vision offers a Media Player Hardware as a Service solution that simplifies digital signage by providing managed hardware with regular upgrades and support. This service ensures high performance for demanding schedules and reduces upfront costs by combining software and hardware into scalable subscriptions. It's designed for organizations seeking a hassle-free digital signage experience with predictable budgeting.

Tags: digital signage, media player, hardware as a service, managed service, subscription

Roact

<https://github.com/briandunnington/Roact>

Roku development library by Brian Dunnington for building video streaming channels and applications.

Tags: roku, framework, development, javascript, streaming

Roku Framework

<https://github.com/karimkawambwa/roku-framework>

Roku app framework to make app creation easier and structured. Under construction

Tags: roku, framework, app-development, streaming-platform

Roku Framework Example

<https://github.com/karimkawambwa/roku-framework-example>

This is a project to show how the roku-framework by Karim Kawambwa is used

Tags: roku, framework, tutorial, hardware-player, example

Roku Lib

<https://github.com/dphang/roku-lib>

Collection of utility functions and helpers for Roku channel development and video streaming apps.

Tags: roku, hardware, utilities, streaming, sdk

Roku Libs

<https://github.com/juliomalves/roku-libs>

BrightScript Utility Libraries.

Tags: roku, brightscript, utility, library, hardware-player

Roku Suite Desktop

<https://github.com/CCecilia/roku-suite-desktop>

Tool suite for Roku channel development.

Tags: roku, channel development, streaming, desktop tool, sdk

RokuFramework

<https://github.com/mrkjfrsn/RokuFramework>

Open source framework for developing Roku channels with reusable components for video streaming applications.

Tags: roku, framework, streaming, sdk, open-source

RokuJSONHelperNode

<https://github.com/T-Pham/RokuJSONHelperNode>

SceneGraph component library for parsing and handling JSON data in Roku channel development.

Rooibos

<https://github.com/georgejecook/rooibos>

simple, flexible, fun brightscript test framework for roku scenegraph apps

Tags: roku, testing, framework, brightscript, scenegraph

Ruffle

<https://github.com/ruffle-rs/ruffle>

A Flash Player emulator written in Rust.

SGPlayer

<https://github.com/libobjc/SGPlayer>

A powerful media play framework for iOS, macOS, and tvOS.

Tags: media player, iOS, tvOS, framework, mobile video

SJVideoPlayer

<https://github.com/changsanjiang/SJVideoPlayer>

An open source media player library for iOS that provides a customizable player interface. It supports multiple backends (AVPlayer, IJKPlayer, etc.) and features like gesture controls and picture-in-picture.

SMPlayer

<https://smplayer.sourceforge.io/>

A free, open source media player for Windows and Linux with built-in codecs. SMPlayer provides a graphical front-end for MPlayer and MPV, including features like subtitles and playback speed control.

Tags: media player, open source, desktop application, video playback

SMTube

<https://github.com/smplayer-dev/smtube>

A lightweight YouTube browser that allows playing YouTube videos in external media players like SMPlayer or mpv, bypassing the web interface.

SUCacheLoader

<https://github.com/DaMingShen/SUCacheLoader>

iOS library for caching video content during streaming to reduce bandwidth and improve playback performance.

Tags: AVPlayer, iOS, video playback, mobile player

Samples

<https://github.com/rokudev/samples>

Collection of sample channels for side-loading on your Roku device

SceneGraphDeveloperExtensions

<https://github.com/rokudev/SceneGraphDeveloperExtensions>

Contribute to rokudev/SceneGraphDeveloperExtensions development by creating an account on GitHub.

SegmentIO Brightscript

<https://github.com/gabek/SegmentIO-Brightscript>

A BrightScript interface to SegmentIO event tracking

Tags: roku, brightscript, tracking, sdk, analytics

Sendbird Live

<https://www.tribuneindia.com/news/impact-feature/top-10-live-streaming-apis-sdks-for-mobile-and-web-apps-590709>

Sendbird Live offers APIs and SDKs for building engaging live streaming experiences in web and mobile applications. It provides features like real-time chat, interactive live video, and scalable infrastructure, enabling developers to create interactive live streaming platforms with ease.

Tags: live streaming, real-time chat, interactive video, API, SDK

Sendbird Live SDKs

<https://sendbird.com/products/live-streaming/sdk>

Sendbird offers live video SDKs for iOS, Android, and JavaScript, enabling developers to host custom private video streaming events within their applications. These SDKs support features like live video streaming, public chat, and moderation, providing a comprehensive solution for engaging and monetizing audiences directly within your app.

Tags: live streaming, video SDK, iOS, Android, JavaScript

Shaka Player

<https://github.com/shaka-project/shaka-player>

An open source JavaScript library for adaptive video streaming, providing a DASH and HLS player with Widevine DRM support for HTML5 browsers.

Tags: javascript, html5, adaptive streaming, DASH, HLS, DRM, Widevine

Shaka Player Embedded

<https://github.com/shaka-project/shaka-player-embedded>

Shaka Player Embedded is a C++ framework that integrates Shaka Player, Google's open-source DASH and HLS player, into native applications. It enables developers to incorporate adaptive streaming capabilities into their applications, supporting various streaming protocols and DRM systems.

Tags: C++, adaptive streaming, DASH, HLS, DRM

Shaka Player Embedded (google)

<https://github.com/google/shaka-player-embedded>

Shaka Player in a C++ Framework.

Srgmediaplayer Apple

<https://github.com/SGSSR/srgmediaplayer-apple>

An advanced media player library, simple and reliable

Stream Video & Audio SDKs

<https://getstream.io/video/sdk/>

Stream provides Video & Audio SDKs that enable developers to implement in-app calling and streaming experiences. Supporting platforms like React, iOS, Android, Flutter, and more, these SDKs facilitate the creation of applications similar to Zoom, Messenger, or Twitch.

Tags: video SDK, audio SDK, in-app calling, live streaming, cross-platform

Stream Video Android SDK

<https://github.com/GetStream/stream-video-android>

Stream Video Android SDK provides versatile Core and Compose UI component libraries that enable developers to build video calling, audio room, and live streaming applications based on WebRTC, running on Stream's global edge network. It includes both low-level video SDKs and reusable UI components, facilitating rapid development of video features in Android apps.

Tags: Android, video SDK, WebRTC, live streaming, UI components

Streamaxia OpenSDK

<https://www.streamaxia.com/>

Streamaxia OpenSDK is a live video streaming library for iOS and Android, enabling developers to add HD and low-latency live streaming capabilities to their apps. It supports broadcasting to any RTMP server or CDN, offering flexibility and ease of integration for live video features.

Tags: live streaming, video SDK, iOS, Android, RTMP

Streamlab

<https://vimond.github.io/streamlab/>

Multi-format stream test tool for the browser. Brings convenience to verifying, inspecting, and troubleshooting adaptive streams and video files. A common interface on top of Shaka Player, HLS.js, and Rx-Player.

Swift-YouTube-Player

<https://github.com/gilesvangruisen/Swift-YouTube-Player>

A Swift library for embedding and controlling YouTube videos in iOS applications via a WKWebView.

TVVLCPlayer

<https://github.com/kodlian/TVVLCPlayer>

TVVLCPlayer lets you integrate easily a powerful video player with playback control views to your tvOS apps.

Tags: tvOS, video player, Apple, mobile player, VLC

Tal

<https://github.com/bbc/tal>

BBC's TV Application Layer framework for building smart TV apps. Cross-platform TV development toolkit.

TimPlayer

<https://github.com/timchunght/timplayerdemo>

TimPlayer is an open-source HTML5 video player that serves as a beautiful alternative to platforms like Vimeo, JWPlayer, VideoJS, YouTube, and Wistia. It is designed to be easy to integrate and customize, offering a sleek user interface and support for various video formats.

Tags: HTML5, video player, open-source, customizable, alternative

URLSessionHLSDownload

<https://github.com/dhairav/URLSessionHLSDownload>

A swift 3 implementation for downloading HLS content and play it back using native AVPlayer

Tags: HLS, iOS, AVPlayer, URLSession, streaming, download

Ukor

<https://github.com/willowtreeapps/ukor>

A Roku build tool with support for build flavors.

Unit Testing Framework

<https://github.com/rokudev/unit-testing-framework>

Tool for automating and testing Roku channels.

VGPlayer

<https://github.com/VeinGuo/VGPlayer>

A simple iOS video player by Vein.

VIMediaCache

<https://github.com/vitoziv/VIMediaCache>

Cache media file while play media using AVPlayer.

VLC

<https://github.com/videolan/vlc>

VLC media player - All pull requests are ignored, please follow https://wiki.videolan.org/Sending_Patches_VLC/

VLC Android

<https://code.videolan.org/videolan/vlc-android>

VLC for Android, Android TV and ChromeOS.

VLC for Android

<https://github.com/videolan/vlc-android>

VLC for Android is the official port of the VLC media player to the Android platform. It supports a wide range of multimedia formats and streaming protocols, offering a versatile solution for media playback on Android devices.

Tags: Android, Media Player, Streaming, Multimedia, Open Source

VLC iOS

<https://code.videolan.org/videolan/vlc-ios>

VLC for iOS is the official port of VLC on the iOS/tvOS platforms.

Tags: mobile, player, ios, tvos, open-source

VTM Decoder VLC Plugin

https://github.com/InterDigitalInc/VTMDecoder_VLCPlugin

VVC (H.266) video decoder plugin for VLC Media Player based on a multithreaded VTM decoder implementation. Includes TS demux plugin with VVC stream format support. Enables VVC playback in VLC with configurable threading and hurry-up mode.

Vapoursynth (GitHub)

<https://github.com/vapoursynth/vapoursynth>

A video processing framework with simplicity in mind

VersaPlayer

<https://github.com/josejuanqm/VersaPlayer>

A versatile video player for iOS, macOS, and tvOS, built on AVPlayer with support for customization and plugins.

Viaccess-Orca VO Player

<https://dashif.org/tools/clients/>

Viaccess-Orca's VO Player is a secure video player offering a unified experience across multiple platforms, including Android, iOS, PC, Mac, STBs, and smart TVs. It supports various streaming protocols like DASH, CMAF, HLS, and MSS, and includes features such as multi-language audio, advertising integration, and offline viewing.

Tags: Viaccess-Orca, VO Player, Secure Video Player, Cross-platform, Streaming

Video Acceleration API (VA-API)

https://en.wikipedia.org/wiki/Video_Acceleration_API

An open-source API that allows applications to utilize hardware video acceleration capabilities, typically provided by the GPU. VA-API supports video decoding, encoding, subpicture blending, and rendering, making it a valuable resource for developers aiming to optimize video processing performance.

Tags: API, video acceleration, hardware, GPU, open-source

Video SDK

<https://www.videosdk.live/>

Video SDK provides real-time video, chat, and audio calling capabilities for web and mobile apps. It offers pre-built integrations and customization options, supporting platforms like JavaScript, React JS, React Native, Android, Flutter, and iOS.

Tags: video calling, SDK, real-time communication, mobile, web

Video SDK Live

<https://github.com/videosdk-live>

Video SDK Live offers tools to help companies create collaborative products with capabilities like live audio/video, cloud recordings, RTMP/HLS streaming, and interaction APIs. Their GitHub repository provides various SDKs and sample projects for platforms including Android, iOS, React, and Flutter.

Tags: Video SDK, Live Streaming, Video Conferencing, Cross-Platform, WebRTC

Video Speed Controller (Chrome Extension)

<https://chrome.google.com/webstore/detail/video-speed-controller/nffaoalbilbmmfgebngppjihopabppdk>

A popular browser extension that allows users to speed up or slow down video playback on any website with HTML5 video.

Video.js

<https://github.com/videojs/video.js/>

Video.js is a full-featured, open-source HTML5 video player that supports all common media formats, including streaming formats like HLS and DASH. It is compatible across desktops, mobile devices, tablets, and web-based Smart TVs. Video.js offers a robust ecosystem of plugins for analytics, advertising, and more, making it highly extensible and customizable.

Tags: HTML5, video player, open-source, streaming, plugins

Video.js Mobile UI

<https://github.com/mister-ben/videojs-mobile-ui>

Video.js Mobile UI is a plugin designed to enhance the mobile user interface of the Video.js player. It provides touch-friendly controls and an improved user experience for mobile devices, ensuring seamless video playback and interaction on smartphones and tablets.

Tags: video.js, mobile UI, video player, plugin, touch controls

VideoPlayView

<https://github.com/MarcinMoskala/VideoPlayView>

A custom Android view that integrates a video player with a loader and placeholder image, enabling quick embedding of video playback in Android apps.

Tags: mobile, android, video-player, custom-view, playback

Videogular

<https://github.com/videogular/videogular>

The HTML5 video player for AngularJS.

Videoplayback iOS

<https://github.com/ustwo/videoplayback-ios>

Swift AVPlayer wrapper using the VIPER architecture. Currently a work in progress

Videoplayer Channel

<https://github.com/rokudev/videoplayer-channel>

SceneGraph version of the SDK1 VideoPlayer Channel

Videoplayer by JingOS

<https://github.com/JingOS-team/videoplayer>

Videoplayer is an open-source video player built with Qt/QML on top of libmpv, developed by the JingOS team. It offers features like double-finger sliding to adjust volume and brightness, automatic playback suspension when switching to the background, and a brand-new UI/UE with JingOS-style design. The player supports keyboard, touchpad, mouse, and screen touch inputs, providing a versatile user experience.

Tags: video player, open-source, Qt, libmpv, JingOS

Videotel Digital VP92+ 4K Digital Signage Media Player

<https://www.videotel.digital.com/products/new-vp92-4k-digital-signage-media-player-network-wifi-capable-access-content-remotely-with-sftp-lan-or-cloud-software-interactive-capabilities-copy>

The VP92+ is a 4K/1080 industrial-grade network digital signage media player capable of playing content locally from USB/SD or remotely via embedded software, SFTP, LAN, or optional CMS subscription-based software. It supports dual HDMI outputs, interactive capabilities, and various content sources, making it suitable for robust digital signage setups.

Tags: digital signage, 4K, network media player, interactive, dual HDMI

Vimeo PlayerKit

<https://github.com/vimeo/PlayerKit>

An open source iOS video player framework by Vimeo for easily integrating Vimeo videos and playback functionalities into apps.

Vonage Video API (formerly TokBox OpenTok)

<https://www.vonage.com/communications-apis/video/>

Vonage Video API allows developers to create custom video experiences on any device, including mobile, web, or desktop. Built on WebRTC industry standards, it supports features like group chat, video stream customization, and encrypted recordings.

Tags: video API, WebRTC, SDK, mobile, Vonage

Vscode Brightscript Language

<https://github.com/rokucommunity/vscode-brightscript-language>

A Visual Studio Code extension for Roku's BrightScript language

XOS Media Player

<https://labs.acmi.net.au/our-open-source-media-player-for-displaying-fleets-of-video-8d518c0c97dc>

XOS Media Player is an open-source media player designed for displaying fleets of video. It is a Python wrapper around VLC that fetches content and reports playback status to a message broker over the network. Delivered as a Docker image, it is designed to be deployed on a fleet of Raspberry Pi 4s and x86 machines, ensuring scalability and robustness.

Tags: media player, open-source, VLC, Docker, Raspberry Pi

Xine

<https://xine.sourceforge.net/>

An open source multimedia playback engine and player for Unix-like systems, featuring support for CD/DVD playback and a wide range of codecs.

Tags: open-source, multimedia, player, unix, codec-support, playback-engine

ZFPlayer

<https://github.com/renzifeng/ZFPlayer>

Support customization of any player SDK and control layer

Ziggeo Mobile SDKs

<https://ziggeo.com>

Ziggeo offers cloud-based video technology with mobile SDKs for iOS and Android, providing asynchronous video APIs and tools to deliver enterprise-grade WebRTC capabilities. These SDKs enable developers to integrate video recording and

playback functionalities into mobile applications.

Tags: video SDK, WebRTC, iOS, Android, video recording

Zype Roku Scenegraph

<https://github.com/zype/zype-roku-scenegraph>

Contribute to zype/zype-roku-scenegraph development by creating an account on GitHub.

afterglow

<https://github.com/moay/afterglow>

An open source HTML5 video player that builds on top of Video.js, offering a simpler integration (just adding data attributes to video tags) and a default aesthetic out-of-the-box. afterglow aims to make including a responsive, skinnable video player on websites quick and easy.

Tags: html5, video player, javascript, web, responsive

api.video Live Streaming SDKs

<https://docs.api.video/sdks/livestream>

api.video offers live streaming SDKs for mobile platforms, including React Native, Flutter, Swift, and Android. These SDKs allow developers to integrate live streaming capabilities into their mobile applications, providing features like real-time video broadcasting and adaptive bitrate streaming.

Tags: live streaming, SDK, React Native, Flutter, Swift, Android

api.video Player SDKs

<https://api.video/catalog/player-sdk/>

api.video offers a suite of player SDKs designed for seamless integration of video playback into mobile applications. These SDKs support both VOD and live streaming, provide customizable UI elements, and handle adaptive bitrate streaming to ensure optimal playback quality across various network conditions. They are available for platforms including Flutter, React Native, Android, and iOS.

Tags: video player, SDK, mobile, VOD, live streaming

elements/packages/mux-player

<https://github.com/muxinc/elements/tree/main/packages/mux-player>

The official Mux-flavored video player web component. The player UI is built on Media Chrome, with the mux-video element driving the core playback logic for Mux streams.

iOS 360 Videos

<https://github.com/nytimes/ios-360-videos>

NYT360Video plays 360-degree video streamed from an AVPlayer on iOS.

ijkplayer

<https://github.com/bilibili/ijkplayer>

An open source cross-platform media player library based on FFmpeg. ijkplayer is widely used on Android and iOS for its stable performance and was originally developed by Bilibili.

jPlayer

<https://github.com/jplayer/jPlayer>

A jQuery-based media library that can be used to build web audio and video players. jPlayer abstracts the HTML5 <video> and <audio> elements and provides a consistent API, with a Flash fallback (for older browsers) and skinnable UI via CSS.

Tags: web, javascript, player, html5, jquery, media, audio, video

mpv

<https://mpv.io/>

mpv is a free, open-source, and cross-platform media player that supports a wide variety of media file formats, audio and video codecs, and subtitle types. Designed for the command line, it offers powerful scripting capabilities for extensive customization. mpv provides high-quality video output with support for OpenGL, Vulkan, and D3D11, and leverages hardware decoding APIs for efficient playback.

Tags: media player, open-source, cross-platform, command-line, customizable

mpv-android

<https://github.com/mpv-android/mpv-android>

An Android port of the mpv media player. It leverages mpv's core for high quality video playback on Android devices, supporting hardware decoding and gesture controls.

mpv.net

<https://github.com/stax76/mpv.net>

A modern Windows user interface for the mpv media player, adding a GUI and settings dialog to the powerful mpv core.

streamlit-WebRTC

<https://github.com/whitphx/streamlit-webrtc>

A Streamlit component that enables real-time video/audio streams in web apps. It allows building interactive computer vision or conferencing apps by integrating WebRTC with Streamlit.

switch media universal player

<https://www.switch.tv/mediahq/universal-player/>

Switch Media's customizable video player framework and UI components for web and mobile applications.

vPlayer

<https://circuitdigest.com/news/vplayer-an-opensource-video-player-that-more-than-just-a-screen>

vPlayer is an open-source, smartwatch-sized 1.69" touchscreen device powered by the ESP32-S3. It offers continuous video playback from an SD card, live camera monitoring, and IoT project capabilities. With WiFi, Bluetooth, USB-C, and an SD card slot, it's a versatile tool for various applications.

Tags: open-source, video player, ESP32, IoT, touchscreen

video-react

<https://github.com/video-react/video-react>

A web video player built for the HTML5 world using React.

Tags: react, html5, web, player, javascript

xgplayer

<https://github.com/bytedance/xgplayer/>

xgplayer is an HTML5 video player with a parser that saves traffic. It is designed to be lightweight and efficient, providing a smooth playback experience with support for various video formats.

Tags: HTML5, video player, lightweight, efficient, parser

µPlayer

<https://liliputing.com/hardware-accelerated-video-playback-now-possible-on-the-pinephon/>

µPlayer is a simple GTK4-based video player application that leverages the latest version of GStreamer to support hardware-accelerated video playback. It's available as a Flatpak and can be installed on devices like the Librem 5 and potentially other Linux phones, providing an efficient and user-friendly media playback experience.

Tags: video player, hardware acceleration, GTK4, GStreamer, Linux

µPlayer: Lightweight Video Player for Linux Phones

<https://source.puri.sm/Librem5/mplayer>

µPlayer is a lightweight video player designed for Linux-based smartphones like the PinePhone and Librem 5. It utilizes GStreamer for hardware-accelerated video playback, ensuring smooth performance on devices with limited resources. µPlayer features a simple GTK4 interface and supports a wide range of video formats.

Tags: video player, Linux phone, hardware acceleration, GStreamer, GTK4

Standards & Industry

AV1 Bitstream & Decoding Specification (1.0.0 with Errata 1)

<https://aomediacodec.github.io/av1-spec/av1-spec.pdf>

The official bitstream and decoding process specification for the AOMedia Video 1 (AV1) codec. Defines how AV1 video is encoded bit-by-bit and decoded.

Tags: codec, specification, AV1, technical standard, video encoding

AV1 Rtp Spec

<https://github.com/AOMediaCodec/av1-rtp-spec>

Current draft (HTML): <https://aomediacodec.github.io/av1-rtp-spec/>

AV1 Video Codec

<https://aomedia.org/av1/>

AV1 is an open-source, royalty-free video codec developed by the Alliance for Open Media, which includes companies like Amazon, Netflix, Google, Microsoft, Cisco, and Mozilla. AV1 is approximately 30% more efficient than HEVC and VP9, significantly reducing bandwidth and delivery costs. It is suitable for both low-latency streaming and 4K content, with growing adoption in both OTT and real-time applications.

Tags: AV1, video codec, open-source, AOMedia, compression

AVC/HEVC in ISOBMFF - ISO/IEC 14496-15

<https://www.iso.org/standard/83336.html>

Defines the carriage of AVC/HEVC in ISOBMFF

AVC/h264 Video Coding Standard - ISO/IEC 14496-10

<https://www.iso.org/standard/83529.html>

Defines the compressed video format for AVC/h264

AVIXA AV Standards

<https://www.avixa.org/standards/>

AVIXA is the leading resource for AV standards, providing standards that define processes and requirements to promote reliable performance of AV systems. These standards can increase competitive advantage by supporting technology design and procedures that focus on reliability, competency, and success.

Tags: AV, standards, performance, reliability, competency

AVstandards

<https://github.com/digitensions/AVstandards>

AVstandards is a collaborative collection of audiovisual file standards and technical documents. It serves as a valuable resource for professionals involved in audiovisual and digital preservation, offering access to various standards and supporting documents.

Tags: audiovisual, file standards, digital preservation, technical documents, collaborative

AWS Elemental Live

<https://aws.amazon.com/elemental-live/>

AWS Elemental Live is a live video encoder that processes real-time video streams for broadcast and streaming delivery. It offers adaptive video encoding and centralized management for live streaming solutions.

Tags: live streaming, video encoding, AWS, broadcast, adaptive bitrate

Academy Color Encoding System (ACES)

<https://acescentral.com/>

A standardized color management and interchange system for motion pictures and television. ACES defines comprehensive workflows for color grading and consistent color reproduction across devices.

Tags: color, color-management, standards, workflow, video-processing

Advanced Television Systems Committee (ATSC)

<https://www.atsc.org/>

ATSC is an international nonprofit organization that develops technical standards for digital terrestrial television and data broadcasting, including the ATSC 3.0 standard for next-generation TV.

Tags: television, digital, broadcasting, standards, ATSC 3.0

Advanced Video Coding (AVC) - H.264/MPEG-4 Part 10

https://en.wikipedia.org/wiki/Advanced_Video_Coding

An in-depth overview of the H.264/MPEG-4 Part 10 standard, also known as Advanced Video Coding (AVC). This resource provides detailed information on the features, design, and applications of AVC, which is widely used for video compression in various media formats.

Tags: H.264, AVC, video compression, MPEG-4, standards

Advanced Video Coding (AVC) Standard

<https://www.itu.int/rec/T-REC-H.264>

Also known as H.264 or MPEG-4 Part 10, AVC is a widely used video compression standard that supports high-definition video and is fundamental in various streaming and broadcasting applications. This resource provides comprehensive information on AVC.

Tags: AVC, H.264, video compression, MPEG-4

Apple Got It Wrong: Encoding Specs for HEVC in HLS

<https://www.streamingmedia.com/Articles/ReadArticle.aspx?ArticleID=121878>

Adding HEVC to your HLS streams is looking like a no brainer, but if you decide to do so, you may not want to take Apple's HEVC encoding recommendations verbatim. You'll deliver noticeably higher quality video if you follow the advice detailed below.

Audacity

<https://audacity.sourceforge.net/>

Audacity is an open-source audio editor available for all major platforms. It offers a range of features for recording and editing

audio, making it a valuable tool for developers working with audio components in video projects.

Tags: audio editing, open-source, cross-platform, recording, editing

Awesome Broadcasting

<https://github.com/ebu/awesome-broadcasting>

A curated list of amazingly awesome open source resources related to broadcast technologies

Awesome Video: A Curated List of Streaming Video Tools and Resources

<https://github.com/krzemienski/awesome-video>

Awesome Video is a curated list of streaming video tools, frameworks, libraries, and learning resources. It covers various aspects of video technology, including DASH, AV1, and video containers, providing valuable information for developers interested in video streaming and encoding.

Tags: video streaming, tools, frameworks, libraries, learning resources

Broadcasting Technologies

<https://github.com/DexxMedialabs/Broadcasting-Technologies->

A curated list of open-source resources related to broadcast technologies, including tools and libraries for video streaming, encoding, and processing. This repository serves as a comprehensive resource for developers looking to explore and implement various broadcasting standards and technologies.

Tags: broadcasting, video streaming, encoding, open source, resources

CMAF Specification (ISO/IEC 23000-19:2018)

<https://www.iso.org/standard/71975.html>

The Common Media Application Format (CMAF) standard defines a container format (based on fragmented MP4) for segmented media delivery that can be used for both HLS and DASH. Understanding CMAF is key to creating streams that work across different streaming protocols with a single media file format.

Tags: CMAF, ISO, streaming standard, container format, media specification

CTA | WAVE Project

<https://cta.tech/Resources/Standards/WAVE-Project>

The WAVE (Web Application Video Ecosystem) Project, hosted by the Consumer Technology Association (CTA)®, aims to improve how internet-delivered commercial video is handled on consumer electronics devices and to make it easier for content creators to distribute video to those devices.

Cable Bible

<https://github.com/amiaopensource/cable-bible>

A guide to cables and connectors used for audiovisual tech

Caniuse

<https://github.com/Fyrd/caniuse>

Raw browser/feature support data from caniuse.com.

CinePaint

<http://www.cinepaint.org/>

CinePaint, also known as Film Gimp, is an open-source program for deep paint manipulation and image processing, such as motion picture frame-by-frame retouching. It's particularly useful for developers involved in film and video post-production.

Tags: image processing, open-source, film, retouching, post-production

Common Encryption - ISO/IEC 23001-7

<https://iso.org/standard/84637.html>

Defines encryption of media supported for receiving devices

Common Media Library

<https://github.com/streaming-video-technology-alliance/common-media-library>

Developed by the Streaming Video Technology Alliance, the Common Media Library is a JavaScript library aimed at standardizing media playback across different platforms. It provides a consistent API for developers to implement media playback functionalities, ensuring compatibility and reducing fragmentation in the streaming ecosystem.

Tags: media playback, JavaScript, streaming, standardization, library

Common Media Library (CML)

<https://web.archive.org/web/20231210195321/https://www.businesswire.com/news/home/20231019229734/en/Streaming-Video-Technology-Alliance-Announces-First-Public-Open-Source-Repository-Common-Media-Library/>

Developed by the SVTA's Players and Playback Working Group, the CML is an open-source project aimed at building a standard set of player functionalities deployable across different JavaScript players. It helps reduce individual player

development efforts and ensures consistent functionality.

Tags: open-source, media, library, player, standardization

DASH-IF IOP v3.2 Change Summary (vs v3.1)

<https://dashif.org/docs/DASH-IF-IOP-v3.2-diff-3.1.pdf>

Document detailing changes between DASH-IF Interoperability Points versions 3.1 and 3.2 for implementers.

Tags: dash, specification, standards, technical document, interoperability

Dashif Content Protection List

https://dashif.org/identifiers/content_protection/

A list of content protection schemes recognized by the DASH Industry Forum for securing DASH streaming content.

Digital Subband Video 2 - Open Source Wavelet Codec

<https://github.com/LMP88959/Digital-Subband-Video-2>

Digital Subband Video 2 (DSV2) is an open-source video codec utilizing multiresolution subband analysis, also known as wavelet transform, for compression. It supports various chroma subsampling formats and features adaptive quantization, in-loop filtering, and quarter-pixel motion compensation, offering improved efficiency over its predecessor.

Tags: video codec, wavelet transform, open-source, compression, motion compensation

Discover MPEG

<https://mpeg.chiariglione.org/about.html>

An official resource providing comprehensive information about MPEG standards, including a full list of standards produced, their purposes, and ongoing extensions. This resource is essential for developers looking to stay updated on MPEG standardization activities.

Tags: MPEG standards, video technology, standardization, developer resources

Dolby Vision

https://en.wikipedia.org/wiki/Dolby_Vision

Dolby Vision is a suite of technologies developed by Dolby Laboratories for high dynamic range (HDR) video. It encompasses content creation, distribution, and playback, utilizing dynamic metadata to optimize picture quality based on display capabilities. Dolby Vision supports up to 12-bit color depth and peak brightness of 10,000 nits, enhancing the visual experience across various devices.

Tags: Dolby Vision, HDR, video technology, dynamic metadata, high brightness

Dolby Vision HDR Technology

<https://www.dolby.com/technologies/dolby-vision/>

Dolby Vision is a proprietary HDR format developed by Dolby Laboratories. This resource provides detailed information on the Dolby Vision technology, including its benefits, technical specifications, and implementation guidelines for content creators and device manufacturers.

Tags: HDR, Dolby Vision, video technology, display standards, content creation

Dolby Vision for Content Creators | Dolby Laboratories

<https://www.dolby.com/us/en/technologies/dolby-vision/dolby-vision-for-creative-professionals.html>

The Dolby Vision integrated workflow gives you the tools needed to efficiently create wide color gamut and high dynamic range content and ensures that the look you create in the color suite stays true when experienced across a multitude of devices. With Dolby Vision high dynamic range imaging, you get bolder highlights and incredible contrast. That means greater sharpness, depth, and more detailed shadows to expand your storytelling possibilities like never before.

EBU Evaluations of Multichannel Audio Codecs

<https://tech.ebu.ch/docs/tech/tech3324.pdf>

EBU technical document evaluating and comparing multichannel audio codec standards for broadcast and streaming applications.

EBU HDR Technology & Innovation

<https://tech.ebu.ch/hdr>

The European Broadcasting Union (EBU) offers a suite of technical documents and resources focused on HDR technology and innovation. These include guidelines for measuring HDR monitors, UHDTV HDR color bars, EOTF test charts, and baseline HDR camera painting controls. Additionally, the EBU provides results from video quality analyses of popular studio codecs and organizes workshops to explore live HDR production aspects.

Tags: EBU, HDR, video quality, technical guidelines, live production

EBU R128 Introduction - Florian Camerer

<https://www.youtube.com/watch?v=iuEtQqC-Sqo>

Florian Camerer gives an introduction to the European Broadcasting Union's R128 Broadcast Standard and speaks in general about perceived loudness, peak norma...

Encoding Guidelines by Academy Software Foundation

<https://github.com/AcademySoftwareFoundation/EncodingGuidelines>

This project provides guidelines for encoding frames to movie files for review purposes. It includes test suites to help select suitable encoding profiles for VFX/Animation media reviews and aims to develop best practices for encoding parameters.

Tags: encoding guidelines, VFX, animation, media review, best practices

Encrypted Media

<https://github.com/w3c/encrypted-media/>

W3C specification for Encrypted Media Extensions (EME) API enabling playback of protected content in web browsers.

Encrypted Media Extensions (W3C Recommendation)

<https://www.w3.org/TR/encrypted-media/>

The W3C specification for Encrypted Media Extensions (EME), which defines a browser API for playback of encrypted content using content decryption modules (CDMs).

Tags: drm, web standards, encryption, browser api, content protection

End-to-End Guidelines for UHD Phase A Implementation

<https://web.archive.org/web/20260113061642/https://ultrahdforum.org/phasea-guidelines-description/>

The Ultra HD Forum's Phase A Guidelines describe end-to-end processes for creating a linear UHD service using technologies that were commercially available by 2016. These guidelines include descriptions and workflow diagrams of real-world deployments of UHD/HDR services for major television events, synergies between Phase A and Phase B technologies, and important standards supporting Phase A workflows. They serve as a foundational resource for implementing UHD services.

Tags: UHD, HDR, video workflows, broadcast standards, video production

Enhanced RTMP (E-RTMP)

<https://veovera.org/docs/enhanced/enhanced-rtmp-v2>

Enhanced RTMP (E-RTMP) is an advancement of the Real-Time Messaging Protocol (RTMP) developed by the Veovera Software Organization (VSO) in collaboration with industry stakeholders, including Adobe, YouTube, and Twitch. E-RTMP introduces support for modern audio and video codecs such as AC-3, E-AC-3, Opus, FLAC, VP8, VP9, HEVC, and AV1 with HDR support. It also includes features like multitrack capabilities, adaptive bitrate streaming support, and improved timestamp precision, enhancing RTMP's functionality for real-time and on-demand streaming while maintaining backward compatibility.

Tags: RTMP, streaming, protocol, E-RTMP, Veovera

FCC Closed Captioning Guidelines

<https://www.fcc.gov/consumers/guides/closed-captioning>

US Federal Communications Commission guidelines on closed captioning requirements for video programming, covering quality, accuracy, and legal obligations.

Tags: accessibility, legal, standards, captions, regulations

FFV1

<https://github.com/FFmpeg/FFV1>

The FFV1 lossless video codec specification.

FFmpeg Decklink HDR Metadata Support

<https://lists.ffmpeg.org/pipermail/ffmpeg-devel/2023-July/312425.html>

A patch for FFmpeg's Decklink output module that adds support for HDR metadata. This enhancement allows for the output of HDR content with proper metadata, ensuring accurate color representation and compliance with HDR standards.

Tags: FFmpeg, HDR, Decklink, metadata, video encoding

Frequently Asked Questions on High Dynamic Range and Hybrid Log-Gamma

https://downloads.bbc.co.uk/rd/pubs/papers/HDR/BBC_HDRTV_FAQ.pdf

BBC R&D guide answering common questions about HDR and HLG technology. Technical reference for HDR implementation.

Generative Face Video Coding Techniques and Standardization Efforts: A Review

<https://arxiv.org/abs/2311.02649>

This project offers a comprehensive survey on Generative Face Video Coding (GFVC) techniques and their standardization efforts. It explores how GFVC can achieve high-quality face video communication in ultra-low bandwidth scenarios, discussing challenges, applications, and future trends in the field.

Tags: GFVC, video coding, video standards, video technology, video streaming

Guidelines for Implementation: DASH-IF Interoperability Points

<https://dashif.org/docs/DASH-IF-IOP-v4.3.pdf>

DASH Industry Forum's implementation guidelines document specifying interoperability points for DASH-based adaptive streaming systems.

Guidelines for UHD Phase B Implementation

<https://web.archive.org/web/20260212223017/https://ultrahdforum.org/phaseb-guidelines-description/>

The Ultra HD Forum's Phase B Guidelines introduce and demystify next-generation UHD technologies that operators are exploring for future enhanced UHD services. These guidelines cover dynamic HDR metadata systems, dual-layer HDR technologies, high frame rate (HFR), next-generation audio (NGA), and content-aware encoding. They provide detailed descriptions of technologies like Dolby Vision, SL-HDR, and MPEG-H Audio, offering valuable insights for developers and industry professionals.

Tags: UHD, HDR, Dolby Vision, SL-HDR, MPEG-H Audio

HDMI 2.1: features, specs and news about the latest HDMI standard

<https://www.whathifi.com/advice/what-hdmi-21-everything-you-need-to-know>

The gateway to a super high-definition future is ever-nearing

HDR Web Video Creator's Guide

<https://github.com/igarrison/HDRWebVideoCreatorsGuide>

An in-depth guide by Ian Garrison detailing the process of creating HDR10 videos for web platforms like YouTube, covering camera specifications, software recommendations, encoding techniques, and best practices for delivering high-quality HDR content.

Tags: HDR10, video creation, YouTube, encoding, best practices

HDR10

<https://en.wikipedia.org/wiki/HDR10>

HDR10 is an open high-dynamic-range video standard introduced by the Consumer Electronics Association. It is widely adopted and includes static metadata to enhance picture quality. HDR10 utilizes the Perceptual Quantizer (PQ) transfer function, 10-bit color depth, and Rec. 2020 color primaries, providing improved contrast and color accuracy in video content.

Tags: HDR10, HDR, video standard, static metadata, Rec. 2020

HDR10+ Resources

<https://hdr10plus.org/resources/>

HDR10+ Technologies LLC offers a range of resources, including white papers and webinars, detailing the advantages and implementation of HDR10+ technology. These materials provide insights into dynamic metadata, live sports broadcasting, and gaming applications.

Tags: HDR10+, dynamic metadata, video technology, live sports, gaming

HEVC/H.265 Encoding Guidelines

<https://academysoftwarefoundation.github.io/EncodingGuidelines/EncodeHvc.html>

Comprehensive guidelines on encoding video using the HEVC/H.265 standard, including support for HDR content. This resource is valuable for developers seeking to implement efficient and standards-compliant video encoding workflows.

Tags: HEVC, H.265, video encoding, HDR, standards

HLG vs PQ Systems for HDR Television

<https://www.displaydaily.com/article/display-daily/hlg-vs-pq-systems-for-hdr-television>

Article explaining hlg vs pq in depth.

HandBrake HDR Encoding Documentation

<https://handbrake.fr/docs/en/latest/technical/hdr.html>

HandBrake is an open-source video transcoder that supports HDR encoding. Its documentation provides detailed information on encoding HDR10, HDR10+, and Dolby Vision content using various encoders. This resource is valuable for developers seeking guidance on implementing HDR encoding in their video processing workflows.

Tags: HandBrake, HDR encoding, video transcoding, open-source, documentation

High Efficiency Video Coding (HEVC)

<https://www.itu.int/rec/T-REC-H.265>

HEVC, also known as H.265 and MPEG-H Part 2, is a video compression standard that offers 25% to 50% better data compression compared to its predecessor, H.264. It supports resolutions up to 8K UHD and includes features beneficial for HDR content, making it valuable for developers working on advanced video streaming and encoding solutions.

Tags: video compression, HEVC, H.265, HDR, 8K

High Efficiency Video Coding (HEVC) - MPEG-H Part 2

<https://www.mpeg.org/standards/mpeg-h/>

HEVC, also known as H.265 and MPEG-H Part 2, is a video compression standard that provides significantly improved compression efficiency compared to its predecessors. It supports resolutions up to 8K UHD and is widely used in various applications, including streaming, broadcasting, and storage.

Tags: HEVC, H.265, video compression, MPEG-H, ISO/IEC 23008-2

Home | Streaming Video Alliance

<https://www.streamingvideoalliance.org/>

Making Streaming Video Better Streaming video is exploding in popularity. Consumers are watching more video online across a myriad of devices. But, the streaming experiences, across providers, can be wildly different from each other which ultimately hurts adoption. The problem is a lack of colla...

How Do I Become an ANSI Member

https://www.ansi.org/membership/how_to_join/how_3

Information about joining ANSI, the American National Standards Institute, which develops and maintains technical standards across various industries.

Tags: standards, industry-forum, specification, membership

IEC TR 62935:2016 - Measurement Methods for High Dynamic Range Video

<https://webstore.iec.ch/en/publication/25463>

This IEC Technical Report introduces the concept of High Dynamic Range (HDR) video, lists related standards and activities, provides information about HDR in the marketplace, and proposes areas of HDR measurement that could be standardized.

Tags: HDR, measurement methods, video standards, IEC, technical report

IMF: A Prescription for Versionitis

<https://medium.com/netflix-techblog/imf-a-prescription-for-versionitis-e0b4c1865c20>

the emerging Interoperable Master Format standard

ISO - ISO/IEC 23009-1:2019 - Information technology — Dynamic adaptive streaming over HTTP (DASH) — Part 1: Media presentation description and segment formats

<https://www.iso.org/standard/79329.html>

ISO catalogue entry for a later edition of ISO/IEC 23009-1 (MPEG-DASH Part 1), covering the media presentation description and segment formats with subsequent amendments.

ISO Base Media File Format Reference Software

<https://github.com/MPEGGroup/isobmff>

Reference software implementation for the ISO Base Media File Format, providing code for encoding and decoding multimedia files that conform to the ISO/BMFF standard.

ISO/IEC 14496-12 (MPEG-4 Part 12) ISO Base Media File Format

<https://www.iso.org/standard/74428.html>

The ISO base media file format standard (used by MP4, MOV, etc.), defining the general structure for time-based media files. It's a foundation for MP4 and fragmented MP4 (CMAF).

Tags: ISO, MPEG-4, file format, standard, technical specification

ISO/IEC TR 23008-14:2018

<https://www.iso.org/standard/72655.html>

This ISO/IEC technical report provides guidance on processing high dynamic range (HDR) and wide color gamut (WCG) video content. It offers recommended guidelines for operating AVC or HEVC video coding systems adapted for compressing HDR/WCG video for consumer distribution applications. The document includes processing steps for converting RGB linear light representation video signals into Y'CbCr signals using the PQ transfer function, applicable to various bit depths and color formats.

Tags: ISO, HDR, WCG, video coding, PQ transfer function

ISO/IEC TR 23008-15:2018

<https://www.iso.org/standard/72656.html>

This ISO/IEC Technical Report reviews approaches for processing and coding HDR/WCG video content. It provides methods for operating AVC or HEVC video coding systems adapted for compressing HDR/WCG video for consumer distribution applications. The document includes a review of video usability information (VUI) indicators and supplemental enhancement information (SEI) messages applicable for HDR/WCG video, as well as descriptions of processing steps for converting from 4:4:4 RGB linear light representation to video signals with ICTCP color representation and PQ transfer function, or with Y'CbCr color representation and HLG transfer function.

Tags: ISO, HDR, WCG, video coding, HLG transfer function

ITU Guidance for HDR Television Production

<https://www.itu.int/hub/publication/r-rep-bt-2408-4-2021/>

The International Telecommunication Union (ITU) provides operational practices to ensure optimal and consistent use of HDR in television production, focusing on the Perceptual Quantization (PQ) and Hybrid Log-Gamma (HLG) methods as specified in Recommendation ITU-R BT.2100.

Tags: ITU, HDR, television production, PQ, HLG

ITU-R BT.2100 - Image Parameter Values for High Dynamic Range Television

<https://www.itu.int/rec/R-REC-BT.2100>

The International Telecommunication Union's recommendation BT.2100 specifies the image parameter values for High Dynamic Range (HDR) television. It defines aspects such as color space, transfer functions, and reference levels to standardize HDR content production and distribution.

Tags: HDR, ITU-R, video standards, color space, television

ITU-R BT.2390 - High Dynamic Range Television for Production and International Programme Exchange

<https://www.itu.int/hub/publication/r-rep-bt-2390-8-2020/>

This ITU Report provides background information on HDR in general, and for the perceptual quantization (PQ) and hybrid log-gamma (HLG) HDR signal parameters specified in Recommendation ITU-R BT.2100. It serves as a comprehensive resource for understanding HDR television production and international programme exchange.

Tags: HDR, television production, international exchange, ITU, technical report

ITU-T P.1204.3 Reference Implementation

https://github.com/Telecommunication-Telemedia-Assessment/bitstream_mode3_p1204_3

This project offers an open-source reference implementation of the ITU-T P.1204.3 standard, a bitstream-based short-term video quality prediction model. It estimates video quality scores on a segment level using full bitstream data, aiding in the assessment of video quality.

Tags: video quality, ITU-T P.1204.3, bitstream analysis, open-source, quality prediction

Immersive Virtual Production With SMPTE 2110

<https://github.com/videofeedback/Immersive-Virtual-Production-With-SMPTE-2110>

This project provides a white paper detailing the integration of SMPTE 2110 standards into immersive virtual production workflows. It offers insights into creating synchronized virtual production environments without significant infrastructure investments, addressing challenges in pre-production, production, and post-production stages.

Tags: SMPTE 2110, virtual production, video standards, video technology, video streaming

Inkscape

<https://www.inkscape.org/>

Inkscape is an open-source vector graphics editor available for all major platforms. It provides tools for creating and editing vector graphics, which can be useful for designing graphical elements in video projects.

Tags: vector graphics, open-source, cross-platform, design, editing

Inside MPEG's Ambitious Plan to Launch 3 Video Codecs in 2020

<https://www.streamingmedia.com/Articles/Editorial/Featured-Articles/Inside-MPEGs-Ambitious-Plan-to-Launch-3-Video-Codecs-in-2020-134694.aspx>

The pace of innovation is getting faster and the demands on video codecs are getting greater. MPEG's three-part plan answers questions of royalties, licensing, and computational efficiency. Meet VVC, MPEG-5 Part 1 (EVC), and MPEG-5 Part 2 (LCEVC).

Internet Protocol Media Experience (IPMX)

<https://en.wikipedia.org/wiki/IPMX>

IPMX is a suite of open standards and specifications that provide an interoperable way to transport compressed and uncompressed video, audio, and related metadata over data networks within multi-vendor IP audiovisual systems. It is based on SMPTE 2110 and AES67 standards, tailored for professional AV workflows.

Tags: IPMX, video, audio, networking, interoperability

LCEVC: Low Complexity Enhancement Video Coding

<https://www.lcevc.org/>

LCEVC (Low Complexity Enhancement Video Coding) is an MPEG-5 Part 2 standard that enhances existing video codecs by adding an efficient enhancement layer. This approach improves compression efficiency and reduces encoding/decoding complexity, making it suitable for live streaming and broadcasting applications. LCEVC can be implemented through software

updates, leveraging existing hardware acceleration for graphics processing.

Tags: LCEVC, video coding, MPEG-5, compression efficiency, video streaming

Luma HDRv

<https://computergraphics.on.liu.se/lumahdrv/>

Luma HDRv is an open-source project that provides libraries and applications for encoding and decoding HDR video. It utilizes Google's VP9 codec and the Matroska container format, offering tools for HDR video compression and playback. This project is beneficial for developers working on HDR video encoding and decoding solutions.

Tags: HDR, video encoding, video decoding, VP9, Matroska

MPAI-EEV: AI-Based End-to-End Video Coding Standardization

<https://arxiv.org/abs/2309.07589>

MPAI-EEV is an initiative by the Moving Picture, Audio, and Data Coding by Artificial Intelligence (MPAI) group to develop standards for end-to-end optimized neural video coding. This project aims to utilize data-trained neural coding technologies to compress high-fidelity video data efficiently, moving beyond traditional hybrid frameworks. It represents a significant step towards integrating AI into video coding standards.

Tags: MPAI-EEV, AI, video coding, standardization, neural networks

MPEG About

<https://www.mpegstandards.org/about-mpeg/>

Overview page providing information about the Motion Picture Experts Group organization and its role in developing multimedia standards.

MPEG Common media application format (CMAF) for segmented media - ISO/IEC 23000-19

<https://www.iso.org/standard/85623.html>

ISO/IEC 23000-19, the Common Media Application Format (CMAF) specification — a segmented media profile that lets a single encode serve both HLS and DASH delivery.

MPEG Essential Video Coding (EVC) Test Model

<https://github.com/MPEGGroup/MPEG-EVC-ETM>

The MPEG Essential Video Coding (EVC) Test Model repository provides the reference software for the EVC standard, offering a foundation for developers to understand and implement this video coding standard.

Tags: MPEG, EVC, video coding, reference software, standards

MPEG Event message track format for the ISO base media file format - ISO/IEC 23001-18

<https://www.iso.org/standard/82529.html>

ISO/IEC 23001-18, the Event Message Track Format specification — defines how timed event messages (emsg) are carried as tracks in ISO base media files.

MPEG Explorations

<https://www.mpeg.org/standards/Explorations/>

MPEG Explorations focus on investigating new areas requiring compression standards and other multimedia technologies. They cover topics like Video Coding for Machines, Audio Coding for Machines, and Haptics Support. The Explorations page provides access to meeting documents, calls for proposals, and reports, offering valuable resources for developers interested in emerging multimedia technologies.

Tags: MPEG explorations, emerging technologies, video coding, audio coding, haptics

MPEG File Format Discussions

<https://github.com/MPEGGroup/FileFormat>

This GitHub repository hosts discussions and developments related to the MPEG file format, providing insights into the structure and implementation of MPEG standards.

Tags: MPEG, file format, video technology, standards, encoding

MPEG High Efficiency Image File Format (HEIF)

<https://nokiatech.github.io/heif/>

MPEG specification for the High Efficiency Image File Format, a standard for storing and managing high-quality images with advanced compression capabilities.

MPEG ISOBMFF 'MP4' - ISO/IEC 14496-12

<https://www.iso.org/standard/83102.html>

ISO/IEC 14496-12, the ISO Base Media File Format specification — defines the structural 'MP4' container that most modern media files and streaming segments are built on.

MPEG Industry Forum

https://en.wikipedia.org/wiki/MPEG_Industry_Forum

The MPEG Industry Forum (MPEGIF) is a non-profit consortium dedicated to furthering the adoption of MPEG standards, such as MPEG-4, MPEG-4 AVC/H.264, MPEG-7, and MPEG-21. The forum engages in promoting these standards, developing certification programs, organizing educational events, and collaborating on new de facto MPEG standards. MPEGIF has played a significant role in facilitating the widespread adoption of MPEG-4 AVC/H.264 as the industry's standard video compression technology.

Tags: MPEG, video compression, MPEGIF, industry standards, video technology

MPEG Meetings

<https://www.mpegstandards.org/meetings/>

Directory of MPEG organization meetings where standards development, voting, and technical discussions take place.

MPEG Scene Description for Media

<https://github.com/MPEGGroup/Scene-Description>

Work related to ISO/IEC 23090-14 Scene Description, providing tools and discussions for describing scenes in MPEG media.

Tags: MPEG, scene description, ISO/IEC 23090-14, media, standards

MPEG Standards Documentation

<https://www.mpeg.org/standards/>

The official repository of MPEG standards, providing comprehensive documentation on various media coding standards, including MPEG-1, MPEG-2, MPEG-4, and the latest MPEG-I. This resource is essential for developers seeking in-depth understanding and implementation details of MPEG standards.

Tags: MPEG standards, video coding, media compression, documentation

MPEG Standards Overview by Leonardo Chiariglione

<https://leonardo.chiariglione.org/public-life/standards/mpeg/>

An extensive overview of MPEG standards authored by Leonardo Chiariglione, detailing the development and applications of various MPEG standards such as MPEG-1, MPEG-2, and MPEG-4. This resource is invaluable for developers seeking historical context and technical details of MPEG standards.

Tags: MPEG standards, video technology, historical overview, technical details

MPEG Standards Video Tutorials

<https://mpeg.chiariglione.org/video-tutorials.html>

A collection of video tutorials covering various MPEG standards, including MPEG Green Metadata, MPEG Multimedia Preservation Application Format (MPAF), MPEG-G 3D Audio, and more. These tutorials provide in-depth insights into the standards, aiding developers in understanding and implementing MPEG technologies.

Tags: MPEG standards, video tutorials, video technology, developer resources

MPEG Technical Requirements

<https://www.mpeg.org/structure/technical-requirements/>

This resource provides detailed information on the technical requirements and ongoing developments within MPEG, including explorations into new areas like AI-based coding and immersive media. It's valuable for developers looking to stay updated on the latest advancements in MPEG standards.

Tags: technical requirements, MPEG, video coding, audio coding, standards development

MPEG V-PCC Volumetric Video Streaming Starter Kit

<https://github.com/Shiyuang-scu/MPEG-V-PCC-Volumetric-Video-Streaming-Starter-Kit>

A comprehensive introduction to volumetric video streaming using MPEG's Video-based Point Cloud Compression (V-PCC) standard. This repository includes tutorials, code samples, and links to relevant resources, making it valuable for developers exploring 3D video streaming technologies.

Tags: MPEG, V-PCC, volumetric video, streaming, 3D video

MPEG Video Tool Library

<https://mpeg.chiariglione.org/standards/mpeg-c/video-tool-library.html>

Part of the MPEG-C standard, the Video Tool Library provides reference software for various video coding tools, including those for High Efficiency Video Coding (HEVC), aiding developers in implementing and testing video coding technologies.

Tags: video coding, HEVC, reference software, MPEG-C, video tools

MPEG Video Tool Library (VTL)

<https://www.iso.org/standard/73398.html>

The MPEG Video Tool Library (VTL) is a standardized collection of functional units (FUs) that can be used to build complete decoding solutions for various MPEG video coding standards, including MPEG-4 Simple Profile, MPEG-4 AVC, and HEVC Main Profile. It provides a modular approach to video decoding, facilitating the development and implementation of video

processing applications.

Tags: MPEG, video decoding, video processing, standards, VTL

MPEG home page

<https://www.mpegstandards.org>

Official website of MPEG, the international standardization body responsible for multimedia compression and streaming standards.

MPEG point cloud compression

<https://mpeg-pcc.org>

Technical specification for MPEG point cloud compression standards used in encoding and decoding three-dimensional point cloud data for efficient storage and transmission.

MPEG – The Moving Picture Experts Group

<https://www.mpeg.org/>

MPEG develops standards for coded representation of digital audio, video, 3D graphics, and genomic data. Since 1988, it has produced standards like AVC, HEVC, and VVC, facilitating the creation of an industry worth hundreds of billions of USD.

Tags: MPEG, video, audio, coding, standards

MPEG-2 Transport Stream Packet Analyser

<https://github.com/daniep01/MPEG-2-Transport-Stream-Packet-Analyser>

An MPEG-2 Transport Stream packet analyser that enables decoding and low-level analysis of ISO/DVB/AVCHD transport streams. It is useful for fault-finding, equipment and system testing, software development, and learning about digital television systems.

Tags: MPEG-2, transport stream, packet analyser, digital television, open-source

MPEG-2: Generic Coding of Moving Pictures and Associated Audio Information

<https://www.mpeg.org/standards/mpeg-2/>

MPEG-2 (ISO/IEC 13818) is a suite of standards for digital television. It defines the generic coding of moving pictures and associated audio information, widely used for broadcast and DVD applications.

Tags: MPEG-2, digital television, video coding, audio coding, broadcast

MPEG-4 Part 10 Advanced Video Coding (AVC)

<https://www.mpeg.org/standards/MPEG-4/10/>

MPEG-4 Part 10, also known as Advanced Video Coding (AVC) or H.264, is a widely used video compression standard that provides good video quality at substantially lower bit rates than previous standards. It is extensively used for video streaming, recording, and compression, making it a fundamental tool for developers in the video technology industry.

Tags: MPEG-4, AVC, H.264, video compression, standards

MPEG-4: Coding of Audio-Visual Objects

<https://www.mpeg.org/standards/mpeg-4/>

MPEG-4 (ISO/IEC 14496) is a suite of standards for multimedia for the fixed and mobile web. It provides a comprehensive framework for coding audio-visual objects, enabling efficient storage, transmission, and manipulation of multimedia content.

Tags: MPEG-4, audio-visual coding, multimedia, compression, mobile web

MPEG-5 Standard

<https://www.mpeg.org/standards/mpeg-5/>

MPEG-5 is a standard that collects video coding parts, including Essential Video Coding (EVC) and Low Complexity Enhancement Video Coding (LCEVC). It aims to provide efficient and flexible video coding solutions for various applications, balancing compression efficiency and computational complexity. This standard is beneficial for developers seeking advanced video coding techniques.

Tags: video coding, EVC, LCEVC, compression, standards

MPEG-5: Essential Video Coding (EVC)

<https://www.mpeg.org/standards/MPEG-5/1/>

MPEG-5 Part 1, known as Essential Video Coding (EVC), aims to provide a standardized video coding solution addressing business needs in use cases such as video streaming. It offers a balance between coding efficiency and computational complexity.

Tags: MPEG-5, EVC, video coding, streaming, compression

MPEG-7 Library

<https://mpeg7.joanneum.at/mpeg-7-library/references/>

The MPEG-7 Library is a tool for describing multimedia content using the MPEG-7 standard. It supports low- and mid-level

feature descriptions such as shots, camera motion, color features, and faces. This library is useful for applications involving content analysis, retrieval, and multimedia metadata management.

Tags: MPEG-7, metadata, content analysis, retrieval, multimedia

MPEG-7: Multimedia Content Description Interface

<https://www.mpeg.org/standards/mpeg-7/>

MPEG-7 (ISO/IEC 15938) is a suite of standards for the description and search of audio, visual, and multimedia content. It provides a rich set of tools to describe multimedia content, facilitating efficient searching, filtering, and management.

Tags: MPEG-7, content description, multimedia, search, metadata

MPEG-DASH: Dynamic Adaptive Streaming over HTTP

<https://www.mpeg.org/standards/mpeg-dash/>

MPEG-DASH (ISO/IEC 23009) is a suite of standards providing a solution for the efficient and easy streaming of multimedia using existing HTTP infrastructure. It enables adaptive streaming of video content over the internet by dynamically adjusting the quality of a video stream to match network conditions.

Tags: MPEG-DASH, adaptive streaming, HTTP, video streaming, network conditions

MPEG-DASH: The Standard for Multimedia Streaming Over Internet

<https://www.mpegstandards.org/wp-content/uploads/whitepapers/html/overview.html>

MPEG-DASH (Dynamic Adaptive Streaming over HTTP) is a suite of standards providing a solution for efficient and easy streaming of multimedia using existing HTTP infrastructure. It enables high-quality streaming of media content over the Internet, dynamically adapting to network conditions and user device capabilities.

Tags: MPEG-DASH, streaming, HTTP, adaptive, multimedia

MPEG-G Genomic Information Representation

<https://mpeg-g.org/>

MPEG-G is a suite of standards developed by MPEG for the representation of genomic information. It provides effective and interoperable solutions for genomic information processing applications, including compression, storage, and transport of genomic data.

Tags: genomic data, compression, MPEG-G, standard, bioinformatics

MPEG-H Audio a Next Generation Audio (NGA) codec - ISO/IEC 23008-3

<https://www.iso.org/standard/83525.html>

Defines the compressed audio format MPEG-H Audio a Next Generation Audio (NGA) codec

MPEG-H High Efficiency Video Coding (HEVC)

<https://www.mpeg.org/standards/MPEG-H/2/>

MPEG-H Part 2, also known as High Efficiency Video Coding (HEVC) or H.265, is a video compression standard designed to substantially improve coding efficiency compared to its predecessor, H.264/AVC. HEVC supports higher resolutions, including 4K and 8K, and provides better compression, making it essential for developers focusing on high-quality video streaming and encoding.

Tags: MPEG-H, HEVC, H.265, video compression, standards

MPEG-H: High Efficiency Video Coding (HEVC)

<https://en.wikipedia.org/wiki/MPEG-H>

MPEG-H Part 2, also known as High Efficiency Video Coding (HEVC) or H.265, is a video compression standard that doubles the data compression ratio compared to its predecessor, H.264/MPEG-4 AVC. It supports higher resolutions and improved video quality for streaming applications.

Tags: MPEG-H, HEVC, H.265, video compression, streaming

MPEG-I: Immersive Video

<https://www.mpeg.org/standards/Explorations/7/>

MPEG-I is a suite of standards aimed at digitally representing immersive media, including 360-degree video and virtual reality content. It defines the coding and delivery mechanisms necessary to provide high-quality immersive experiences, addressing aspects such as depth estimation, view synthesis, and quality metrics.

Tags: MPEG-I, immersive video, 360-degree, VR, media

MPEG: What Happened?

<https://www.streamingmedia.com/Articles/ReadArticle.aspx?ArticleID=141678>

At the end of last month, MPEG co-founder Leonardo Chiariglione announced the 'MPEG is closed.' That's not quite true, but it is undergoing a reorganization. So what does that mean for the organization and the new codec standards it is bringing out this year?

MPEGGroup GitHub Repositories

<https://github.com/MPEGGroup>

The official GitHub organization for the Moving Picture Experts Group (MPEG) hosts various repositories related to MPEG standards, including file format discussions, conformance frameworks, and reference software.

Tags: MPEG, video standards, file formats, reference software, open-source

Matroska Specification

<https://www.matroska.org/>

The open specification for the Matroska multimedia container, which is the basis for MKV files. It defines how video, audio, subtitle, and metadata tracks are stored in a single file and is fully open and extensible.

Tags: container, specification, mkv, multimedia, file format

Media Coding Industry Forum (MC-IF)

<https://www.mc-if.org/>

The Media Coding Industry Forum (MC-IF) is an open industry forum dedicated to furthering the adoption of MPEG standards, initially focusing on Versatile Video Coding (VVC). MC-IF works to establish these standards as well-accepted and widely used, benefiting consumers and the industry by promoting interoperability and sharing implementation guidelines.

Tags: media coding, standards, VVC, interoperability, MPEG

Media Source Extensions (W3C Recommendation)

<https://www.w3.org/TR/media-source/>

The W3C specification for Media Source Extensions (MSE), an API that allows JavaScript to feed media to the HTML5 video or audio tag for adaptive streaming.

Tags: w3c, specification, adaptive streaming, html5, api, web standard

Moving Picture Experts Group (MPEG)

<https://mpeg.chiariglione.org/>

The official forum that develops international standards for compression, decompression, processing and coded representation of moving pictures, audio, and their combination (e.g., MPEG-2, MPEG-4, DASH).

Tags: standards, compression, video-standards, international-standards, multimedia

NBCU-UHD-HDR-SDR-Resources-Table-of-Contents

<https://github.com/digitaltvGuy/NBCU-UHD-HDR-SDR-Resources-Table-of-Contents>

A comprehensive repository summarizing various UHD resources authored by NBCUniversal, including guidelines and LUTs for single-master production workflows, objective color metrics for broadcasting, and conformance files for video format identifiers.

Tags: UHD, HDR, SDR, production workflows, color metrics

NBCUniversal UHD HDR-SDR Single Master Production Workflow Recommendation LUTs

<https://github.com/digitaltvGuy/NBCUniversal-UHD-HDR-SDR-Single-Master-Production-Workflow-Recommendation-LUTs>

This repository provides Look-Up Tables (LUTs) recommended by NBCUniversal for a single master production workflow accommodating both HDR and SDR content. It serves as a valuable resource for professionals aiming to streamline their production processes while maintaining high-quality output across different display standards.

Tags: HDR, SDR, production workflow, LUTs, video technology

ORI Encoding Guidelines

<https://academysoftwarefoundation.github.io/EncodingGuidelines/>

The Open Review Initiative (ORI) Encoding Guidelines, part of the Academy Software Foundation, aim to establish best practices for encoding video media, particularly for VFX and animation media review. The guidelines provide detailed instructions on HDR encoding, including PQ10 and HLG formats, and offer practical examples using tools like FFmpeg. This resource is essential for developers looking to implement HDR encoding in their workflows.

Tags: HDR encoding, VFX, animation, FFmpeg, best practices

ORI Encoding Guidelines: HDR Encoding

https://academysoftwarefoundation.github.io/EncodingGuidelines/enctests/HDR_Encoding.html

Part of the Academy Software Foundation's Open Review Initiative, this resource provides detailed guidelines on encoding HDR content, covering formats like PQ10 and HLG, and includes frame preparation and FFMPEG encoding instructions.

Tags: HDR encoding, PQ10, HLG, FFMPEG, video standards

ORI Encoding Guidelines: HDR Reference Source Test Media

https://academysoftwarefoundation.github.io/EncodingGuidelines/enctests/sources/hdr_sources/README.html

This resource offers a collection of source clips for testing HDR media, primarily sourced from Netflix's Open Source Content site, along with scripts for downloading and converting frames for encoding tests.

Tags: HDR test media, video encoding, test clips, video standards

Official Registration Authority for the ISO/BMFF family of standards

<https://mp4ra.org/>

Official registry and authority for managing code registrations and technical definitions within the ISO Base Media File Format family of standards.

Open-Source Project for MPEG-5 EVC (Essential Video Coding)

<https://research.samsung.com/news/Open-Source-Project-for-MPEG-5-EVC-Essential-Video-Coding>

Samsung Research has released open-source implementations of the MPEG-5 Essential Video Coding (EVC) standard, including an encoder (XEVE) and decoder (XEVD). These tools aim to provide better visual quality and alleviate royalty costs associated with video codecs.

Tags: MPEG-5 EVC, video coding, encoder, decoder, open-source

Publicly Available MPEG Standards

<https://alexzambelli.com/blog/2022/01/13/publicly-and-freely-available-mpeg-standards/>

A curated list of MPEG standards that are publicly and freely available, compiled by Alex Zambelli. This resource is invaluable for developers seeking access to MPEG standards without purchase barriers.

Tags: MPEG standards, free resources, video technology

RIST Forum

<https://www.rist.tv/>

The RIST forum manage the Reliable Internet Stream Transport (RIST), for transporting live video over unmanaged networks.

Real-HDRV

<https://github.com/yungsyu99/Real-HDRV>

Real-HDRV is a large-scale real-world benchmark dataset designed for HDR video reconstruction from sequences with alternating exposures. It comprises 500 LDR-HDR video pairs, covering diverse scenes and motion patterns, and includes high-quality HDR labels. Alongside the dataset, the project proposes an end-to-end network with a two-stage alignment strategy for HDR video reconstruction. This resource is invaluable for developers and researchers focusing on HDR video reconstruction and related technologies.

Tags: HDR video, video reconstruction, benchmark dataset, alignment network, video processing

Rec. 2020

https://en.wikipedia.org/wiki/Rec._2020

ITU-R Recommendation BT.2020, commonly known as Rec. 2020, defines various aspects of ultra-high-definition television (UHDTV), including picture resolutions, frame rates, bit depths, and color primaries. It serves as a foundational standard for UHDTV systems, providing guidelines for production and international program exchange.

Tags: Rec. 2020, UHDTV, video standard, color primaries, bit depth

Rec. 2100

https://en.wikipedia.org/wiki/Rec._2100

Rec. 2100 is a recommendation by the International Telecommunication Union (ITU) that defines parameters for high dynamic range (HDR) television. It specifies aspects such as resolution, frame rates, bit depth, and color space, providing a standardized framework for HDR content production and distribution. Rec. 2100 is essential for ensuring consistency and compatibility in HDR video technology.

Tags: Rec. 2100, HDR, video standards, ITU, color space

SBTVD Forum

https://en.wikipedia.org/wiki/SBTVD_Forum

The SBTVD Forum is a non-profit organization responsible for the development and deployment of the Brazilian Digital Television System (SBTVD), also known as ISDB-Tb. The forum comprises over 80 member companies from various sectors of the television industry, including broadcasters, manufacturers, universities, and regulatory agencies. It develops technical standards covering all aspects of digital television transmission and reception in Brazil.

Tags: digital TV, ISDB-Tb, broadcasting, standards, Brazil

SCTE

<https://github.com/jamesfining/scte>

Python library to work with SCTE standards.

SDVoE Alliance

<https://sdvoe.org/>

The SDVoE Alliance is a consortium of pro AV manufacturers, system designers, integrators, and technology managers working together to accelerate the replacement of point-to-point connectivity and the matrix switch with Ethernet-based AV

distribution.

Tags: AV, Ethernet, distribution, interoperability, standards

SMPTE 2110

https://en.wikipedia.org/wiki/SMPTE_2110

SMPTE 2110 is a suite of standards from the Society of Motion Picture and Television Engineers that describes how to send digital media over an IP network. It is intended for use within broadcast production and distribution facilities, emphasizing quality and flexibility over bandwidth efficiency.

Tags: SMPTE, video, IP, networking, broadcast

SMPTE HDR, 4K, and Beyond Webcast

<https://www.smpte.org/webcast-events/hdr-4k-and-beyond>

The Society of Motion Picture and Television Engineers (SMPTE) offers educational resources, including webcasts and courses, focusing on HDR, 4K, and advanced video technologies. These resources provide insights into the latest standards, implementation strategies, and technological advancements in the field of high dynamic range and ultra-high-definition video.

Tags: HDR, 4K, SMPTE, video standards, education

SMPTE ST 2084:2014 - High Dynamic Range Electro-Optical Transfer Function

<https://ieeexplore.ieee.org/document/7291452>

The Society of Motion Picture and Television Engineers (SMPTE) standard ST 2084:2014 defines the Electro-Optical Transfer Function (EOTF) for High Dynamic Range (HDR) content. This standard is crucial for ensuring consistent HDR video reproduction across different display devices.

Tags: HDR, SMPTE, EOTF, video standards, display technology

SMPTE Standards

<https://www.smpte.org/standards/overview>

The Society of Motion Picture and Television Engineers (SMPTE) is a globally recognized standards organization that has published over 800 technical standards for broadcast, filmmaking, digital cinema, and more. SMPTE standards ensure high-quality content delivery and interoperability across various media formats. Key standards include the Material eXchange Format (MXF), Interoperable Master Format (IMF), and the SMPTE ST 2110 suite for professional media over IP networks.

Tags: SMPTE, media standards, broadcast, digital cinema, interoperability

SMPTE Standards Projects

<https://www.smpte.org/projects>

The Society of Motion Picture and Television Engineers (SMPTE) develops and publishes technical standards for the media and entertainment industry, covering areas such as broadcast, filmmaking, digital cinema, and audio recording. Their standards are essential for ensuring interoperability and quality in video streaming and encoding technologies.

Tags: video standards, media technology, broadcasting, digital cinema, audio recording

SMPTE Virtual Course on HDR

<https://www.smpte.org/virtual-course/hdr>

The Society of Motion Picture and Television Engineers (SMPTE) offers a virtual course on High Dynamic Range (HDR). This course provides in-depth knowledge about HDR technology, including its implementation, workflows, and best practices. It's designed for professionals seeking to enhance their understanding of HDR in the context of video production and broadcasting.

Tags: SMPTE, HDR, virtual course, video production, broadcasting

SRT Alliance

<https://www.srtalliance.org/>

Home page for the SRT protocol

SRT Cookbook

<https://srtlab.github.io/srt-cookbook/>

More in-depth technical documentaion on the SRT protocol and library.

Society of Motion Picture and Television Engineers (SMPTE)

<https://www.smpte.org/>

An organization that develops and maintains standards for cinema and broadcast, including many video formats (e.g., SMPTE ST 2110 for IP video, HDR formats, etc.).

Tags: standards, specification, video-format, professional-organization, technical-standards

Streaming Video Alliance (SVA)

<https://www.svta.org/>

A global technical association that publishes best practices, specifications, and guidelines to address challenges in streaming video (e.g., open caching, QoE, ad tech).

Tags: standards, industry-association, best-practices, specifications, streaming-guidelines

Streaming Video Technology Alliance (SVTA) Projects

<https://www.svta.org/svta-projects/>

The Streaming Video Technology Alliance (SVTA) undertakes various projects aimed at improving streaming video technologies. These projects cover areas such as common token standards and streaming media tracing standards, contributing to the development of best practices and standards in the streaming industry.

Tags: streaming video, technology standards, best practices, media tracing, token standards

Techniques for Establishing and Maintaining Audio Loudness for Digital Television

<https://www.atsc.org/wp-content/uploads/2015/03/Techniques-for-establishing-and-maintaining-audio-loudness.pdf>

ATSC technical standard documenting methods for measuring, establishing, and maintaining consistent audio loudness levels in digital television.

Telecommunication-Telemedia-Assessment

<https://telecommunication-telemedia-assessment.github.io/>

A collection of repositories focused on methods, tools, and solutions for assessing telecommunication and telemedia technology. Key aspects include video quality models, databases, and tools, providing valuable resources for developers working on video quality assessment.

Tags: video quality, assessment, models, databases, tools

The GIMP

<https://www.gimp.org/>

The GIMP is an open-source image manipulation program available for all major platforms. It offers a wide range of tools for image editing, making it a valuable resource for developers working on video projects that require image processing.

Tags: image editing, open-source, cross-platform, manipulation, design

The H.264 Sequence Parameter Set

<https://www.cardinalpeak.com/the-h-264-sequence-parameter-set>

This is a follow-up to my World's Smallest H.264 Encoder post. I've received several emails asking about precise details of things in two entities in the H.264 bitstream: the Sequence Parameter Set (SPS) and the Picture Parameter Set (PPS). Both entities contain information that an H.264 decoder needs to decode the video data, for example,...

The Real-Time Messaging Protocol Explained

<https://www.wowza.com/blog/rtmp-streaming-real-time-messaging-protocol>

Explanation of the Real-Time Messaging Protocol, a protocol used for streaming audio and video content over the internet.

UCSHDRVideoComp

<https://github.com/ratnajitmukherjee/UCSHDRVideoComp>

A novel HDR video compression algorithm that compresses HDR video frames into codec-suitable YUV files, compatible with 10-bit video codecs like x264, x265, and AV1. It focuses on uniform color space-based HDR video compression.

Tags: HDR, video compression, color space, x264, x265

UWA HDR Vivid Standards

<https://uhd-world-association.com/uwa-resources/>

The UHD World Association (UWA) has developed the HDR Vivid Standards, which provide technical requirements and test methods for various devices and applications involved in HDR video technology. These standards aim to ensure consistent technical indicators across the industry, promoting innovation, compatibility, and cohesive growth. The standards cover aspects such as system integration, post-production, display devices, player devices, and live encoder devices.

Tags: HDR Vivid, UWA, standards, video technology, guidelines

Ultra HD Forum Guidelines

<https://uhdf.svta.org/guidelines/>

Industry guidelines produced by the Ultra HD Forum, covering end-to-end workflows for Ultra HD content (4K, HDR, Wide Color Gamut, Next-Gen Audio) including distribution and streaming considerations. These guidelines help ensure interoperability and best practices for advanced video formats in streaming and broadcast.

Tags: standards, 4K, HDR, ultra HD, workflow, guidelines, interoperability

Understanding High Dynamic Range (HDR) by The American Society of Cinematographers

<https://theasc.com/reports/understanding-high-dynamic-range-hdr>

The American Society of Cinematographers offers an in-depth white paper on understanding HDR, discussing the technical

aspects and implications of HDR in cinematography, including dynamic range definitions and camera capabilities.

Tags: ASC, HDR, cinematography, dynamic range, camera technology

Unsupervised HDR Image and Video Tone Mapping via Contrastive Learning

<https://github.com/cao-cong/UnCLTMO>

This project presents a unified framework (IVTMNet) for unsupervised image and video tone mapping, utilizing contrastive learning to improve the training process. It introduces a novel latent code to measure similarity between tone-mapped results and employs spatial-feature-enhanced and temporal-feature-replaced modules to enhance information exchange and temporal consistency. The project also provides a large-scale unpaired HDR-LDR video dataset, making it a valuable resource for developers working on HDR video tone mapping and related applications.

Tags: HDR video, tone mapping, contrastive learning, unsupervised learning, video processing

Use DirectX with Advanced Color on High/Standard Dynamic Range Displays

<https://github.com/MicrosoftDocs/win32/blob/docs/desktop-src/direct3darticles/high-dynamic-range.md>

A detailed guide from Microsoft on utilizing DirectX for Advanced Color scenarios, including HDR and wide color gamut, providing instructions for rendering HDR content, adapting to display capabilities using tone mapping, and capturing HDR screen content.

Tags: DirectX, HDR, WCG, tone mapping, screen capture

VESA DisplayHDR Standard

<https://vesa.org/featured-articles/vesa-defines-new-standard-to-help-speed-pc-industry-adoption-of%E2%80%A8high-dynamic-range-technology-in-laptop-and-desktop-monitor-displays/>

The Video Electronics Standards Association (VESA) has defined the DisplayHDR standard, the industry's first open HDR specification with a transparent testing methodology. This standard specifies HDR quality levels, including luminance, color gamut, bit depth, and rise time, to facilitate the adoption of HDR in laptop and desktop monitors. It establishes three distinct levels of HDR system performance to cater to various market needs.

Tags: VESA, DisplayHDR, HDR, standards, monitor displays

Versatile Video Coding (VVC) - MPEG-I Part 3

<https://www.mpeg.org/standards/mpeg-i/>

VVC, also known as H.266 and MPEG-I Part 3, is a video compression standard designed to achieve higher compression efficiency and versatility for a wide range of applications, including streaming, broadcasting, and immersive media. It supports resolutions up to 16K and various color formats.

Tags: VVC, H.266, video compression, MPEG-I, ISO/IEC 23090-3

Versatile Video Coding (VVC) Standard

<https://www.mpeg.org/standards/MPEG-I/3/>

Part of the MPEG-I suite, VVC (ISO/IEC 23090-3) is the latest video compression standard offering improved compression efficiency and support for a wide range of applications, including 4K and 8K video streaming. This resource provides detailed information and whitepapers on VVC.

Tags: VVC, video compression, MPEG-I, high efficiency

Versatile Video Coding (VVC) Test Model

https://vcgit.hhi.fraunhofer.de/jvet/VVCSoftware_VTM

The VVC Test Model (VTM) is the reference software for the Versatile Video Coding (VVC) standard, also known as H.266. VVC aims to provide significant compression efficiency improvements over previous standards. The VTM serves as a reference for developers implementing VVC in their applications.

Tags: VVC, H.266, video coding, compression, reference software

Versatile Video Coding Standard: A Review

<https://arxiv.org/abs/2106.14245>

A comprehensive review paper detailing the Versatile Video Coding (VVC) standard, covering coding tools, consumer deployment, and implementation aspects. This resource is beneficial for developers and researchers interested in the technical details and applications of VVC.

Tags: VVC, H.266, video coding, research

Video Coding - BBC R&D

<https://www.bbc.co.uk/rd/projects/video-coding>

BBC video encoding R&D home page

Video Electronics Standards Association (VESA)

<https://www.vesa.org/>

VESA is a technical standards organization for computer display standards. It has issued several widely used standards,

including DisplayHDR, DisplayPort, and the Flat Display Mounting Interface. VESA's standards are crucial for ensuring compatibility and performance across various display technologies and devices.

Tags: display standards, DisplayPort, DisplayHDR, VESA, computer displays

Video Quality Experts Group (VQEG)

<https://www.its.bldrdoc.gov/vqeg/vqeg-home.aspx>

International standards group focused on video quality assessment methodologies and research.

Tags: standards, quality, video-assessment, industry-group, specification

Video Services Forum (VSF)

<https://www.vsf.tv/>

VSF is an international association dedicated to interoperability, quality metrics, and education for media networking technologies. It provides forums to identify issues and promotes interoperability by contributing to the development of standards.

Tags: video, services, forum, networking, interoperability

W3C Audiovisual Media Formats for Browsers Community Group

<https://github.com/w3c/avmedia-formats-cg>

This GitHub repository is the official space for the W3C Audiovisual Media Formats for Browsers Community Group. The group focuses on discussing and developing support for modern audiovisual formats within the W3C framework, aiming to synchronize open and proprietary solutions to enhance web-based media experiences.

Tags: W3C, audiovisual, media formats, web standards, browsers

W3C Media & Entertainment Interest Group

<https://www.w3.org/blog/news/archives/9075>

A World Wide Web Consortium group focused on media and entertainment. It examines use cases and requirements for web technologies in the media industry (like streaming, broadcasting integration, etc.) and often produces recommendations or liaisons with other standards (like W3C for HTML5 media, or CTA WAVE).

Tags: standards, w3c, web-standards, media-technology, streaming-standards

W3C Media Working Group Repositories

<https://www.w3.org/groups/wg/media/tools/>

The W3C Media Working Group maintains several repositories related to media technologies on the web, including specifications and tools for media playback, encryption, and streaming. These resources are crucial for developers working on web-based media applications.

Tags: W3C, media, web standards, streaming, encryption

XEVD: eXtra-fast Essential Video Decoder

<https://github.com/mpeg5/xevd>

XEVD is the counterpart decoder to XEVE, providing an open-source implementation of the MPEG-5 EVC standard. It enables efficient decoding of EVC-encoded video streams.

Tags: MPEG-5, EVC, video decoding, open-source, video codec

XEVE: eXtra-fast Essential Video Encoder

<https://github.com/mpeg5/xeve>

XEVE is an open-source video encoder implementing the MPEG-5 Essential Video Coding (EVC) standard. It offers improved visual quality for video calls and addresses licensing complexities associated with other codecs.

Tags: MPEG-5, EVC, video encoding, open-source, video codec

latest HLS Spec

<https://tools.ietf.org/html/draft-pantos-hls-rfc8216bis-08>

The official IETF specification document for HTTP Live Streaming, defining the protocol standards and technical requirements.

mediacommon Go Package

<https://pkg.go.dev/github.com/bluenvoron/mediacommon>

A Go package providing definitions and functions shared between gortsplib, gohlslib, and MediaMTX, including codec and format utilities, as well as bit reader and writer functionalities.

Tags: Go, package, codec, format, utilities

uAVS3

<https://github.com/uavs3/uavs3d>

uAVS3 is an open-source and cross-platform AVS3 encoder and decoder. The decoder (uAVS3d) and encoder (uAVS3e)

support the AVS3-Phase2 baseline profile. uAVS3d can be compiled for Windows, Linux, macOS, iOS, and Android, while uAVS3e can be compiled for Windows and Linux. These tools are valuable for developers working with the AVS3 standard, offering efficient encoding and decoding capabilities.

Tags: AVS3, video encoding, video decoding, open-source, cross-platform

xAVS2 & dAVS2

<https://github.com/pkuvcl/xavs2>

xAVS2 and dAVS2 are open-source encoder and decoder published by Peking University Video Coding Laboratory (PKU-VCL) based on the AVS2-P2/IEEE 1857.4 video coding standard. These tools are offered under either version 2 of the GNU General Public License (GPL) or a commercial license. They are valuable for developers working with the AVS2 standard, providing efficient encoding and decoding solutions.

Tags: AVS2, video encoding, video decoding, open-source, PKU-VCL